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# NUMERICAL AND DEVELOPMENT APPENDIX TO NEURAL BELLMAN OPERATORS

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This appendix retains the economic-inference, safeguarded-selection, localization, certified-readout, deployment, recursive-query and vector-control sections without textual deletion. It records their original scientific evidence and verification boundaries. Complete numerical catalogues remain alongside the exact qualification duals. The new evidence section reports the prespecified independent calibration/evaluation, all paired intervals and width components, price-fee breakpoints, and exact multi-date regression examples. Core proofs remain in the Technical Supplement; the original broader applications retain their unchanged companion editions.

## APPENDIX A: COMPLETE-POLICY QUALIFICATION AND ENDOGENOUS OCCUPATION

The first-date result holds the continuation policy fixed. A complete policy changes both later actions and the states at which they are taken. We now retain both changes. This extension concerns the original transition kernels on the continuous state space, rather than a replacement finite-state economy.

Fix a feasible reference policy $\tau$ and its bounded original-law values $V_t^\tau$. In the expectation model of Section 2, define

$$A_t^\tau(x,a) = c_t(x,a) + \beta_t P_t^a V_{t+1}^\tau(x) - V_t^\tau(x). \tag{A.1}$$

Here $a$ can denote a whole-cell feasible action rule, including acquisition and repair. At every date let $\mathcal{C}_t$ be a finite measurable acquisition partition. A menu label $a \in \mathcal{A}_{tC}$ denotes a measurable rule $x \mapsto a(x) \in A_t(x)$ for all $x \in C$. A parameter vector $\theta \in \Theta$ specifies the menu choice in every cell at every date; ties follow the declared measurable convention. The vector may share coordinates across dates and

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regimes. Let $\Theta_\epsilon = \{\theta \in \Theta : F(\theta) \le U + \epsilon\}$, where $U = F(\widehat{\theta})$ for a feasible attained vector. The qualification functional may use the candidate's own future. Convexity is not required for the transfer; a claimed exclusion must nevertheless have a valid certificate for this actual functional.

Write $v = (t, C, a)$ for a menu event. Retain any family $\mathcal{H}_\epsilon$ of finite event sets for which a verified calculation proves

$$\Theta_\epsilon \cap \bigcap_{v \in K} \{\theta : \pi_\theta \text{ selects } v\} = \varnothing, \qquad K \in \mathcal{H}_\epsilon. \tag{A.2}$$

A numerical infeasibility status alone is not such a certificate. The full-menu support duals in Section V give one implementation when labels and features yield the stated linear constraints. With nonlinear own-future dependence, those particular duals cannot simply be reused; verified outer relaxations or direct interval exclusion are required. Only strict separation excludes an event, so threshold equality remains included.

For every menu event and successor cell $D$, obtain uniform original-law enclosures

$$\begin{aligned} \ell_{tCa} \le A_t^\tau(x, a(x)) \le u_{tCa}, &\qquad x \in C, \\ \underline{p}_{tCaD} \le P_t(D \mid x, a(x)) \le \overline{p}_{tCaD}, &\qquad x \in C. \end{aligned} \tag{A.3}$$

The bounds must cover acquisition boundaries, feasible repairs and numerical error. Initial cell probabilities $\mu_C$ are known here; certified initial-probability constraints can instead be included in the feasible set. Averages under the reference policy are not substitutes for the uniform bounds in (A.3).

Define $\mathcal{P}_\epsilon$ by nonnegative occupation variables $m_{tC}$, $q_{tCa}$, transition flows $z_{tCaD}$, and $0 \le y_{tCa} \le 1$, subject to

$$m_{0C} = \mu_C, \qquad \sum_a q_{tCa} = m_{tC}, \qquad \sum_a y_{tCa} = 1,$$

$$q_{tCa} \le y_{tCa}, \qquad \sum_D z_{tCaD} = q_{tCa}, \qquad m_{t+1,D} = \sum_{C,a} z_{tCaD},$$

$$\underline{p}_{tCaD} q_{tCa} \le z_{tCaD} \le \overline{p}_{tCaD} q_{tCa},$$

$$\sum_{v \in K} y_v \le |K| - 1 \qquad (K \in \mathcal{H}_\epsilon). \tag{A.4}$$

Every $\mu_C$ is nonnegative and $\sum_C \mu_C = 1$. Thus each $m_t$ is a probability vector. These are outer occupation constraints, not a claim that the acquired cell process is Markov. In particular, they allow the within-cell state law to depend on all earlier actions.

THEOREM A.1—Recursive common-parameter cost transfer: *Under the preceding boundedness, feasibility and enclosure conditions, put*

$$L_\epsilon = \min_{\mathcal{P}_\epsilon} \sum_{t,C,a} B_t \ell_{tCa} q_{tCa}, \qquad U_\epsilon = \max_{\mathcal{P}_\epsilon} \sum_{t,C,a} B_t u_{tCa} q_{tCa}. \tag{A.5}$$

*Every $\theta \in \Theta_\epsilon$ satisfies*

$$L_\epsilon \le J_0^{\pi_\theta} - J_0^{\tau} \le U_\epsilon.$$

*The conclusion allows changes in every date's action and continuation, together with their induced original-law occupations. Valid additional compatibility or transition constraints can only tighten the enclosure. If $\epsilon$-dependent exclusions are retained monotonically, the enclosures are nested as qualification is strengthened. Certified dual lower and upper bounds suffice; an optimizer's reported optimum does not.*

*For two complete-policy classes with reference-cost interval $[l_{jk}, u_{jk}]$ on a simultaneous event, their cost difference is enclosed on the same event by*

$$[\, l_{jk} + L_j - U_k,\ u_{jk} + U_j - L_k \,].$$

*No probability union over the parameter vectors is added after that event. A qualified bound on a reference policy's loss against $V_0^*$ can also be enlarged by $U_\epsilon$, retaining the original optimum as the benchmark.*

The proof in Supplement E uses a telescoping identity under the candidate policy's occupation law, then embeds each complete candidate in (A.4). Independent choices at different cells and dates are not treated as jointly attainable. As in the first-date assignment bound, the relaxation itself can have unattainable points. The economic conclusion is an enclosure, not a worst-policy realization claim.

### A.1. *Nonlinear recursion, temporal selves and games*

For a monetary evaluation $\rho_t$ satisfying the monotonicity and cash-invariance conditions in Section 2, let $\mathcal{T}_t^\pi$ be the fixed-policy operator. Suppose, uniformly over the states and policies in the declared class,

$$d_t^- \le \mathcal{T}_t^\pi V_{t+1}^\tau - V_t^\tau \le d_t^+.$$

There is no linear-occupation identity for an arbitrary nonlinear $\rho_t$. Its correct counterpart is the following recursive order statement.

PROPOSITION A.2—Recursive order and deviation bounds: *If $e_T^- = e_T^+ = 0$ and $e_t^\pm = d_t^\pm + \beta_t e_{t+1}^\pm$, then $e_t^- \le V_t^\pi - V_t^\tau \le e_t^+$ for every class member. More generally, on invariant recursive-value domains, if a monotone fixed-policy aggregator is $\lambda_t$-Lipschitz in its continuation argument and its reference residual is at most $b_t$ in absolute value, then*

$$\|V_0^\pi - V_0^\tau\|_\infty \le \sum_{t<T} \left( \prod_{j<t} \lambda_j \right) b_t.$$

*For a dynamic game, apply the one-sided sandwich of Proposition 1 to each player's best-response problem with the other players' complete strategies fixed. If that account is at most $r_{it}$ in every subgame, the attained profile has unilateral cost-improvement incentive at most $r_{it}$ there. This is an approximate equilibrium certificate, not an equilibrium-existence assertion. Date-specific temporal selves are treated as distinct decision makers, with their stated objectives and future strategies, rather than as one artificially time-consistent planner.*

Endogenous preference states belong in the augmented state and transition before these hypotheses are checked. A diffusion approximation contributes its certified one-step operator error at every date, propagated with the same continuation moduli; an unproved weak-discretization rate is not assumed. The broader economic applications and their original hypotheses remain intact. The expectation identity, nonlinear order bound and strategic deviation certificate are distinct results, not interchangeable interpretations of one simulation table.

## APPENDIX B: SIGNED TRANSITION QUALIFICATION FOR RECURSIVE PREFERENCES

The transport account can use more information than an absolute error radius. A numerical transition may have a known direction of bias, and the implemented action may have a more precise upper error bound than the whole feasible menu. We retain these distinctions. The resulting qualification applies to the original recursive evaluation at every date, not only to expected cost with a fixed continuation. It does not change an expectation objective into a risk objective or identify a static tail-risk criterion with its nested counterpart.

Retain the augmented economic state, original feasible actions and measurable complete policies of Section 2. Suppose bounded original values satisfy backward recursion with

$$Q_t(v;x,a) = c_t(x,a) + \beta_t \rho_{t,x,a}(v), \qquad T_t v(x) = \inf_{a \in A_t(x)} Q_t(v;x,a),$$

where each original conditional evaluation is monotone and cash invariant: $\rho(v + k) = \rho(v) + k$ for constants $k$. Discounts $\beta_t \geq 0$ are deterministic, $B_0 = 1$, and $B_t = \prod_{s<t} \beta_s$. The state includes the preference variables required by the original model. The relevant bounded functions are in the evaluation's domain, and the original Bellman optimum and supplied feasible policies are well defined. For each common parameter $\theta$, store bounded $f_{t,\theta}$, a complete feasible $\pi_\theta$, and numerical action values $\widehat{Q}_{t,\theta}$ on the original feasible actions.

Put $\widehat{T}_{t,\theta}(x) = \inf_a \widehat{Q}_{t,\theta}(x,a)$ and define the signed discrepancy

$$D_{t,\theta}(x,a) = Q_t(f_{t+1,\theta};x,a) - \widehat{Q}_{t,\theta}(x,a).$$

Assume uniform, verified bounds

$$\begin{aligned} \ell_{t,\theta} \leq \widehat{T}_{t,\theta}(x) - f_{t,\theta}(x) \leq u_{t,\theta}, \\ 0 \leq \widehat{Q}_{t,\theta}(x, \pi_{t,\theta}(x)) - \widehat{T}_{t,\theta}(x) \leq \gamma_{t,\theta}, \\ b^-_{t,\theta} \leq D_{t,\theta}(x,a) \quad (a \in A_t(x)), \\ D_{t,\theta}(x, \pi_{t,\theta}(x)) \leq b^{\pi,+}_{t,\theta}. \end{aligned} \tag{A.6}$$

Here $b^{-}_{t,\theta}$ and $b^{\pi,+}_{t,\theta}$ may be signed. A bound confined to sampled actions cannot replace the all-action lower bound. Original-owner repairs, unsampled feasible actions, integration error and arithmetic error must be included in the relevant inequalities. Let $d^{-}_{\theta} \leq g - f_{T,\theta} \leq d^{+}_{\theta}$ throughout the terminal domain.

THEOREM A.3—Signed, recursive original-optimum qualification: *Under the stated conditions, every supplied complete policy satisfies, at every initial state,*

$$0 \leq V_0^{\pi_\theta} - V_0^{*} \leq \mathcal{S}_\theta,$$
$$\mathcal{S}_\theta = \sum_{t<T} B_t \left\{ u_{t,\theta} - \ell_{t,\theta} + \gamma_{t,\theta} + b^{\pi,+}_{t,\theta} - b^{-}_{t,\theta} \right\} + B_T (d^{+}_{\theta} - d^{-}_{\theta}). \tag{A.7}$$

*Consequently a class for which these inequalities hold uniformly and* $\sup_\theta \mathcal{S}_\theta \leq \varepsilon$ *consists entirely of complete policies with original-recursion loss at most* $\varepsilon$*. Actions, attained continuations and endogenous occupations may change at every date. The assertion is pointwise in the augmented state and does not require independent deployment contexts or a continuous actor.*

*The bound is invariant to arbitrary date-specific, state-independent shifts of the stored continuations when the numerical action values are shifted consistently. If the original-minus-numerical action value is a constant* $k_{t,\theta}$ *independent of state and action, its contribution to* (A.7) *is zero. A bias common to all actions at a given state, but varying across states, does not in general cancel. The coefficient one on* $b^{\pi,+} - b^{-}$ *cannot be reduced uniformly, even for one decision date and deterministic transitions.*

The proof compares both the complete policy and the original optimum with the same stored continuation by recursive order. It does not propagate a linear occupation identity through nonlinear preferences. Setting $b^{-} = -b$ and $b^{\pi,+} = b$ recovers Theorem A.31; a verified signed enclosure can strictly improve it without a new fit or economic observation. With stationary bounded data and $0 \leq \beta < 1$, the corresponding stationary bound is

$$V^{\pi} - V^{*} \leq \frac{u - \ell + \gamma + b^{\pi,+} - b^{-}}{1 - \beta}.$$

The proof and the necessary bias counterexamples are in Supplement I.

### B.1. *Explicit qualification of two nonlinear evaluations*

Wasserstein distance controls expectations of Lipschitz functions. For nonlinear evaluations its price must be derived, not borrowed from the expectation model. For a bounded $K$-Lipschitz continuation $f$ and probability laws $P, \widehat{P}$, define the upper-tail average value at risk and entropic evaluation by

$$\rho_P^{\alpha}(f) = \inf_{z\in\mathbb{R}} \left\{ z + \frac{P(f-z)_+}{1-\alpha} \right\}, \qquad 0 \le \alpha < 1,$$
$$\rho_P^{\lambda}(f) = \lambda^{-1} \log P \exp(\lambda f), \qquad \lambda > 0. \tag{A.8}$$

The limit at $\lambda = 0$ is expectation. The first definition includes atoms; it is not the conditional mean obtained by discarding all mass at a quantile. The variational representation and its treatment of general loss distributions are established results (Rockafellar and Uryasev, 2002). We use these evaluations only when they are the original one-step primitives of the recursive economic objective.

PROPOSITION A.4—Risk-specific transport allowances: *If* $W_1(P, \widehat{P}) \le e$, *then*

$$|\rho_P^{\alpha}(f) - \rho_{\widehat{P}}^{\alpha}(f)| \le \frac{Ke}{1-\alpha},$$
$$|\rho_P^{\lambda}(f) - \rho_{\widehat{P}}^{\lambda}(f)| \le K \exp\{\lambda \operatorname{osc}(f)\}\, e. \tag{A.9}$$

*The entropic expression is invariant to constants added to* $f$. *The factor* $(1-\alpha)^{-1}$ *is necessary uniformly over bounded Lipschitz continuations and probability laws. If a coupling satisfies* $f(Y) \ge f(\widehat{Y})$ *almost surely, the original-minus-numerical risk difference is nonnegative for either evaluation.*

If the additional numerical action-value error is at most $\xi_t$ in absolute value, a symmetric choice in (A.6) is

$$b_t^{-} = -\beta_t \kappa_t e_t - \xi_t, \qquad b_t^{\pi,+} = \beta_t \kappa_t e_t + \xi_t,$$

where $\kappa_t = K_{t+1}/(1-\bar{\alpha}_t)$ for a verified risk-level cap $\alpha \le \bar{\alpha}_t < 1$, or $\kappa_t = K_{t+1} \exp(\bar{\lambda}_t O_{t+1})$ for verified bounds on the entropic parameter and

continuation oscillation. A monotone coupling instead permits $b_t^- = -\xi_t$ and the same upper endpoint. Approximate evaluation of the exponential requires an outward upper bound; machine output is not an exact modulus.

These local risk coefficients enter the dated budget additively:

$$\sum_{t<T} B_t \eta_t + 2 \sum_{t<T} B_{t+1} \kappa_t e_t + 2 \sum_{t<T} B_t \xi_t + B_T (d^+ - d^-) \leq \varepsilon, \tag{A.10}$$

where $\eta_t$ bounds residual span plus greediness. Monotonicity and cash invariance propagate constant error by $\beta_t$, not by a product of all later tail-risk multipliers. The continuation moduli themselves may grow with horizon or preference parameters and must still be established and paid for. Subject to this qualification, the coupled resource program of Section N allocates residual and risk-adjusted transition precision without deleting descendant or representation costs.

*Economic and strategic interpretation.* A larger tail-risk level can require finer transition information for the same original-policy loss allowance. That requirement is part of the original preference primitive, not a favorable reweighting of measured work. When preference levels evolve endogenously, the bounds must cover the augmented state and actual feasible actions. For games and temporal selves, the theorem applies to each fixed-opponent or fixed-future best-response recursion under that decision maker's original objective. Applying nested average value at risk to an originally static terminal-risk problem would change that problem; no such substitution is made. General non-cash-invariant recursive aggregators retain Proposition A.2 and its separate continuation modulus.

## APPENDIX C: DECISION-SCALE PRECISION AND THE COST OF A QUALIFIED NBO

An approximation can save continuation queries and still leave an economic comparison unresolved. The investment experiment in Section P illustrates this distinction: the original policy intervals establish equivalence only at a margin equal to four fifths of the maximal first-date investment charge. We now connect numerical integration, policy admission, statistical resolution and the cost of deployment. The original economic law and the NBO construction remain the benchmark. Neither

a smaller numerical interval nor a new admission rule is silently attributed to the unchanged historical actor.

### C.1. *Risk quadrature from the convexity of the original value*

Fix a current state and feasible action. Let $v(z) = V_1(F(x,a,z))$ on $[-R,R]$, where $R = 1/32$. Proposition 5 implies that $v$ is convex and 15-Lipschitz. This conclusion concerns the true one-date value. The complete terminal controller need not be convex: its original bound $V_1 \le V_1^{\widehat{\pi}} \le V_1 + \varepsilon_1$, $\varepsilon_1 = 2^{-20}$, will be added separately.

Let $z_0, \dots, z_n$ be the endpoints of equal innovation bins, and let $m_i = (z_{i-1} + z_i)/2$. Define $M_n$ to put probability $1/n$ at each midpoint. Define $E_n$ to put probability $1/(2n)$ at each of the two endpoints of every bin, counting an interior endpoint twice. These are numerical measures used for enclosures, not alternative economic transition laws.

PROPOSITION A.5—Convex-order risk enclosure: *For a convex $K$-Lipschitz function $v$ and the uniform law $P$ on $[-R,R]$,*

$$\rho_\alpha^{M_n}(v) \le \rho_\alpha^{P}(v) \le \rho_\alpha^{E_n}(v), \qquad \rho_\alpha^{E_n}(v) - \rho_\alpha^{M_n}(v) \le \frac{KR}{(1-\alpha)n^2}. \tag{A.11}$$

*If midpoint lower values and endpoint upper values have errors at most $e_-$ and $e_+$ respectively, the computed width is at most the last term plus $e_- + e_+$. A terminal controller uniformly within $\varepsilon_1$ of $V_1$ adds only $\beta\varepsilon_1$ to its first-date upper cost. All numerical endpoint arithmetic must be enclosed separately.*

The proof combines the established convex midpoint–trapezoid order with the AVaR threshold representation; neither underlying principle is claimed as new (Alomari, 2016, Rockafellar and Uryasev, 2000). Their use here changes the implementable original-law precision account. The previous full-bin Lipschitz allowance decays as $n^{-1}$. The new width decays as $n^{-2}$ without assuming a differentiable risk-averse value or a fixed maximizing risk density. Both numerical risks are finite weighted tail calculations. Ignoring the half weights at the two outer endpoints would compute a different bound.

COROLLARY A.6—Sharpness on an explicit neural continuation: *Let* $\alpha \in \{1/2, 3/4\}$ *and let* $n$ *be a positive multiple of eight. For* $K, R > 0$, *put* $c_n = R/n$ *and*

$$v_n(z) = K\{\sigma(z - c_n) + \sigma(c_n - z)\} = K|z - c_n|.$$

*This convex two-unit ReLU continuation satisfies*

$$\rho_\alpha^{E_n}(v_n) - \rho_\alpha^{M_n}(v_n) = \frac{KR}{(1-\alpha)n^2}.$$

*Consequently neither the exponent nor the leading constant in the noiseless width bound of Proposition A.5 can be reduced uniformly over convex K-Lipschitz continuations, even on this explicit neural subclass. The conclusion concerns the stated midpoint and endpoint rules, not a lower bound for all integration algorithms or a superiority claim for neural representations.*

The same argument applies to a rectangular product innovation law when $v$ is convex in each coordinate. If coordinate $j$ has radius $R_j$, Lipschitz constant $K_j$ and $n_j$ bins, the product midpoint and product endpoint rules have width at most

$$\frac{1}{1-\alpha} \sum_j \frac{K_j R_j}{n_j^2}.$$

The number of evaluation points still grows as a product of the resolutions. This extension therefore retains the two-innovation cost rather than claiming that the curse of dimension has disappeared. It does not automatically apply to the possibly nonconvex minimum-of-cones representation of the general constructive backend. Convexity of a supplied continuation, or of the original value being enclosed, is a substantive hypothesis.

For the investment model, a midpoint affine support also supplies a feasible density witness for the all-original-action lower quadratic in Theorem 6. The new endpoint upper value and that lower quadratic therefore give

$$\underline{V}_0(x) \le V_0(x) \le J_0^{\widehat{\pi}}(x) \le \overline{J}_0(x).$$

The policy upper and lower costs use the same continuous innovation law. The state-cover residual remains a whole-domain statement, whereas an admission inequality is checked at the queried state. These two quantifiers are not interchangeable.

### C.2. *Admission changes the statistical support*

A stored continuation first proposes an action. A qualified service accepts that action only when its complete-policy bound is at most $\tau$ against the original optimum. Otherwise it pays for the declared conventional action and verifies that action. An exact finite-cover fallback makes this a complete measurable service on the declared dyadic initial domain, rather than a controller defined only on successful sample rows. The fallback is part of the policy, and a rejected proposed action remains in the record.

THEOREM A.7—Qualification and simultaneous resolving power: *Consider $A$ complete services, fixed independently of an evaluation sample, each satisfying $0 \le J^a(x) - V^*(x) \le \tau$ for every state in the declared initial domain. Suppose each reported policy-cost enclosure has width at most $w$. For $N$ independent initial states, let $[d_i^-, d_i^+]$ be the difference of two cost enclosures, intersected with $[-\tau, \tau]$. For any $r > 0$, all $H$ declared pairwise intervals*

$$\left[\frac{1}{N}\sum_i d_i^- - r, \frac{1}{N}\sum_i d_i^+ + r\right] \tag{A.12}$$

*cover their original-law mean contrasts with probability at least $1 - 2H\exp\{-Nr^2/(2\tau^2)\}$. Their half-widths are at most $r + w$, apart from the separately enclosed arithmetic used to report endpoints.*

The global service definition is essential. Success on the observed states alone does not establish the support used in this theorem. Nor may a failed fitted proposal be deleted from a nominally unchanged-policy sample. The present service records its proposal, acceptance decision, final action, numerical refinement and exact-fallback entry. The original R95 policy experiment remains a different, unchanged experiment.

We use $\tau = 1/256$, $w = 1/8192$, $N = 4096$, $H = 12$ across two risk levels, and $r = 1/4096$. Then $2H\exp(-8) < 0.05$, while

$$r + w = \frac{3}{8192} < \frac{1}{2560} = \frac{\epsilon}{2}, \qquad \epsilon = \frac{1}{1280} = \frac{h(1/4)}{100}. \tag{A.13}$$

Thus the sample size and numerical precision are fixed together to resolve an average policy-cost difference at one percent of the largest first-date investment charge. The margin is eighty times smaller than the earlier $1/16$ margin. It is an explicit model-relative purchaser tolerance, not an estimated welfare weight or external monetary calibration. We report which comparisons actually meet it; the design alone implies no favorable sign.

### C.3. *Acquisition, reuse and an economic purchase*

Let $K_a$ be a one-time acquisition charge for service $a$, and $c_a$ its cost per future workload. The acquisition account includes labels, feature search, fitting, validation, serialization and the chosen audit campaign. A fresh-process workload includes loading, identity checks, action proposal, admission, fallback, numerical certification, output and shutdown. Let a workload contain $B$ initial-state decisions and let $d = \mathbb{E}[J^a - J^b]$ be the per-decision original policy-cost contrast, $\lambda \geq 0$ the price of computational work in the same normalized cost units, and $F$ an additional fixed fee difference. For $m$ future workloads, the total normalized difference is

$$m\{Bd + \lambda(c_a - c_b)\} + \lambda(K_a - K_b) + F. \tag{A.14}$$

For the recorded timing workloads $B = 32$; for a single decision $B = 1$. This expression distinguishes a policy-quality loss from the value of saved computation. A price is not inferred from seconds, a coefficient count, or an integration bin.

PROPOSITION A.8—Robust acquisition-and-reuse region: *Suppose the policy contrast has simultaneous upper bound $d^+$, the workload saving $c_b - c_a$ has a valid lower bound $s^-$, and the incremental acquisition and fee have upper bounds $k^+$ and $F^+$. Then service a weakly improves total normalized cost whenever*

$$m(\lambda s^- - Bd^+) \geq \lambda k^+ + F^+. \tag{A.15}$$

*If $\lambda s^- - Bd^+ > 0$ and the right side is nonnegative, the smallest sufficient integer workload count is its ratio rounded upward. If the denominator is nonpositive and the right side is positive, no positive workload count is certified by these bounds. Separately, a purchaser's quality condition is $d^+ \le \epsilon$; it must not be inferred from a positive computational saving.*

The timing repetitions below produce descriptive machine-specific ranges, not probabilistic guarantees for arbitrary future machines. Inserting their extrema in (A.15) gives a sensitivity calculation conditional on that timing envelope. It does not turn five repeated processes into a confidence bound on long-run prices or runtime. The theorem states the economic decision that adequately measured cost bounds would license, while the experimental record states precisely which costs were observed.

## APPENDIX D: INFORMATION, REPRESENTATION, AND THE PRICE OF A COMPLETE POLICY

The validity of a Bellman comparison and the usefulness of its neural representation are different mathematical questions. Original-model qualification answers the first. The approximation class, its acquisition cost, and the work of rejected proposals answer the second. This section sharpens the distinction without changing the policy-construction problem.

### D.1. *An information bound for continuous-law risk*

Let $\mathcal{C}_{K,R}$ be the convex, $K$-Lipschitz real functions on $[-R, R]$, and let $P$ be uniform on this interval. An admissible information procedure chooses at most $N$ locations, possibly as deterministic functions of earlier observed function values. It has no derivative oracle, functional formula, or uncharged continuation labels. Its output is an interval containing $\mathrm{AVaR}_\alpha(f(Z))$ for every $f \in \mathcal{C}_{K,R}$ consistent with its observations. Constants $K, R > 0$ and $0 \le \alpha < 1$ are known.

THEOREM A.9—Deterministic information lower bound: *Every such procedure has an input for which the reported interval has width at least*

$$\frac{KR(1-\alpha)}{6(N+1)^2}. \tag{A.16}$$

*The worst-case absolute error of any point-valued procedure is at least half this quantity. Conversely, the midpoint–endpoint bracket with $n$ cells uses $2n+1$ distinct function values and has width at most $KR/\{(1-\alpha)n^2\}$. Thus, for every fixed $\alpha < 1$, the optimal deterministic information exponent is two.*

The proof constructs two convex continuations with the same adaptive transcript but different tail integrals. It is given in the supplement rather than inferred from the sharpness of a particular quadrature rule. This is an information-class statement, whereas the earlier two-ReLU construction attains the constant of the specified midpoint–endpoint pair. Neither statement establishes a constant optimal uniformly in $\alpha$, a randomized lower bound, or a dimension-independent bound. With $p$ independent coordinates and $n_j$ cells per coordinate, the direct product implementation evaluates $\prod_{j=1}^{p} n_j$ midpoint locations and $\prod_{j=1}^{p}(n_j+1)$ endpoint locations. The additive coordinatewise error estimate must be read together with these counts.

The argument uses classical information-based lower-bound reasoning, not a claim that indistinguishable-input arguments originate here; see Novak (2015). Likewise, the AVaR variational representation is due to Rockafellar and Uryasev (2000). The contribution here is the explicit convex-Lipschitz tail construction, its admissible information class, and its connection to the certified recursive-policy resolution problem. Convexity of a *class* of inputs is not the same as convexity of each input function; the proof does not invoke a theorem requiring a balanced class.

### D.2. *What a neural representation can contribute*

A representation-neutral verifier does not imply representation-neutral approximation cost. The following comparison isolates one elementary source of a neural advantage and one limit to that interpretation.

PROPOSITION A.10—An oblique continuation: *On* $[0,1]^2$, *let* $v(x) = (x_1 + x_2 - 1)_+$. *A single oblique ReLU represents* $v$ *exactly. The best uniform approximation errors from arbitrary additive functions* $a(x_1) + b(x_2)$ *and from polynomials of total degree at most two are, respectively,* $1/4$ *and* $1/16$.

This is a structural example in the continuation class, not a claim that the investment value function equals this ridge, that its action loss equals its value-approximation error, or that every conventional method is separated. A conventional method with the same oblique hinge also represents it exactly. Consequently, the identifiable object is acquired geometry, rather than the name attached to an equivalent formula. In the investment experiment the quadratic arm remains the observed computational winner. The proposition explains why that result neither proves a universal quadratic advantage nor supplies evidence of a neural advantage in that particular application.

For the investment continuations, write a fixed-feature readout as $h_\theta(z) = \phi(z)'\theta$. At acquired sites $z_i$, the original-model computation gives $V(z_i) \in [\ell_i, u_i]$. An interval-aware construction solves

$$\min_{\theta,l,u} \; u - l \quad \text{subject to} \quad \phi(z_i)'\theta + l \le \ell_i, \qquad \phi(z_i)'\theta + u \ge u_i \quad \text{for every } i. \tag{A.17}$$

Nonnegative nonlinear coefficients preserve convexity; the constant coefficient can be fixed because residual span is invariant to an additive constant. Numerical feasibility of this finite program is not itself a certificate on the state domain.

PROPOSITION A.11—Verified readout improvement: *Let* $C(h) = [L(h), U(h)]$ *be a valid whole-domain enclosure of* $V - h$ *for every candidate readout. Starting from* $h_0$, *construct and round a candidate* $h_1$, *recompute both enclosures, and retain* $h_1$ *only when* $U(h_1) - L(h_1) \le U(h_0) - L(h_0)$. *Otherwise retain* $h_0$. *The retained readout has no larger certified residual span than the baseline, regardless of optimization error in* (A.17). *For a monotone, cash-invariant risk functional,*

$$L(h) \le \rho(V \circ F) - \rho(h \circ F) \le U(h) \tag{A.18}$$

*whenever* $F$ *takes values in the covered domain.*

This gives a finite certificate-aware neural readout that is actually executable. It does not turn a nodewise linear program into a globally optimal approximation problem, and it does not guarantee lower rejection rates or shorter service time. Acquisition includes the candidate solve, its validation, and a rejected candidate. The experiment compares this rule with least squares using the same selected features and available labels, and applies the same comparison to the quadratic and spline families. These controls distinguish the value of certificate-oriented fitting from the value of neural geometry.

### D.3. *A workload-consistent purchase frontier*

Let $a$ be a candidate complete policy service and $b$ its comparator. The per-decision policy-cost difference is $d = \mathbb{E}[J^a - J^b]$. A purchased workload contains $B$ initial-state decisions; $c_a, c_b$ are seconds per *whole* workload, and $K_a, K_b$ are one-time acquisition seconds. Let $\lambda \geq 0$ be a price in normalized model-cost units per second and $F \geq 0$ an additional adoption fee. For $m$ workloads, the gain from choosing $a$ is

$$G(m) = m\{\lambda(c_b - c_a) - Bd\} - \lambda(K_a - K_b) - F. \tag{A.19}$$

The factor $B$ is essential: a cost estimate for one initial state cannot be combined directly with the clock of a 32-state process.

THEOREM A.12—Interval purchase decisions: *Suppose jointly valid bounds are* $d \in [d^-, d^+]$, $c_b - c_a \in [s^-, s^+]$, $K_a - K_b \in [k^-, k^+]$, *and* $F \in [F^-, F^+]$. *Then, simultaneously for every integer* $m \geq 1$ *and* $\lambda \geq 0$,

$$G^-(m) = m(\lambda s^- - Bd^+) - \lambda k^+ - F^+ \leq G(m), \tag{A.20}$$

$$G(m) \leq m(\lambda s^+ - Bd^-) - \lambda k^- - F^- = G^+(m). \tag{A.21}$$

*Positive* $G^-$ *certifies adoption; negative* $G^+$ *certifies rejection. Otherwise the available information leaves the decision unresolved. If* $q = \lambda s^- - Bd^+ > 0$ *and* $C = \lambda k^+ + F^+ \geq 0$, *the first strictly certified workload count is* $\lfloor C/q \rfloor + 1$.

If every qualified service has pointwise regret in $[0, \tau]$, the distribution-free choice $[d^-, d^+] = [-\tau, \tau]$ is valid without a statistical sample. Sharper simultaneous cost

intervals may replace it for the *same frozen services*. They may not be transferred from one fitted model to a later refit. Observed clock ranges do not, by themselves, give probability bounds for future clocks: a frontier using those ranges is conditional on their continued applicability. The economic accounting is exact; the empirical strength of its inputs remains an independently assessable question.

## APPENDIX E: PROOFS FOR COMPLETE-POLICY AND POPULATION TRANSFER

### E.1. *Original-law occupation identity*

Fix $\theta \in \Theta_\epsilon$ and use the law generated by its complete feasible policy. Boundedness permits finite summation and conditional expectation. From $B_{t+1} = B_t \beta_t$ and (A.1),

$$\begin{aligned}\sum_{t<T} B_t \mathbb{E}_{\pi_\theta} A_t^\tau(X_t, \pi_{\theta,t}(X_t)) &= \sum_{t<T} B_t \mathbb{E}_{\pi_\theta} c_t(X_t, \pi_{\theta,t}(X_t)) \\ &\quad + B_T \mathbb{E}_{\pi_\theta} V_T^\tau(X_T) - \mathbb{E} V_0^\tau(X_0) \\ &= J_0^{\pi_\theta} - J_0^\tau.\end{aligned}$$

The conditional-expectation step uses the candidate's transition, not the reference occupation. The terminal equality uses the common original terminal cost $V_T^\tau = g$.

Define $y_{tCa}$ as the indicator that this candidate selects label $a$ in cell $C$, $m_{tC} = \Pr_{\pi_\theta}(X_t \in C)$, and $q_{tCa} = m_{tC} y_{tCa}$. Define $z_{tCaD}$ as the joint probability of $X_t \in C$, label $a$, and $X_{t+1} \in D$. Then the initial, conservation and assignment equations in (A.4) follow directly. Integrating the pointwise transition bounds against the candidate's subprobability law on $C$ gives the two inequalities for $z$, including when $q = 0$. An actually realized common parameter cannot realize all the events in a certified forbidden set, so every conflict inequality holds. Consequently this integral point belongs to $\mathcal{P}_\epsilon$. Each stage's advantage expectation is bounded by $\sum_{C,a} \ell_{tCa} q_{tCa}$ and $\sum_{C,a} u_{tCa} q_{tCa}$. Taking extrema over the outer polytope proves Theorem A.1.

The polytope is closed and bounded: $m_t, q_t$ are subprobabilities or probabilities, $0 \le y \le 1$, and $z$ is bounded by its nonnegative row sum. It is nonempty whenever the qualification class is nonempty, in particular when it contains $\widehat{\theta}$. Hence its extrema exist. Adding valid constraints shrinks it. If a smaller tolerance retains all larger-tolerance constraints and possibly more, its lower extremum increases and its

upper extremum decreases. Independent dual feasibility supplies the appropriate weak-duality bound even when the numerical solve is incomplete. Adding and subtracting the two reference costs proves the class comparison. These operations are deterministic on the simultaneous input event.

*Why reference weights are insufficient.* Consider a two-date, zero-stage-cost example. Initially only state $s$ is visited. At date zero, action $L$ moves to state $\ell$ and $R$ to state $r$. At date one the costly action costs one at $r$ and zero at $\ell$. The reference selects $L$; a candidate selects $R$ and then the costly action at $r$. The reference's date-one occupation gives $r$ mass zero, but the candidate's cost difference is one. A first-date formula using the reference continuation sees zero date-zero contrast and omits the later change. The flow constraint instead forces $m_{1r} = 1$ and charges the date-one advantage. This example identifies an error that cannot be repaired by a union bound over cells.

*Shared coefficients remain material.* In the same example, let $\theta \in \{-1, 1\}$. At $s$ action $R$ has score $-\theta$ and $L$ score zero; at $r$ the costly action has score $\theta$ and the free action score zero. Each event is individually possible, but $R$ and the costly action require opposite signs. Every common-parameter policy has cost zero, whereas the unconstrained action relaxation permits cost one. The exact conflict inequality excludes their simultaneous integral choice. Its fractional relaxation can still admit cost $1/2$; this is why the theorem asserts validity and monotone tightening, not exact attainability of every relaxed occupation.

### E.2. *Nonlinear evaluations and strategic deviations*

Assume inductively that $V^{\tau}_{t+1} + e^{-}_{t+1} \leq V^{\pi}_{t+1} \leq V^{\tau}_{t+1} + e^{+}_{t+1}$. Monotonicity and cash invariance give

$$\mathcal{T}^{\pi}_{t} V^{\tau}_{t+1} + \beta_t e^{-}_{t+1} \leq V^{\pi}_{t} \leq \mathcal{T}^{\pi}_{t} V^{\tau}_{t+1} + \beta_t e^{+}_{t+1}.$$

Subtract $V^{\tau}_{t}$ and add the stated residual bounds. The common terminal condition starts the induction. Under a general continuation Lipschitz modulus instead,

triangle inequality gives $\|V_t^\pi - V_t^\tau\|_\infty \le b_t + \lambda_t \|V_{t+1}^\pi - V_{t+1}^\tau\|_\infty$; iteration proves the second formula of Proposition A.2 on the stipulated invariant domains.

For player $i$, fix every other player's complete strategy. The induced controlled kernel and the player's feasible actions define a best-response problem, with the player's own stated recursive objective. Its best-response value is below the cost of the attained action sequence. Apply the one-sided residual sandwich to this problem from every date and state. A bound $r_{it}$ on its difference is exactly a bound on that player's profitable unilateral cost reduction. Applying it simultaneously to every player and subgame establishes the claimed approximate deviation property. This argument does not establish existence of a profile satisfying the premises. For temporal selves, each self is a separate player with its own deviation problem; no identity with a commitment planner's optimum is asserted.

If a numerical kernel $\widetilde{\mathcal{T}}_t^\pi$ differs uniformly from the original kernel by at most $\kappa_t$ on the continuations in use, it adds $\kappa_t$ to the stage residual. Its contribution is therefore at most $\sum_t (\prod_{j<t} \lambda_j)\kappa_t$. To compare a numerical policy with the original optimum, the corresponding optimal-operator approximation must also be charged in the two-sided sandwich. This explicitly accounts for a diffusion approximation without postulating an unproved rate in its time step.

## APPENDIX F: PROOFS FOR JOINT-DOMAIN CONSTRUCTION AND EXPLORATION

### F.1. *One population circuit*

Let $G_t(r) = \mathcal{T}_t f_{t+1}(r)$ on the joint domain. Primitive regularity and the Hausdorff action bound imply that it is $L_t$-Lipschitz by the same comparison as in Theorem 2, now including the context coordinates. For each node, $-e_t \le y_{ti} - G_t(r_{ti}) \le D_t k_t + e_t$. Thus every cone is at least $G_t(r) - e_t$, and a node within $h_t$ gives a cone at most $G_t(r) + D_t k_t + e_t + 2L_t h_t$. The lower envelope has Lipschitz modulus $L_t$ because each of its members does. The original label owner, the feasible repair, and the original action modulus give the selected-policy upper residual $e_t + D_t \zeta_t$ exactly as in the original construction. The argument closes backward from the terminal approximation, and then the original policy sandwich proves the loss account.

Use $|u| = \mathrm{ReLU}(u) + \mathrm{ReLU}(-u)$ and $\min(u, v) = u - \mathrm{ReLU}(u - v)$. Compute all cone values and combine them in a balanced minimum tree. This proves the claimed weight/operation count and depth, up to the fixed affine layers. Lexicographic original-owner selection is a separate measurable comparison operation; the discontinuous actor is not falsely called a continuous ReLU function. Its feasible repair is charged separately. The guarantee holds at every point of every joint domain, so evaluating it along an endogenous sequence uses no distributional inference or independence assumption. The covering exponent and effective construction cost remain explicit.

### F.2. *Adaptive acquisition coverage*

Fix one cover ball $B$. Let $E_i$ be the event that the first $i$ acquisitions miss it. By the conditional lower-probability assumption,

$$\Pr(E_i) = \mathbb{E}[\mathbf{1}_{E_{i-1}} \Pr(Y_i \notin B \mid \mathcal{F}_{i-1})] \leq (1 - a_h) \Pr(E_{i-1}).$$

Induction yields $(1 - a_h)^n$. A union bound over the $M$ fixed balls gives the result. If none is missed, any point is within $h$ of a cover center and that center is within $h$ of an acquired node, hence the radius is $2h$. This argument controls geometry, not label error; any probabilistic label-certification event is included separately in the overall error allocation. Pure exploitation without a conditional lower-probability bound does not satisfy the proposition.

### F.3. *Prediction and fallback work*

Suppose the best action is $a_*$ and $\Delta > 4e$. For every competitor $b$, its estimate exceeds that of $a_*$ by at least $\Delta - 2e > 2e$. The estimate-minimizing action is consequently $a_*$, and its upper confidence endpoint is below every competitor's lower endpoint. Thus failure of the screen implies $\Delta \leq 4e$. Its probability is at most the tie mass plus $C(4e)^\alpha$, truncated at one. Multiplying by the fallback's upper work bound and adding proposal and test overhead proves Proposition A.17. To certify a saving one must compare that upper account with a lower account for the conventional

baseline, not with another upper bound. This distinction is the work analogue of not subtracting policy-loss upper bounds to estimate a cost difference.

## APPENDIX G: PROOF OF THE DATED ACCURACY ALLOCATION

Hölder's inequality gives, for every feasible positive allocation,

$$S = \sum_t (a_t \eta_t^{-p})^{1/(p+1)} (w_t \eta_t)^{p/(p+1)}$$
$$\le \left( \sum_t a_t \eta_t^{-p} \right)^{1/(p+1)} \left( \sum_t w_t \eta_t \right)^{p/(p+1)}.$$

Therefore the objective is at least $S^{p+1} \varepsilon^{-p}$. Substituting (A.124) attains equality and uses the whole error budget. Strict convexity of $\eta \mapsto \eta^{-p}$ gives uniqueness. The discounted expression follows by summing the geometric series with ratio $\beta^{p/(p+1)}$; the undiscounted expression follows from $S \le (T+1) A^{1/(p+1)}$. For the clipped allocation $\eta_t' = \min(\eta_t^*, \eta_{\max})$, the error budget can only decrease and

$$a_t (\eta_t')^{-p} = a_t \max\{(\eta_t^*)^{-p}, \eta_{\max}^{-p}\} \le a_t (\eta_t^*)^{-p} + a_t \eta_{\max}^{-p}.$$

The asserted additive bound follows. Dyadic downward rounding by less than a factor two proves the final factor. Bit complexity, per-date setup and storage costs lie outside this algebraic upper model unless expressly included in $a_t$.

## APPENDIX H: PROSPECTIVE MENU CHOICE AND UNIFORM ECONOMIC SENSITIVITY

### H.1. *Finite-menu selection*

Condition on everything used to construct the finite service menu. Let $C_j(W) \in [0, H]$ be the true cost of service $j$ on a new IID workload $W$, and suppose $l_{ji} \le C_j(W_i) \le u_{ji}$ for each evaluated workload. The endpoints are valid whole-input enclosures, not unqualified point calculations. A bounded-variable exponential inequality and a union bound over the $m$ services give, with probability at least $1 - \delta$,

$$|\overline{C}_j - J_j| \le q_n, \quad q_n = H \sqrt{\log(2m/\delta)/(2n)}, \quad j = 1, \dots, m.$$

For completeness, the conditional moment-generating-function bound for a centered variable in an interval of length $H$ is at most $\exp(\lambda^2 H^2/8)$. Iterating it over the independent draws, applying Markov's inequality and optimizing $\lambda$ proves each tail $\exp(-2nq_n^2/H^2)$. Hence $J_j \in [\bar{l}_j - q_n, \bar{u}_j + q_n]$ simultaneously.

Let service $\widehat{j}$ minimize $\lambda \bar{u}_j + p \cdot w_j + f_j$ for $\lambda \geq 0$, and let $j_*$ minimize its true counterpart. On this event, the selected objective minus the true optimum is at most

$$\begin{aligned}\lambda(\bar{u}_{\widehat{j}} + q_n) + p \cdot w_{\widehat{j}} + f_{\widehat{j}} - \lambda(\bar{l}_{j_*} - q_n) - p \cdot w_{j_*} - f_{j_*} \\ \leq \lambda\{2q_n + \bar{u}_{j_*} - \bar{l}_{j_*}\} \leq \lambda\{2q_n + \max_j(\bar{u}_j - \bar{l}_j)\}.\end{aligned}$$

The same event works for every price and fee vector simultaneously, because it was constructed for the costs before prices were applied. This does not make an outcome-dependent acquisition or construction of new services free of a selection charge. On a separate evaluation stream, simultaneous pairwise intervals give the sharper evaluated regret bound $\max\{0, \max_j u_{\widehat{j},j}\}$ for the calibration-selected service. The stream need not show strict superiority to establish noninferiority at a prespecified margin.

For adaptively generated workloads, iteration of the conditional moment-generating-function bound instead bounds the average predictable cost $n^{-1}\sum_i \mathbb{E}[C_j(W_i) \mid \mathcal{F}_{i-1}]$, for a menu fixed before that sequence. It does not identify a stationary population expectation or the cost under another policy's counterfactual workload distribution. This is the correct distinction between sequential validation and an IID new-task claim.

## H.2. *Price and fee regions*

For a compact joint cost polytope $\mathcal{D}$ and two feasible assignments $x, y$, the largest objective difference is

$$\begin{aligned}\sup_{c \in \mathcal{D}}\{\lambda c \cdot (x - y) + p \cdot (w_x - w_y) + f \cdot (x - y)\} \\ = \lambda h_{\mathcal{D}}(x - y) + p \cdot (w_x - w_y) + f \cdot (x - y).\end{aligned}$$

Nonnegativity of $\lambda$ is used here. The candidate $x$ is optimal for every $c \in \mathcal{D}$ precisely when this expression is nonpositive for every feasible $y$. This proves the polyhedral region in Proposition A.21; a certified support upper bound gives a sufficient subregion. Linear optimization over a compact price-fee polytope reduces uniform verification to its vertices. The boundaries are the stated price-fee breakpoints. If $\mathcal{D}$ includes correlated pairwise constraints they must be retained in its support problem; independently choosing the worst endpoint of each pair can be looser. No step estimates a price, monetary scale or adoption fee from a clock measurement.

# APPENDIX I: SIGNED QUALIFICATION AND NONLINEAR TRANSPORT: PROOFS

## I.1. *Proof of Theorem A.3*

Fix $\theta$ and omit it from the notation. Because $b_t^-$ is valid for every original feasible action, taking infima gives

$$T_t f_{t+1} - f_t \geq \widehat{T}_t - f_t + b_t^- \geq \ell_t + b_t^-.$$

For the actual feasible action, the numerical greediness and actor-specific upper bounds give

$$T_t^\pi f_{t+1} - f_t \leq \widehat{Q}_t(\cdot, \pi_t) - f_t + b_t^{\pi,+} \leq u_t + \gamma_t + b_t^{\pi,+}.$$

Define real constants backwards by

$$a_T = d^-, \quad c_T = d^+, \qquad a_t = \ell_t + b_t^- + \beta_t a_{t+1}, \quad c_t = u_t + \gamma_t + b_t^{\pi,+} + \beta_t c_{t+1}.$$

At the terminal date $f_T + a_T \leq g \leq f_T + c_T$. If $V_{t+1}^* \geq f_{t+1} + a_{t+1}$, monotonicity and cash invariance, including under the infimum, imply

$$V_t^* = T_t V_{t+1}^* \geq T_t f_{t+1} + \beta_t a_{t+1} \geq f_t + a_t.$$

Similarly, $V_{t+1}^\pi \leq f_{t+1} + c_{t+1}$ implies

$$V_t^\pi = T_t^\pi V_{t+1}^\pi \leq T_t^\pi f_{t+1} + \beta_t c_{t+1} \leq f_t + c_t.$$

Induction proves both comparisons. Subtracting at date zero gives $V_0^\pi - V_0^* \leq c_0 - a_0$, which expands to (A.7). Feasibility and optimality give its lower bound

zero. Infima need not be attained: the order comparison holds directly under infima, provided the stipulated original value recursion and feasible policy are defined. Every parameter is treated by the same argument; a supremum of valid bounds provides the stated classwise conclusion. No union over parameter vectors and no reference-policy occupation law is introduced.

For shifts $f'_t = f_t + k_t$, set $\widehat{Q}'_t = \widehat{Q}_t + \beta_t k_{t+1}$. Both numerical residual endpoints shift by $\beta_t k_{t+1} - k_t$; their difference is unchanged. Numerical action differences and original-minus-numerical action values are unchanged by cash invariance. Both terminal endpoints shift by $-k_T$. Thus every term in (A.7) is invariant. If the action-value discrepancy itself is a constant $k_t$, choose $b_t^- = b_t^{\pi,+} = k_t$ and its charge vanishes.

For the stationary assertion, the Bellman and fixed-policy maps are sup-norm contractions with modulus $\beta$: monotonicity and cash invariance imply sup-norm nonexpansiveness of each one-step evaluation. The finite-horizon argument with a bounded terminal comparison then has a terminal remainder of order $\beta^T$. Letting $T$ tend to infinity proves the displayed stationary bound. The asserted constant-shift invariance does not justify shifting an original preference primitive or an action-dependent immediate cost.

### I.2. *Sharpness and the distinction between constant and statewise bias*

For $h > 0$, take one decision date, terminal cost $g(y) = y$, discount one and zero immediate costs. The original successors of actions zero and one are respectively $h$ and zero; the numerical successors are both zero. Store $f_0 = 0$ and $f_1 = g$, and let the numerical tie convention select action zero. All numerical residual and greediness terms vanish. The original-minus-numerical action values lie in $[0, h]$, the selected-action upper endpoint is $h$, and the implemented policy loses exactly $h$. Thus the signed charge $h$ is attained. The symmetric allowance $2h$ remains valid but is strictly larger. The example works for any monotone cash-invariant normalized risk evaluation because its successors are deterministic. Both laws can be singular, so no total-variation convergence is being used.

A statewise common-action bias is different. Let a root action choose state $L$ or $R$ at the next decision date. At that date there is only one feasible action, with original cost $2h$ in $L$ and zero in $R$. Let every numerical cost, stored continuation and terminal cost be zero, with discount one. Break the root numerical tie toward $L$. All numerical residual spans and greediness gaps vanish. At the second date the discrepancy is common to every action at its state, but equals $2h$ or zero depending on that state. The implemented policy loses $2h$. Cancelling the discrepancy merely because it is action independent at each state would falsely report zero loss. The signed uniform range correctly charges $2h$. This is precisely the earlier action's effect on future states; fixing the reference occupation would also lose it.

### I.3. *Proof of Proposition A.4*

Let $(Y, \widehat{Y})$ have any coupling of the two successor laws. For every real threshold $z$, the map $r \mapsto (r - z)_+$ is one-Lipschitz. Hence

$$\left| P(f - z)_+ - \widehat{P}(f - z)_+ \right| \leq \mathbb{E}|f(Y) - f(\widehat{Y})| \leq K\mathbb{E}d(Y, \widehat{Y}).$$

The same bound, divided by $1 - \alpha$, is uniform in $z$ for the two functions minimized in (A.8). Infima of uniformly close functions are close. Taking the infimum over couplings proves the first bound, with approximating couplings sufficient when an optimum is not attained. At $\alpha = 0$ the variational expression equals expectation: the expression is at least $Pf$ and equality holds for any threshold no larger than the bounded loss's infimum.

To verify that the tail factor cannot be uniformly reduced, take $f(y) = y$ on $[0, 1]$, $\widehat{P} = \delta_0$ and $P = (1 - h)\delta_0 + h\delta_1$, with $0 < h \leq 1$ and $\alpha = 1 - h$. Then $K = 1$, $W_1(P, \widehat{P}) = h$, $\rho_P^\alpha(f) = 1$ and $\rho_{\widehat{P}}^\alpha(f) = 0$. The bound $Ke/(1 - \alpha) = 1$ is attained. In particular, a bound uniform up to risk level one cannot be deduced from Wasserstein distance alone with a fixed coefficient.

For the entropic assertion, choose $m \leq f \leq M$. For positive $\lambda$, the mean value theorem gives

$$|\exp(\lambda f(y)) - \exp(\lambda f(\widehat{y}))| \leq \lambda \exp(\lambda M) K d(y, \widehat{y}).$$

Both exponential integrals are at least $\exp(\lambda m)$, and $|\log s - \log t| \leq |s - t| / \min(s, t)$ for positive $s, t$. Combining these facts, dividing by $\lambda$ and taking the infimum over couplings yields $K \exp\{\lambda(M - m)\} e$. Letting $m$ and $M$ approach the continuation's infimum and supremum proves (A.9). Translation invariance follows from the oscillation rather than the absolute level. The expectation limit follows from boundedness and expansion of the exponential integral; its displayed coefficient tends to $K$.

If $f(Y) \geq f(\widehat{Y})$ almost surely, every positive-part integral is weakly larger and so is its minimized variational expression. Exponentiation and logarithm preserve the same order for the entropic evaluation. This gives the signed lower endpoint zero. If $|\widehat{Q} - c - \beta \rho_{\widehat{P}}(f)| \leq \xi$, subtracting the numerical approximation places the action-value discrepancy in $[-\xi, \beta\kappa e + \xi]$ under that ordering, or in $[-\beta\kappa e - \xi, \beta\kappa e + \xi]$ without it. These are bounds for the original risk primitive; no statement is made about replacing one primitive by another.

### I.4. *Nested objectives and the dated resource account*

Nested one-step evaluations and a static terminal evaluation differ even without decisions. With four equiprobable terminal losses $0, 0, 0, 4$, group the first two and last two under two equiprobable intermediate states. At risk level one half, the intermediate evaluations are zero and four, and the nested value is four. Static average value at risk of the four terminal outcomes is two. Consequently a static objective needs its own state augmentation or a different analysis; applying the preceding nested recursion without that step changes the objective.

For an originally nested criterion, the preceding induction propagates a state-independent continuation allowance $a$ as $\beta_t a$. The local Wasserstein-to-risk coefficient $\kappa_t$ enters only the current discrepancy. Substituting the symmetric signed interval into (A.7) gives (A.10); monotone-coupling information replaces the factor two on that transition component by one. This establishes the budget used for dated allocation. It does not bound the cost of obtaining $K_{t+1}$ or the number of labels, innovations or feasible-action searches. In particular, a growth of the verified

continuation modulus with horizon is retained in $\kappa_t$ and cannot be removed by this algebra.

For a fixed-opponent best-response problem the identical induction compares a player's implemented complete strategy with the original best-response optimum. Applying it separately in every declared subgame gives the corresponding unilateral-deviation certificate. Each temporal self likewise retains its own original criterion and fixed future strategies. Monotone but non-cash-invariant aggregators do not satisfy the constant-shift step above and must instead use their certified absolute-error recursion. This establishes the advertised scope without conflating expectation, nested risk, static risk and strategic evaluation.

## APPENDIX J: INFORMATION BOUNDS AND CERTIFICATE-ORIENTED ACQUISITION

### J.1. *Proof of Theorem A.9*

Put $q = -R + 2R\alpha$ and $L = R - q = 2R(1-\alpha)$. Answer the algorithm's queries with

$$f_0(z) = \frac{K}{2L}(z-q)_+^2.$$

This function is convex and nondecreasing. Its derivative is zero to the left of $q$, and equals $K(z-q)/L$ to the right, so its Lipschitz constant on $[-R,R]$ is $K$. After the algorithm terminates, collect all distinct query points in $(q,R)$ and augment them by $q$ and $R$. They partition the tail interval into $s \le N+1$ nonempty intervals of lengths $h_1,\dots,h_s$.

Let $f_1 = f_0$ on $[-R,q]$ and let $f_1$ equal the chord of $f_0$ on each interval of this partition. For endpoints $a < b$ in $[q,R]$, the chord slope is $K(a+b-2q)/(2L)$. These slopes increase along the partition, lie in $[0,K]$, and are compatible with the zero left derivative at $q$. Hence $f_1 \in \mathcal{C}_{K,R}$ and is nondecreasing. The two functions agree at every query point, including any points outside the tail. By induction on the number of queries, a deterministic adaptive algorithm receives exactly the same transcript on $f_0$ and $f_1$, chooses the same next points, and stops with the same output. The partition is constructed after running the algorithm on $f_0$; the argument therefore does not assume nonadaptive sampling.

For any nondecreasing continuous $f$, the quantile representation of AVaR gives

$$\mathrm{AVaR}_\alpha(f(Z)) = \frac{1}{L}\int_q^R f(z)\,dz.$$

This also holds at $\alpha = 0$ and in the presence of the flat portion of these functions. On an interval $[a,b]$ of length $h$, the chord minus the quadratic equals $K(z-a)(b-z)/(2L)$. Its integral is $Kh^3/(12L)$. Consequently,

$$\begin{aligned}\mathrm{AVaR}_\alpha(f_1(Z)) - \mathrm{AVaR}_\alpha(f_0(Z)) &= \frac{K}{12L^2}\sum_{i=1}^{s} h_i^3 \\ &\geq \frac{K}{12L^2}\frac{(\sum_{i=1}^{s} h_i)^3}{s^2} \geq \frac{KL}{12(N+1)^2} = \frac{KR(1-\alpha)}{6(N+1)^2}.\end{aligned}$$

The first inequality follows from convexity of the cube. A valid interval returned on the common transcript must contain both values. A point estimate has error at least half their separation on one input. The midpoint–endpoint upper bound already proved in the article uses $n$ midpoint values and $n+1$ distinct endpoint values. Taking $n = \lfloor (N-1)/2 \rfloor$ for $N \geq 3$ gives the stated upper order. The constants on the two sides have different $\alpha$ dependence; the conclusion concerns the exponent at fixed $\alpha$, not uniform constant optimality. *Q.E.D.*

The construction deliberately permits exact function values to the lower-bound algorithm. Thus any finite-precision procedure with no additional information cannot evade the bound by using more accurate arithmetic. Conversely, this lower bound does not price arithmetic, optimization, or the work of acquiring a continuation formula; those costs enter the service ledger. Providing the entire piecewise representation of $f$ for free would change the information model and invalidate this comparison.

### J.2. *Proof of Proposition A.10*

For $g(x) = a(x_1) + b(x_2)$ the corner difference

$$g(1,1) - g(1,0) - g(0,1) + g(0,0)$$

is zero. For $v$ it equals one. If $\|v - g\|_\infty \le e$, the triangle inequality gives $1 \le 4e$. Equality is attained by $g(x) = (x_1 + x_2)/2 - 1/4$, because with $s = x_1 + x_2 \in [0,2]$ the error is $1/4 - s/2$ for $s \le 1$, and $s/2 - 3/4$ for $s \ge 1$.

For a total-degree-two polynomial, restrict to $x_1 = x_2 = (u+1)/2$, $u \in [-1,1]$. If $p$ approximates $u_+$ with error at most $e$, then $r(u) = 2p(u) - u$ approximates $|u|$ with error at most $2e$. Symmetrizing $r$ does not increase the error and produces an even polynomial $au^2 + c$. The linear functional

$$\mathcal{L}f = f(1/2) - \tfrac{1}{4}f(1) - \tfrac{3}{4}f(0)$$

annihilates every such polynomial, has sum of absolute weights two, and satisfies $\mathcal{L}|u| = 1/4$. Therefore its approximation error for $|u|$ is at least $1/8$, and $e \ge 1/16$. The polynomial

$$p(x) = \tfrac{1}{2}\{u + u^2 + 1/8\}, \qquad u = x_1 + x_2 - 1,$$

attains the bound: $|u| - u^2$ ranges from zero to $1/4$. Finally, $v$ is precisely one ReLU with weight $(1,1)$ and offset $-1$. *Q.E.D.*

The distinction between function and decision error matters. Cash shifts do not affect an action choice, and even a nonconstant approximation error may preserve the minimizer. No lower bound for policy regret follows from this proposition without a further decision construction. In particular, the proposition is not used to reinterpret the R96 quadratic winner as a neural winner.

### J.3. *Proof of Proposition A.11*

By definition, $L(h) \le V(z) - h(z) \le U(h)$ on the entire covered domain. Substitute $z = F(\omega)$, use monotonicity of $\rho$, and then use cash invariance to obtain (A.18). After the numerical candidate is rounded, the whole-domain enclosure is recomputed for those *actual* coefficients. The selection rule compares two valid enclosure widths and chooses the smaller one. This proves the width statement independently of the linear-program solver's termination tolerance. *Q.E.D.*

For completeness, a general finite-site extension makes the distinction between node and domain certificates explicit. Suppose $V$ and $h$ have known Lipschitz

constants $L_V$ and $L_h$, the sites have covering radius $\delta$, and $V(z_i) \in [\ell_i, u_i]$. Then

$$\inf_z (V-h)(z) \geq \min_i \{\ell_i - h(z_i)\} - (L_V + L_h)\delta,$$

$$\sup_z (V-h)(z) \leq \max_i \{u_i - h(z_i)\} + (L_V + L_h)\delta.$$

Indeed, choose a nearest site and apply the Lipschitz inequality to $V - h$. Directed rounding must enclose the displayed arithmetic. The implementation retains the stronger inherited convex-cell residual construction rather than replacing it by this generic bound. Its linear program is only a candidate generator; both the baseline and candidate are evaluated by the same whole-domain validator.

The LP fixes the intercept to zero, bounds the two affine slopes in $[-10^4, 10^4]$, and bounds nonlinear coefficients in $[0, 10^4]$. These are explicit acquisition restrictions, not claimed characterizations of the best neural approximation. The two residual endpoints are unrestricted. Setting all readout coefficients to zero and choosing endpoints to enclose all acquired value intervals is feasible, so a solver failure is an execution failure, not evidence that the mathematical program is infeasible. The retained nonnegative nonlinear coefficients remain convex after dyadic rounding. All attempted feature fits, rank rejections, selection ties, and LP iterations are retained in construction receipts.

### J.4. *Proof of Theorem A.12*

Subtract total candidate cost $m\{B\mathbb{E}J^a + \lambda c_a\} + \lambda K_a + F$ from comparator cost $m\{B\mathbb{E}J^b + \lambda c_b\} + \lambda K_b$. This gives (A.19). Monotonicity of this affine expression in each interval argument, using $m, B \geq 0$ and $\lambda \geq 0$, gives (A.20)–(A.21). The strict inequality $mq > C$ is equivalent to $m > C/q$ when $q > 0$, yielding the first integer stated in the theorem. *Q.E.D.*

If $q \leq 0$ and $C \geq 0$, this sufficient lower-bound test never certifies adoption at a positive workload count. That is an unresolved or negative *comparison*, not a failure of the research program. An upper bound strictly below zero certifies rejection for the specified prices and workload. An equality at zero is kept distinct from either strict conclusion. When each service satisfies $0 \leq J^a(x) - J^*(x) \leq \tau$, subtraction gives $-\tau \leq J^a(x) - J^b(x) \leq \tau$ pointwise and hence after expectation.

No independence assumption is needed for this deterministic bound. For sample-based refinements, independence concerns initial-state draws; the paired services intentionally share each initial state and their deterministic original-law integration. The factor $B$ cannot be dropped merely because the services share these draws.

## APPENDIX K: RETAINED RISK OCCUPATIONS AND COMPUTATIONAL CONSEQUENCES

### K.1. *Exact risk occupations and safe kernel exclusion*

Consider now a finite-state implementation for which the original one-step risk has a finite affine representation

$$\rho_{t,s,a}(v) = \max_{j \in \mathcal{J}_{t,s,a}} \{q_{t,s,a,j}^{\mathsf{T}} v - \psi_{t,s,a,j}\}.$$

The supplied complete policy is fixed before the following maximization. Write $r_{ts} = r_t(s, \pi_t(s))$, $k_{tsj} = k_t^f(s, \pi_t(s), q_j)$ and $q_{tsj} = q_{t,s,\pi_t(s),j}$. For an initial probability vector $\mu$, let $\mathcal{Y}^\pi$ consist of nonnegative masses satisfying

$$m_{0s} = \mu_s, \qquad \sum_j y_{tsj} = m_{ts}, \qquad m_{t+1,z} = \sum_{s,j} y_{tsj} q_{tsj,z}. \tag{A.22}$$

These are risk-distorted occupations, not frequencies under the physical law. Initial integration by $\mu$ is outside the conditional value function; a different original initial risk aggregation needs its own evaluation.

PROPOSITION A.13—Penalized occupation identity and dual witness: *For the preceding finite representation,*

$$\mu^{\mathsf{T}}(V_0^\pi - f_0) = \max_{\mathcal{Y}^\pi} \Bigg\{ \sum_{t<T,s} B_t r_{ts} m_{ts} - \sum_{t<T,s,j} B_{t+1} k_{tsj} y_{tsj} + B_T \sum_s (g_s - f_{Ts}) m_{Ts} \Bigg\}. \tag{A.23}$$

*Any feasible occupation gives a lower bound on this fixed-policy value. Any $U$ with $U_T \geq g - f_T$ and*

$$U_{ts} \geq r_{ts} + \beta_t (q_{tsj}^{\mathsf{T}} U_{t+1} - k_{tsj}) \quad \textit{for every } t, s, j$$

*gives an upper bound $\mu^{\mathsf{T}} U_0$. Matching witnesses certify equality using rational arithmetic when the input is rational. The optimal-policy recursion retains the order $\inf_a \sup_q$; it is not obtained by minimizing jointly over policy actions and risk kernels.*

A stored continuation can also exclude risk kernels, not just action events. Fix a feasible retained witness $q_0$ and a verified set $\mathcal{D}$ containing the successor error vector to be evaluated. Define its support function $h_{\mathcal{D}}(d) = \sup_{z \in \mathcal{D}} d^{\mathsf{T}} z$. Then the strict certificate

$$h_{\mathcal{D}}(q - q_0) < k^f(q) - k^f(q_0) \tag{A.24}$$

excludes $q$ from the maximizing affine pieces throughout $\mathcal{D}$. Equality is retained. A verified upper support bound and lower bound on the penalty difference suffice. For a box $L \le z \le U$ the support is

$$\sum_i (q_i - q_{0i})_+ U_i + \sum_i \min\{q_i - q_{0i}, 0\} L_i.$$

For $\operatorname{osc}(z) \le s$, it is $s\|q - q_0\|_1/2$. The error set must be justified before exclusions are used; defining it from the same unverified pruned computation is circular. Unpruned signed or state-dependent certificates supply one valid starting set.

### K.2. *Preference primitives and computational implications*

For average value at risk, $q_i \le p_i/(1-\alpha)$, $q_i \ge 0$ and $\sum_i q_i = 1$, with zero penalty on that capped simplex. Hence $k^f(q) = \rho_p^\alpha(f) - q^{\mathsf{T}} f$. Quantile atoms and zero-probability outcomes are retained (Rockafellar and Uryasev, 2002). For entropic evaluation, put

$$p_f(dy) = \frac{e^{\lambda f(y)} p(dy)}{\int e^{\lambda f}\, dp}.$$

Its centered penalty is $k^f(q) = \lambda^{-1} \mathrm{KL}(q \| p_f)$, and therefore

$$\mathcal{E}^f(z) = \lambda^{-1} \log \int e^{\lambda z}\, dp_f. \tag{A.25}$$

The change of measure is a computational identity, not a change to the original economic law or its risk parameter. Supplement 2 proves the representation and its

support conditions. A finite affine occupation program is exact only when the stated finite representation is exact; discretizing the entropic dual without an enclosure does not meet that condition.

For a fixed finite representation, backward certificate evaluation uses at most a constant multiple of

$$\sum_{t,s,a} \sum_{j \in \mathcal{J}_{t,s,a}} |\operatorname{supp}(q_{t,s,a,j})|$$

arithmetic operations once residuals and penalties are available. It stores successive value layers rather than expanding descendant histories. Kernel construction, original-law integration, neural-circuit evaluation, support certificates, rational bit lengths and stored witnesses are additional costs. Risk-envelope enumeration can itself be large. This bound does not remove state-dimension dependence, certify an unverified continuous-state aggregation, or prove a speed advantage over another implementation.

For fixed penalties, increasing residuals by nonnegative local allowances $e_{ts}$ increases the fixed-policy upper value by no more than

$$\max_{\mathcal{Y}^\pi} \sum_{t,s} B_t e_{ts} m_{ts}, \tag{A.26}$$

including a terminal allowance at $t = T$. This gives an occupation-sensitive sufficient precision budget in place of $\sum_t B_t \max_s e_{ts}$. The budget prices allowable numerical errors; an application-specific price of computation remains a separate input. Shared-parameter restrictions must be retained before taking an envelope over policies, as in Section W.

*Interpretation.* The new mechanism depends on the stored continuation through $k^f$ and on the original preference through its risk envelope. Equivalent neural and non-neural representations have identical centered guarantees. A representation-specific computational benefit requires a matched execution, not a validity theorem. The existing held-out ReLU result, warning records, unresolved economic comparisons, and adverse vector horizon evidence therefore remain unchanged. The exact finite diagnostics below test the nonlinear certificate and its independent

implementation; they are not a substitute for the original large-vector comparison or new economic observations.

## APPENDIX L: RETAINED UNIFORM HORIZON AND VECTOR-CONTROL ACCOUNT

### L.1. *A fixed-accuracy horizon account for the original scalar and vector economies*

The continuous two-date implementation is not the limit of the constructive argument. The original scalar model and its retained two-control, two-innovation extension in Section BT admit a horizon-uniform primitive account under nested AVaR. For the vector extension, actions satisfy $a_1, a_2 \geq 0$, $a_1 + a_2 \leq A(x)$; the second exposure column interchanges $1/2$ and $1/4$, the action cost is $a_1^2 + a_2^2 + 4(a_1^4 + a_2^4) + a_1 a_2/4$, and an independent common-direction shock of radius $1/64$ is added. These are the retained original vector primitives, not a new model selected to obtain a horizon bound.

PROPOSITION A.14—Uniform risk regularity and constructive accuracy: *Consider either economy at any finite horizon, with $\beta = 15/16$ and a prespecified risk level $\alpha_t \in \{1/2, 3/4\}$ at each date. Then the original value functions are convex and $27/d$-Lipschitz in the $\ell^1$ norm. Their Bellman objectives have joint state curvature at least $107/(128d)$ and action curvature at least $7/4$ on the feasible graph. No continuous differentiability of a risk-averse value function is asserted.*

*The own-future cone construction of Theorem 2 may use the same modulus $27/d$ at every date. Let its state grid have $n_x$ intervals per coordinate; let the action net have resolution $m$ on the original capacity interval or capacity simplex; and let each innovation coordinate have $n_z$ equal bins, where $n_z$ is a multiple of four. With exact labels for the finite-bin risk calculation, the charged continuous-law integration allowance and exact feasible repairs, the constructed complete policy satisfies*

$$J_0^{\pi} - V_0^{*} \leq 16 \left\{ \frac{27}{n_x} + \frac{1319}{512m} + \frac{1215}{512n_z} \right\}. \tag{A.27}$$

*The bound holds for every horizon with these same resolutions. Acquisition and final action-rounding allowances are added according to (9); reserving any positive share of the target tolerance for them gives finite sensor and action precisions independent of the horizon.*

For an approximation allowance $\varepsilon_0 > 0$ before acquisition, sufficient choices are

$$n_x \geq \frac{1296}{\varepsilon_0}, \qquad m \geq \frac{3957}{32\varepsilon_0}, \qquad n_z \geq \frac{3645}{32\varepsilon_0}, \quad 4 \mid n_z.$$

These conservative sufficient resolutions are not a practical recommendation. They make explicit what was absent from a node-count statement alone: fixed original-optimum accuracy can be obtained with resolutions independent of the horizon in these original discounted economies.

Put $S = (n_x + 1)^d$, let $A_m = m + 1$ in the scalar case and $A_m = (m+1)(m+2)/2$ in the vector case, and let $s \in \{1, 2\}$ be the innovation dimension. A dense stored implementation uses $O(TdS^2 A_m n_z^s)$ rational arithmetic operations, plus risk sorting and output. Original-owner compilation replaces continuation evaluation by

$$O\Big(T\{dS + SA_m n_z^s\, d[2^d + \log(n_x + 1)]\}\Big)$$

operations, plus the same primitive, risk-sorting and output charges. The state and innovation costs remain explicit. Fixed precision is not presumed: a common-denominator realization has rational bit length $O((T+1)\log D)$ for a fixed polynomial-size primitive denominator $D$. Thus operation counts are linear in the horizon at fixed accuracy and dimension, while bit-operation costs also pay for growing exact numbers. This constructive result does not assert that the largest vector run has been independently executed, or that its conservative worst-case resolutions are economically competitive.

### L.2. *What is established by each part of the method*

Markov risk dynamic programming, conditional-risk duality and policy iteration are established results (Ruszczyński, 2010, 2014). Feature-based risk evaluation and structured policy improvement are also established research directions (Ruszczyński and Yang, 2025). We do not claim these objects, or elementary dual centering, as new. Table L.I separates the prior mathematical tool from the conclusion obtained here.

Three levels must remain distinct. The bounded-function risk theorem is analytical and applies under its stated dual and selection assumptions. The finite affine-risk

TABLE L.I

ESTABLISHED TOOLS AND THE ADDITIONAL QUALIFICATION OBJECT

| Tool | Additional information retained | Conclusion in this paper |
|---|---|---|
| Risk Bellman recursion | Actual stored continuation and complete feasible controller | State-dependent bounds against the original optimum, not a surrogate objective. |
| Residual comparison | Signed local discrepancies and reference-centered dual penalties | A policy upper inequality and an all-original-action lower inequality; harmless offsets need not be charged twice. |
| Finite occupation duality | Risk-distorted flows and penalties attached to the same continuation | Exact finite comparison with the original $\inf_a \sup_q$ order. |
| Affine support domination | An independently verified error set | Strict safe exclusions; no exclusion at equality and no circular construction of the error set. |
| Convex risk optimization | Primitive curvature, imperfect multiplier residuals and continuous-law bin bounds | Computable whole-action lower witnesses and full-controller upper costs, Theorem 6. |
| Feature-based learning | Frozen labels, coefficients, selector and common original-optimum target | A reproducible comparison of representation, policy cost and verification work, not a presumption of neural superiority. |

occupation result has an exact rational implementation. The continuous investment construction has outward numerical enclosures and primitive-specific convexity. Neither of the latter two implementations establishes numerical tractability for all instances of the first. The NBO contribution includes constructive continuations, their attained-policy qualification and their resource account; a validity theorem need not privilege a neural parameterization for that research question to remain meaningful.

## APPENDIX M: A COMMON OPERATOR BEYOND THE OBSERVED CATALOGUE

A coefficient vector checked at finitely many tasks need not work at a new task. The constructive NBO has a stronger route when the economic primitives provide joint regularity. We state the necessary regularity on states and contexts rather than assuming successful extrapolation by a fitted nine-coefficient readout.

Let $r = (s, x)$ range over an invariant compact subset $K_t$ of $\mathbb{R}^{p_t}$, where $s$ contains declared economic contexts and any endogenous preference variables. Actions and transitions are defined on this augmented domain. Suppose the primitive moduli, Hausdorff-continuous feasible menus and measurable repairs in Theorem 2 hold jointly in $r$, with constants $L_t, D_t, K_t^A$. Let an effective cover have $S_t$ nodes and radius $h_t$. At each node obtain its own-future minimized label with lower error $-e_t$ and upper error $D_t k_t + e_t$. Repair displacement is bounded by $K_t^A \|r - r_i\|_1 + \zeta_t$.

THEOREM A.15—Population construction with one shared neural circuit: *The single function*

$$f_t(r) = \min_{i \le S_t} \{y_{ti} + L_t \|r - r_{ti}\|_1\}$$

*has an exact affine–ReLU circuit, a measurable original-owner actor and the bounds* (6) *uniformly on* $K_t$. *Consequently* (7)*, with the joint-domain constants, holds for every initial context and every admissible evolution remaining in those domains. Contexts may be generated endogenously and adaptively. No independence assumption on their deployment sequence is needed for this uniform deterministic assertion.*

*The circuit uses* $O(p_t S_t)$ *nonzero scalar operations and weights, with absolute-value gates and a balanced minimum tree of depth* $O(\log S_t)$. *If the joint domains have effective* $\ell^1$ *covering numbers at most* $C_t h^{-k_t}$, *this yields* $S_t \le C_t h_t^{-k_t}$. *It is the explicitly shared joint-domain circuit, not a collection of task-specific networks. The original-owner and repair obligations remain necessary.*

This is a constructive population theorem, not a claim that the small feature maps in the historical experiments satisfy its approximation bound. The covering dimension $k_t$, the ambient readout cost $p_t$, acquisition, integration and label work are charged separately. A low intrinsic covering dimension must be estab-

lished for the declared economic family. The same cone also has a native min-plus implementation: exact equivalence of two encodings is not neural superiority.

PROPOSITION A.16—Acquisition coverage under persistent exploration: *Choose $M$ deterministic radius-$h$ balls whose centers cover $K$. If, before each acquisition $i$, every such ball has conditional probability at least $a_h > 0$ of containing the next node, then after $n$ acquisitions the probability that the acquired nodes fail to form a radius-$2h$ cover is at most $M(1-a_h)^n \le M\exp(-na_h)$. Thus $n \ge [\log M + \log(1/\delta)]/a_h$ is sufficient. The assertion permits an adaptive acquisition law satisfying the conditional lower bound; it does not silently replace such a law by IID sampling. On the coverage event, Theorem A.15 applies with $h_t = 2h$, provided the own-future labels and primitive moduli have their stated simultaneous certification.*

### M.1. *From approximation to predictor work*

The certificate's validity and a proposal's value are different. Consider a complete finite action menu at a query, with true objective $Q$ and estimates having simultaneous radius at most $e$. Select a minimum estimated objective and close only when its upper endpoint does not exceed every competitor's lower endpoint. Let $\Delta$ be the true best-to-second-best gap, with $\Delta = 0$ for ties.

PROPOSITION A.17—A distributional work consequence of uniform accuracy: *The preceding test closes whenever $\Delta > 4e$. Suppose, uniformly over the relevant policy-induced query laws, the tie probability is at most $p_0$ and $\Pr(0 < \Delta \le z) \le Cz^\alpha$ for $z > 0$. If proposal and certification work are at most $h(e)$ and a protected fallback uses at most $W_+$, then*

$$\mathbb{E}W \le h(e) + W_+ \min\{1, p_0 + C(4e)^\alpha\}.$$

*If the always-conventional comparator has expected work at least $W_-$, a positive saving follows when the displayed upper bound is below $W_-$. Comparing two upper bounds is not sufficient. The assertion is componentwise for nonnegative operation ledgers, or for a separately declared nonnegative resource-price vector. It is not a wall-clock assertion without a valid mapping from the ledger to time.*

Uniformity over induced query laws is essential: a margin observed under a baseline state distribution does not establish the bound after the policy changes. The tie term cannot be discarded. A representation-specific empirical comparison must measure $h(e)$, achieved accuracy and the relevant query distribution for each representation. The constructed neural circuit provides an explicit member of this comparison; protected fallback alone supplies no evidence about neural training reliability. Proofs are in Supplement F.

## APPENDIX N: ACCURACY ALLOCATION WITH ENDOGENOUS SUCCESSOR COST

The separable allocation in Proposition A.62 applies when its coefficients are fixed independently of the allowances being optimized. A dense successor circuit generally has a size that depends on its own allowance. Charging that size at the preceding date couples the two choices. We now optimize this dependency explicitly.

Let $\eta = (\eta_0, \dots, \eta_{T-1})$ be positive dated residual allowances. Primitive analysis of a specified stored-continuation implementation supplies the resource upper model

$$F(\eta) = F_0 + \sum_{j=1}^{J} c_j \prod_{t<T} \eta_t^{-a_{jt}}, \qquad c_j > 0, \quad a_{jt} \geq 0. \tag{A.28}$$

The coefficients, exponents and setup term $F_0 \geq 0$ are fixed before choosing $\eta$. A term $c_t \eta_t^{-p_t} \eta_{t+1}^{-s_{t+1}}$ charges date-$t$ label acquisition and evaluation of its stored successor. Compilation, terminal evaluation and storage-related work may supply other terms. An operation ledger is scalarized only at declared nonnegative resource prices; the model does not estimate monetary prices or energy use.

The feasible accuracy set is

$$\mathcal{A} = \{\eta : 0 < \eta_t \leq \bar{\eta}_t, \quad \sum_{t<T} w_t \eta_t \leq \varepsilon\}, \qquad w_t > 0, \quad \bar{\eta}_t > 0, \quad \varepsilon > 0. \tag{A.29}$$

For the original expected-cost sandwich, $w_t = B_t$. The caps specify where the implementation's error and resource bounds hold. A coordinate with no resource-dependent term can be fixed separately; assume here that every coordinate has a positive exponent in at least one term.

THEOREM A.18—Coupled accuracy optimization with an exact certificate: *The upper-model minimization $F_* = \min_{\mathcal{A}} F(\eta)$ has a unique positive minimizer. It is a convex program in $\eta$ and, after $\eta_t = \exp(x_t)$, also has a convex exponential representation. Standard geometric-programming methods can propose an allocation, but their numerical status is not its certificate.*

*For any feasible $\eta^0$, write $g = \nabla F(\eta^0)$. Let $\lambda \geq 0$ and $\nu_t \geq 0$ satisfy*

$$g_t + \lambda w_t + \nu_t \geq 0 \qquad (t < T). \tag{A.30}$$

*Then the independently checkable bounds are*

$$D(\eta^0, \lambda, \nu) = F(\eta^0) - g \cdot \eta^0 - \lambda\varepsilon - \sum_t \nu_t \bar{\eta}_t,$$

$$D(\eta^0, \lambda, \nu) \leq F_* \leq F(\eta^0). \tag{A.31}$$

*At the optimizer, the Karush–Kuhn–Tucker multipliers make these bounds equal. For any candidate, the choice $\nu = 0$ and $\lambda = \max_t\{-g_t / w_t\}$ supplies a valid, possibly loose, lower bound. With integer exponents and rational inputs and candidates, every check is rational arithmetic. For other exponents, certified enclosures of the function and derivatives are required.*

*Put $P = \max_j \sum_t a_{jt}$. Rounding each allowance down by less than a factor two preserves feasibility and gives*

$$F(\eta^{\mathrm{dyad}}) - F_0 \leq 2^P \{F(\eta^0) - F_0\}.$$

*The exponent is the total degree of a coupled term, not merely the exponent of the date at which that term is paid.*

The proof in Supplement CC includes attainment, strict convexity and the certificate. Convex duality and geometric programming are established techniques (Boyd and Vandenberghe, 2004, Boyd et al., 2007); their role here is to preserve the successor dependency in the NBO's original accuracy account. The certificate bounds the optimum of an explicitly stated resource upper model. It is not a lower bound on the work of every algorithm solving the economic problem.

COROLLARY A.19—A constructive horizon bound with coupled evaluation: *Suppose* $w_t = 1$, $\varepsilon/T \le \min\{1, \min_t \bar{\eta}_t\}$, $J \le KT$, $c_j \le C$, *and the total degrees are bounded by a fixed* $P$. *Then a feasible stored-continuation implementation satisfies*

$$F_* \le F_0 + KC\,T^{P+1}\varepsilon^{-P}.$$

*For a dense successor account with label exponent* $p$ *and evaluation exponent* $s$, *the adjacent-date term has degree* $p + s$. *Thus a valid polynomial horizon account charges* $p + s$, *rather than silently holding the successor cost constant. Worsening regularity, growing cover dimension, bit lengths and setup remain explicit inputs; none is bounded by this corollary without its stated hypotheses.*

*A priced two-date allocation.* Consider the exact upper model

$$F(\eta_0, \eta_1) = \eta_0^{-1} + (\eta_0\eta_1)^{-1} + 2\eta_1^{-2}, \qquad \eta_0 + \eta_1 \le \tfrac{1}{2}, \quad \bar{\eta}_t = 1.$$

The equal allocation costs 52. The current numerical appendix gives a rational feasible allocation, its independently recomputed lower certificate and its gap. The example isolates the accuracy dependency algebraically; its resource units are not fitted runtime measurements. It also provides a regression test that rejects a negative multiplier, an infeasible budget or a forged lower bound. In the original economy the same construction applies only after its coefficients have been derived from the actual label, action, integration and stored-circuit accounts.

*Policy and resource certificates.* Qualification determines which policy outputs meet the economic accuracy requirement. The coupled program determines how a specified implementation distributes the corresponding dated allowances. These are different optimizations. A tight resource certificate cannot replace an original-law policy certificate, and a tight policy certificate does not determine hardware time. Keeping both accounts explicit permits constructive improvement without a claim of dimension-free or universally faster computation.

# APPENDIX O: PROSPECTIVE CHOICE AND ECONOMICALLY INTERPRETABLE COMPARISONS

Exact ranking on a completed catalogue and preference before a new workload are different claims. We connect them by a population selection bound and a separately frozen evaluation, retaining the original investment economy and every comparator.

PROPOSITION A.20—Population regret for a fixed service menu: *Condition on construction of $m$ complete services, each with original-law cost in $[0, H]$. For $n$ independent new workloads, obtain valid pathwise cost enclosures $[l_{ji}, u_{ji}]$. Write $\bar{l}_j, \bar{u}_j$ for endpoint means and $q_n = H\sqrt{\log(2m/\delta)/(2n)}$. Select the service minimizing $\lambda \bar{u}_j + p \cdot w_j + f_j$, with a fixed tie rule. Here $\lambda \geq 0$, $p \geq 0$, and the nonrandom resource and fee inputs are declared separately. With probability at least $1 - \delta$, simultaneously for every such price-fee vector, its expected objective regret is at most*

$$\lambda\{2q_n + \max_j(\bar{u}_j - \bar{l}_j)\}.$$

*A separate evaluation stream with simultaneous pairwise intervals $[l_{jk}, u_{jk}]$ gives cost regret at most $\max\{0, \max_k u_{\widehat{j},k}\}$ for any calibration-selected service $\widehat{j}$. This comparison is with the best expected-cost member of the declared menu, not the original Bellman optimum; the original-optimum account remains the separate policy sandwich.*

## O.1. *The frozen independent evaluation*

The prospective protocol preceding the retained evaluation fixes the nine complete two-date investment controllers: the zero controller and both ReLU and Bernstein readouts, under both legacy and calibrated qualification, at acquisition sizes eight and sixteen. Coefficients, complete action tables, constraints and original continuous innovation laws are unchanged. There are 8,192 calibration bins and 131,072 distinct evaluation bins, with forty bits per coordinate. Selection uses the smallest calibration conditioned upper-endpoint mean, with lexical ties, and is fixed before evaluation.

Both independently written whole-bin evaluators run on every policy and bin; rational midpoint fixtures check their hulls. The two estimators are cash cost and analytically conditioned cash cost. Each stream has 36 unordered pairs for each estimator. The empirical-Bernstein error account used in the original study assigns $4\exp(-11)$ to each interval, yielding $288\exp(-11) < 0.005$ per stream and $576\exp(-11) < 0.01$ jointly. No fitting-success event is inferred. The 139,264 paired workload rows count once, rather than once per policy or per comparison. Reconstructing the frozen streams during the current build adds no independent observations.

TABLE O.I

INDEPENDENT EVALUATION: SELECTED FINE CALIBRATED BERNSTEIN MINUS COMPARATOR

| Comparator | Lower endpoint | Upper endpoint |
|---|---|---|
| Zero | −0.177441 | −0.170788 |
| Legacy ReLU, size 8 | −0.010137 | −0.005786 |
| Legacy Bernstein, size 8 | −0.087292 | −0.081642 |
| Legacy ReLU, size 16 | −0.009460 | −0.005046 |
| Legacy Bernstein, size 16 | −0.086152 | −0.080530 |
| Calibrated ReLU, size 8 | −0.009056 | −0.004812 |
| Calibrated Bernstein, size 8 | −0.004823 | −0.000716 |
| Calibrated ReLU, size 16 | −0.003559 | 0.000573 |

Original-law conditioned cost. The table retains the prespecified R85 evaluation checkpoint; the current reproduction regenerates it from exact integer moments. Endpoints are rounded outwards to six decimal places. Selection uses 8,192 calibration rows, not the 131,072 evaluation rows. The simultaneous evaluation family contains 72 intervals across cash and conditioned estimators. The margin is 1/256 in normalized cost units.

Calibration selects the fine calibrated Bernstein controller. In the retained evaluation, its upper expected-cost regret is less than 0.000573, below the prespecified 1/256 noninferiority margin. Seven of its eight comparisons have negative upper endpoints. Its comparison with fine calibrated ReLU crosses zero and supports equivalence at that margin, not strict basis superiority. The resolution comparison within calibrated Bernstein is signed. The crossed basis-by-qualification effect is a separate factorial estimand, not a change in the underlying economy. Full endpoint

moments, all 144 stream-estimator contrasts, sampling and enclosure components, and all comparisons are retained with the reproduction record.

The margin is a declared acceptable normalized loss. It is not an estimated monetary adoption fee. This evaluation does not replace the historical recursive-query workload, and does not assert that the nine acquired controllers are a prospectively validated neural query proposal on every new regime or horizon. The population construction in Section M supplies an explicit broader guarantee under its own joint-domain hypotheses; the two empirical objects are not conflated.

### O.2. *Uniform price-fee sensitivity*

Let $\mathcal{X}$ be a finite set of feasible service assignments, including the original capacity constraints. Let $\mathcal{D}$ be a nonempty compact polytope containing their joint normalized cost vector. For an assignment $x$, let $w_x$ be its complete resource vector, $\lambda \geq 0$ the value of normalized policy loss, $p \geq 0$ the resource prices, and $f$ the assignment fee vector. All coordinates refer to explicit units. Put $h_{\mathcal{D}}(v) = \sup_{c \in \mathcal{D}} c \cdot v$.

PROPOSITION A.21—Uniform purchase region: *Assignment $x$ is optimal for every cost vector in $\mathcal{D}$ exactly on the region*

$$\mathcal{R}_x = \bigcap_{y \in \mathcal{X}} \{(\lambda, p, f) : \lambda h_{\mathcal{D}}(x - y) + p \cdot (w_x - w_y) + f \cdot (x - y) \leq 0\}. \tag{A.32}$$

*This region is polyhedral. Rational support-function upper certificates give an independently checkable sufficient region. Uniformity over a compact polytope of price-fee inputs can be checked at its vertices. The supporting equalities give the breakpoints at which a guaranteed choice can change. A point outside the sufficient region is not automatically evidence that the candidate is inferior.*

For a single selected service and comparator $k$, a simultaneous cost upper bound $u_{\widehat{j},k}$ gives the transparent sufficient condition

$$f_{\widehat{j}} - f_k + p \cdot (w_{\widehat{j}} - w_k) \leq -\lambda u_{\widehat{j},k}.$$

The numerical appendix prints every selected-service breakpoint, including negative ones. In particular, the fine-versus-coarse calibrated Bernstein policy com-

parison supports a positive extra-fee allowance even at zero computation price, whereas a comparison whose upper cost endpoint is positive cannot justify a positive fee solely from policy quality. A resource-driven choice is distinguished from a policy-cost-driven one. Application-specific prices, energy costs and monetary scales require independent calibration; none is manufactured from the normalized tolerance or the favorable timing repetition.

*Evidence identities.* The inherited experiments have distinct observation units. The retained R67 description records 212 service returns and 48 neural fitting warnings; the later training, validation, reuse and vector catalogue records 188 complete processes. The attained-policy factorial study instead has nine controllers and its own independent path rows. The current complete-menu calculation has 2,548 qualification certificates and no new fits. Counts from different catalogues, including the 336-service description in the advisory report, are not added together or assigned to this evaluation without a source-matched manifest. The finer-bin experiment does not erase the older adverse timing and horizon results. Every inherited active section is retained, and the reply identifies the relevant experiment for each contested assertion.

## APPENDIX P: A FROZEN CONTINUATION COMPARISON UNDER RECURSIVE RISK

We apply the preceding construction to $d = 2$ at both declared risk levels. There are four arms: a fitted oblique ReLU continuation, an additive linear-spline continuation, a projected-quadratic continuation, and conventional online evaluation of the original one-date risk Bellman problem. The last arm pays for that calculation; it is not supplied with a free optimal-value oracle. Every fitted label is itself computed from the original terminal primitive and a paid all-action certificate. The terminal continuation is the known $g$, so this last-date fit is consistent with the own-future construction rather than training against an externally supplied optimal continuation.

### P.1. *Frozen design and common accuracy*

Each risk level has 1,024 training states, 128 calibration states and 2,048 evaluation states. The three streams are domain-separated before execution. States are drawn from the declared uniform 20-bit dyadic initial-state distribution; the state dynamics and innovations between decisions remain continuous. Each fitted arm uses the same labels, 32 fixed nonlinear features, an unrestricted affine part and nonnegative nonlinear coefficients. Least squares uses the same solver tolerance and iteration cap; readouts are stored on a 30-bit dyadic grid. The source, serialized coefficients and calibration selector are checked before evaluation. No fitting or reselection occurs on the evaluation stream.

The selector minimizes calibration mean cost-interval midpoints with a fixed lexicographic tie rule. It chooses online risk-DP at both risk levels. All twelve pairwise comparisons across the two levels were specified in advance, including those not involving the selected arm. The independent observation is an initial state, not a shock bin or a repeated solver query.

A common own-continuation gap is insufficient for a fair policy comparison. We require both an own-continuation gap at most $1/64$ and an original-optimum complete-policy loss bound at most $1/16$ at every queried initial state. Fitted arms pay for the extra original-continuation lower witness in Theorem 6; online risk-DP reuses its already computed witness. Integration may refine from 128 to 4,096 bins, but the proposed first action cannot change. All failures are retained rather than silently replaced by conventional actions. The terminal controller is common to every arm and has the uniform allowance proved above.

The ReLU fits have the smallest training RMSE and the narrowest whole-square residual brackets in both cases. These are distinct observations: training errors alone do not give a whole-domain certificate. The quadratic design has effective rank six although all three fitted arms have 35 stored coefficients; matching feature counts is therefore not matching approximation spaces or statistical degrees of freedom. No rank-matched representation superiority is inferred from this comparison.

TABLE P.I

FITTED CONTINUATIONS AND WHOLE-SQUARE RESIDUAL ENCLOSURES

| $\alpha$ | Continuation | Rank | Training RMSE | $V_1 - f$ on the whole square |
|---|---|---|---|---|
| 0.5 | Quadratic | 6 | 0.04122 | [-0.13508, 0.52318] |
| 0.5 | ReLU | 35 | 0.01668 | [-0.04373, 0.08665] |
| 0.5 | Spline | 35 | 0.03979 | [-0.11690, 0.50298] |
| 0.75 | Quadratic | 6 | 0.04166 | [-0.11726, 0.53606] |
| 0.75 | ReLU | 35 | 0.01670 | [-0.04599, 0.09043] |
| 0.75 | Spline | 35 | 0.03984 | [-0.10431, 0.45189] |

Each fit uses the same 1,024 labels and 32 nonlinear features plus an affine part. The quadratic feature matrix has rank six, not 35; the experiment matches nominal features and labels, not independent parameter rank. The residual enclosures cover every state in the continuous square.

## P.2. *A common-policy comparison of certificate accounts*

Let $[l, u]$ be the verified whole-square bracket for $V_1 - f$, $E = \max\{|l|, |u|\}$, and $\delta(x)$ the own-continuation all-action gap. Two valid original-optimum bounds for the same complete policy are

$$B_{\text{norm}}(x) = \delta(x) + \beta(2E + \varepsilon_1), \qquad B_{\text{signed}}(x) = \delta(x) + \beta(u - l + \varepsilon_1).$$

Monotonicity and cash invariance prove these inequalities directly. The joint local-support bound is the paid bound in (21), not a new policy. Table P.II re-accounts for every existing fitted-arm row under all three bounds. At the fixed whole-square cover, neither scalar account meets $1/16$ on any of these rows, whereas the joint certificate meets it on all 12,288 fitted-arm rows. Thus keeping the economically relevant local continuation information matters for qualification, even though it does not establish a neural policy-cost advantage.

This re-accounting was performed after the primary evaluation, with unchanged data, actors and targets. It is a deterministic certificate comparison, not a prospectively registered comparison of optimally tuned centered and uncentered solvers. The scalar accounts pay for the global residual cover; the joint account pays for the original-value witness. Their measured preparation and query work remain

TABLE P.II

THREE VALID CERTIFICATE ACCOUNTS FOR THE SAME FROZEN FITTED POLICIES

| $\alpha$ | Continuation | Supremum norm | Signed scalar | Joint support | Qualified |
|---|---|---|---|---|---|
| 0.5 | Quadratic | 0.981430 | 0.617589 | 0.005501 | 2048/2,048 |
| 0.5 | ReLU | 0.162986 | 0.122750 | 0.021836 | 2048/2,048 |
| 0.5 | Spline | 0.943607 | 0.581660 | 0.012112 | 2048/2,048 |
| 0.75 | Quadratic | 1.005610 | 0.612983 | 0.006422 | 2048/2,048 |
| 0.75 | ReLU | 0.170090 | 0.128429 | 0.022145 | 2048/2,048 |
| 0.75 | Spline | 0.847773 | 0.521923 | 0.008098 | 2048/2,048 |

Entries are maximum original-optimum policy-loss bounds across the same 2,048 rows. Supremum-norm and signed-scalar bounds use the same frozen whole-square residual cover. Joint support uses the paid original-value witness. No row satisfies 1/16 with either scalar account; every row does with the joint certificate. This is post-evaluation deterministic re-accounting, not a prospective runtime ablation or an optimality claim for any certificate-construction algorithm.

recorded separately, so certificate width is not converted into an assumed runtime saving.

### P.3. *Original policy loss and economic accuracy*

All 16,384 arm-state records satisfy both qualification targets. The largest original-optimum loss bound among them is less than 0.022145, below the common 0.0625 target. This is a statement about every recorded initial state, not an empirical proof of that same tolerance on all possible initial states. The separate residual cover qualifies the whole continuous state square.

The estimand is the initial-distribution mean of each complete policy's original nested-risk value. It is not average realized cash expenditure or a monetary welfare estimate. The fixed margin $1/16$ equals four fifths of the maximal first-date investment charge $h(1/4) = 5/64$ in this normalized model. It is a declared numerical economic tolerance; judging it small for a particular application requires an external economic calibration.

All twelve simultaneous empirical-Bernstein intervals lie strictly inside $[-1/16, 1/16]$. Thus all four complete policies are equivalent at that prespecified

TABLE P.III

ORIGINAL NESTED-RISK COST AND QUALIFICATION AT 2,048 EVALUATION STATES PER RISK LEVEL

| $\alpha$ | Continuation | Mean cost enclosure | Maximum loss bound | Failures |
|---|---|---|---|---|
| 0.5 | Online risk-DP | [0.625634, 0.632502] | 0.003435 | 0 |
| 0.5 | Quadratic | [0.625754, 0.632622] | 0.005501 | 0 |
| 0.5 | ReLU | [0.626391, 0.633259] | 0.021836 | 0 |
| 0.5 | Spline | [0.625792, 0.632660] | 0.012112 | 0 |
| 0.75 | Online risk-DP | [0.625663, 0.632531] | 0.003435 | 0 |
| 0.75 | Quadratic | [0.625808, 0.632676] | 0.006422 | 0 |
| 0.75 | ReLU | [0.626600, 0.633468] | 0.022145 | 0 |
| 0.75 | Spline | [0.625805, 0.632674] | 0.008098 | 0 |

Cost enclosures here contain numerical uncertainty only. Simultaneous sampling intervals for all paired differences appear in Table BQ.I. The maximum loss bound is against the original Bellman optimum; all displayed endpoints are rounded outwards. A failure is a failed own-continuation or original-optimum target, with every row retained.

tolerance on the declared initial distribution. None of the intervals excludes zero, so none establishes strict superiority. In particular, the narrower ReLU residual bracket does not translate here into an identified improvement over online risk-DP. The numerical mean-cost brackets in Table P.III are not themselves sampling confidence intervals; the full paired intervals and their sampling components are reported in the appendix.

The numerical component includes original-law integration, terminal observation and optimization, and outward arithmetic. Initial-state discretization is part of the declared evaluation distribution, not an unreported approximation to a continuous uniform initial distribution. There is no transition-law substitution by a state grid, and therefore no hidden state-aggregation allowance in these costs. Sampling and numerical uncertainty are combined as proved in the supplement. The protocol does not permit changing the margin or sample size after examining the evaluation outcomes.

TABLE P.IV

MEASURED WORK OF THE COMPLETED EVALUATION ATTEMPT

| $\alpha$ | Continuation | Seconds | Terminal proposals | All-action witnesses |
|---|---|---|---|---|
| 0.5 | Online risk-DP | 32.509 | 112,721,920 | 2,048 |
| 0.5 | Quadratic | 12.254 | 13,631,488 | 4,096 |
| 0.5 | ReLU | 11.797 | 13,631,488 | 4,096 |
| 0.5 | Spline | 11.442 | 13,631,488 | 4,096 |
| 0.75 | Online risk-DP | 30.770 | 112,721,920 | 2,048 |
| 0.75 | Quadratic | 11.746 | 13,631,488 | 4,096 |
| 0.75 | ReLU | 11.237 | 13,631,488 | 4,096 |
| 0.75 | Spline | 11.523 | 13,631,488 | 4,096 |

Every arm processes all 2,048 initial states. Timings include first-action proposals, qualification, the original-value witness, complete-policy cost enclosures and CSV output. They exclude preparation, the independent audit, diagnostic screening and failed earlier execution starts, all reported separately. Counts are instrumented work units, not every arithmetic instruction. This is a single x86-64 execution, not a cross-architecture speed guarantee.

## P.4. *Verification work and the value of storage*

In the completed execution, the fitted arms require approximately 11.2–12.3 seconds per 2,048-state risk-level workload, compared with approximately 30.8–32.5 seconds for online risk-DP. These measurements include the extra original-optimum witness required of the fitted arms. Preparation for the entire two-risk-level, six-fit comparison takes approximately 8.79 seconds. A separately coded rational audit takes approximately 89.98 seconds and checks the continuous-risk kernels and retained evidence; that verification is a separate paid campaign cost, not zero-cost validation.

These measurements identify a benefit of stored continuations on this workload, not a uniquely neural advantage. Spline and quadratic continuations obtain similar execution reductions, and the fitted arms are not rank matched. The counts identify the repeated terminal computations avoided by storage; they do not identify the cost of every machine instruction. The reported timings are from one x86-64 execution. Two earlier tool-interrupted starts are recorded separately, without recoverable

complete cost ledgers. Accordingly the table is not a claim about the aggregate cost of every attempted run, nor a cross-architecture, energy or asymptotic scalability result.

Whole-bin screening excludes between 738 and 746 of 4,096 candidates at $\alpha = 1/2$, and between 2,053 and 2,062 at $\alpha = 3/4$. After paying for unpruned original-value bounds and the strict test, the screened calculation is slightly slower in every displayed case. The exact unchanged tail values validate the exclusions; they do not establish a computational gain. No-cut rows and all denominators are retained in the appendix. The experiment therefore separates a valid geometric certificate, the amount it removes and the net value of constructing it.

The two-date scalar-control experiment does not replace the inherited two-control, two-innovation horizon frontier. Proposition A.14 now gives explicit fixed-accuracy resolutions and a horizon-uniform construction for both the scalar and original vector economies. The general NBO construction and joint-domain theorem remain available under their hypotheses, while a full independent largest-vector implementation remains a separate numerical obligation. This distinction preserves the general research program without borrowing scalability evidence from a different problem.

## APPENDIX Q: FROZEN NEURAL CONTINUATIONS AND POLICY-COST PRECISION

The optimal-advantage estimator in Section BV makes a policy comparison informative, but its conditional optimal-value queries are expensive. A fitted continuation can replace those queries without changing the estimand. The necessary certificate concerns its Bellman residual across states, not its distance from an unknown optimizer. This places the original NBO continuation, rather than only a proposed action, inside the evaluation calculation.

### Q.1. *Residual oscillation is the relevant range*

Consider the bounded expected-cost model with common terminal cost $g$, discounts $\beta_t \geq 0$, and a common initial-state law. Write $b_0 = 1$ and $b_t = \prod_{j<t} \beta_j$. Fix bounded measurable continuations $h_0, \ldots, h_T$, conditionally on their construction.

Define

$$Q_t^h(x,a) = c_t(x,a) + \beta_t P_t^a h_{t+1}(x),$$
$$r_t^h(x) = \inf_{a \in A_t(x)} Q_t^h(x,a) - h_t(x),$$
$$a_t^{h,\pi}(x) = Q_t^h(x, \pi_t(x)) - \inf_{a \in A_t(x)} Q_t^h(x,a).$$

The infimum is over the original continuous feasible set. Suppose verified constants give $0 \le a_t^{h,\pi} \le \eta_t^\pi$, and likewise for $\nu$, throughout the true state domain. Let $\omega_t$ bound the oscillation of $r_t^h$, and $\omega_T$ that of $g - h_T$. Set

$$\Omega_h = \sum_{t<T} b_t \omega_t + b_T \omega_T, \qquad H_h^\pi = \sum_{t<T} b_t \eta_t^\pi.$$

These constants include acquisition, action rounding and arithmetic allowances wherever those operations affect the implemented policy or the verified functions. A small empirical residual alone is not such a certificate.

THEOREM A.22—Frozen-continuation comparison: *Along the actual trajectory of $\pi$, put*

$$Z_h^\pi = \sum_{t<T} b_t \left[ Q_t^h(X_t^\pi, \pi_t(X_t^\pi)) - h_t(X_t^\pi) \right] + b_T (g - h_T)(X_T^\pi). \tag{A.33}$$

*For any coupling of the two original trajectory laws, $D_h = Z_h^\pi - Z_h^\nu$ satisfies*

$$\mathbb{E} D_h = \mathbb{E}[J_0^\pi - J_0^\nu], \tag{A.34}$$
$$-\Omega_h - H_h^\nu \le D_h \le \Omega_h + H_h^\pi. \tag{A.35}$$

*Moreover, at every initial state,*

$$0 \le J_0^\pi - V_0^* \le \Omega_h + H_h^\pi. \tag{A.36}$$

*Adding an arbitrary state-independent constant to each $h_t$ changes neither $D_h$ nor these oscillation bounds. With $h_t = V_t^*$, the result reduces to optimal-advantage centering.*

The proof is in Supplement, Section BY.1.

The terminal correction in (A.33) is essential unless $h_T = g$. Conditional expectation, not pathwise equality with the realized cash-cost difference, supplies (A.34). The result uses the actual kernels and costs; it assumes neither a correct statistical transition model estimated from data nor a convex neural fit. The pointwise policy certificate also applies conditionally after an adaptive choice among valid continuations. The independent-sample confidence statement below requires its additional sampling conditions.

For $M$ conditionally fixed comparisons, suppose independent trajectory rows yield outward enclosures $[L_i, U_i]$ of $D_h$, intersected with its valid support. Write $R_h = 2\Omega_h + H_h^\pi + H_h^\nu$. A simultaneous interval with failure probability at most $\alpha$ is

$$\left[\overline{L} - R_h\sqrt{\frac{\log(2M/\alpha)}{2n}},\quad \overline{U} + R_h\sqrt{\frac{\log(2M/\alpha)}{2n}}\right]. \tag{A.37}$$

This follows by the bounded-sum inequality applied to both endpoint samples and a union bound. The numerical and sampling width allocations of Corollary A.64 apply with $R_h$. Fitting, uniform residual qualification, conditional integration and any subsequent selection are distinct paid operations. In particular, a cache that adapts across rows changes the controller's history: its rows cannot be treated as independent evaluations of one frozen Markov policy. A fixed cache or an independent episode-reset design is sufficient to retain the stated sampling construction.

### Q.2. *Conditional integration of an actual neural continuation*

A shallow rational affine–ReLU continuation has the form

$$h(y) = c + \ell^\top y + \sum_{j=1}^{K} v_j (w_j^\top y + u_j)_+.$$

Its readout weights may have either sign. In the original investment laws the next state, conditional on $(x, a)$, is affine in one or two independent centered uniform innovations. A neuron therefore has projected center $z$ and nonnegative projection radii $A, B$. Write $z_+ = \max\{0, z\}$.

PROPOSITION A.23—Exact conditional ReLU moments: *For independent uniform projections of radii $A, B > 0$, the conditional means of a neuron are*

$$\mathcal{M}_1(z; A) = \frac{(z + A)_+^2 - (z - A)_+^2}{4A}, \tag{A.38}$$

$$\mathcal{M}_2(z; A, B) = \frac{1}{24AB}\big[(z + A + B)_+^3 - (z + A - B)_+^3 - (z - A + B)_+^3 + (z - A - B)_+^3\big]. \tag{A.39}$$

*A zero projection radius is removed; if all radii vanish the mean is $z_+$. Consequently $P^a h$ is a finite exact rational calculation for rational primitives, states, actions and weights. For $n$ trajectories and at most $K$ neurons per date, its arithmetic-operation count is $O(nTKd)$, exclusive of fitting, trajectory construction and uniform qualification. No optimal-value descendant query occurs in this calculation.*

The proof is in Supplement, Section BY.2.

The operation count does not bound rational bit complexity. Near-zero projected radii can also make a naive floating evaluation unstable; directed rounding or rational evaluation, including the zero-radius branches, is required. The executed moment catalogue checks the rational identities against an independently written polygon integral. It is a conditional-integration experiment, not a new trained-policy cost comparison. The formulas do not imply that the achieved residual oscillation of an arbitrary fitted network is small.

### Q.3. *Qualifying a fitted continuation without a Bellman tree*

Uniform qualification need not query $V^*$. Suppose $Q_t^h$ is Lipschitz in state and action with constants $L_{x,t}, L_{a,t}$, and $h_t$ has constant $L_{h,t}$, all in $\ell^1$. In the investment capacity simplex, an action feasible at $y$ can be repaired continuously at $x$ with displacement at most $\kappa\|x - y\|_1$, where $\kappa = 1/(8d)$. Thus a valid residual Lipschitz constant is

$$L_{r,t} = L_{x,t} + \kappa L_{a,t} + L_{h,t}.$$

Let a finite state cover have fill radius $\rho$, and let $[l_j, u_j]$ enclose $r_t^h$ at every cover point. Then

$$\min_j l_j - L_{r,t}\rho \le r_t^h(x) \le \max_j u_j + L_{r,t}\rho \tag{A.40}$$

throughout the cube. At a cover point, a feasible action mesh of fill radius $s$ gives a lower infimum bound by subtracting $L_{a,t}s$ from the smallest action-value lower endpoint; the smallest upper endpoint is an upper bound. Exact moments can supply those action endpoints. None of these steps requires convexity of $h$ or a true-value curvature minorant for $Q^h$.

For $N_t$ state-cover points and $A_t$ actions per point, this qualification costs $O(\sum_t N_t A_t K_{t+1} d)$ arithmetic operations, plus certification of the moduli and any actor-specific greedy-gap bound. Its cover can still be expensive in high dimension. The result replaces a recursive optimal-value oracle by an explicit approximation-and-qualification tradeoff; it does not make that qualification free. The residual-cover proof, the action-mesh argument and the terminal allowance are given in the supplement. Value-function control variates have an established variance-reduction literature (Munos, 2006); the additional requirement here is an original-law, whole-domain certificate for the fitted NBO continuation and the actual acquired policy.

## APPENDIX R: FITTING A CONTINUATION FOR QUALIFICATION

A continuation used to evaluate a policy should be fitted to the error that determines the evaluation cost. The residual-cover bound in (A.40) supplies such an objective. We now make its optimization constructive. The future continuation is held fixed at each backward step; neither an optimal-value oracle nor success of unrestricted hidden-layer training is assumed.

Fix a date and suppress its subscript. Let $x_1, \dots, x_N$ be a state cover of radius $\rho$ in the $\ell^1$ metric, and suppose

$$y_j^- \le (\mathcal{T}h_+)(x_j) \le y_j^+.$$

These are verified bounds on the original continuous-action infimum, not only on a proposal menu. Write $h_\theta = \sum_{k=1}^K \theta_k \phi_k$, where the fixed features have verified

Lipschitz constants $L_k$ and $|\theta_k| \le B_k$. The constant feature is allowed with $L_k = 0$. Let $C$ be a verified Lipschitz constant of $\mathcal{T}h_+$, including state-dependent feasibility repair. With $F_{jk} = \phi_k(x_j)$, define

$$\mathcal{W}(\theta) = \max_j (y_j^+ - F_j\theta) - \min_j (y_j^- - F_j\theta) + 2\rho\left(C + \sum_k L_k|\theta_k|\right). \tag{A.41}$$

THEOREM A.24—Qualification-optimal readout: *For every feasible* $\theta$, $\operatorname{osc}(\mathcal{T}h_+ - h_\theta) \le \mathcal{W}(\theta)$. *Minimizing* (A.41) *is a finite linear program with* $2K+2$ *variables and* $2N+2K$ *inequality constraints, apart from variable bounds. For probability vectors* $p, q$ *on the nodes, set* $\lambda_k = 2\rho L_k$ *and*

$$\mathcal{D}(p,q) = p^\top y^+ - q^\top y^- + 2\rho C - \sum_k B_k\big(|[F^\top(q-p)]_k| - \lambda_k\big)_+. \tag{A.42}$$

*Then* $\mathcal{D}(p,q) \le \inf_\theta \mathcal{W}(\theta)$, *with equality for some* $p, q$. *Thus any feasible readout and exact probability vectors certify an achieved optimization gap of at most* $\mathcal{W}(\theta) - \mathcal{D}(p,q)$.

The proof is in the Technical Supplement. It constructs the primal constraints and dual support function explicitly. Rationalizing a solver's proposed readout and probability vectors gives a checkable certificate; a convergence flag is not the certificate. Label intervals, features, moduli, coefficient bounds and the cover remain part of the numerical contract.

Applying the theorem backwards gives a fitted continuation at every date. Its achieved ranges $\mathcal{W}_t$, together with the terminal oscillation and the actual implemented greedy-gap bounds, can be substituted directly into Theorem A.22. This connects the attained fitting objective to both original-optimum regret and the range of the original-law cost observation. Each readout minimizes its stated one-step qualification bound conditional on the fixed future. It does not minimize the joint multi-date range, the true policy loss, or a hidden-feature objective. The next backward step recomputes its labels and moduli from the continuation actually returned.

The label construction in (A.40) uses exact conditional moments for a shallow ReLU future and a verified continuous-action cover. Its work is charged in addition to the linear program, coefficient rationalization, modulus calculation and dual

check. A conventional basis with the same number of coefficients receives the same labels and the same objective; it uses its own valid feature moduli. This is a comparison of qualified approximations, not an assertion that different bases span the same functions or that a neural encoding has an intrinsic advantage.

## APPENDIX S: REPRESENTATION CALIBRATION AND OPTIMIZATION-STABLE POLICIES

The attained continuation in Section Q is an economic object, not merely an optional action proposal. Its qualification must charge the represented function rather than an avoidable featurewise overestimate. We make this comparison explicit and then pass from an attained fit to a class of optimization outputs with the same implemented policy. All statements in this section retain the original two-date investment law, fixed terminal payoff, acquired cells and feasible action menus.

### S.1. *A common affine calibration*

Write a two-state additive continuation as

$$h_\theta(x) = \theta_0 + \sum_{j=1}^{4} \theta_{1j}\psi_j(x_1) + \sum_{j=1}^{4} \theta_{2j}\psi_j(x_2).$$

For the ReLU basis, $\psi_j(s) = (s - (j-1)/4)_+$. For the Bernstein basis, $\psi_j(s) = \binom{4}{j} s^j (1-s)^{4-j}$; the omitted zeroth coefficient on each coordinate is zero. Both representations have nine coefficients and use the same coefficient box. Define

$$L_R(\theta) = \max_{i=1,2} \max_{k=1,\dots,4} \left| \sum_{j=1}^{k} \theta_{ij} \right|, \tag{A.43}$$

$$L_B(\theta) = 4 \max_{i=1,2} \max_{j=1,\dots,4} |\theta_{ij} - \theta_{i,j-1}|, \qquad \theta_{i0} = 0. \tag{A.44}$$

These are state Lipschitz moduli for the $\ell^1$ distance. The first is exact for the represented piecewise-affine function; the second is a derivative-control bound and need not be exact for a general polynomial.

Suppose the fixed-label Bellman value at nodes $x_i$ belongs to $[y_i^-, y_i^+]$, every state is within $\ell^1$ distance $\rho$ of a node, and $G$ bounds the Bellman target's state modulus.

With $\phi_i^\top \theta = h_\theta(x_i)$, put

$$F(\theta) = \max_i (y_i^+ - \phi_i^\top \theta) - \min_i (y_i^- - \phi_i^\top \theta) + 2\rho\{G + L(\theta)\}. \tag{A.45}$$

Here the labels, $G$, the cover and the feature map are fixed before this readout is optimized. A change in the fitted future defines a new fixed-label problem.

PROPOSITION A.25—Affine-exact qualification: *The moduli in* (A.43)–(A.44) *give valid residual enclosures and the residual-oscillation upper bound* (A.45)*. The objective is convex and admits a finite linear-programming representation. For every common affine continuation* $h(x) = a + ux_1 + vx_2$*, whenever its two coefficient representations are feasible, both moduli equal* $\max\{|u|, |v|\}$*. Thus, for common labels, node cover and target modulus, the two qualification objectives agree on this affine function.*

PROOF: A ReLU coordinate has the displayed cumulative slopes on its four affine intervals. The dual norm of $\ell^1$ is $\ell^\infty$, so their largest coordinate magnitude gives the state modulus. A Bernstein derivative is a convex combination of $4(\theta_{ij} - \theta_{i,j-1})$ with degree-three Bernstein weights. This proves its bound. Residuals change by at most $(G + L)\rho$ between a node and a covered state, proving the enclosure and its oscillation bound. Each term in (A.45) is a maximum of finitely many affine functions. For an affine function, the ReLU coefficients are $(u, 0, 0, 0)$ and $(v, 0, 0, 0)$; the Bernstein coefficients are $(u/4, u/2, 3u/4, u)$ and the analogous vector for $v$. All relevant slopes or derivative controls are $u$ or $v$. *Q.E.D.*

This calibration does not identify the two non-affine function classes, equalize every numerical conditioning effect, or remove all conservatism from Bernstein controls. It isolates a particular qualification difference. For example, the old coefficientwise modulus charges $ux_1$ as $|u|$ in its ReLU encoding but as $10|u|$ in its Bernstein encoding. The calibrated modulus charges both as $|u|$. The factorial experiment below retains the old fits and evaluates the new fits separately; the calibration is not an ex post correction of their recorded costs.

## S.2. *Exact certificates for action stability*

Fix the terminal-label qualification problem and an attained coefficient vector $\widehat{\theta}$, with $U = F(\widehat{\theta})$. On each first-date observation cell $C$, let $a_C$ be its attained robust greedy action, evaluated at the declared cell representative. The terminal policy is fixed by the common terminal payoff. For each different action $b$ in the same robust menu, exact conditional integration gives an affine contrast

$$g_{Cb}(\theta) = Q^{h_\theta}(x_C, b) - Q^{h_\theta}(x_C, a_C) = c_{Cb} + d_{Cb}^\top \theta.$$

The switch problem minimizes $F(\theta)$ on the coefficient box subject to $g_{Cb}(\theta) \le 0$. Let $D_{Cb}$ be any verified lower bound on its infimum. If the constraint is infeasible, its infimum is $+\infty$. Define $D_C = \min_{b \neq a_C} D_{Cb}$ and

$$\mathcal{S}_\epsilon = \{C : D_C > U + \epsilon\}, \qquad \epsilon \ge 0. \tag{A.46}$$

Strict inequality excludes ties as well as strict reversals.

THEOREM A.26—Optimization-to-policy transfer: *Every feasible readout satisfying $F(\theta) \le U + \epsilon$ selects exactly $a_C$ on every cell in $\mathcal{S}_\epsilon$. In particular this holds for every $\epsilon$-optimal solution of the fixed-label qualification problem, because its optimum is at most $U$.*

*Let $q_\epsilon$ be the initial-law probability of the cells outside $\mathcal{S}_\epsilon$. Suppose the total cash cost of every such complete feasible controller belongs to $[0, H]$. With the terminal policy fixed as above,*

$$\sup_{F(\theta) \le U + \epsilon} |J^{\pi_\theta} - J^{\pi_{\widehat{\theta}}}| \le H q_\epsilon. \tag{A.47}$$

*If all cells are certified, every readout in that sublevel gives the same complete policy and the same original-law cost. Under the uniform initial law on $[0,1]^2$ and an $n$-by-$n$ equal-cell cover, $q_\epsilon$ is exactly the fraction of uncertified cells; boundary assignments do not change that probability.*

PROOF: If $b$ weakly displaced $a_C$, the readout would be feasible for its switch problem and hence $F(\theta) \ge D_{Cb} \ge D_C$. This contradicts $F(\theta) \le U + \epsilon < D_C$. Couple the attained and alternative controllers with the same initial state and innovations.

On a certified initial cell their first actions agree, their next states agree, and their fixed terminal-policy actions agree. Their cash costs are identical there. Elsewhere the absolute cash-cost difference is at most $H$, giving (A.47) by integration. *Q.E.D.*

The theorem does not replace failed switch certificates by successful ones: a nonpositive surplus is unresolved. Nor does a terminal-label certificate automatically propagate through newly fitted label problems at a longer horizon. The numerical result here concerns exactly the two-date construction for which only $h_1$ determines the first-date action and the terminal actor is common.

The switch bounds have an exact dual representation. Write $L(\theta) = \max_j |s_j^\top \theta|$, coefficient limits $|\theta_k| \le B_k$, and $\alpha = 2\rho$. For probability vectors $p, q$, nonnegative multipliers $\xi_j^+, \xi_j^-$ with $\sum_j (\xi_j^+ + \xi_j^-) \le \alpha$, and $\nu \ge 0$, set

$$z = \sum_i (q_i - p_i)\phi_i + \sum_j (\xi_j^+ - \xi_j^-)s_j + \nu d_{Cb},$$

$$D(p, q, \xi, \nu) = p^\top y^+ - q^\top y^- + 2\rho G + \nu c_{Cb} - \sum_k B_k |z_k|. \tag{A.48}$$

Every such number is a valid $D_{Cb}$. The proof and independent exact replay are given in Appendix BD. A floating optimizer supplies proposed multipliers only; their nonnegativity, normalization, total mass and lower-bound value are verified rationally.

## S.3. *Simultaneous economic conclusions for optimizer classes*

Suppose the frozen attained-policy comparison supplies a simultaneous interval $[l_{jk}, u_{jk}]$ for $J^{\pi_j} - J^{\pi_k}$. Apply Theorem A.26 separately to their fixed-label sublevels, with radii $v_j = H_j q_j$ and $v_k = H_k q_k$. Then, on the same event,

$$J^{\pi_\theta} - J^{\pi_\vartheta} \in [l_{jk} - v_j - v_k,\ u_{jk} + v_j + v_k] \tag{A.49}$$

simultaneously for every pair of readouts in those sublevels. No union bound over an uncountable optimizer class is needed: the extension is deterministic on the original finite-policy event. When both covers are entirely stable the interval is unchanged. Sharp controller-choice and deployment-allocation results in Section AE

can therefore be applied directly to these qualified optimizer classes after the stated enlargement. This transfers attained-cost evidence without identifying numerical fitting success with either population training reliability or a new economic observation.

## APPENDIX T: ECONOMIC ACCURACY OF APPROXIMATELY OPTIMIZED CONTINUATIONS

A qualification tolerance need not preserve every acquired action to preserve an economic conclusion. The bound in Theorem A.26 charges the full cost range on each uncertified cell. We now retain the particular actions that remain possible and the direction of their economic effects. This gives a cost guarantee for an entire optimization sublevel, including approximately optimal readouts, without treating an inconclusive switch certificate as an observed policy failure.

Fix a finite first-date acquisition partition $\mathcal{C}$, its initial-law masses $p_C$, a robust finite action menu on each cell, and a common feasible continuation policy $\tau$ after that date. These objects are held fixed throughout the comparison. The horizon may be any finite horizon for which the conditional costs below are enclosed. Use the fixed-label qualification objective $F$, attained readout $\widehat{\theta}$, value $U = F(\widehat{\theta})$, reference action $a_C$, and switch lower bounds $D_{Cb}$ from Section S. For $\epsilon \geq 0$, define

$$\mathcal{A}_C(\epsilon) = \{a_C\} \cup \{b \neq a_C : D_{Cb} \leq U + \epsilon\}. \tag{A.50}$$

The weak inequality deliberately retains ties. Let $\pi_\theta$ use a minimizing critic action at each cell representative and then $\tau$. Write $Q^\tau(x,b) = c_0(x,b) + \beta_0 P_0^b J_1^\tau(x)$, and, on every positive-mass cell, define the original-law contrast

$$\Delta_C(b) = \mathbb{E}[Q^\tau(X_0,b) - Q^\tau(X_0,a_C) \mid X_0 \in C].$$

Suppose a finite calculation provides $\ell_{Cb} \leq \Delta_C(b) \leq u_{Cb}$ for every retained action, with both endpoints zero for $a_C$. Set

$$\underline{v}_\epsilon = \sum_C p_C \min_{b \in \mathcal{A}_C(\epsilon)} \ell_{Cb}, \qquad \overline{v}_\epsilon = \sum_C p_C \max_{b \in \mathcal{A}_C(\epsilon)} u_{Cb}. \tag{A.51}$$

Zero-mass cells make no contribution.

THEOREM A.27—Actionwise optimization-to-cost transfer: *Under the preceding conditions, every feasible $\theta$ satisfying $F(\theta) \leq U + \epsilon$ obeys*

$$\underline{v}_\epsilon \leq J^{\pi_\theta} - J^{\pi_{\hat{\theta}}} \leq \overline{v}_\epsilon. \tag{A.52}$$

*In particular the conclusion holds for every $\epsilon$-optimal readout. If all complete costs are in $[0, H]$ and the actionwise intervals are intersected with $[-H, H]$, then*

$$-Hq_\epsilon \leq \underline{v}_\epsilon \leq 0 \leq \overline{v}_\epsilon \leq Hq_\epsilon,$$

*where $q_\epsilon$ is the initial mass of cells retaining a nonreference action. For two such optimizer classes, an attained-policy simultaneous interval $[l_{jk}, u_{jk}]$ gives, on the same event, the classwide interval*

$$[\, l_{jk} + \underline{v}_j - \overline{v}_k, \quad u_{jk} + \overline{v}_j - \underline{v}_k \,]. \tag{A.53}$$

*These conclusions require no probability bound over the readouts in the sublevels.*

PROOF: Any critic-minimizing action $b \neq a_C$ satisfies $g_{Cb}(\theta) \leq 0$, so $D_{Cb} \leq F(\theta) \leq U + \epsilon$. It therefore belongs to (A.50). Conditioning on the initial cell expresses the policy-cost difference as $\sum_C p_C \Delta_C(b_C(\theta))$, which is between the endpoints in (A.51). The infimum of $F$ is at most $U$, giving the optimization conclusion. A cell retaining only $a_C$ contributes zero; the signed support bounds every other contribution. Finally add and subtract the two attained costs to obtain (A.53). All steps after the attained-policy event are deterministic. *Q.E.D.*

The cellwise extrema relax the common-coefficient restriction and need not be attained by one readout. The theorem is consequently not a sharp joint characterization of the optimizer class. Its useful feature is different: it can retain an identified economic comparison when some actions remain uncertified and the binary $Hq_\epsilon$ allowance is too large. The original Bellman optimum and the constructive accuracy results are unchanged. A reference expected-loss upper bound can also be enlarged by $\overline{v}_\epsilon$; the actionwise comparison itself is not a new bound obtained by replacing the original optimum with a menu optimum.

For the executed two-date economy, the conditional contrasts are computed from the original stage costs, transitions and fixed terminal actor. Common innovations

preserve their actual marginal laws. Centered polynomial differences retain cancellation, and rectangle queries enclose every crossed acquired cell. The final innovation is integrated analytically. Appendix BF proves whole-bin validity, deterministic refinement convergence under vanishing arithmetic error, and the work bound. It also separates integration accuracy from the independent statistical interval to which (A.53) is applied.

## APPENDIX U: ECONOMICALLY SUFFICIENT OPTIMIZATION ACCURACY

The qualification objective and the economic loss have different units. A numerical stopping rule should therefore be connected to the decision it supports, rather than chosen by equating their tolerances. The actionwise transfer supplies this connection for every tolerance in an interval, not only for a finite list of optimizer outputs.

For construction $j$, fix the labels, attained value $U_j$, action-switch bounds and continuation policy of Theorem A.27. Fix $e_j > 0$ and enclose every action with $D_{jCb} \le U_j + e_j$. Write $[v_j^-(\epsilon), v_j^+(\epsilon)]$ for its class-cost band. The only possible changes of this band on $[0, e_j]$ occur at

$$\mathcal{K}_j = \{0, e_j\} \cup \{D_{jCb} - U_j : 0 < D_{jCb} - U_j \le e_j\}.$$

At a knot the weak switch inequality is already active. Thus the band is constant on each interval between successive knots, with the left endpoint included and the right endpoint excluded; the final endpoint is evaluated separately if it is itself a switch knot. No interpolation across a knot is permitted.

Let $[l_{jk}, u_{jk}]$ be an attained-cost interval on the original simultaneous event. Define

$$L_{jk}(\epsilon, \eta) = l_{jk} + v_j^-(\epsilon) - v_k^+(\eta), \tag{A.54}$$

$$U_{jk}(\epsilon, \eta) = u_{jk} + v_j^+(\epsilon) - v_k^-(\eta). \tag{A.55}$$

For an additional charge $\tau \ge 0$ for construction $k$, and a declared economic equivalence margin $m > 0$, put

$$\mathcal{R}_{jk}(\tau) = \{(\epsilon, \eta) : L_{jk}(\epsilon, \eta) > \tau\},$$

$$\mathcal{E}_{jk}(m) = \{(\epsilon,\eta) : -m < L_{jk}(\epsilon,\eta),\ U_{jk}(\epsilon,\eta) < m\}.$$

Both sets are restricted to $[0,e_j] \times [0,e_k]$. A common-tolerance comparison is one diagonal slice; different constructions need not receive identical tolerances.

THEOREM A.28—Economic stopping frontier: *On the attained-policy simultaneous event, every pair of feasible readouts satisfying*

$$F_j(\theta) \le U_j + \epsilon, \qquad F_k(\vartheta) \le U_k + \eta$$

*has cost difference in* $[L_{jk}(\epsilon,\eta), U_{jk}(\epsilon,\eta)]$*, simultaneously for all tolerances in the stated rectangle. Consequently* $\mathcal{R}_{jk}(\tau)$ *certifies a strictly beneficial replacement by* $k$ *after its additional charge, and* $\mathcal{E}_{jk}(m)$ *certifies a cost difference strictly within the margin for the entire pair of readout classes.*

*These two sufficient regions are downward closed in their tolerance coordinates and are finite unions of knot rectangles with the stated endpoint conventions. Their maximal extent relative to the supplied actionwise bands can be obtained without a further optimizer call or cost observation. In particular, if a feasible numerical output has a verified primal–dual gap at most* $\epsilon$*, it belongs to the corresponding class and may be returned whenever its gap pair is in the required region. Choosing the stopping tolerance after examining the simultaneous intervals does not require an additional probability allowance.*

PROOF: As a tolerance increases, its possible-action set expands. Hence $v_j^-$ is nonincreasing and $v_j^+$ is nondecreasing. Each set changes only at the displayed switch knots, where equality retains the newly possible actions. Theorem A.27 gives (A.54)–(A.55) for every readout and every tolerance by the same deterministic argument. The lower endpoint decreases and the upper endpoint increases in either tolerance. This proves downward closure and the finite-rectangle representation, including the strict economic tests at their boundaries.

If $d_j \le \inf F_j$ is a verified dual value and $F_j(\theta) - d_j \le \epsilon$, then $F_j(\theta) \le \inf F_j + \epsilon \le U_j + \epsilon$. The same argument applies to $k$. All statements therefore hold for any stopping output satisfying these verified inequalities on the original event. Nothing in this implication requires a union bound over the outputs or tolerances. *Q.E.D.*

A boundary of a sufficient region is not an identified failure boundary of the true optimizer class. The cellwise relaxation can combine actions that no common coefficient vector selects. Nor does a strict region necessarily include its supremum tolerance: at an exact switch knot, the added action can destroy the desired strict economic inequality. Appendix BG gives an exact finite algorithm, checks both sides of every knot, and separates this stopping guarantee from a distributional claim about neural fitting success.

## APPENDIX V: COMMON READOUTS AND ECONOMIC POLICY QUALIFICATION

An approximately optimized continuation is one function. Its coefficients determine the action in every observation cell. Separate action-switch tests deliberately forget part of this structure: an alternative may defeat the reference action without winning the complete feasible menu, and individually possible choices need not be jointly generated by the same readout. We retain these restrictions when transferring the original NBO qualification account to an economic decision. The attained continuation, economic primitives, action menu and policy-cost observations are unchanged.

### V.1. *Winning actions and compatible choices*

Fix the finite acquisition partition $\mathcal{C}$, its original-law masses $p_C$, and a robust finite action menu $\mathcal{A}_C$. Each action in $\mathcal{A}_C$ is feasible throughout $C$. Let $\Theta = \prod_{j=1}^{d}[-B_j, B_j]$ be the declared readout box. With fitted labels and continuation policy $\tau$ held fixed, the implemented first-date rule minimizes

$$q_{Cb}(\theta) = c_{Cb} + v_{Cb}^{\mathsf{T}}\theta, \qquad b \in \mathcal{A}_C,$$

using the original deterministic tie order. This is the fixed-feature own-future readout used in the preceding qualification construction, not an additional policy-value oracle. Write $F$ for its qualification objective, $\widehat{\theta}$ for the attained feasible readout, $U = F(\widehat{\theta})$, and

$$\Theta_\epsilon = \{\theta \in \Theta : F(\theta) \le U + \epsilon\}, \qquad \epsilon \ge 0.$$

The reference action in $C$ is $a_C$. The closed winner region

$$W_{Cb} = \{\theta \in \Theta : q_{Cb}(\theta) \leq q_{Ca}(\theta) \text{ for every } a \in \mathcal{A}_C\} \tag{A.56}$$

contains every readout that actually selects $b$. It can additionally contain ties resolved in favor of another action. Using this closed enlargement preserves the original tie rule and is conservative at its boundaries.

Suppose $D_{Cb}$ is a verified lower bound on $\inf_{W_{Cb}} F$. A separately certified empty winner region has bound $+\infty$. Retain the reference action and every alternative with $D_{Cb} \leq U + \epsilon$ in $\mathcal{A}_C(\epsilon)$. An inherited reference-only switch bound remains valid and may be intersected with the full-menu bound by taking their maximum.

A joint event $H$ specifies actions $(C, b)$ at distinct cells. Suppose

$$D_H \leq \inf\{F(\theta) : \theta \in \bigcap_{(C,b)\in H} W_{Cb}\}$$

is independently certified. If $D_H > U + \epsilon$, no member of $\Theta_\epsilon$ can select all the actions in $H$. Equality does not exclude the event. The collection of such verified events is denoted by $\mathcal{H}_\epsilon$; it need not contain every incompatible event.

Let $Q^\tau(x, b)$ be the actual original-law cost of taking $b$ and then following the common fixed continuation $\tau$. For each positive-mass cell, suppose numerical integration gives

$$\ell_{Cb} \leq \mathbb{E}[Q^\tau(X, b) - Q^\tau(X, a_C) \mid X \in C] \leq u_{Cb}. \tag{A.57}$$

Set $\ell_{Ca_C} = u_{Ca_C} = 0$. Zero-mass cells contribute zero. All acquisition ambiguity, continuous-law integration and arithmetic allowances belong to these endpoints. The preceding actionwise calculation provides precisely these inputs.

### V.2. *A computable economic bound*

For $z = (z_{Cb})$, define the assignment relaxation

$$\mathcal{Z}_\epsilon = \Big\{ z \geq 0 : \sum_{b \in \mathcal{A}_C(\epsilon)} z_{Cb} = 1 \quad (C \in \mathcal{C}), \\ \sum_{(C,b) \in H} z_{Cb} \leq |H| - 1 \quad (H \in \mathcal{H}_\epsilon) \Big\}. \tag{A.58}$$

Absent variables have value zero. Every policy induced by one readout in $\Theta_\epsilon$ defines a feasible zero–one assignment. Fractional assignments, and some unattainable zero–one assignments, can remain in $\mathcal{Z}_\epsilon$.

For any weights $w_{Cb}$, take numbers $\alpha_C \in \mathbb{R}$ and $\lambda_H \geq 0$ satisfying

$$\alpha_C + \sum_{H \ni (C,b)} \lambda_H \geq w_{Cb} \quad \big(b \in \mathcal{A}_C(\epsilon)\big), \tag{A.59}$$

and define

$$V(w; \alpha, \lambda) = \sum_C \alpha_C + \sum_{H \in \mathcal{H}_\epsilon} (|H| - 1)\lambda_H.$$

The dual need not be numerically optimal. Its feasibility and value can be checked exactly.

THEOREM A.29—Compatible optimizer-class cost bound: *Under the preceding feasibility, winner, joint-event and conditional-cost conditions, let $V_\epsilon^+$ be any feasible dual value for $w_{Cb} = p_C u_{Cb}$, and let $V_\epsilon^-$ be any feasible dual value for $w_{Cb} = -p_C \ell_{Cb}$. Then every $\theta \in \Theta_\epsilon$ satisfies*

$$-V_\epsilon^- \leq J(\pi^\theta) - J(\pi^{\widehat{\theta}}) \leq V_\epsilon^+. \tag{A.60}$$

*The exact assignment optima cannot give a wider interval than the full-menu cellwise extrema, which cannot give a wider interval than the inherited reference-only actionwise bound. Intersecting independently valid computed bounds preserves this ordering even when numerical dual proposals are not optimal.*

*For two constructed controllers $A, B$, suppose the attained-cost event gives $L \le J(\pi^{\widehat{\theta}_A}) - J(\pi^{\widehat{\theta}_B}) \le U_0$. Then all members of their respective qualified classes satisfy*

$$L - V_A^- - V_B^+ \le J(\pi^{\theta_A}) - J(\pi^{\theta_B}) \le U_0 + V_A^+ + V_B^-. \tag{A.61}$$

*The implication holds on that same event simultaneously for all allowances and all valid certificates, including certificates chosen after inspecting the covered intervals. It adds no failure probability for the continuum of readouts. A feasible numerical output with verified dual value $d \le \inf_\Theta F$ and $F(\theta) - d \le \epsilon$ belongs to $\Theta_\epsilon$.*

PROOF: Fix $\theta \in \Theta_\epsilon$. Its chosen action at each cell lies in the closed winner region. A singleton bound strictly larger than $U + \epsilon$ therefore cannot exclude that action. Similarly, selecting all members of a joint event in $\mathcal{H}_\epsilon$ would imply $F(\theta) \ge D_H > U + \epsilon$, a contradiction. Its choice indicators $z^\theta$ consequently belong to (A.58).

The common continuation and the original initial law give the exact decomposition

$$J(\pi^\theta) - J(\pi^{\widehat{\theta}}) = \sum_C p_C \sum_b z^\theta_{Cb} \mathbb{E}[Q^\tau(X, b) - Q^\tau(X, a_C) \mid X \in C].$$

Apply the upper inequality in (A.57). Multiplying (A.59) by $z^\theta_{Cb}$, summing, and applying the nonnegative joint-event multipliers bounds the resulting expression by $V_\epsilon^+$. Apply the same argument to the negative lower endpoints for the other inequality in (A.60).

Removing all joint-event restrictions gives independent cell simplices. Their support is the sum of the cellwise maximal weights. Adding valid restrictions can only reduce that support. Full-menu exclusions retain a subset of the alternatives admitted by a reference-only comparison, proving the second ordering. Intersections preserve containment because each input bound contains the same cost difference.

Subtract the two instances of (A.60) and add the attained-policy interval to obtain (A.61). On the attained-cost event this entire argument is deterministic and holds for every valid collection of inequalities. Taking any number of allowances or any coefficient class therefore requires no additional probabilistic event. Finally, $F(\theta) \le d + \epsilon \le \inf_\Theta F + \epsilon \le U + \epsilon$. *Q.E.D.*

A positive lower endpoint in (A.61) supports replacing $A$ by $B$ for every additional normalized charge below that endpoint. A strict interval contained in $(-m, m)$ gives equivalence at the declared margin $m$. These statements concern actual policy costs throughout the stated optimization classes. They do not identify a probability that an unrestricted fitting algorithm will enter those classes.

### V.3. *Qualification with one common coefficient vector*

The fixed-label calibrated objective is

$$F(\theta) = \max_i (y_i^+ - \phi_i^\mathsf{T}\theta) - \min_i (y_i^- - \phi_i^\mathsf{T}\theta) + 2\rho\big(G + \max_r |r_r^\mathsf{T}\theta|\big).$$

A winner or joint-winner event adds linear inequalities $d_j^\mathsf{T}\theta + c_j \le 0$. The Numerical Appendix derives an exact lower certificate from two probability vectors, a derivative-budget multiplier and nonnegative event multipliers. All inequalities in a joint event use the same $\theta$ and the same qualification objective. A positive coefficient-box support separation certifies an empty event; a solver's infeasibility flag alone does not.

The complete-menu calculation and joint-event calculations can be stopped at any finite budget. Unchecked events add no exclusion, and inconclusive calculations retain the existing bound. Conversely, a valid new exclusion can immediately improve the economic assignment problem without any additional policy-cost observations. The arithmetic and integration costs required to obtain that exclusion remain separately chargeable.

## APPENDIX W: SHARED POLICY PATTERNS AND SHARP OCCUPATION BOUNDS

A common continuation restricts an entire contingent plan, including its prescriptions at states that are not reached. Pairwise exclusions need not retain that restriction after the occupation variables are relaxed. The following construction preserves the complete pattern before relaxing the state distribution. It strengthens Theorem A.1, rather than replacing the original Bellman problem with an unrelated finite-policy benchmark.

Use the original continuous-law model, reference policy, acquisition cells and whole-cell feasible action rules of Section A. A *pattern* $\sigma$ assigns one menu label $\sigma(t,C)$ to every dated cell. Let $\Sigma_\epsilon$ be the patterns actually induced by members of $\Theta_\epsilon$, with the stipulated tie convention. This is a finite set even when the parameter set is infinite. For validity, it is sufficient to retain a finite superset $\overline{\Sigma}_\epsilon$ containing every actual pattern and satisfying all previously certified conflicts. A numerical search that finds some patterns does not establish such coverage. Conversely, asserting that a retained pattern is attained requires a qualified parameter witness; a fractional solution is not that witness.

For each retained pattern introduce a weight $\lambda_\sigma \geq 0$, cell masses $m^\sigma_{tC}$ and transition masses $z^\sigma_{tCD} \geq 0$. Define the disaggregated flow polytope by

$$\sum_\sigma \lambda_\sigma = 1, \qquad m^\sigma_{0C} = \lambda_\sigma \mu_C,$$

$$\sum_D z^\sigma_{tCD} = m^\sigma_{tC}, \qquad m^\sigma_{t+1,D} = \sum_C z^\sigma_{tCD},$$

$$\underline{p}_{tC\sigma(t,C)D} m^\sigma_{tC} \leq z^\sigma_{tCD} \leq \overline{p}_{tC\sigma(t,C)D} m^\sigma_{tC}. \tag{A.62}$$

All initial probabilities, transition enclosures and advantage enclosures are the original-law inputs in (A.3). In particular, the transition restrictions cover the whole cell and its feasible repair, not just an acquired point. Put

$$L^{\mathrm{pat}}_\epsilon = \min \sum_{\sigma,t,C} B_t \ell_{tC\sigma(t,C)} m^\sigma_{tC},$$

$$U^{\mathrm{pat}}_\epsilon = \max \sum_{\sigma,t,C} B_t u_{tC\sigma(t,C)} m^\sigma_{tC}, \tag{A.63}$$

where both extrema are over (A.62).

THEOREM A.30—Pattern-consistent transfer and its approximation error: *Suppose the qualification class is nonempty and the preceding coverage and enclosure conditions hold. Then every $\theta \in \Theta_\epsilon$ satisfies*

$$L_\epsilon \leq L^{\mathrm{pat}}_\epsilon \leq J^{\pi_\theta}_0 - J^{\tau}_0 \leq U^{\mathrm{pat}}_\epsilon \leq U_\epsilon.$$

*The outer pair $(L_\epsilon, U_\epsilon)$ is that of Theorem A.1 with the same conflicts and transition inputs. Valid exclusion of additional whole patterns can only improve the new pair. Certified lower and upper dual bounds remain sufficient for implementation.*

*Suppose further that the retained patterns are exactly the attainable patterns, and each has a qualification witness. Define*

$$\omega_t = \max_{C,a}(u_{tCa} - \ell_{tCa}), \qquad M_t = \max_{C,a} \max\{|\ell_{tCa}|, |u_{tCa}|\},$$

$$\delta_t = \max_{C,a} \sum_{D} (\overline{p}_{tCaD} - \underline{p}_{tCaD}), \qquad e_t = \min\left\{2, \sum_{s<t} \delta_s\right\},$$

*with $e_0 = 0$, and put $\mathcal{E} = \sum_{t<T} B_t(\omega_t + M_t e_t)$. If $j_-$ and $j_+$ are the minimum and maximum true class-cost differences, respectively, then*

$$j_- - \mathcal{E} \le L_\epsilon^{\mathrm{pat}} \le j_- \le j_+ \le U_\epsilon^{\mathrm{pat}} \le j_+ + \mathcal{E}. \tag{A.64}$$

*In particular, exact finite-state transitions and advantages give the exact class-cost extrema, attained by qualified patterns. More generally, any refinement sequence for which the displayed $\mathcal{E}$ tends to zero yields convergence of both bounds to the corresponding true extrema. The stated transition-width condition, not refinement alone, licenses this conclusion.*

The full proof appears in Supplement CB. It does not assume that the cell process of the continuous economy is Markov. Conditional cell-to-cell probabilities exist under each fixed policy at each date; the uniform enclosures contain those probabilities regardless of the within-cell distribution. Disaggregation prevents an optimization relaxation from buying one parameter pattern initially and a different pattern after observing its induced state.

*Why complete patterns add information.* The two-date example in Supplement E has two qualified signs of one shared coefficient. One sign visits the right-hand state and then selects its free action. The other sign selects the costly action there but never visits that state. Every actual policy therefore has cost zero. The pair-conflict flow relaxation permits mass 1/2 on the right-hand state and its costly action; its upper bound is 1/2. Disaggregation by the two attainable patterns gives upper

bound zero. This strict improvement requires no new policy-cost sample, changed economic primitive or successful nonconvex fit.

*Computation and scope.* The number of complete patterns can be exponential in the number of dated cells. Enumeration is therefore a finite exact implementation, not a general polynomial-time claim. One may begin with an outer covering of parameter space, retain every unresolved pattern and refine only economically material branches. Every discarded branch needs a valid exclusion for the actual own-future qualification functional. Missing a branch invalidates the bound; retaining an unresolved branch only weakens it. The state-flow width in (A.64) and the global-pattern relaxation are distinct approximation costs. This distinction preserves both the original continuous economy and the original-optimum policy sandwich while making the added qualification work explicit.

## APPENDIX X: ORIGINAL-LAW QUALIFICATION UNDER TRANSPORT APPROXIMATION

The occupation bounds retain a complete policy and its induced state distribution. Their cellwise convergence result, however, requires shrinking transition-probability widths. Moving a deterministic successor by an arbitrarily small distance need not make those widths small. We develop a complementary certificate that tests the transition approximation against the actual stored continuation. It applies to singular as well as diffuse transition laws and does not require a continuous greedy actor.

Work in the original finite-horizon expected-cost model, with compact metric state spaces, nonempty feasible action correspondences, bounded costs and terminal cost $g$. All kernels below are probability kernels with finite first moments. Minima may be replaced by infima provided a measurable feasible approximate minimizer is supplied. Write $B_0 = 1$ and $B_t = \prod_{s<t} \beta_s$, where $\beta_s \geq 0$. A parameter $\theta$ determines all stored continuations $f_{t,\theta}$ and a complete feasible policy $\pi_\theta$. Parameters, actions and continuations can be shared across dates and economic contexts; the state includes every endogenous variable needed by the original model.

Let $P_t(\cdot \mid x,a)$ denote the original transition law and $\widehat{P}_t(\cdot \mid x,a)$ a numerical approximation. The latter is an integration device, not a substituted economic primitive. Suppose that, uniformly on the declared domains and qualification class,

$$\mathrm{Lip}(f_{t+1,\theta}) \le K_{t+1}, \quad W_1(P_t(\cdot \mid x,a), \widehat{P}_t(\cdot \mid x,a)) \le e_t,$$
$$\left| \widehat{Q}_{t,\theta}(x,a) - c_t(x,a) - \beta_t \widehat{P}_t^a f_{t+1,\theta}(x) \right| \le \xi_t. \tag{A.65}$$

Here $W_1$ uses the stated metric on successor states, and $\widehat{Q}$ includes any additional certified quadrature and arithmetic approximation. In particular, $\xi_t$ is not set to zero merely because a computation returns a finite number. The bounds must hold for every original feasible action needed by the Bellman comparison, including any acquisition-boundary repair.

Define $\widehat{T}_{t,\theta}(x) = \inf_{a \in A_t(x)} \widehat{Q}_{t,\theta}(x,a)$. For each $\theta$ obtain constants $\ell_{t,\theta}, u_{t,\theta}$ and $\gamma_{t,\theta} \ge 0$ such that

$$\ell_{t,\theta} \le \widehat{T}_{t,\theta}(x) - f_{t,\theta}(x) \le u_{t,\theta},$$
$$\widehat{Q}_{t,\theta}(x, \pi_{t,\theta}(x)) - \widehat{T}_{t,\theta}(x) \le \gamma_{t,\theta} \tag{A.66}$$

throughout the original state domain. A finite action search must include its certified gap to the full feasible menu in $\gamma$ or in the Bellman enclosures; a minimum over sampled actions alone is not the displayed infimum. Let $d_\theta^- \le g - f_{T,\theta} \le d_\theta^+$ uniformly, and put

$$b_t = \beta_t K_{t+1} e_t + \xi_t, \qquad \mathcal{R}_\theta = \sum_{t<T} B_t \{ u_{t,\theta} - \ell_{t,\theta} + \gamma_{t,\theta} + 2b_t \} + B_T (d_\theta^+ - d_\theta^-).$$

THEOREM A.31—Transport-qualified original-optimum policy bound: *Under* (A.65)–(A.66), *every complete feasible candidate satisfies, at every initial state,*

$$0 \le V_0^{\pi_\theta} - V_0^* \le \mathcal{R}_\theta. \tag{A.67}$$

*Thus any common-parameter qualification class for which $\mathcal{R}_\theta \le \varepsilon$ is a class of complete policies with original-law loss at most $\varepsilon$. The result permits changes at every date and all corresponding changes in occupation. It requires neither a fixed continuation after the first*

*date, independence of deployment contexts, nor Lipschitz continuity of the implemented policy.*

*The certificate is invariant to replacing each stored continuation by $f_{t,\theta} + k_{t,\theta}$, where the shifts are state independent, and shifting its approximate action values consistently. If the numerical transition is obtained by a measurable successor quantizer $q_t$ with $d(y, q_t(y)) \leq h_t$ almost surely, then $e_t = h_t$ is valid even when the two kernels are mutually singular. Its additional contribution is at most*

$$2 \sum_{t<T} B_{t+1} K_{t+1} h_t. \tag{A.68}$$

*In particular, the transport part vanishes along any refinement for which this explicit sum tends to zero. No total-variation convergence is required. Convergence of the full certificate additionally requires the displayed residual, action-search, terminal and arithmetic contributions to vanish.*

The full proof is in Supplement CD. It compares the candidate and the original Bellman optimum separately to the same stored continuation, before subtracting their bounds. Applying a baseline occupation law to a different complete policy is neither necessary nor valid. The use of a uniform continuation modulus is essential: a small transport distance alone does not control integration of an arbitrary discontinuous value function.

*What the representation contributes.* For the constructed affine–ReLU continuations, a valid Lipschitz modulus follows either from the cone construction or from verified layer norms. The resulting $K_{t+1}$ prices transition approximation in the policy certificate. The exact same function represented by a native minimum circuit has the same mathematical guarantee. The result identifies the role of the neural continuation without making a method-neutral verifier evidence of universal neural speed. Wasserstein robustness and finite-model approximation have established analyses (Zhou et al., 2024, Saldi et al., 2015); our use is the uniform, translation-invariant qualification account for shared NBO continuations and their complete original-law policies.

COROLLARY A.32—Discounted stationary qualification: *In a stationary bounded-cost problem with $0 \le \beta < 1$, let $f$ be bounded and $K$-Lipschitz, and let a feasible stationary policy satisfy the preceding inequalities with constants $\ell, u, \gamma, e, \xi$. Then*

$$0 \le V^{\pi} - V^{*} \le \frac{u - \ell + \gamma + 2(\beta K e + \xi)}{1 - \beta}.$$

*This is an original-optimum certificate, not a claim of horizon-independent construction cost. The cost of acquiring a sufficiently accurate $f$, certifying $K$, integrating the original kernel and representing all relevant contexts remains part of the resource account.*

### X.1. *Joint allocation of residual and transition accuracy*

The coupled successor-cost program can now allocate two distinct sources of precision. If an implementation certifies residual-plus-greediness allowances $\eta_t$ and transport allowances $e_t$, its sufficient budget is

$$\sum_{t<T} B_t \eta_t + 2 \sum_{t<T} B_{t+1} K_{t+1} e_t + 2 \sum_{t<T} B_t \xi_t + B_T (d^{+} - d^{-}) \le \varepsilon. \tag{A.69}$$

When fixed, verified continuation moduli and primitive resource estimates yield a positive-monomial upper model in $(\eta, e)$, Theorem A.18 applies to its positive-weight coordinates. Coordinates with zero budget weight require separate caps and treatment; they do not satisfy that theorem's positive-weight premise. A modulus depending on an optimized parameter is not held fixed without a class-uniform upper certificate. This preserves the cost of transition precision rather than hiding it in the number of acquired states.

### X.2. *Nonlinear and strategic evaluations*

For a cash-invariant monotone one-step risk evaluation, the same signed recursion applies when a separately established bound on the risk-evaluation discrepancy replaces $b_t$. Wasserstein distance and cash invariance alone do not imply such a bound. General monotone Lipschitz aggregators use the absolute-error recursion of Proposition A.2, with their own continuation moduli, rather than an unsupported linear expectation identity. For games, the transport account enters each

fixed-opponent best-response certificate separately. Preference states and temporal selves retain the augmented-state and unilateral-deviation interpretation already stated in Section A.1. The new result supplements, rather than removes, the original diffusion, recursive-preference and game constructions.

*A singular-transition diagnostic.* Take one decision date, $\beta = 1$, terminal cost $g(x) = x$ on $[0,1]$, and actions 0 and 1. Action 0 has zero immediate cost and true successor $h > 0$; action 1 costs $h/2$ and has successor zero. The numerical kernel places both successors at zero. With $f_0 = 0$ and $f_1 = g$, its greedy choice is action 0, its residual span is zero and its true regret is $h/2$. Omitting the transition charge would falsely certify zero regret. The transport certificate is $2h$ and converges to zero, while the total-variation distance for action 0 is one for every $h > 0$. Appendix BM gives exact rational checks. This diagnostic is not a new observation in the investment experiment.

## APPENDIX Y: SEQUENTIAL QUALIFICATION OF POLICY COSTS

A fixed-sample interval does not by itself justify repeated inspection until a controller appears acceptable. Nor do cost observations from changing controllers identify the last controller's cost. We extend the frozen-continuation identity to predictable episodes and distinguish these targets. This uses established time-uniform concentration principles (Howard et al., 2020); confidence sequences have also been used for gated deployment in contextual bandits (Karampatziakis et al., 2021). Here the observations are verified, continuous-law NBO cost contrasts, not importance-weighted bandit rewards.

Let $\mathcal{F}_{i-1}$ contain all information before episode $i$. Conditional on it, choose two complete controllers, a common initial law, a continuation $h_i$ fixed through the episode, and valid coefficients of Theorem A.22. The resulting cost contrast $D_i$ has conditional mean $\Delta_i$ equal to the original-law expected cost difference and predictable support $[a_i, b_i]$. Let $[L_i, U_i] \subseteq [a_i, b_i]$ enclose $D_i$ for the entire continuous innovation cell. Write $V_n = \sum_{i \le n} (b_i - a_i)^2$. For fixed positive numbers $\lambda_j, w_j$, $1 \le$

$j \le J$, with $\sum_j w_j \le 1$, define

$$B_n = \min_{1 \le j \le J} \left\{ \frac{\log(2/(\alpha w_j))}{\lambda_j} + \frac{\lambda_j V_n}{8} \right\}. \tag{A.70}$$

THEOREM A.33—Predictable original-law cost certificate: *Under the preceding conditional-law, support and enclosure conditions, with probability at least* $1 - \alpha$*, simultaneously for every* $n \ge 1$*,*

$$\frac{\sum_{i \le n} L_i - B_n}{n} \le \frac{1}{n} \sum_{i \le n} \Delta_i \le \frac{\sum_{i \le n} U_i + B_n}{n}. \tag{A.71}$$

*Each endpoint may also be intersected with the corresponding average predictable support endpoint. The assertion holds at every almost surely finite stopping time. If the complete controller pair and initial-law/reset contract stay fixed, then* $\Delta_i = \Delta$*, even when the qualified continuation changes predictably between episodes. Otherwise the target is the average conditional cost difference, not the last controller's cost.*

The proof applies conditional Hoeffding bounds to each endpoint and Ville's inequality to a finite collection of nonnegative supermartingales. It requires no independent or identically distributed deployment contexts. It does require the stated conditional transition laws. Within-episode cache adaptation is allowed only when the controllers are treated as history-dependent algorithms and the greedy-gap qualification covers every memory state that can occur. Repeating the same program identifier while carrying a different cache state does not establish a fixed conditional target.

The implementation uses a finite dyadic collection of $\lambda_j$ and exact rational upper bounds for the logarithms. These give conservative, computable versions of (A.70). It reserves the support and target identity before accepting an episode. A failed numerical evaluation consumes the full previously qualified support interval and its actual work; it is not deleted after inspecting its outcome. A failure of the support certificate itself stops qualification.

COROLLARY A.34—Costed replacement after optional stopping: *For a fixed target pair, let* $[l_N, u_N]$ *be the interval at a finite stopping time* $N$*. Let a declared deployment count*

*be $m \geq 1$, a one-time replacement and qualification charge be $F \geq 0$, and the incremental per-deployment charge be $c$, all in the same units as the policy cost. On the simultaneous event of Theorem A.33, the net incremental cost lies in*

$$[l_N + F/m + c,\ u_N + F/m + c].$$

*Consequently $u_N + F/m + c < 0$ certifies strict net improvement. Inclusion of $[l_N, u_N]$ in a declared equivalence band certifies equivalence at that band, not strict superiority. New adaptively selected controller pairs may be qualified in separate epochs with conditional error allocations whose sum is at most the overall error budget.*

A wall-clock reading is not a monetary charge. Measured setup, validation, rejected evaluations and successful retrievals are retained separately; prices are explicit decision inputs. The number of qualification episodes is not automatically the number of deployments over which a setup cost is amortized. These distinctions let the original investment decision use a valid stopping rule without silently changing either its estimand or its units.

## APPENDIX Z: ORIGINAL-LAW QUALIFICATION OF CONSTRUCTED POLICIES

A continuation can have a verified fitting objective without its implemented policy having an identified economic advantage. The distinction is resolved by evaluating the same complete policy that the construction returns. We now connect the qualification-optimal readouts of Section R to independent original-law cost observations. Neither the fitted future nor the acquired actor is replaced during evaluation. The comparison concerns the constructed NBO policies, rather than a different recursive controller supplied with the same neural label.

### Z.1. *Conditional integration and numerical observations*

In the two-state, two-date investment economy, write $a_t = \pi_t(X_t)$ and $\beta = 15/16$. A cash observation and a conditionally integrated observation are

$$C^\pi = c(X_0, a_0) + \beta c(X_1, a_1) + \beta^2 g(X_2), \tag{A.72}$$

$$Y^\pi = c(X_0, a_0) + \beta c(X_1, a_1) + \beta^2 \bar{g}(X_1, a_1), \tag{A.73}$$

where $\bar{g}(x,a) = P^a g(x)$. If $m(x,a)$ is the deterministic part of the transition, the original common, opposite-signed innovation gives the exact identity

$$\bar{g}(x,a) = g(m(x,a)) + \frac{5}{3072}. \tag{A.74}$$

The shortage term depends on the sum of the coordinates and is unchanged by that innovation. The quadratic state and difference terms supply the variance correction. Consequently,

$$\mathbb{E}Y^\pi = \mathbb{E}C^\pi = \mathbb{E}J_0^\pi(X_0), \qquad \mathrm{Var}(Y^\pi - Y^\sigma) \le \mathrm{Var}(C^\pi - C^\sigma)$$

when both policies use the same initial state and innovations. The variance inequality concerns the exact observations; it does not assert that every finite-sample numerical interval must be narrower.

The state cube is invariant. On that cube, $0 \le a \le 1/4$, a stage cost is at most $103/64$, and the terminal cost is at most $37/16$. Both observations therefore lie in

$$[0, H], \qquad H = \frac{103}{64}(1+\beta) + \frac{37}{16}\beta^2 = \frac{21097}{4096}. \tag{A.75}$$

A paired difference has the signed support $[-H, H]$. Negative common-path gains are not clipped. No original-optimum value query is required to compute either observation for a frozen acquired actor.

An initial or innovation bin is a continuous cell, not a point mass. The evaluator encloses every true state in the bin, every acquired cell reached by that state enclosure, and every resulting feasible action. When an enclosure crosses an acquisition boundary, all intersected actions enter the continuation enclosure. A second implementation evaluates the same primitives using integer-directed arithmetic and expanded polynomials. The reported endpoints are the hull of the two valid enclosures, intersected with the proved support. Arithmetic agreement is a check on the implementations; validity follows from the enclosure rules and the primitive identities, not from agreement alone.

### Z.2. *Precision and the decision criterion*

The following statement applies to any fixed finite collection of qualified policies, not only to the two feature bases used below. All construction information is fixed before the independent evaluation rows are drawn. Dependence among policies on a common row is allowed.

THEOREM A.35—Original-law cost qualification: *For each of $K$ declared estimands, let $Z_i$ be its original-law observation on independent row $i$, and suppose a deterministic full-bin calculation returns*

$$a \le L_i \le Z_i \le U_i \le b, \qquad i = 1, \dots, N.$$

*The inclusion may instead hold for the original conditional expectation within the sampled bin. Put $R = b - a$, let $\bar{L}, \bar{U}$ denote sample means, and let $s_L^2, s_U^2$ be the unbiased endpoint sample variances. For $N \ge 2$ and $\ell > 0$, define*

$$r_L = \sqrt{\frac{2s_L^2 \ell}{N}} + \frac{7R\ell}{3(N-1)}, \qquad r_U = \sqrt{\frac{2s_U^2 \ell}{N}} + \frac{7R\ell}{3(N-1)}.$$

*With probability at least $1 - 4Ke^{-\ell}$, simultaneously for the declared collection,*

$$\mathbb{E}Z_i \in [\bar{L} - r_L, \bar{U} + r_U]. \tag{A.76}$$

*Before rounding the final endpoints, its width is exactly*

$$(\bar{U} - \bar{L}) + \sqrt{\frac{2s_L^2 \ell}{N}} + \sqrt{\frac{2s_U^2 \ell}{N}} + \frac{14R\ell}{3(N-1)}. \tag{A.77}$$

*Upward square roots and outward final rounding preserve coverage. The first term includes all retained numerical and acquisition uncertainty; the middle terms reflect observed endpoint dispersion; the last term is the finite-support penalty. A declared full-width target is met only when their computed sum meets it.*

The proof in the Technical Supplement applies the empirical-Bernstein inequality of Maurer and Pontil (2009) to the lower and upper endpoint variables. The statistical inequality is established work. The contribution here is its application to the

original-law costs of the actual constructed actors, with executable whole-bin observations, independent arithmetic and an explicit precision account. Refining only a simulation bin cannot remove the finite-support or sampling terms. Conversely, an inaccurate uniform regret certificate need not prevent an accurate comparison of two actual policy costs.

Precision, equivalence and improvement are different conclusions. For a cost-difference interval $[l, u]$ and a predeclared margin $\delta$, equivalence requires $-\delta < l \leq u < \delta$. Noninferiority of the first policy requires $u < \delta$; strict superiority requires $u < 0$. An interval can meet its width target and fail equivalence because it identifies a material difference. These decisions are made from exact endpoints, not from rounded table entries.

COROLLARY A.36—Priced selection and adoption: *On the simultaneous event of Theorem A.35, let $[L_j, U_j]$ qualify the cost of policy $j$, and let $w_j$ be its declared nonnegative resource charge. A minimizer $\widehat{j}$ of $U_j + w_j$ satisfies, for every policy $j$,*

$$J_{\widehat{j}} + w_{\widehat{j}} \leq J_j + w_j + (U_j - L_j).$$

*The inequality holds for every assignment of resource prices on the same event. If the gain from replacing an installed controller is in $[l, u]$, then $m$ deployments, monetary scale $v \geq 0$, qualification charge $q \geq 0$ and replacement fee $f \geq 0$ give expected net benefit in*

$$[mvl - q - f,\ mvu - q - f].$$

*In particular, when $l > 0$ and $v > 0$, every $m > (q + f)/(vl)$ is certified beneficial.*

PROOF: The first assertion follows from $J_{\widehat{j}} \leq U_{\widehat{j}}$, the minimizing property, and $U_j \leq J_j + U_j - L_j$. These deterministic inequalities hold for all nonnegative prices once all cost intervals cover. The second assertion is multiplication by $mv$ and subtraction of $q + f$. *Q.E.D.*

The resource charges in this result must be specified, not inferred from normalized losses. An already paid construction and a prospective purchase have different charges. When only a shared qualification service has been timed, charging that

entire service to a proposed replacement is conservative; inventing a per-policy marginal qualification time is not.

### Z.3. *The constructed-controller experiment*

The fixed catalogue consists of the four previously constructed NBO actors and the admissible zero-investment incumbent. ReLU and degree-four Bernstein continuations each have nine coefficients and are constructed on eight- or sixteen-cell coordinate covers. At a fixed cover, the two bases use the same acquisition contract and primitive information. The actor arrays, actual fitted futures and exact optimization certificates are unchanged. The zero-investment policy supplies an economically interpretable installed reference; it is not substituted for the strong conventional query competitors in the preceding study.

Before execution, the protocol fixes $N = 524{,}288$, forty-bit continuous bins, the sample stream, the full-width target $1/256$, and the separate two-sided equivalence margin $1/256$. All five marginal costs and ten paired differences are reported for both observations in (A.72)–(A.73). Thus $K = 30$ and $\ell = 10$ give a joint failure bound below $1/100$. Reusing a row for another controller or estimator does not increase the independent sample count. The declared independence model concerns the simulated rows; a fixed pseudorandom seed establishes reproducibility, not mathematical independence.

*Outcomes of the fixed precision design.* Across both estimators, 20 of the twenty paired intervals meet the declared full-width target. Their widths range from 0.000936 to 0.002285. Exactly 4 meet the separate equivalence criterion. Neither a failed width target nor a failed equivalence comparison is omitted.

The conditionally integrated comparisons identify a positive gain over the zero-investment incumbent for 4 of the four constructed policies. These are gains of complete implemented controllers, not reductions in a qualification objective.

At the two matched acquisition resolutions, the ReLU-minus-Bernstein comparisons identify 2 lower-cost ReLU outcomes, 0 higher-cost outcomes, and 0 unresolved signs. This finite comparison is not a universal representation ranking.

TABLE Z.I

ORIGINAL-LAW COSTS OF THE CONSTRUCTED CONTROLLERS

| Controller | Expected cost | Gain over zero incumbent |
|---|---|---|
| Zero | [0.773561, 0.780120] | – |
| R8 | [0.607716, 0.613107] | [0.165305, 0.167552] |
| B8 | [0.684034, 0.690159] | [0.089044, 0.090443] |
| R16 | [0.607022, 0.612384] | [0.165995, 0.168280] |
| B16 | [0.682910, 0.689022] | [0.090168, 0.091580] |

R and B denote the original nine-coefficient ReLU and Bernstein constructions; the number is the coordinate cover size. All policies are frozen before the new evaluation rows. The displayed observations integrate the last innovation analytically without changing expected cost. Endpoints are rounded outward. Paired gain intervals, not differences of marginal cost intervals, determine gain signs.

TABLE Z.II

PAIRED FEATURE AND ACQUISITION COMPARISONS

| First minus second | Expected-cost difference | Full width | Equivalent |
|---|---|---|---|
| R8 − B8 | [-0.077568, -0.075801] | 0.001767 | No |
| R16 − B16 | [-0.077159, -0.075368] | 0.001791 | No |
| R8 − R16 | [0.000227, 0.001191] | 0.000963 | Yes |
| B8 − B16 | [0.000661, 0.001599] | 0.000937 | Yes |

A negative interval identifies lower cost for the first controller. Equivalence is determined from exact endpoints at the predeclared two-sided margin 1/256. The separate full-width target is also 1/256. Matched-basis comparisons hold the acquisition contract fixed; cross-resolution comparisons do not.

The construction and the evaluation now refer to the same complete controllers. The resulting cost intervals, rather than a reduction in the residual-fitting objective, determine the adoption regions in Corollary A.36. Comparisons between the feature bases identify only what their paired intervals support. The earlier unresolved and adverse query-method comparisons remain distinct evidence and are preserved in the Numerical and Development Appendix.

Both interval implementations evaluate every new trajectory and every acquired-action rectangle. Additional rational midpoint checks test predetermined rows. The phase ledger separates controller loading, lookup preparation, binary64 evaluation,

integer evaluation, exact statistical arithmetic, output and the complete parent process. The independent replay recomputes the entire new enclosure catalogue without replacing its original clocks. This evaluates the two-date constructed-policy problem; it is not an independent rerun of the older, much larger vector Bellman tree.

## APPENDIX AA: QUALIFICATION, REPRESENTATION, AND ATTAINED ECONOMIC COST

The preceding attained-controller experiment used a coefficientwise qualification bound. Its ReLU–Bernstein ordering is an observation about those specified constructions. We now examine whether that ordering persists after the two bases receive the same qualification charge on their common affine functions. This directly tests a source of representation-dependent conservatism while preserving the original nonlinear investment economy and the Neural Bellman Operator construction.

### AA.1. *A prospective factorial comparison*

The design fixes two bases, two qualification rules and two acquisition resolutions. The eight constructed controllers and the installed zero controller are evaluated on the same original-law paths. Each basis has nine coefficients with absolute bounds eight, the same node covers, thirty-three-action search menus and downward action quantum 1/4096. The calibrated rule replaces the coefficientwise modulus by (A.43) or (A.44), both in residual qualification and in the regularity allowance passed to the previous fitted date. Both own-future continuations are constructed backwards. An exact primal–dual gap of at most 1/65536 qualifies each new fit. The coefficient proposal is quantized at forty bits; no hidden-feature optimizer or outcome-selected basis is introduced.

A source manifest was committed before construction, and the complete actor identities were committed before generating 524,288 new four-coordinate continuous-bin rows. The bin precision is forty bits. Two estimators, cash cost and exact last-innovation conditional expectation, evaluate the same expected costs under the original uniform initial law and continuous innovations. For each esti-

mator the frozen family contains nine marginal costs, thirty-six paired differences and two basis-by-qualification interactions. All ninety-four estimands share a single failure allowance bounded by $376\exp(-11) < 1/100$. Shared paths and the two estimators do not multiply the number of independent rows.

The whole-bin enclosures are independently computed in outward binary64 arithmetic and directed-integer arithmetic. Their hull, intersected with the proved support, is used for inference. An exact rational implementation additionally checks 512 midpoint trajectories per controller. All nine controllers, bins, endpoints, exact sample moments and disagreements are part of the retained protocol. This revision re-evaluates every bin and every inference quantity; it neither regenerates those observations as a second sample nor substitutes replay time for the original service clock.

TABLE AA.I

BASIS COMPARISON BEFORE AND AFTER QUALIFICATION CALIBRATION

| Qualification | $n$ | ReLU minus Bernstein | Lower cost |
|---|---|---|---|
| Legacy | 8 | [-0.077633, -0.075733] | R |
| Legacy | 16 | [-0.077232, -0.075307] | R |
| Calibrated | 8 | [0.003571, 0.004688] | B |
| Calibrated | 16 | [0.000955, 0.002013] | B |

All entries are simultaneous paired original-law cost intervals from the same prospective R80 family, with exact last-innovation conditional integration. Positive values favor Bernstein. Equal coefficient counts and affine-exact modulus charging do not make the two nonlinear function classes identical. The old fits are evaluated, not overwritten.

The calibration improves the expected cost of each basis at each resolution. In particular, the fine ReLU controller gains between 0.005159 and 0.006341 relative to its legacy construction. The matched basis ordering nevertheless reverses: calibrated Bernstein has lower cost than calibrated ReLU at both resolutions. At $n = 16$ the ReLU-minus-Bernstein interval is $[0.000955, 0.002013]$, rounded outward. That strict ordering coexists with equivalence at the declared margin $1/256$; a small positive

TABLE AA.II

ACTUAL CONTROLLER GAINS FROM CALIBRATED QUALIFICATION

| Basis | $n$ | Legacy minus calibrated cost |
|---|---|---|
| Relu | 8 | [0.000457, 0.001653] |
| Relu | 16 | [0.005159, 0.006341] |
| Bernstein | 8 | [0.080964, 0.082772] |
| Bernstein | 16 | [0.082601, 0.084405] |

Each comparison retains its basis, coefficient count and bounds, robust action menus, observation resolution and original economic primitives. Both attained future fits are reconstructed under the stated qualification rule. The displayed gains are not timing estimates or claims about unrestricted neural fitting.

difference can satisfy both statements. The corresponding coarse-resolution interval does not establish equivalence at that margin.

TABLE AA.III

PREDECLARED BASIS-BY-QUALIFICATION INTERACTIONS

| $n$ | Change in Bernstein-minus-ReLU cost |
|---|---|
| 8 | [-0.082254, -0.079371] |
| 16 | [-0.079209, -0.076297] |

The interaction is (calibrated Bernstein minus calibrated ReLU) minus (legacy Bernstein minus legacy ReLU). These four-term contrasts were included in the source-frozen family before the new cost rows were generated. They are not obtained by treating the four marginal estimates as independent.

The predeclared interactions isolate the change in the basis comparison. They do not establish that the bases span identical functions, that the Bernstein derivative-control bound is globally exact, or that every other numerical difference has been removed. The positive operational conclusion is more specific: charging the function's derivative structure changes the attained original-law policy cost, and a policy ranking cannot be attributed to a representation name without examining its qualification rule. The historical ReLU gain and the new reversed matched ranking remain separate valid observations.

## AA.2. *Precision of the economic comparison*

The target was a paired total width of 1/256, fixed before seeing the new rows. Appendix BE reports every interval and its width components, including every marginal interval whether or not its width falls below that paired target. Variance reduction by conditional integration affects the variance component, not the bounded-support penalty or the acquisition contract. For example, in the fine calibrated basis comparison the conditional interval's width consists of approximately 0.001008605 from the finite-support term, 0.000048152 from endpoint sampling-variance allowances, and $2.69 \times 10^{-11}$ from the numerical enclosure. The support term is therefore the principal remaining contribution in that comparison.

The numerical component includes the implemented handling of continuous-bin width, acquisition ambiguity and arithmetic propagation. It is not represented as a pure floating-point error estimate. Likewise, a small numerical component does not justify dropping the finite-sample support term. These diagnostics explain the achieved precision without inferring significance from raw mean differences or treating the inherited R67 policy-cost rows as observations of these newly constructed actors.

## AA.3. *From a fitted readout to a qualified optimizer class*

For each new terminal-label qualification problem the action-switch catalogue covers every first-date observation cell and all thirty-two alternative robust actions. The 20,480 rational dual bounds are independently recomputed using the antiderivative and box-infimum calculations of Appendix BD; no switch optimization is rerun to replace an unfavorable certificate.

At zero extra objective allowance, the two fine-resolution constructions and coarse Bernstein have complete first-date action stability. Every exact minimizer of each of these fixed-label qualification problems therefore gives its attained complete policy, not merely a similar fitted value. The original-law intervals in Table AA.V extend to these optimizer classes without additional probability expenditure. The same conclusion holds for the positive dyadic sublevels displayed in Table AA.IV.

TABLE AA.IV

FIRST-DATE ACTION STABILITY OF QUALIFICATION SUBLEVELS

| Actor | Cells | $\epsilon = 0$ | 1/65536 | 1/4096 | Uniform allowance |
|---|---|---|---|---|---|
| CR8 | 64 | 28 | 28 | 28 | – |
| CB8 | 64 | 64 | 64 | 55 | 1/32768 |
| CR16 | 256 | 256 | 248 | 195 | 1/262144 |
| CB16 | 256 | 256 | 243 | 200 | 1/8388608 |

Counts use the strict inequality D greater than U plus epsilon. U is the exact objective of the attained feasible reference fit. The last column is a dyadic allowance strictly below the smallest positive switch surplus, when such a surplus is certified. A dash is unresolved, not evidence of instability. The independent verifier checks every one of the 20,480 action-switch duals.

TABLE AA.V

COST COMPARISONS UNIFORM OVER EXACT QUALIFICATION MINIMIZERS

| Complete optimizer classes | Original-law cost difference |
|---|---|
| CB16 minus CB8 | [-0.003286, -0.002241] |
| CB16 minus CR16 | [-0.002013, -0.000955] |
| CB8 minus CR16 | [0.000748, 0.001812] |

CB8, CR16 and CB16 have fully stable first-date covers at epsilon zero. Their terminal actors are fixed by the common terminal payoff. Consequently each stated exact-minimizer class gives the attained complete policy and inherits its interval without enlargement. This is a deterministic transfer of the same finite statistical family, not new sampling over optimizers.

The standard 1/65536 allowance leaves some fine-grid cells unresolved. Theorem A.26 then supplies an explicit enlargement based on their uniform initial mass. This conservative enlargement need not preserve a small cross-method sign. Coarse ReLU also retains unresolved cells even at zero allowance. None is relabeled as a discovered instability, and exact-minimizer invariance is not a claim that an arbitrary fitting routine reaches the requisite sublevel. The new statement addresses optimizer-independent policy stability under explicit attained-objective conditions; it does not estimate a hidden-layer success probability.

### AA.4. *Controller choice and complete costs*

Both cash and conditional marginal and pair intervals can be used together on their original simultaneous event. Exact shortest-path closure gives consistent cost potentials and the pure-choice regret bounds of the inherited decision theorem. The four-term interactions remain separately reported; sharpness of the pair-constraint calculation is not asserted after adding those extra restrictions. The complete nine-controller closure and attaining potentials are deposited in the revision audit.

For the fine calibrated controllers, the positive ReLU-minus-Bernstein interval determines a nonempty region of additional Bernstein charges for which its lower policy cost outweighs its extra charge. It also determines the opposite region in which ReLU is preferred after that charge. All such statements are conditional on an explicitly supplied conversion of computation, installation and deployment into normalized cost units. The exact boundary values may be tightened by the common pair-constraint closure; no new simulation is needed for that deterministic calculation.

The construction work is recorded rather than presumed equal. Legacy clocks include a label-loss ablation that the calibrated services do not execute. Their difference therefore does not identify an end-to-end speed gain from calibration. Exact action and expected-feature evaluation counts are reported alongside the clocks, and the original cost-inference, barrier and publication services retain separate measured costs. The present result supplies an identified policy-cost and conditional purchase comparison in the original economy. It does not extrapolate a two-date fitted-continuation calculation into a measured long-horizon speed result for the recursively queried vector tree.

## APPENDIX AB: ECONOMIC STOPPING REGIONS IN THE INVESTMENT ECONOMY

The four calibrated constructions in Section AA permit an exact application of the stopping theorem on $0 \leq \epsilon \leq 2^{-12}$. The full 20,480-switch catalogue leaves 976 actions for economic integration. All four declared dyadic levels are retained. Two independent arithmetic implementations reevaluate every paired box and reproduce the frozen records exactly. The tolerance calculation has 118 model strata;

a separate enumeration checks all 12,732 pair-stratum tests. Appendix BG records this computation and the earlier interrupted execution separately.

At allowance $2^{-16}$, the fine-grid ReLU-minus-Bernstein class-cost band is $[0.000496, 0.002503]$, with endpoints rounded outward. The corresponding binary-mass band is $[-0.421558, 0.424526]$. Thus the same attained-policy evidence identifies a Bernstein saving throughout the approximately optimized classes once unresolved actions are priced individually. Every additional Bernstein deployment charge below 0.000496 preserves the positive uniform net gain in normalized model units. This is not a calibrated installation fee.

TABLE AB.I

CERTIFIED COMMON OPTIMIZATION TOLERANCES

| Pair | Second has lower cost | Equivalent within $2^{-8}$ |
|---|---|---|
| R8–B8 | $\varnothing$ | $\varnothing$ |
| R8–R16 | $\varnothing$ | $\varnothing$ |
| R8–B16 | $\varnothing$ | $\varnothing$ |
| B8–R16 | $[0, 0.000049063]$ | $[0, 2^{-12}]$ |
| B8–B16 | $[0, 0.000213074]$ | $[0, 0.000034382]$ |
| R16–B16 | $[0, 0.000067491]$ | $[0, 0.000177245]$ |

Every tolerance in each displayed closed interval is certified. Decimal endpoints are rounded inward, not estimates of the exact boundary. Empty means this sufficient interval test is inconclusive even at zero excess. Exact rational frontiers, boundary inclusion and heterogeneous-tolerance strips are deposited. R denotes ReLU and B Bernstein.

The exact equal-tolerance boundary for that ordering is approximately 0.0000674917368 and is excluded: a switch enters at equality. Its exact rational value, not the decimal approximation, governs a stopping rule. Cost equivalence within $1/256$ is a different question. Coarse Bernstein and fine ReLU satisfy this equivalence throughout $[0, 2^{-12}]$, with final class-cost band $[-0.001622, 0.003879]$. Both statements survive nonzero optimization error; neither is a universal representation ordering.

All four classes retain a positive cost saving against the zero controller throughout the integrated tolerance domain. Coarse ReLU nevertheless has inconclusive pairwise class comparisons already at zero excess. It and all opposite-ordering tests are not removed. The new frontier reuses the existing simultaneous cost event and supplies deterministic coverage for a continuum of verified stopping outputs. It creates neither new observations nor an estimate of fitting-success frequencies. Complete phase work, all discrete refinement bands, all eighteen frontier targets and both fixed-reference comparisons are retained in the appendix.

## APPENDIX AC: ECONOMIC VALUE OF COMMON-READOUT QUALIFICATION

The new calculation refines the economic accuracy of the original attained NBO controllers. It does not fit another controller or change a conditional cost observation. The four calibrated readouts, their frozen labels, terminal actors, robust 33-action menus, acquisition grids and original continuous laws are those of the preceding factorial investment study. The original actionwise integrations and simultaneous attained-policy intervals remain the paid inputs.

### AC.1. *A complete, fixed qualification design*

At acquisition sizes eight and sixteen, every alternative admitted by the reference-only barrier at excess objective at most $2^{-12}$ receives a full-menu qualification problem. This gives 976 problems: 832 for coarse ReLU, nine for coarse Bernstein, 78 for fine ReLU and 57 for fine Bernstein. Each event includes the other 32 feasible menu actions, with a common coefficient vector and the same objective.

The joint-event design permits at most 512 pairs per model. When at most 128 alternatives remain at the largest allowance, all remaining alternatives seed the pair catalogue. Otherwise the minimum-lower and maximum-upper conditional-cost alternatives in each cell seed it. Cross-cell pairs are ranked by the product of their maximum absolute cost endpoints, with lexicographic tie handling, and the first 512 are retained. The complete roster contains 1572 pairs; the small Bernstein model contributes all 36 of its possible pairs. This rule is deterministic postprocessing of

an already covered cost event, not a newly randomized validation design. A small implementation pilot preceded the source lock and is disclosed separately.

All six declared allowances are evaluated: zero and $2^{-k}$ for $k = 16, 15, 14, 13, 12$. The output preserves singleton bounds, pair bounds, positive box separations, retained alternatives, active conflicts and both economic duals. Every one of the 104 infeasible optimization statuses has an independent positive rational separation certificate. A numerical status alone is never used to remove an event.

TABLE AC.I

ORIGINAL-LAW COST DEVIATIONS AT EXCESS QUALIFICATION OBJECTIVE $2^{-12}$

| Model | Single | Joint | Allowed | Cuts | Actionwise | Compatible |
|---|---|---|---|---|---|---|
| R8 | 832 | 512 | 691 | 351 | [-0.010033, 0.012680] | [-0.010023, 0.008293] |
| B8 | 9 | 36 | 9 | 14 | [-0.001080, 0.000900] | [-0.001032, 0.000792] |
| R16 | 78 | 512 | 61 | 140 | [-0.001168, 0.001291] | [-0.001009, 0.001053] |
| B16 | 57 | 512 | 56 | 204 | [-0.001311, 0.001311] | [-0.001012, 0.001000] |

R and B denote calibrated ReLU and Bernstein readouts; the suffix is the acquisition size in each state coordinate. Bounds concern each optimizer class relative to its attained controller. Single and joint are qualification-problem counts; allowed and cuts apply at the displayed tolerance. Both interval endpoints are rounded outwards. Every tolerance, including unchanged bounds, appears in the appendix.

The full-menu test gives a strict contraction of the inherited interval in eight of the 24 model–allowance records. Shared-coefficient restrictions give a further strict contraction in fifteen records. Some smaller-tolerance records already have no alternative action or no useful selected conflict and remain unchanged. The table and appendix retain both outcomes. For coarse ReLU at the largest allowance, the inherited deviation interval [-0.010033, 0.012680] becomes [-0.010023, 0.008293]. This is a reduction in the uncertainty about a class of possible policies, not another measured gain of the attained reference policy.

## AC.2. *A new classwide refinement decision*

All six pairwise comparisons at the largest allowance appear in Table AC.II. They add the compatible deviation bounds to the same attained-cost intervals used in

the preceding study. The result is simultaneous over the declared coefficient classes, rather than a comparison of two numerical certificate values.

TABLE AC.II

ALL PAIRED OPTIMIZER-CLASS COMPARISONS AT EXCESS OBJECTIVE $2^{-12}$

| First minus second | Actionwise interval | Compatible interval | Sign |
|---|---|---|---|
| R8–B8 | [-0.007360, 0.018447] | [-0.007242, 0.014012] | unresolved |
| R8–R16 | [-0.006468, 0.019812] | [-0.006221, 0.015267] | unresolved |
| R8–B16 | [-0.005006, 0.021441] | [-0.004685, 0.016754] | unresolved |
| B8–R16 | [-0.001622, 0.003879] | [-0.001336, 0.003612] | unresolved |
| B8–B16 | [-0.000149, 0.005497] | [0.000210, 0.005089] | positive |
| R16–B16 | [-0.001523, 0.004614] | [-0.001054, 0.004077] | unresolved |

Intervals use the same R80 simultaneous attained-controller cost event and the same original economic law. Positive means that the first controller class has higher cost throughout the certified pair of classes. An interval containing zero remains unresolved. These rows add no fitting attempt or independent observation.

For coarse versus fine Bernstein, the previous optimizer-class interval at allowance $2^{-12}$ is [-0.000149, 0.005497]. It does not identify a signed ordering. The common-readout interval is [0.000210, 0.005089], whose lower endpoint is strictly positive. Thus the finer Bernstein controller has lower original-law cost for every pair of readouts in the two stated sublevels. Every additional normalized refinement charge below 0.000210 is supported by this classwide saving. This conclusion holds throughout smaller allowances by containment of the qualification classes, not by interpolation between the six printed tolerances.

The refinement decision is distinct from the matched basis comparison. At allowance $2^{-16}$ the fine ReLU-minus-Bernstein interval is [0.000549, 0.002455]. It retains the identified Bernstein advantage under calibrated qualification while narrowing the preceding actionwise band. The earlier legacy qualification advantage of ReLU remains part of the factorial evidence. The paper therefore identifies the qualification rule and its verified optimization accuracy as economically relevant, rather than choosing a basis label and suppressing the opposite ranking.

The complete catalogue also retains coarse-ReLU comparisons that remain unresolved, and larger-allowance fine-basis comparisons whose intervals cross zero. An interval strictly inside the declared 1/256 margin supports the corresponding equivalence statement; crossing zero alone does not. The new support calculation supplies no population fitting-success probability and does not convert a normalized cost unit into an estimated currency amount. The original priced-selection theorem can instead use the verified interval together with a separately supplied loss scale, deployment count and refinement charge.

## AC.3. *Complete incremental work and independent arithmetic*

The recorded scientific interpreter, including imports, reads, all qualification and assignment problems, durable outputs and shutdown, takes 104.699 seconds on the stated ARM runner. Singleton, joint-event and assignment phases are recorded separately. These measurements price the additional qualification calculation. They are neither a cold-start cost for rebuilding the original controllers nor a matched ReLU-versus-Bernstein timing experiment. Original fitting and integration clocks remain unchanged and must be added when those inputs have not already been purchased.

A separate rational implementation reconstructs the feature antiderivatives, all winner comparisons, qualification supports, pair-event separations, conflict rosters and 16548 assignment inequalities. It checks all 2,548 qualification certificates and all 36 policy-class comparisons without invoking the production optimizer. Original R77/R80 continuous-bin and actionwise kernel verification is separately replayed in the full publication build. The distinction between new exact certificate verification, inherited full-bin re-integration and independent cost sampling is explicit: this study adds no new fitting attempt, integration bin or independent cost row.

The economic improvement is consequently a more informative decision from the original NBO continuation and its previously qualified evidence. Full-menu winner regions remove alternatives that cannot be selected; common-coefficient restrictions remove jointly impossible selections; conditional integration prices what remains. Each step has its own proof, numerical certificate and work account. The original

broader results and the difficult vector-horizon observations retain their separate assumptions and evidence.

## APPENDIX AD: A FROZEN SHARED PROPOSAL ON UNSEEN WORKLOADS

The joint-domain construction in Section M establishes a mathematical shared-operator guarantee. A trained proposal's incremental numerical value is a different question. We test that value in the original investment model without changing its continuous innovation law, feasible action cap or protected Bellman certificate. This experiment adds a prospective comparison to the inherited recursive-query and complete-policy studies; it does not substitute one of those objects for another.

### AD.1. *Construction, selection and evaluation*

The protocol is fixed before fitting. The training stream contains 128 states in each of dimensions two and eight, with two remaining dates. Protected adaptive search at tolerance 1/256 supplies feasible action labels, normalized by the original action cap. Both approximation families receive the same fourteen features: eight zero-padded state coordinates, the overall, even-coordinate and odd-coordinate means, the mean square, the mean adjacent product, and the dimension divided by eight. The neural model is one width-32 ReLU network shared across both dimensions. The nonneural comparator is one complete quadratic on the same features. Each family has one attempted fit and fixed optimization settings, with no hyperparameter search. All coefficients, training rows and fitting diagnostics are retained before calibration.

Calibration has 64 new states in each training dimension. The five services are the adaptive baseline, bisection, a one-date analytic proposal, the shared ReLU proposal and the shared quadratic proposal. Each runs at tolerance 1/64 using the unchanged protected oracle. Selection minimizes their total action-value queries, with the prespecified tie order. A terminal query is already included in that total and is not added a second time. Calibration selects the adaptive baseline: its 990 queries compare with 999 for bisection, 1008 for the analytic proposal, 996 for ReLU

and 999 for the quadratic proposal. This choice is committed before the evaluation states are generated.

Evaluation contains 128 fresh states in each of dimensions two, four and eight. Dimension four is absent from both training and calibration. All services receive the identical states; three fresh-process repetitions preserve the complete timing distribution. The saved coefficients and evaluation inputs also define the cross-architecture replay. Repetitions and machine replay do not add independent states. All streams are reproducible seeded numerical workloads. A seed alone is not an assumption establishing population randomness; Proposition A.20 and the coverage theorem state the separate assumptions needed for probabilistic extrapolation.

## AD.2. *Results and their economic interpretation*

The two fitting attempts return finite coefficients without a recorded warning. All held-out services meet the common certificate tolerance. The adaptive service attains the smallest total query count in each evaluation dimension; at dimension eight it ties bisection and the quadratic proposal. Its empirical query regret against the declared menu is therefore zero in each dimension. This is not a zero expected-regret theorem and not a comparison with the original Bellman optimum.

The ReLU proposal uses 987, 1017 and 987 queries in dimensions two, four and eight, respectively, against 975, 993 and 969 for the adaptive service. The predictor is genuinely shared and is evaluated on unseen states and an unseen dimension, but it does not reduce verification work in this experiment. The protected first partition already resolves many workloads before any model call. In the remaining workloads, a model's proposed query can fail to improve the protected baseline's allocation. All such calls and refinements are charged. This result identifies the distinction between a valid common operator, a successfully returned fit and a cost-saving proposal; none implies the next.

Complete process times include interpreter startup, coefficient loading, computation, serialization and shutdown. The numerical appendix reports every repetition and peak resident memory on each recorded architecture, rather than only the fastest execution. Logical dense-model arithmetic and disjoint oracle queries remain

TABLE AD.I

SHARED PROPOSALS ON THE HELD-OUT ORIGINAL-LAW WORKLOAD

| $d$ | Service | Nonterminal | Terminal | Model calls | Mean queries | Excess |
|---|---|---|---|---|---|---|
| 2 | Adaptive | 325 | 650 | 0 | 7.61719 | 0 |
| 2 | Bisection | 332 | 664 | 0 | 7.78125 | 21 |
| 2 | Analytic | 333 | 666 | 0 | 7.80469 | 24 |
| 2 | ReLU | 329 | 658 | 69 | 7.71094 | 12 |
| 2 | Quadratic | 334 | 668 | 69 | 7.82812 | 27 |
| 4 | Adaptive | 331 | 662 | 0 | 7.75781 | 0 |
| 4 | Bisection | 333 | 666 | 0 | 7.80469 | 6 |
| 4 | Analytic | 338 | 676 | 0 | 7.92188 | 21 |
| 4 | ReLU | 339 | 678 | 75 | 7.94531 | 24 |
| 4 | Quadratic | 334 | 668 | 75 | 7.82812 | 9 |
| 8 | Adaptive | 323 | 646 | 0 | 7.57031 | 0 |
| 8 | Bisection | 323 | 646 | 0 | 7.57031 | 0 |
| 8 | Analytic | 325 | 650 | 0 | 7.61719 | 6 |
| 8 | ReLU | 329 | 658 | 67 | 7.71094 | 18 |
| 8 | Quadratic | 323 | 646 | 67 | 7.57031 | 0 |

Each row covers the same 128 states in its dimension. All services meet tolerance 1/64. Mean queries counts nonterminal and terminal queries once. Excess is total queries above the best declared service in that held-out dimension, not population or policy-cost regret. Dimension four is absent from training and calibration. Three repetitions and the second machine do not multiply the sample size.

separate accounts. They are not measured CPU instructions, energy expenditures or monetary costs. Since the calibration objective prices only action-value queries, its selected service need not minimize process time or a different priced resource vector. The price-region analysis in Section O.2 gives the appropriate conditional decision rule once those prices and complete resource accounts are specified.

*Protection is not prediction success.* The diagnostic catalogue separately records prediction-free returns, attempted screens, screen closures, protected refinements, invalid predictions, feasibility repairs and zero-progress probes. No failed fit or invalid prediction is omitted; the prescribed safe fallback remains a service outcome,

not successful neural prediction. The two fits in this experiment do not erase the inherited nonconvergence warnings or estimate their population frequency. Original-law policy-cost conclusions continue to come from the separately frozen nine-controller evaluation in Section O, not from this query-proposal table.

## APPENDIX AE: DEPLOYMENT CHOICE UNDER QUALIFIED COST UNCERTAINTY

The attained continuation is an input to a complete economic decision, not merely to another approximation score. The same qualified costs can support a choice of acquisition precision across deployments. We retain the original NBO actors and draw a controller before observing the initial state or innovations. This implements a mixture of complete policies, not an average of their actions, coefficients or information sets. Its expected original-law cost is the corresponding mixture of the qualified costs. Minimax-regret allocation has an established decision-theoretic interpretation (Manski, 2004); the result below concerns the finite cost set supplied by NBO qualification, conditional on its simultaneous coverage event.

Let $\mathcal{C}$ and $d(j,i)$ be the consistent difference-constraint set and distances of Theorem A.49, stated and proved in the Numerical and Development Appendix. For a probability vector $p \in \Delta_q$ and per-deployment charges $w$, set $H_{ji}(w) = d(j,i) + w_i - w_j$, for $i,j = 1,\dots,q$. The charges include all costs that actually vary linearly with deployment shares; additional installation costs are treated separately.

THEOREM A.37—Sharp deployment allocation: *The largest expected regret of allocating complete controllers with probabilities $p$ is*

$$\mathcal{R}(p,w) = \max_{\theta\in\mathcal{C}} \left\{ \sum_i p_i(\theta_i + w_i) - \min_j (\theta_j + w_j) \right\} = \max_j \sum_i p_i H_{ji}(w). \tag{A.78}$$

*Thus the optimal allocation and its value $V(w)$ are the solution of a finite linear program. For any rational $p, \lambda \in \Delta_q$, the computable numbers*

$$L = \min_i \sum_j \lambda_j H_{ji}(w), \qquad U = \max_j \sum_i p_i H_{ji}(w)$$

*satisfy $L \le V(w) \le U$. There exist optimal primal and dual vectors with $L = U$. On the original coverage event, every implemented allocation has expected catalogue regret at*

*most $U$, simultaneously for all $p, w$, including choices made after examining the qualified intervals. The worst-case bound in* (A.78) *is attained by one of the cost vectors already constructed in Theorem A.49.*

This is a decision calculation from the purchased observations, not a new experiment or an assertion that uncertainty reduction changes the true controller costs. Pure choices remain available as vertices of $\Delta_q$. Consequently $V(w) \le \min_i \Gamma_i(w)$, with strict improvement possible when no pure controller is uniformly best. The comparator is the best single qualified controller at the true cost vector. The original Bellman-optimum guarantees and the empirical comparisons between fitted representations remain separate.

*An exact acquisition-fee region.* Let $A$ and $B$ denote the fine and coarse ReLU controllers. The qualified gain is $J_B - J_A \in [l, u]$, where $0 < l < u$. All other controllers are dominated by $B$ when their charges are equal. Suppose only $A$ has the incremental per-deployment charge $f$. For $l < f < u$, the fine-controller share and sharp expected regret are

$$p_A^*(f) = \frac{u - f}{u - l}, \qquad V(f) = \frac{(f - l)(u - f)}{u - l}. \tag{A.79}$$

For $f \le l$, use $A$ alone; for $f \ge u$, use $B$ alone. At the fee midpoint, allocation halves the best pure worst-case regret: $(u - l)/4$ rather than $(u - l)/2$. This is an exact decision bound, not a newly measured expected-cost reduction. In the retained experiment these bounds are approximately 0.0002405736 and 0.0004811472, respectively; Table AZ.I gives outward endpoints from the original rational intervals.

Maintaining both controllers need not be free. If a nontrivial allocation incurs an additional normalized cost $\kappa$, not incurred by either pure choice, its advantage survives precisely when

$$\kappa < \frac{\min\{f - l, u - f\}^2}{u - l}. \tag{A.80}$$

The full installed-subset account and a deterministic finite-cohort implementation are proved in Appendix AZ. They charge every selected controller and do not turn shared experimental qualification into an unmeasured cold-start saving. For $m$

deployments and monetary loss scale $v > 0$, the two-controller setup allowance is $mv$ times the right side. This gives an economically explicit frontier conditional on the actual fee, loss scale and deployment count, without inventing their calibration.

## APPENDIX AF: FROM RETURNED POLICIES TO ECONOMIC RESOURCE DECISIONS

An all-state upper bound and a comparison of actual policy costs answer different economic questions. We now connect the policies returned by a construction frontier to direct cost inference and then to a choice of implementation resources. The construction data select the policies; independently generated evaluation paths assess them. This separation permits a valid comparison of first-crossing policies with different resolutions.

### AF.1. *Simultaneous inference after construction*

Let $\mathcal{H}$ denote all construction information: primitive inputs, numerical labels, fitted continuations, returned policies and complete certificate ladders. Conditional on $\mathcal{H}$, fix a finite catalogue of implemented-policy contrasts. Each implementation includes its observation rule, original witness or nearest-node selector, true-state feasible repair and action quantization. For a contrast $i$, let $D_i$ be the difference in expected discounted costs under a specified initial law. A common-innovation path score evaluates the two policies on their own endogenous trajectories. It is not the difference between two regret bounds.

Suppose the initial and innovation uniforms are partitioned into equal bins. Independent uniform bin indices are drawn, and interval simulation encloses the score for every continuous realization inside the drawn bins. All unresolved state, selector and repair branches remain in the enclosure. If $[A_i, B_i]$ is a deterministic support interval, clip the path endpoints to that support. Write $\underline{m}_i, \overline{m}_i$ for outward sample-mean bounds on the lower and upper endpoint samples and $v_{i,L}, v_{i,U}$ for upper bounds on their unbiased sample variances. For $n \geq 2$ define

$$R_i(v) = \sqrt{2v\ell/n} + \frac{7(B_i - A_i)\ell}{3(n-1)}, \qquad \ell \geq \log(4K/\alpha), \tag{A.81}$$

where $K$ bounds the number of sampled contrasts.

THEOREM A.38—Construction-conditional economic intervals: *Under the stated independent-bin model, conditional on any valid construction history $\mathcal{H}$, with probability at least $1-\alpha$ all catalogue contrasts satisfy*

$$D_i \in [l_i, u_i] := [\max\{A_i, \underline{m}_i - R_i(v_{i,L})\}, \min\{B_i, \overline{m}_i + R_i(v_{i,U})\}]. \tag{A.82}$$

*The same coverage holds unconditionally. Subsequent selection among these covered contrasts, or evaluation at any deterministic implementation-price vector, needs no additional union bound. This conclusion applies only to policies and initial laws included in the catalogue; it does not create direct-cost observations for every unstudied tolerance or initial state.*

The proof is in Supplement, Section 1.10.

In particular, selecting the first policy to meet a specified certificate is permitted when that selection uses construction records rather than the subsequent evaluation paths. Repeated construction clocks are not independent draws from a neural-optimizer population. The statistical randomness here belongs to the evaluation design. A fixed pseudorandom stream implements a reproducible experiment; it is not itself a proof of the independent-bin assumption.

### AF.2. *Replacement-fee identification*

Write $D = J^W - J^F$. If witness is installed, replacing it by FVI changes total cost by $k - D$, where $k \geq 0$ is the replacement fee. If FVI is installed, the change is $k + D$. Thus the two directional gains before fees are $D$ and $-D$.

COROLLARY A.39—The break-even fee frontier: *On an event where $D \in [l, u]$, the two directional gains belong to $[l, u]$ and $[-u, -l]$. The unknown absolute cost difference satisfies*

$$|D| \in [\operatorname{dist}(0, [l, u]), \max\{|l|, |u|\}]. \tag{A.83}$$

*For every $k \geq \max\{0, u, -l\}$ neither replacement strictly lowers expected cost. By contrast, $k \geq \max\{0, l, -u\}$ only removes a* certified *strictly profitable replacement. The two thresholds generally differ. Strict inequalities in the first criterion certify that both replacements strictly increase cost.*

The proof is in Supplement, Section 1.11.

The fee frontier is more informative than a conclusion at one generous charge. The earlier 1/64 replacement decision is retained with its original interpretation and confidence account. The new analysis reports the two directional gain intervals and both thresholds instead of selecting a favorable fee after seeing the data. There is no welfare calibration hidden in the units: a decision maker must supply the economically relevant charge, or use the full frontier to see what remains unidentified.

### AF.3. *Observation costs on the same bit grid*

Let $\mathcal{B} = \{4, \dots, 16\}$ be the available bit precisions. For one fixed returned controller, write $J_b$ for the expected cost of its actual $b$-bit, robustly repaired and quantized implementation. Use $b_0 = 16$ as a common reference and obtain simultaneous intervals for $D_b = J_b - J_{b_0}$. The identity interval at $b = b_0$ is exactly $[0, 0]$. Let $s_b = db \sum_{t<T} B_t$ be the discounted number of coordinate-bits purchased; the net cost is $J_b + \lambda s_b$.

THEOREM A.40—Certified choice of actual net implementation cost: *Suppose all $D_b \in [l_b, u_b]$ simultaneously. For any $\lambda \geq 0$ define*

$$L(\lambda) = \min_b \{l_b + \lambda s_b\}, \qquad U(\lambda) = \min_b \{u_b + \lambda s_b\}, \tag{A.84}$$

$$\widehat{b}(\lambda) = \min \arg\min_b \{u_b + \lambda s_b\}.$$

*Then, simultaneously for all $\lambda \geq 0$,*

$$0 \leq J_{\widehat{b}} + \lambda s_{\widehat{b}} - \min_{b \in \mathcal{B}} (J_b + \lambda s_b) \leq U(\lambda) - L(\lambda). \tag{A.85}$$

*The lower and upper functions are finite piecewise-affine envelopes. All changes in their active lines occur at nonnegative intersections of catalogue lines, with exact comparisons deciding ties. The smallest selected precision $\widehat{b}(\lambda)$ is weakly decreasing in $\lambda$.*

The proof is in Supplement, Section 1.12.

This result makes a statement about *actual* net cost within the declared bit catalogue. It differs from minimizing a uniform all-state loss bound. Both criteria are

useful: the latter gives distribution-free protection relative to the full-information optimum, while (A.85) uses a specified initial law to compare feasible implementations of one returned policy. Neither criterion gives an optimum over arbitrary sensor technologies or all policies.

For a fair comparison the conventional actor receives its own acquisition allowance. Suppose its nearest-node policy has a certified selected-policy account at a reported state, obtained from the graph modulus $L_t^{\mathrm{gr}}$, and its actual interpolated critic has modulus $L_t^f$. Transporting the same selected node action to the true state, then using cell-wide capacity repair, adds at most

$$(L_t^{\mathrm{gr}} + L_t^f + 2D_t K_t^A) r(b) + D_t \Delta_t. \tag{A.86}$$

Indeed, transport the feasible node pair directly to the true state, bound its node distance by the reported-state distance plus $r(b)$, and bound $f_t(\widehat{x}) - f_t(x)$ by $L_t^f r(b)$. Robust repair adds $D_t(2K_t^A r(b) + \Delta_t)$. This proves (A.86) without assuming actor continuity. For a witness critic $L_t^f = L_t^{\mathrm{gr}}$, it reduces to the preceding witness allowance. The empirical comparison therefore evaluates the all-state-account minimizer, the interval upper-net-cost selector, and a descriptive midpoint selector on the *same* integer bit and price grid. The midpoint selector is not called the true optimum.

## APPENDIX AG: CERTIFICATE-GATED NEURAL QUERIES

A prediction of the optimal action and a prediction of a useful numerical query are different objects. Inserting a learned action before examining an already available certificate charges an extra continuation query even when the two boundary actions suffice. We instead make the demand for another query a mathematical decision and let a trained NBO selector determine its location only after that demand has been established. The underlying continuation recursion and the economic optimum are unchanged.

At a recursive state, start with the two feasible action endpoints. Compute the global lower bound from the parabolic minorants of Lemma A.50, including the capacity remainder, and the smallest upper endpoint. Return immediately when

their difference is at most the requested tolerance. Otherwise choose a partition interval with smallest current minorant. A learned action is used only if it lies in the central half of that interval; it then replaces the conventional proposed split. Outside that region the safeguarded parabolic split is used. The rounded split must pass the one-fifth interior check before evaluation. A prediction is evaluated at most once at each recursive state and is reused only as a proposed location. It supplies neither a value endpoint nor a lower bound.

THEOREM A.41—Gated prediction and complete-policy validity: *Under the assumptions of Theorem A.51, the gated service returns the same original-optimum and actual-policy guarantees for every supplied prediction, independently of fitting error. In real arithmetic, with endpoint-plus-capacity width at most $5\delta/8$, a sufficient number of endpoint evaluations at a state is*

$$A_r(\delta, x) = 4 + \left\lceil 5c(x)\sqrt{H_r/(3\delta)} \right\rceil .$$

*Each requested evaluation either reduces the unresolved action partition or terminates the query; no initial interior evaluation is mandatory. If the boundary certificate suffices, no predictor is evaluated at that state. If every prediction is declined at every recursive query, the value endpoints and implemented actions coincide with those of the conventional parabolic service using the same arithmetic and tie rules.*

The proof is in Supplement, Section 1.8.

This theorem establishes robustness to the numerical content of a fitted prediction, not that learning is free or always useful. It also does not state that an arbitrary routed transcript has fewer queries than the adaptive conventional transcript. The executed comparison records both possibilities. Unlike a policy catalogue with a stored reference action at every state, neither service constructs a conventional state lattice before it can return a certified policy.

## AG.1. *State dimension, working memory, and reliability*

For the original scalar model, the primitive bounds $dG_r \le 27$ and $dM_r \le 54$ imply

$$H_r \le \frac{3369}{256}, \qquad \Lambda_r \le \frac{1319}{128}, \qquad \frac{\beta M_{r-1} d}{6144} \le \frac{135}{16384}.$$

Thus the action and scalar-innovation budgets at fixed remaining horizon and tolerance are independent of the number of state coordinates. Every primitive evaluation still requires $O(d)$ arithmetic. A width-$w$ one-hidden-layer ReLU prediction on the recorded $d+5$ features requires $O(dw)$ work and stored parameters; a full quadratic prediction on these features uses $O(d^2)$ terms. Neither cost is included for free in the Bellman recursion.

Let $q_r(\delta)$ be the dyadic quadrature count and let $P_r(d)$ be the cost of one predictor evaluation. Apart from the separately identified exact-arithmetic recovery, a conservative work recursion is

$$W_r(d,\delta) \le CdA_r(\delta) + P_r(d) + A_r(\delta) q_r(\delta) W_{r-1}(d, \delta/(4\beta)).$$

The predictor term is absent on a boundary-certified query. This expression removes the exponential state lattice, not the horizon or innovation-tree dependence. A depth-first scalar implementation can retain the current action partition and one innovation child at a time. The executed batched implementation instead reports its maximum state batch and process peak memory. A zero count of stored state-lattice nodes does not mean zero storage.

The deterministic certificate is valid for any supplied finite network weights. A failure to fit and return such weights is a separate service event. Our reliability estimand is the return count and worst measured complete work over the preregistered finite seed catalogue, with every repetition retained. Numerical workload paths are not optimizer replications. The real-arithmetic cap and the exact-rational precision condition establish algorithmic termination at admissible tolerances; recorded processor seconds are descriptive, not a hardware-uniform worst-case bound.

### AG.2. *An explicit economic purchase criterion*

The decision maker buys a callable controller rather than a table of successful sample decisions. Fix a physical capital numeraire for the normalized state, a conversion $m > 0$ dollars per unit of the model's loss index, a computation charge $p \geq 0$ dollars per measured second, and a fixed deployment fee $f \geq 0$ dollars. For $n$ uses under a specified initial-state law, the monetary objective is

$$\mathcal{C}(\pi) = nm\,\mathbb{E}[J_0^{\pi}(X_0)] + pW(\pi) + f.$$

These are declared preference and service-price parameters, not estimated economic primitives. The policy theorem bounds the expected loss under the original continuous law for every such initial-state law. The finite dyadic workload below measures execution; its average cost is not substituted for that expectation.

If two policies have an independently valid expected-gain interval $[l, u]$ for $\mathbb{E}[J_0^{\pi_0} - J_0^{\pi_1}]$, and their incremental measured service charge is $\Delta = p(W_1 - W_0) + f_1 - f_0$, then the monetary net gain lies in

$$[nml - \Delta,\ nmu - \Delta].$$

This follows by multiplication by the positive numeraire conversion and subtraction of the same service charge. A positive lower endpoint supports purchase; a negative upper endpoint rules it out at the stated prices. Uniform regret certificates alone do not supply a directional expected-gain interval. They instead support a worst-case accuracy requirement imposed before selecting the least expensive eligible service.

Finally, a forecast of amortized query work is distinguishable from a measured complete service. With a fixed model and exactly repeated query workload, if training requires $B$ operations and deployment requires $D_L$ rather than $D_C$ operations per workload, then learning uses fewer total operations after $k$ repetitions precisely when $D_L < D_C$ and $k > B/(D_C - D_L)$. If $D_L \geq D_C$ and $B \geq 0$, no repetition of that unchanged workload produces an operation saving. This arithmetic criterion includes training and cannot be applied to unrelated future states without a new assumption. The executed catalogue reports its own full process clocks, not a favorable hypothetical amortization.

## APPENDIX AH: WHEN A PREDICTED QUERY COMPLETES THE CERTIFICATE

The safeguarded theorem is insensitive to the quality of a supplied prediction. A separate question is when that prediction is sufficient to finish verification. The answer must concern both the selected action and the unresolved lower bound. Proximity to a minimizer alone does not make a coarse lower certificate accurate.

PROPOSITION A.42—Witness accuracy and certificate localization: *Let $f$ be a differentiable, $H$-semiconcave objective with $H \geq 0$ on $[0, c]$, let $a^*$ minimize $f$, and suppose $|f'| \leq \Lambda$. An evaluated proposal $\widehat{a}$ lies in $[0, \underline{c}]$, where $0 < \underline{c} \leq c$ and $c - \underline{c} \leq \Delta$. Each endpoint interval is a valid enclosure of $f$ and has width at most $w$. Suppose the partition containing $\widehat{a}$ covers $[0, \underline{c}]$ and has maximum interval length $h$. Let $L$ be its parabolic lower certificate with the capacity allowance $\Lambda\Delta$, and let $U$ be the smallest upper endpoint, retaining earlier valid bounds as in the query algorithm. With $e = |\widehat{a} - a^*|$ and $\lambda = |f'(a^*)|$,*

$$U - L \leq 2w + \Lambda\Delta + \frac{Hh^2}{8} + \lambda e + \frac{He^2}{2}. \tag{A.87}$$

*For an interior minimizer $\lambda = 0$. At a boundary the linear term cannot generally be removed. Thus the right side at most $\delta$ is a sufficient certificate-return condition, not an assumption that a trained prediction meets it.*

*If old and new valid endpoint records give $(L, U)$ and $(L', U')$ after one additional query, with the running maximum lower bound and minimum upper bound, then*

$$(U - L) - (U' - L') = (U - U') + (L' - L) \geq 0. \tag{A.88}$$

*An added prediction may have zero certified progress. The same identities apply to a conventional query.*

The proof is in Supplement, Section 1.9.

This proposition isolates what a fitted representation can contribute without crediting it with a conventional solution. It can provide a small-error witness and change which unresolved interval is refined. The global partition, endpoint arithmetic and capacity account still have to support that witness. The maximum-width sufficient condition is deliberately conservative; the implemented minimum-minorant rule can stop earlier when only a subset of the partition remains competitive.

### AH.1. *A complete accuracy contract and its resource dependence*

For $T$ dates and target $\epsilon$, write $S_T = \sum_{t<T} \beta^t$. The recorded service chooses a common local tolerance by rounding $\epsilon/(2S_T)$ to binary64 and verifies the final outward allowance. Under $dG_r \le 27$, a uniform per-date acquisition allowance at forty bits is

$$\alpha_b = \left(8 + \frac{297\beta}{16} + 27\right) 2^{-40}.$$

The service certifies $(\delta + \alpha_b)S_T \le \epsilon$. Each descendant request is $\delta/(4\beta)$, rounded as in the actual code and included in its outward endpoint guard. At the smallest request in the two- and three-date catalogues, the declared thirty-two-bit action quantum satisfies the exact-recovery condition $\delta_r > 8\Lambda_r 2^{-32}$. The supplement checks this inequality with rationals for every horizon and target, rather than inferring it from a successful workload.

For fixed remaining horizon and accuracy, the bounds on action endpoints and scalar-innovation bins are independent of state dimension in this normalized economy. The factors $dG_r$ and $dM_r$ are bounded, while every transition and cost still reads $d$ coordinates. A width-$w_\theta$ one-layer ReLU predictor has $O(dw_\theta)$ parameters and evaluation work; a dense quadratic on the same features has $O(d^2)$ terms. These statements identify the absence of a $(n+1)^d$ stored state table. They do not count an arbitrary-precision rational operation as a unit-cost floating operation and do not suppress the product of descendant query counts as horizon grows.

The two fitted implementations and their conventional controls therefore share one original-optimum contract, but need not share a transcript, a returned policy or a runtime. The completed experiments charge their distinct transcripts. A finite-catalogue reliability statement concerns all specified seeds and repetitions, including fitting warnings. It is not a population statement about optimizer success.

## APPENDIX AI: MATCHED READOUTS AND COMPLETE RECOVERY

For $x \in [0,1]^d$, the two fixed feature maps used in the comparison are

$$\phi^{\mathrm{R}}_{jh}(x) = \frac{1}{d}(1 - |4x_h - j|)_+, \qquad \phi^{\mathrm{B}}_{jh}(x) = \frac{1}{d}\binom{4}{j} x_h^j (1 - x_h)^{4-j}, \quad 0 \le j \le 4,$$

with $1 \le h \le d$. Both are nonnegative partitions of unity with $5d$ box readout coefficients. The first is an additive hat basis with an exact ReLU realization; the second is a degree-four Bernstein basis. Both purchase the same verified boundary contexts and approximate action labels, and optimize the same certificate loss. The label-loss control changes the objective, not its hat features. The achieved rational optimization certificate is evaluated after quantization.

### AI.1. *Complete recovery under a validation law*

The validation design draws each coordinate as $U/2^{16}$, with $U$ uniform on $\{0, \dots, 2^{16} - 1\}$. These states are representable by the twenty-four-bit policy acquisition contract, but are not uniform over that finer grid. The following result applies to the declared independent context law and recovery. It does not assert that on-policy recursive states have that law.

PROPOSITION A.43—A finite catalogue of complete work: *Condition on a frozen finite family of predictors and a predictor-independent validation law. Suppose a failed screen invokes a deterministic recovery whose root-query count is at most $K(z)$, and every root query costs at most $W(z)$ under the declared descendant and arithmetic contract. If $0 \le W(z)K(z) \le C_{\max}$, then with probability at least $1 - \alpha$, simultaneously for $M$ fixed predictors,*

$$\mathbb{E}[\textit{recovery work}_\theta] \le \frac{1}{n}\sum_{i=1}^{n} W(z_i)K(z_i)\ell_\tau(z_i, \theta) + C_{\max}\sqrt{\frac{\log(M/\alpha)}{2n}}.$$

*Training, context acquisition, prediction, loading, identity checks and output costs are additional terms in complete work. Replacing $W$ by one gives a root-query result, not a bound on elapsed process time.*

*Proof.* See Supplement, Section BX.1.

The executed validation uses the root-query specialization and records descendant queries separately. It does not fit a runtime model and declare it a valid worst-case endpoint cost. The predictable-deployment result in Section AJ adds an explicit context-transfer premise and a complete operation account.

### AI.2. *An additional vector witness*

In the two-control economy a trained, projected and downward-rounded proposal can supply a new evaluated upper witness. Its insertion leaves the triangular lower cover unchanged. The original corner endpoints remain in that cover.

PROPOSITION A.44—Optional vector witnesses: *Suppose a triangular action cover has a valid global lower bound $L$ and feasible evaluated witnesses with upper bound $U$. Adding any finite number of feasible, correctly evaluated witnesses and replacing $U$ by the minimum of all their upper bounds preserves containment of the original feasible infimum. Every returned witness with $U - L \le \delta$ has original Bellman regret at most $\delta$. The conclusion is independent of the training method and of whether an added witness is a vertex of the lower cover.*

*Proof.* See Supplement, Section BX.2. The vector benchmark changes the trained proposal or the choice of a triangle to refine, not the authority of the global lower cover. Its training objective remains action-label loss.

## APPENDIX AJ: FROM VALIDATED READOUTS TO COMPLETE DEPLOYMENT

The fixed-feature construction and its optimization certificate identify an achieved numerical object. The next issue is its use after training. Independent validation need not imply that the states subsequently visited by a controller are independent. We give a work account that keeps those two sources of information separate and charges the descendants of a failed screen.

### AJ.1. *Predictable selection and changing context laws*

Condition on a frozen family $\Theta$ of $M$ readouts. Write

$$f_\theta(z) = K(z)\ell_\tau(z,\theta), \qquad 0 \le N_\theta(z) \le f_\theta(z) \le K_{\max},$$

where the fallback and its endpoint cap are the ones specified for the context. Let $Z_1^v, \dots, Z_n^v$ be independent validation contexts with law $P$, independent of readout

construction. Set

$$u_\theta = \min\left\{K_{\max},\ \frac{1}{n}\sum_{i=1}^{n} f_\theta(Z_i^v) + K_{\max}\sqrt{\frac{\log(M/\alpha)}{2n}}\right\}. \tag{A.89}$$

During deployment, let $\mathcal{F}_{b-1}$ contain construction, validation, and all observations before block $b$. The selected readout $\theta_b$ is $\mathcal{F}_{b-1}$-measurable. The new context $Z_b$ can depend on this history. Suppose a declared predictable number $\rho_b \in [0,1]$ satisfies

$$\|\mathcal{L}(Z_b \mid \mathcal{F}_{b-1}) - P\|_{\mathrm{TV}} \le \rho_b, \qquad \|P-Q\|_{\mathrm{TV}} = \sup_A |P(A)-Q(A)|. \tag{A.90}$$

This is a distributional condition, not a consequence of matching a model hash.

Use an additive operation model. Let $F_0$ bound fitting, serialization and validation work. Let $h_b$ bound loading, identity checks, prediction, previously purchased endpoints, acquisition and durable output at block $b$. Finally let a nonrandom $c_b$ bound the complete descendant and numerical-recovery work of one additional root endpoint. The budgets must cover all admissible histories, not merely the largest observed runtime. Assume the deployed work satisfies

$$W_B \le F_0 + \sum_{b=1}^{B}\{h_b + c_b N_{\theta_b}(Z_b)\}.$$

This includes a complete-action mesh fallback when that is the fallback validated by the experiment. Substitution of a different fallback requires its own cap and loss contract.

THEOREM A.45—Validated work under predictable deployment: *For a fixed deployment window $B$ and $0<\alpha,\gamma<1$, with probability at least $1-\alpha-\gamma$,*

$$W_B \le F_0 + \sum_{b=1}^{B} h_b + \sum_{b=1}^{B} c_b \min\{K_{\max}, u_{\theta_b} + K_{\max}\rho_b\} + K_{\max}\sqrt{\frac{1}{2}\log(1/\gamma)\sum_{b=1}^{B} c_b^2}. \tag{A.91}$$

*The readout can be selected using the validation sample and the preceding deployment history. The conclusion does not require the deployment contexts to be independent. Under*

*the same local feasibility and certificate premises, this selection does not change the original-optimum policy guarantee.*

*Proof.* See Supplement, Section BW.7.

This theorem gives a prospective route from training to complete work rather than interpreting a training loss as a timing result. A smaller root-query mean is useful when its descendant-weighted savings exceed setup and per-block overhead. The $c_b$ terms expose the continuation-tree cost. Setting them to one would prove a root-query statement, not a full-service statement. The R69 independent-context experiment below uses its declared mesh recovery; its adaptive recursive deployment records are not claimed to satisfy an empirically verified version of (A.90).

## AJ.2. *Policy identity, declined reuse, and adoption*

Cost inference can be combined with a deployment clock only after identifying the controller whose cost was estimated. The release checks the complete readout arrays, tolerance, acquisition precision, action rounding and original kernels. Consider a service processing $B$ blocks of $m$ trajectories each. In $D$ predetermined blocks it declines an incompatible readout and uses the benchmark $\nu$; otherwise it uses a fixed $\pi$. Suppose each block has the same initial law and economic primitives, and the decline schedule is independent of the trajectories. Let $[l, u]$ be a simultaneous interval for $\Delta = \mathbb{E}[J^\pi - J^\nu]$.

PROPOSITION A.46—Complete-service adoption region: *Let $W_\pi, W_\nu$ be complete work observations under the same boundary. If $v \geq 0$ values one normalized expected-cost unit, $p \geq 0$ prices one work unit, and $f \geq 0$ is the incremental installation fee, then the net cost difference belongs to*

$$[\, vm(B-D)l + p(W_\pi - W_\nu) + f,\ vm(B-D)u + p(W_\pi - W_\nu) + f \,] \tag{A.92}$$

*on the original simultaneous event. The statement holds for every such price triple on that event. With observed work saving $S = W_\nu - W_\pi > 0$, strict adoption is supported whenever*

$$pS > f + vm(B-D)u.$$

*If work is instead bounded by valid intervals, replacing $W_\pi - W_\nu$ by its interval difference preserves the statement.*

*Proof.* See Supplement, Section BW.8.

The executed reuse design has $B = 64$, $m = 8$, and one predetermined identity rejection, so its normal-policy multiplier is 504. This does not make an observed prefix clock a completed service, or an observed timing range a confidence interval for future hardware. The paired complete receipts and the conservative envelope across the observed repetitions are both reported. The inputs $v, p, f$ remain declared decision parameters; the experiment estimates neither an installation fee nor the monetary value of normalized investment losses. The policy-cost margin supplies an economically interpretable eligibility condition, while the full adoption region keeps computational savings and policy quality separate.

## APPENDIX AK: TRAINING, REUSE, AND POLICY-COST PRECISION

The experiment links the constructive readout to the actual controlled economy. It compares representation and training loss at a common information budget, validates a frozen readout on independent contexts, measures a longer deployment window, and evaluates the returned controllers at a prespecified policy-cost margin. The original continuous laws, state-dependent investment capacity and Bellman target are unchanged.

### AK.1. *Design and information balance*

The scalar tasks are $(d, T, \varepsilon) = (2, 2, 1/128)$ and $(8, 3, 1/64)$, with twenty-four-bit state acquisition and thirty-two-bit downward action rounding. Four fitted proposals receive the same forty-eight training states, approximate action labels and verified boundary endpoints per nonterminal date. The certificate-aware hat–ReLU and degree-four Bernstein readouts have exactly $5d$ box-constrained coefficients and the same objective, penalty and optimizer budget. An action-label readout on the same hat features separates the objective from the representation. A quadratic label fit retains its explicitly different parameter count. Adaptive parabolic search

and upper-chord minimization are prediction-free controls; unused training is not charged to them.

The hat basis has an exact finite ReLU realization. It also has a conventional piecewise-linear interpretation, so this comparison does not count two encodings of that basis as independent algorithms. The Bernstein comparison holds the coefficient count and purchased information fixed. All fitted proposals use the same final certificate. Every returned certificate-aware iterate is quantized and receives an exact rational value of the objective and box dual gap from Proposition A.54. The achieved gap is reported rather than assuming that an optimizer's success flag proves optimality.

The protocol fixes two fitting seeds and four process repetitions. It contains eighty cold comparisons, sixteen independent-validation services, fifty-six reuse services, two paired-inference services and thirty-four vector services: 188 complete processes. All returned. This is finite-catalogue evidence for the complete verifier, not a population claim about fitting success. Four clock repetitions are not four new training samples. The construction, receipts and original source freeze predate this manuscript integration; the publication replay adds no scientific observations.

The complete cold-start results are reported in Supplement Table AQ.I. At a fixed purchased-information and coefficient budget, the hat–ReLU readouts close 48 and 50 of 256 small-task validation contexts for the two seeds, compared with 25 and 21 for Bernstein. The achieved exact box dual gaps are at most $10^{-7}$ across the certificate-aware catalogue. This is a method-specific improvement in the declared context screen, not a uniform complete-runtime advantage. All methods and both tasks remain in the supplementary validation table.

Cold-start work remains an exacting comparison. In the two-state task the median complete hat–ReLU process takes about 0.465 seconds, against 0.365 for adaptive search. In the eight-state task the respective medians are about 1.618 and 0.666 seconds; both use 2,594 policy-query evaluations in the displayed workload. These counts exclude separately reported training queries. Thus the certified objective is now an executed arm, but its execution is not an identified complete-work improvement. The recorded fitting, prediction and descendant counts locate the cost rather than attributing every successful return to a successful predictor.

## AK.2. *Prospective validation and reuse*

Each fitted readout is frozen before a new predictor-independent context stream drawn uniformly from the sixteen-bit dyadic state grid is generated. There are 256 contexts per nonterminal date. The validation calculation executes the declared complete-action mesh recovery whenever the pre-query screen fails. Its clipped loss bounds the additional root queries on every recorded context. The supplementary table gives the achieved loss, actual recovery queries, simultaneous validation upper bound and descendant counts for every task, date, method and seed. These 6,144 method–context–date evaluations validate the stated context law and fallback; they are not relabeled as independent on-policy recursive states.

Each reuse process trains and serializes once and then serves sixty-four predetermined blocks of eight trajectories. At block index 16 it deliberately receives a stale acquisition identifier and uses the adaptive benchmark. Loading, hashing, identity validation, rejection, queries, cash-path enclosures, durable files and process shutdown are included. A stable observed crossing must begin by block index 48 and remain below the matched adaptive prefix through index 63. This requirement is fixed before execution.

Three of the eight hat–ReLU processes in the two-state task satisfy that stable-prefix criterion; four have smaller final complete times than their matched adaptive process. The Bernstein and quadratic rules also exhibit small-task crossings. None of the learned processes wins the eight-state reuse comparison. In the small task, the adaptive complete times range from about 3.17 to 7.13 seconds, while hat–ReLU ranges from about 3.32 to 5.07 seconds. Consequently, a favorable matched crossing cannot be interpreted as uniform amortized superiority. The operation counts and every unfavorable repetition remain in the record.

The deployment theorem states what additional context-law and descendant-cost information would be sufficient for a prospective work bound. The experiment does not substitute its identity check for a distributional drift bound. Likewise, the sixty-four-block window is not extrapolated to an unobserved stopping horizon.

TABLE AK.I

SIXTY-FOUR-BLOCK PROSPECTIVE REUSE

| $(d, T)$ | Rule | Policy queries | Median (s) | Range (s) | Wins | Stable |
|---|---|---|---|---|---|---|
| 2,2 | A | 10057 | 3.621 | [3.169, 7.129] | – | – |
| 2,2 | B | 10392–10502 | 3.521 | [3.320, 3.572] | 4/8 | 4/8 |
| 2,2 | H | 10332–10447 | 3.746 | [3.320, 5.075] | 4/8 | 3/8 |
| 2,2 | Q | 9982–9992 | 3.621 | [3.120, 4.473] | 4/8 | 3/8 |
| 8,3 | A | 218656 | 24.914 | [24.762, 25.216] | – | – |
| 8,3 | B | 220282–220579 | 26.444 | [26.068, 26.921] | 0/8 | 0/8 |
| 8,3 | H | 220498–222019 | 26.469 | [25.915, 27.568] | 0/8 | 0/8 |
| 8,3 | Q | 223929–226193 | 27.872 | [26.964, 28.973] | 0/8 | 0/8 |

A complete-time win is relative to the adaptive process with the same repetition index. Stable means an observed cumulative crossing by index 48 that persists through index 63. The last summary and shutdown belong to complete clocks, not prefix clocks. The stale identity at index 16 is declined in every process. The same adaptive repetition is shared across matched seeds; these comparisons are not independent trials.

## AK.3. *A cost comparison at the declared precision*

The central inference change is the Bellman-centered observation of Theorem A.63. It estimates the actual expected cost difference by evaluating optimal advantages along each actual policy trajectory. The conditional Bellman expectations are evaluated with a tighter tolerance and paid for separately. The estimator is not the difference between two policy-loss upper bounds, and its realized observation is not asserted equal to a same-shock cash-cost difference.

For each scalar task the protocol fixes 1,024 new common-path rows, forty-eight-bit continuous bins and evaluation tolerance $\varepsilon/32$. The policies are adaptive search, certificate-aware hat–ReLU, certificate-aware Bernstein, and the original R67 ReLU controller at its pinned fitting seed. That historical controller is loaded unchanged. The predeclared loss margin is $1/256$, with numerical and sampling half-width allocations of $1/512$ each. No new row is added after inspecting a sign or a failed comparison. There are twelve paired contrasts across the two tasks on one centered-inference event with failure probability at most 0.01.

TABLE AK.II

BELLMAN-CENTERED EXPECTED POLICY-COST DIFFERENCES

| $(d, T)$ | Pair | Simultaneous interval | Numerical width | Radius | Margin |
|---|---|---|---|---|---|
| 2,2 | A–B | [-0.000683, 0.000709] | 0.000194 | 0.000599 | Yes |
| 2,2 | A–H | [-0.000683, 0.000708] | 0.000194 | 0.000599 | Yes |
| 2,2 | A–R | [-0.000658, 0.000728] | 0.000188 | 0.000599 | Yes |
| 2,2 | B–R | [-0.000670, 0.000715] | 0.000187 | 0.000599 | Yes |
| 2,2 | H–B | [-0.000695, 0.000695] | 0.000193 | 0.000599 | Yes |
| 2,2 | H–R | [-0.000670, 0.000715] | 0.000187 | 0.000599 | Yes |
| 8,3 | A–B | [-0.001603, 0.001603] | 0.000812 | 0.001197 | Yes |
| 8,3 | A–H | [-0.001603, 0.001603] | 0.000812 | 0.001197 | Yes |
| 8,3 | A–R | [-0.001583, 0.001613] | 0.000801 | 0.001197 | Yes |
| 8,3 | B–R | [-0.001583, 0.001613] | 0.000802 | 0.001197 | Yes |
| 8,3 | H–B | [-0.001603, 0.001603] | 0.000812 | 0.001197 | Yes |
| 8,3 | H–R | [-0.001583, 0.001613] | 0.000802 | 0.001197 | Yes |

Pair first minus second; R is the unchanged R67 ReLU readout at seed 6603. Every task uses 1,024 new common-path rows, not that number per pair. All intervals are outward-rounded. Margin denotes containment in the prospectively specified interval $[-1/256, 1/256]$, not strict superiority. The numerical term is a full width; the sampling term is a half-width.

All twelve centered intervals lie within $[-1/256, 1/256]$. They therefore establish two-sided equivalence at the declared normalized-loss margin, including comparisons involving the unchanged historical neural controller. Every interval still contains zero: strict cost superiority is not identified. Both precision allocations are met. This replaces the earlier inability to assess the declared margin by a positive, original-law eligibility result, without changing the economic estimand or assuming a fitted value is exact.

For adaptive-minus-hat–ReLU, the two-state interval is approximately $[-0.00068263, 0.00070764]$ and the eight-state interval is approximately $[-0.00160291, 0.00160253]$, with outward display rounding. The corresponding cash-cost intervals on the same new rows have widths about 0.423 and 0.529. The centered widths are about 0.00139 and 0.00321. This is an identified precision

improvement for the same expected-cost target, not an identified neural speed advantage. The complete shared inference processes cost about 9.03 and 408.63 seconds, respectively. Conditional Bellman integration, rather than state-bin width, dominates their numerical enclosure budget.

The two controllers' common initial-value terms cancel in expectation. All compatible true states within a sampled bin remain covered. No new inference trajectory in this catalogue has an unresolved acquisition branch. The raw conditional-value width, propagated bin width, acquisition allowance and final arithmetic diagnostics are reported separately; they are not assumed statistically independent, and pre-intersection component widths are not falsely presented as an additive decomposition of the tighter final interval.

### AK.4. *Two controls and the economic service*

The vector study compares simplicial search, largest-edge uniform refinement, a trained two-output hat readout and a trained two-output quadratic proposal at two state dimensions and horizons two and three. Both learned rules purchase the same twelve teacher states per nonterminal date. Their training objective is action-label loss, not the scalar convex certificate loss. All receive the same triangular lower certificate and feasible rounding rule under two independent continuous innovations. The horizon-four hat proposal transfers its three-date model and is identified as a transfer, not as a four-date fitted model.

Supplement Table AQ.II reports every two-control workload.

The vector records contain actual continuous-bin policy paths, complete clocks, peak resident memory and descendant counts. Eight paths per ordinary service, and one per horizon-four transfer service, are workload diagnostics rather than a precision-designed population comparison. The experiment adds a trained vector comparison, but it does not identify a vector expected-cost ranking or remove continuation-tree growth. Derivative branch-and-bound is not an executed arm of this catalogue.

For the scalar deployment decision, the release verifies that the cost-inference readouts coincide with the relevant cold and reuse readouts at seed 6901. Param-

eter matching is necessary for pairing these records. Under the additional same-numerical-policy premise, Proposition A.46 gives their complete-service adoption region, with normal-policy multiplier 504 and observed work differences. Cross-platform prediction portability is separately discussed in the supplement. The supplementary table reports the work savings and critical price ratios, including cases in which savings are negative. The loss margin is a prospective numerical design choice, not an estimated installation charge. Equivalence at that margin makes policy quality an eligible comparison; adoption depends on declared prices and complete work, including newly purchased qualification.

### AK.5. *Earlier evidence and independent reconstruction*

The original R67 catalogue remains a separate, unchanged companion, together with all preceding adverse studies. It records 212 returns, no neural complete-reuse winner, unresolved ordinary cost intervals, forty-eight neural fitting warnings and a severe horizon-four vector expansion. Those observations are not overwritten by the new loss, new inference rows or different reuse window. The previous accumulation-aliasing counterexample and corrected execution also remain unchanged.

The new publication audit verifies every R69 receipt and stored-file identity, recomputes the achieved rational training certificates, checks every validation loss and executes independently written rational path calculations. A second interval implementation uses exact full-bin state and cost bounds, not only representative points, for unambiguous acquired paths. Its coverage and every diagnostic disagreement are recorded explicitly. This provides independent reconstruction of the sampled cash-path calculation and independent final inference arithmetic. It is not represented as an independent generation of every optimal-advantage endpoint or of the largest vector Bellman descendant tree.

## APPENDIX AL: ACHIEVED READOUTS AND INDEPENDENT VALIDATION

The two certificate-aware feature families use the same purchased states, action labels, endpoint enclosures, penalty and iteration budget. The replay verifies equality of every training-array payload at a fixed task and seed before comparing the

methods. The label-loss readout shares the hat features but not the objective. The quadratic arm shares information but has a different parameter count. A finite-dimensional hat function has both a conventional piecewise-linear description and an exact ReLU realization; changing the encoding does not create a new independent method.

TABLE AL.I: Achieved convex readout certificates

| $(d,T,r)$ | Rule | Seed | Weights | $F(w)$ | $D(w)$ | $F(w)/\tau^2$ |
|---|---|---|---|---|---|---|
| 2,2,2 | B | 1 | 10 | 0.0187 | 1.72e-08 | 7.35e+04 |
| 2,2,2 | B | 2 | 10 | 0.0161 | 2.56e-08 | 6.33e+04 |
| 2,2,2 | H | 1 | 10 | 0.0183 | 9.8e-08 | 7.19e+04 |
| 2,2,2 | H | 2 | 10 | 0.0154 | 3.26e-08 | 6.05e+04 |
| 8,3,2 | B | 1 | 40 | 0.00909 | 2.47e-12 | 1.89e+04 |
| 8,3,3 | B | 1 | 40 | 0.0112 | 0 | 2.34e+04 |
| 8,3,2 | B | 2 | 40 | 0.00891 | 5.94e-10 | 1.85e+04 |
| 8,3,3 | B | 2 | 40 | 0.00883 | 1.09e-09 | 1.84e+04 |
| 8,3,2 | H | 1 | 40 | 0.00908 | 8.07e-10 | 1.89e+04 |
| 8,3,3 | H | 1 | 40 | 0.0112 | 0 | 2.34e+04 |
| 8,3,2 | H | 2 | 40 | 0.00882 | 8.81e-10 | 1.83e+04 |
| 8,3,3 | H | 2 | 40 | 0.00877 | 4.43e-10 | 1.82e+04 |

These are achieved objective and optimization-gap values, not optimizer flags or hidden-layer convergence claims. The complete rational fractions are retained in DETAILS70.json. Repeated services of the same readout are listed once here and remain separately charged in their receipts.

The objective and dual gap are evaluated on the returned quantized weights using rational contexts and features. The optimization certificate remains valid whether an approximate optimization routine reports success or a warning. The table concerns the achieved convex box problem, not unrestricted hidden-layer optimization. Raw weights, endpoints, labels and exact fractions are retained. Previously reported neural-fitting warnings belong to the unchanged earlier catalogue and are not removed by the successful computation of a new objective.

*Exact context law and counting.* Training and validation coordinates are uniform on the sixteen-bit dyadic grid, not on the continuous state cube or the full twenty-four-bit acquisition grid. Matching methods share these validation draws. The 6,144 method–context–date evaluations are not pooled as 6,144 independent states: every stated validation bound uses its own 256-row record. The separate cost-inference experiment retains the original continuous initial and innovation laws.

TABLE AL.II: Independent-context work-loss validation

| $(d,T,r)$ | Rule | Seed | Returns | Root mean | Upper | All recovery queries |
|---|---|---|---|---|---|---|
| 2,2,2 | H | 1 | 48 | 13.81 | 16.442 | 17680 |
| 2,2,2 | H | 2 | 50 | 13.68 | 16.315 | 17510 |
| 2,2,2 | B | 1 | 25 | 15.34 | 17.000 | 19635 |
| 2,2,2 | B | 2 | 21 | 15.61 | 17.000 | 19975 |
| 2,2,2 | L | 1 | 42 | 14.21 | 16.973 | 18190 |
| 2,2,2 | L | 2 | 47 | 13.88 | 16.482 | 17765 |
| 2,2,2 | Q | 1 | 45 | 14.01 | 16.707 | 17935 |
| 2,2,2 | Q | 2 | 52 | 13.55 | 16.282 | 17340 |
| 8,3,2 | H | 1 | 21 | 15.61 | 17.000 | 19975 |
| 8,3,3 | H | 1 | 23 | 15.47 | 17.000 | 462088 |
| 8,3,2 | H | 2 | 13 | 16.14 | 17.000 | 20655 |
| 8,3,3 | H | 2 | 21 | 15.61 | 17.000 | 469565 |
| 8,3,2 | B | 1 | 25 | 15.34 | 17.000 | 19635 |
| 8,3,3 | B | 1 | 23 | 15.47 | 17.000 | 462088 |
| 8,3,2 | B | 2 | 11 | 16.27 | 17.000 | 20825 |
| 8,3,3 | B | 2 | 20 | 15.67 | 17.000 | 471130 |
| 8,3,2 | L | 1 | 9 | 16.40 | 17.000 | 20995 |
| 8,3,3 | L | 1 | 6 | 16.60 | 17.000 | 486452 |
| 8,3,2 | L | 2 | 5 | 16.67 | 17.000 | 21335 |
| 8,3,3 | L | 2 | 13 | 16.14 | 17.000 | 481419 |
| 8,3,2 | Q | 1 | 8 | 16.47 | 17.000 | 21080 |
| 8,3,3 | Q | 1 | 21 | 15.61 | 17.000 | 464894 |
| 8,3,2 | Q | 2 | 10 | 16.34 | 17.000 | 20910 |
| 8,3,3 | Q | 2 | 4 | 16.73 | 17.000 | 493938 |

Each row has 256 new contexts; seed suffix 1 or 2 denotes 6901 or 6902. The upper bound is independently recomputed from an exact sample sum and an outward radius. It applies to the declared complete-action mesh recovery and context law, not automatically to an adaptive on-policy stream. Recovery-query totals include descendants.

The validation family contains twenty-four task–date–method–seed records with 256 contexts per record. The logarithm allowance twelve covers the stated finite family by a union bound. Reusing a context stream across methods induces dependence between estimators but does not invalidate that bound. The same stream is not counted as additional independent information for each method. The deterministic work inequality is checked on every context. The distributional upper endpoint is recomputed from the exact sample sum, an eighty-digit square root with an explicit expansion, and outward conversion to the printed numerical endpoint.

The full-action recovery mesh in this validation design has a declared root cap $K$. Its actual descendant queries are separately recorded. These observations do not validate a different adaptive fallback by definition. In Theorem A.45, a distributional transfer allowance and a complete endpoint-work cap are additional premises. The theorem therefore handles predictable choices and dependent contexts without asserting that the recorded recursive state stream is independent.

## APPENDIX AM: PRECISION AND THE CONTINUOUS-LAW ESTIMAND

The centered observation is the sum of optimal Bellman advantages along an actual policy trajectory. Its expectation equals policy loss. Subtracting two such observations cancels the common initial optimal value in expectation. It does not cancel cash-flow risk path by path, and no empirical fitted value is substituted for the optimal value. Each conditional Bellman evaluation uses its own tolerance, action covering and continuous-innovation integration account, whose work is charged to the inference service.

The twelve paired intervals form one finite family over the two declared scalar tasks. With logarithm allowance twelve, $2 \cdot 12 \exp(-12) < 0.01$. For example, a rational partial sum of the exponential series already proves $\exp(12) > 2400$; no rounded logarithm is needed for the coverage claim. The ordinary cash-cost estimator uses

TABLE AM.I

PAID INFERENCE WORK AND RAW ENCLOSURE COMPONENTS

| $(d, T)$ | Rule | Evaluator queries | Arm (s) | Conditional | State-bin | Ambiguous |
|---|---|---|---|---|---|---|
| 2,2 | A | 101226 | 1.98 | 0.000113 | 2.3e-13 | 0 |
| 2,2 | B | 101226 | 1.91 | 0.000113 | 2.3e-13 | 0 |
| 2,2 | H | 101226 | 1.91 | 0.000113 | 2.3e-13 | 0 |
| 2,2 | R | 101226 | 1.83 | 0.000113 | 2.3e-13 | 0 |
| 8,3 | A | 2921243 | 100.74 | 0.000595 | 3.7e-13 | 0 |
| 8,3 | B | 2921234 | 102.56 | 0.000596 | 3.7e-13 | 0 |
| 8,3 | H | 2921234 | 102.59 | 0.000596 | 3.7e-13 | 0 |
| 8,3 | R | 2921465 | 101.36 | 0.000597 | 3.7e-13 | 0 |

Component columns are mean full widths before valid regret-range intersections. They do not add to the final narrower contrast width. Arm clocks are internal diagnostics; complete shared inference is charged as a whole: 9.03 seconds, 408.63 seconds. State-bin widths include propagated arithmetic. No acquisition ambiguity is discarded.

a separate empirical-Bernstein family on the same rows. A joint claim about both families requires the sum of their tail budgets, not an assumption that the estimators are independent.

The mean conditional-value and state-bin component widths are recorded before intersecting with valid local regret ranges. The intersection can substantially reduce the resulting interval. Accordingly, those diagnostic widths need not sum to the final centered width, and they are not interpreted as independent errors. The state-bin column includes propagated state arithmetic. The actual final interval width is the mean final numerical width plus the two sampling radii, before any final support clipping.

For every path, the numerical action used by the controller is fixed before the more accurate evaluation. Its feasibility is checked on the true-state box as well as on the acquired state. An unresolved acquisition branch must receive a valid remaining support; it cannot be discarded. None is encountered in the new inference catalogue. The absence of sampled ambiguity is an observation, not a theorem excluding acquisition boundaries from the model.

## APPENDIX AN: INDEPENDENT FULL-BIN RECONSTRUCTION

The new independent implementation imports no NBO numerical kernel. It uses exact rational intervals for the continuous initial and innovation bins, twenty-four-bit acquired-state comparisons, true-state feasibility, stage costs, transitions and terminal costs. It treats the two cumulative endpoints as separate immutable rational numbers. In a transition coordinate, the deterministic function

$$f(u,v) = \frac{1}{16} + \frac{u}{2} + \frac{v}{8} + \frac{u(1-v)}{16}$$

has derivatives $1/2 + (1-v)/16 \geq 0$ and $1/8 - u/16 \geq 0$ on the state cube. Thus evaluation at the lower–lower and upper–upper corners bounds its range without invoking the production interval library. Quadratic costs use exact interval squares; shortage uses the monotone positive-part map followed by an interval square. Both shocks keep their original widths and marginal laws.

The implementation checks that its complete bin enclosure lies inside the archived cash-cost enclosure. A second, independently written rational midpoint calculation checks acquisition and cumulative cost at a represented point. The midpoint test is not substituted for the full-bin test. Across 3,706 physical path archives there are 38,722 recorded path rows; 736 distinct payloads contain 14,242 unique rows. Repeated timing copies are hash-checked and compared, but their duplicate rational evaluation is not counted again. The independent full-bin replay covers all 14,242 unique rows, with no unresolved branch in this executed catalogue.

A separate exact-sum and high-precision calculation checks every final centered confidence interval from its stored advantage endpoints. This independently verifies final inference arithmetic, not the generation of those optimal-advantage endpoints. The regression suite compares primitive identities and small complete value problems with separate rational formulas. It does not independently regenerate the largest multivariate Bellman descendant tree. The release states these different coverage levels rather than assigning an undifferentiated independent-verification label to the whole program.

### AN.1. *Prediction portability and retained disagreements*

An initial strict replay stopped because one Bernstein validation prediction was not bitwise equal to a recomputed value. That failed attempt, its original audit source and traceback are retained on their own branch and in the publication inputs. The failure does not by itself identify an invalid Bellman bound. Nor does a successful rerun on another processor prove the earlier numerical result bitwise reproducible.

The current audit checks the saved actual action on its declared lattice and recomputes its strong-convexity chord with exact rational arithmetic. A discrepancy with a recomputed readout is additionally checked against the declared action-rounding and prediction allowances, not accepted by an arbitrary numerical closeness threshold. The original scientific endpoints, actions, models and source freeze are unchanged. Numerical parameter identity, callable-policy identity and successful certificate verification are distinct statements.

In particular, equality of saved readout arrays is necessary but not sufficient for transferring a cost estimate to an implementation on a different arithmetic runtime. The complete-service purchase region assumes the same numerical callable policy or an additional valid implementation-transfer bound. Matching processor metadata or passing a finite replay is not silently promoted to an all-state policy-identity theorem.

## APPENDIX AO: COMPLETE WORK AND CONDITIONAL ADOPTION

TABLE AO.I: Conditional reuse-stage adoption thresholds

| $(d,T)$ | Rule | Repeat | Saving (s) | Loss upper | $p/v$ threshold |
|---|---|---|---|---|---|
| 2,2 | H | 0 | 0.000 | 0.3441 | 1383.7120 |
| 2,2 | H | 1 | 0.350 | 0.3441 | 0.9821 |
| 2,2 | H | 2 | -0.201 | 0.3441 | – |
| 2,2 | H | 3 | 3.457 | 0.3441 | 0.0996 |
| 2,2 | B | 0 | -0.201 | 0.3440 | – |
| 2,2 | B | 1 | 0.601 | 0.3440 | 0.5727 |

| $(d, T)$ | Rule | Repeat | Saving (s) | Loss upper | $p/v$ threshold |
|---|---|---|---|---|---|
| 2,2 | B | 2 | -0.401 | 0.3440 | – |
| 2,2 | B | 3 | 3.808 | 0.3440 | 0.0904 |
| 8,3 | H | 0 | -1.153 | 0.8079 | – |
| 8,3 | H | 1 | -1.301 | 0.8079 | – |
| 8,3 | H | 2 | -1.399 | 0.8079 | – |
| 8,3 | H | 3 | -1.555 | 0.8079 | – |
| 8,3 | B | 0 | -1.306 | 0.8079 | – |
| 8,3 | B | 1 | -1.606 | 0.8079 | – |
| 8,3 | B | 2 | -1.302 | 0.8079 | – |
| 8,3 | B | 3 | -1.955 | 0.8079 | – |

Seed 6901; zero incremental installation fee and common or previously paid qualification. The loss upper is 504 times the corresponding policy-cost upper endpoint. A displayed positive-saving threshold describes the adoption region only when the numerical callable policy is the one evaluated by inference. Saved parameter equality is checked, but cross-platform numerical equality is an additional requirement. A dash means no positive work saving in that matched observation. Newly purchased qualification work must be added before claiming a first-purchase saving. Prices are declared inputs, not estimated currency values.

*Qualification is not free.* The displayed reuse thresholds condition on a common or previously paid qualification calculation. The process receipts cover the complete reuse service, not the separate policy-cost inference experiment. For a first purchase, let $C_q$ be additional qualification work not paid by the benchmark alternative. In Proposition A.46, use $W_\pi = W_\pi^{\text{run}} + C_q$ and $W_\nu = W_\nu^{\text{run}}$; the observed saving becomes $S - C_q$. This is a change of the explicit accounting boundary, not an additional theorem assumption about free inference. Charging the entire observed shared inference process gives $C_q$ approximately 9.03 seconds in the small task and 408.63 seconds in the larger task. Every displayed matched saving then becomes negative. Thus the current receipts do not identify a positive qualification-plus-reuse work saving under that conservative first-purchase boundary. An allocated marginal inference time or a longer repayment horizon was not separately executed.

The machine-readable table retains both boundaries, and the general price region remains valid with the actual charged work and installation fee.

The multiplier 504 follows from sixty-four blocks of eight trajectories with one predetermined fallback block. In that block the service uses the benchmark itself, giving zero expected policy-cost difference. In the other blocks it uses the fixed candidate. The cost interval can therefore be scaled by 504 under the policy-identity premise of Proposition A.46. A positive work saving is an observed matched process difference, not a confidence bound for future latency. The complete timing range also records all unfavorable repetitions and does not authorize selection of a favorable timing result as a population estimate.

The parent receipt includes child launch, imports, fitting or loading, all policy and path calculations, serialization, child exit and log flushing. Parent receipt bookkeeping belongs to the group clock. Training, inference and publication replay have different boundaries and are not assigned zero cost when excluded from a displayed sub-account. The machine record includes the permitted CPU, thread counts, available frequency and governor data. Processor frequency was not locked by the experiment. Fine timing differences consequently remain descriptive even when operation counts reproduce exactly.

The original normalized investment units are retained. The loss margin is fixed by the numerical protocol; the value of a loss unit, compute price and installation fee are purchaser inputs. This produces a transparent conditional adoption region, not an estimated monetary welfare result. Its noninferiority and equivalence statements concern the actual expected-cost target and do not imply strict neural superiority.

## APPENDIX AP: PRESERVATION AND SCOPE

Every active R67 theorem statement is retained verbatim in the current theoretical sections, and every proof is retained in the article or this supplement. The original empirical article and its full supplement are unchanged companion documents. They retain all earlier conventional winners, unresolved direct-cost comparisons, fitting warnings, vector horizon costs and the accumulation-aliasing counterexample. The broader NBO development remains available under its original diffusion,

recursive-preference, temporal-self and game hypotheses; those hypotheses are not replaced by the convex expected-cost premises of the current query theorem.

The new frozen R69 catalogue is integrated once. Publication replay, repeated clocks, repeated random streams, duplicate path payloads and regenerated tables do not increase its number of independent economic observations. The final source and content map records the location and hash of each retained item. Compilation, regression tests and independent numerical calculations are evidence for their stated checks, not an external editorial decision or a formal proof-checker certificate.

## APPENDIX AQ: COMPLETE COLD-START AND VECTOR WORKLOADS

TABLE AQ.I

COMPLETE COLD-START SERVICES AT A COMMON INFORMATION BUDGET

| $(d,T)$ | Rule | Processes | Policy queries | Median (s) | Range (s) |
|---|---|---|---|---|---|
| 2,2 | A | 4 | 597 | 0.365 | [0.364, 0.366] |
| 2,2 | B | 8 | 607–612 | 0.465 | [0.465, 0.466] |
| 2,2 | H | 8 | 592–612 | 0.465 | [0.465, 0.466] |
| 2,2 | L | 8 | 592–597 | 0.440 | [0.414, 0.466] |
| 2,2 | Q | 8 | 597 | 0.415 | [0.415, 0.967] |
| 2,2 | U | 4 | 597 | 0.390 | [0.364, 0.415] |
| 8,3 | A | 4 | 2594 | 0.666 | [0.615, 0.666] |
| 8,3 | B | 8 | 2594 | 1.618 | [1.618, 1.619] |
| 8,3 | H | 8 | 2594 | 1.618 | [1.618, 1.669] |
| 8,3 | L | 8 | 2594 | 1.518 | [1.517, 1.568] |
| 8,3 | Q | 8 | 2594 | 1.518 | [1.467, 1.519] |
| 8,3 | U | 4 | 2594 | 0.641 | [0.615, 0.666] |

A: adaptive; U: upper-chord minimizer; H: certificate-aware hat–ReLU; B: certificate-aware Bernstein; L: label-loss hat–ReLU; Q: quadratic. Queries concern the policy workload; training and validation queries are additional. Four process repetitions per seed are descriptive clocks, not independent fits or confidence bounds. All complete times include launch and shutdown.

TABLE AQ.II

TWO-CONTROL AND TWO-INNOVATION COMPLETE SERVICES

| $(d, T)$ | Rule | Processes | Queries | Median (s) | Peak KiB |
|---|---|---|---|---|---|
| 2,2 | H | 2 | 98 | 0.47 | 65520 |
| 2,2 | Q | 2 | 99 | 0.49 | 65524 |
| 2,2 | S | 2 | 82 | 0.39 | 65164 |
| 2,2 | U | 2 | 82 | 0.36 | 65328 |
| 2,3 | H | 2 | 5634 | 5.83 | 66004 |
| 2,3 | Q | 2 | 5914 | 5.93 | 66256 |
| 2,3 | S | 2 | 4541 | 3.70 | 65184 |
| 2,3 | U | 2 | 9039 | 7.93 | 65016 |
| 2,4 | H | 1 | 872033 | 62.01 | 65872 |
| 2,4 | S | 1 | 592658 | 41.37 | 65256 |
| 8,2 | H | 2 | 91 | 0.46 | 65780 |
| 8,2 | Q | 2 | 91 | 0.46 | 65864 |
| 8,2 | S | 2 | 75 | 0.36 | 65252 |
| 8,2 | U | 2 | 75 | 0.36 | 65336 |
| 8,3 | H | 2 | 3495 | 5.93 | 66028 |
| 8,3 | Q | 2 | 3495 | 5.85 | 66208 |
| 8,3 | S | 2 | 2645 | 3.55 | 65168 |
| 8,3 | U | 2 | 3278 | 4.60 | 65544 |

S: simplicial; U: largest-edge uniform refinement; H and Q: trained two-output hat and quadratic proposals. Ordinary services have eight continuous-bin paths. Horizon-four services have one path, and H transfers its three-date readout. These are workload records, not precision-designed estimates of population policy costs. Peak memory is the maximum observed resident set within the listed processes.

The tables retain every declared method and task. They are relocated intact from the article to keep its reading path focused; no row, adverse comparison or timing repetition is removed.

## APPENDIX AR: NEW REUSE RECORDS AND INDEPENDENT ARITHMETIC

The new scientific source commitment binds all five new scientific modules and their inherited Python dependencies, together with the exact task and timeout

catalogue. It precedes execution. The numerical study contains no newly fitted predictor and no new independent policy-cost sample. Its roots are predetermined localized states, not a uniform or stationary deployment distribution.

TABLE AR.I: Every new complete-process repetition

| Task | Rule | Rep. | Seconds | Queries | Hits | RSS |
|---|---|---|---|---|---|---|
| S2/2 | Fresh | 1 | 1.166251 | 490 | 0 | 97,804 |
| S2/2 | Exact | 1 | 0.715207 | 115 | 25 | 97,944 |
| S2/2 | Geometric | 1 | 0.665186 | 55 | 29 | 97,808 |
| S2/2 | Exact | 2 | 0.765338 | 115 | 25 | 97,816 |
| S2/2 | Geometric | 2 | 0.668814 | 55 | 29 | 97,816 |
| S2/2 | Fresh | 2 | 1.216814 | 490 | 0 | 97,804 |
| S2/2 | Geometric | 3 | 0.668796 | 55 | 29 | 97,800 |
| S2/2 | Fresh | 3 | 1.115930 | 490 | 0 | 97,928 |
| S2/2 | Exact | 3 | 0.765269 | 115 | 25 | 97,932 |
| S8/3 | Fresh | 1 | 2.739717 | 1,610 | 0 | 97,936 |
| S8/3 | Exact | 1 | 1.921036 | 959 | 7 | 97,944 |
| S8/3 | Geometric | 1 | 1.115856 | 401 | 13 | 97,936 |
| S8/3 | Exact | 2 | 1.867363 | 959 | 7 | 97,932 |
| S8/3 | Geometric | 2 | 1.216160 | 401 | 13 | 97,936 |
| S8/3 | Fresh | 2 | 2.772324 | 1,610 | 0 | 97,932 |
| S8/3 | Geometric | 3 | 1.166799 | 401 | 13 | 97,932 |
| S8/3 | Fresh | 3 | 2.721673 | 1,610 | 0 | 97,944 |
| S8/3 | Exact | 3 | 1.867344 | 959 | 7 | 97,932 |
| V2/2 | Fresh | 1 | 2.422182 | 397 | 0 | 97,932 |
| V2/2 | Exact | 1 | 1.219324 | 125 | 17 | 97,932 |
| V2/2 | Geometric | 1 | 0.915594 | 61 | 21 | 97,932 |
| V2/2 | Exact | 2 | 1.216086 | 125 | 17 | 97,944 |
| V2/2 | Geometric | 2 | 0.919073 | 61 | 21 | 97,936 |
| V2/2 | Fresh | 2 | 2.470714 | 397 | 0 | 97,936 |
| V2/2 | Geometric | 3 | 0.915826 | 61 | 21 | 97,928 |
| V2/2 | Fresh | 3 | 2.471993 | 397 | 0 | 97,932 |
| V2/2 | Exact | 3 | 1.216277 | 125 | 17 | 97,936 |

| Task | Rule | Rep. | Seconds | Queries | Hits | RSS |
|---|---|---|---|---|---|---|
| V2/3 | Fresh | 1 | 21.474661 | 25,534 | 0 | 98,072 |
| V2/3 | Exact | 1 | 14.203954 | 17,023 | 3 | 98,076 |
| V2/3 | Geometric | 1 | 9.388726 | 11,349 | 5 | 98,056 |
| V2/3 | Exact | 2 | 14.556513 | 17,023 | 3 | 98,056 |
| V2/3 | Geometric | 2 | 9.793062 | 11,349 | 5 | 98,008 |
| V2/3 | Fresh | 2 | 22.620244 | 25,534 | 0 | 98,064 |
| V2/4 | Fresh | 1 | 347.971704 | 5,085,947 | 0 | 99,212 |
| V2/4 | Exact | 1 | 168.526652 | 2,542,973 | 2 | 99,216 |
| V2/4 | Geometric | 1 | 84.068560 | 1,271,487 | 3 | 99,212 |

Each row is one separately launched and completed process. No clock repetition is a new fitting seed or an independent policy-cost observation. All stage and source identities remain in its receipt.

TABLE AR.II: Paid fills, rejected records and lookup overhead

| Task | Rule | Rep. | Checks | Stored | Stale | Fill (s) | Lookup (s) |
|---|---|---|---|---|---|---|---|
| S2/2 | Fresh | 1 | 0 | 0 | 0 | 0.5554 | 0.0000 |
| S2/2 | Exact | 1 | 25 | 4 | 3 | 0.1301 | 0.0055 |
| S2/2 | Geometric | 1 | 30 | 2 | 1 | 0.0642 | 0.0039 |
| S2/2 | Exact | 2 | 25 | 4 | 3 | 0.1349 | 0.0056 |
| S2/2 | Geometric | 2 | 30 | 2 | 1 | 0.0641 | 0.0040 |
| S2/2 | Fresh | 2 | 0 | 0 | 0 | 0.5678 | 0.0001 |
| S2/2 | Geometric | 3 | 30 | 2 | 1 | 0.0621 | 0.0041 |
| S2/2 | Fresh | 3 | 0 | 0 | 0 | 0.5363 | 0.0000 |
| S2/2 | Exact | 3 | 25 | 4 | 3 | 0.1344 | 0.0051 |
| S8/3 | Fresh | 1 | 0 | 0 | 0 | 2.1140 | 0.0000 |
| S8/3 | Exact | 1 | 7 | 5 | 4 | 1.2850 | 0.0036 |
| S8/3 | Geometric | 1 | 14 | 2 | 1 | 0.5343 | 0.0029 |
| S8/3 | Exact | 2 | 7 | 5 | 4 | 1.2600 | 0.0039 |
| S8/3 | Geometric | 2 | 14 | 2 | 1 | 0.5502 | 0.0029 |
| S8/3 | Fresh | 2 | 0 | 0 | 0 | 2.1810 | 0.0000 |
| S8/3 | Geometric | 3 | 14 | 2 | 1 | 0.5306 | 0.0033 |
| S8/3 | Fresh | 3 | 0 | 0 | 0 | 2.1113 | 0.0000 |

| Task | Rule | Rep. | Checks | Stored | Stale | Fill (s) | Lookup (s) |
|---|---|---|---|---|---|---|---|
| S8/3 | Exact | 3 | 7 | 5 | 4 | 1.2652 | 0.0038 |
| V2/2 | Fresh | 1 | 0 | 0 | 0 | 1.8126 | 0.0001 |
| V2/2 | Exact | 1 | 17 | 4 | 3 | 0.5964 | 0.0040 |
| V2/2 | Geometric | 1 | 22 | 2 | 1 | 0.3040 | 0.0032 |
| V2/2 | Exact | 2 | 17 | 4 | 3 | 0.6196 | 0.0042 |
| V2/2 | Geometric | 2 | 22 | 2 | 1 | 0.3143 | 0.0033 |
| V2/2 | Fresh | 2 | 0 | 0 | 0 | 1.8519 | 0.0000 |
| V2/2 | Geometric | 3 | 22 | 2 | 1 | 0.3170 | 0.0033 |
| V2/2 | Fresh | 3 | 0 | 0 | 0 | 1.8495 | 0.0000 |
| V2/2 | Exact | 3 | 17 | 4 | 3 | 0.6215 | 0.0041 |
| V2/3 | Fresh | 1 | 0 | 0 | 0 | 20.8660 | 0.0000 |
| V2/3 | Exact | 1 | 3 | 3 | 2 | 13.5845 | 0.0009 |
| V2/3 | Geometric | 1 | 6 | 2 | 1 | 8.7833 | 0.0013 |
| V2/3 | Exact | 2 | 3 | 3 | 2 | 13.9392 | 0.0011 |
| V2/3 | Geometric | 2 | 6 | 2 | 1 | 9.1646 | 0.0012 |
| V2/3 | Fresh | 2 | 0 | 0 | 0 | 21.9832 | 0.0000 |
| V2/4 | Fresh | 1 | 0 | 0 | 0 | 347.3480 | 0.0000 |
| V2/4 | Exact | 1 | 2 | 2 | 0 | 167.8846 | 0.0006 |
| V2/4 | Geometric | 1 | 3 | 1 | 0 | 83.4527 | 0.0006 |

Stored is the maximum number of retained witnesses. Stale counts discarded records at the declared context invalidation. Fill and lookup are internal subclocks, not the complete process boundary. The lookup account includes record-load validation. Repeated storage, hashing, output and shutdown are additionally included in complete receipts.

## AR.1. *Complete timing and retained records*

The process receipt includes creation, Python imports, original Bellman fills, all recursive descendants, cache retrieval, rational checks, repeated serialization, final output and process exit. The driver writes a separate receipt after that boundary. Summing child block clocks is not the complete process time. Every block's body prefix excludes imports and has its own exact definition in the source. The complete receipts, rather than those prefixes, determine the reported timing comparison.

No processor frequency lock is claimed. The start and end records include the selected CPU, frequency fields where the environment exposes them, interpreter version and platform. A cyclic mode order limits a simple fixed-order confound but does not create randomized hardware-frequency inference. Close clock differences require more evidence than the prescribed finite repetitions. Bellman action counts are consequently reported separately from seconds. They are invariant descriptive counts for these transcripts, not predictions of another implementation's throughput.

The 128-record cap bounds logical retained witnesses. It does not bound total process resident memory: a fresh recursive fill can retain batched descendants, interpreter state, interval work arrays and output buffers. The largest process memory remains in the table. The executed cache operates at workload roots only. The four-date pressure is not hidden by counting a later root hit as though the first fill used no descendants.

## AR.2. *Independent checks and trusted premises*

The additional admission audit matches all 708 returns to the prescribed state, control dimension, horizon, tolerance and context epoch. Cache reloads in this catalogue use the same process and numerical model. Arbitrarily supplied external intervals are not made valid by deserialization or a file hash.

For a returned hit, the independent checker locates the earlier paid source record with the same context, horizon, source state, action and endpoints. It recomputes the original regularity constants, state distance, action displacement and interval expansion in exact rational arithmetic. It checks the returned action's lattice and capacity and that the serialized binary endpoints contain the exact expanded interval. A narrow stored interval alone would not prove that a source fill encloses the original value: the inherited verifier and its primitive proof remain explicit premises.

For conditional ReLU integration, the second implementation reconstructs the original projected drift and shocks independently. It does not call the divided-power formula. It clips the support rectangle against the affine neuron's positive half-plane

and uses oriented area and first moments. Degenerate projections reduce to one-dimensional integration or a deterministic positive part. All rational discrepancies are fatal to the new arithmetic audit and are retained as evidence if encountered.

The earlier R70 independent cash-path audit and final inference arithmetic are unchanged. The historical full descendant trees are not certified by pretending that a file hash is a mathematical proof. Neither the new rational transfer checks nor the polygon integrations independently regenerate all those trees. The release states this boundary in its machine-readable audit and response.

### AR.3. *Development probe and publication preservation*

Before the new source commitment, a vector development probe queried $(0.4, 0.6)$ at tolerance 0.04 and horizons two, three and four. The first two returned. The probe was stopped at its 35-second process limit while computing horizon four. Its log is retained separately and is not merged with the later 900-second scientific catalogue or used as a favorable timing observation.

Every R70 active section is assigned to exactly one current document and checked byte for byte. All inherited formal statements and proofs are thereby retained, not reconstructed from a favorable summary. The complete old main article and supplement remain companion PDFs. New tables are derived from the committed receipts and independent audit, and all four current documents are rebuilt from a clean source directory. Visual inspection is documented against the exact PDF hashes actually inspected. Compilation and tests do not constitute external mathematical acceptance.

## APPENDIX AS: Complete continuation-qualification records

The fixed design was committed before any new construction service ran. Its four points are the product of covers with $8^2$ or $16^2$ state nodes and two nine-coefficient bases. The terminal continuation is the original $g$; each preceding date is fitted against the continuation actually returned at the following date. The stage label is an interval containing the full continuous-action Bellman infimum. The coefficient

box, quantum, feature knots, action-cover size and optimization tolerance were fixed in advance.

TABLE AS.I

EVERY ATTAINED READOUT AND ITS FIXED-LABEL LEAST-SQUARES ABLATION

| $n$ | Basis | Date | $\mathcal{W}$ upper | LS bound upper | Dual gap |
|---|---|---|---|---|---|
| 8 | ReLU | 0 | 2.38478 | 3.59860 | 0.000000007 |
| 8 | ReLU | 1 | 3.21372 | 4.91701 | 0.000000014 |
| 8 | Bernstein | 0 | 1.92163 | 5.25637 | 0.000000000 |
| 8 | Bernstein | 1 | 3.67363 | 9.63856 | 0.000000000 |
| 16 | ReLU | 0 | 1.63091 | 1.99849 | 0.000000014 |
| 16 | ReLU | 1 | 2.24222 | 2.75521 | 0.000000014 |
| 16 | Bernstein | 0 | 1.59492 | 2.78996 | 0.000000000 |
| 16 | Bernstein | 1 | 3.02915 | 5.10081 | 0.000000000 |

LS is bounded least squares on the identical features and exact fixed-future interval midpoints. Its bound is evaluated by the same residual-qualification formula. It is not a separately propagated controller. The primal and dual calculations are exact rational checks, independently reconstructed without relying on the optimizer status.

The ReLU basis consists of the constant and, in each coordinate, the identity and positive parts at $1/4$, $1/2$ and $3/4$. The conventional basis consists of the constant and the four nonconstant degree-four Bernstein polynomials per coordinate. Their valid feature moduli are respectively one and four. Having the same number of coefficients does not make their function classes or approximation moduli identical. Each basis receives its own actual fixed-future labels and the same construction rule; the label-loss ablation within each stage shares those exact labels and features.

Every linear-program output is rounded into its declared coefficient box and checked with rational probability multipliers. The table's gap is the achieved primal value minus the exact dual lower bound, not the solver's reported duality tolerance. Constant recentering changes the continuation gauge without changing this objective. The predeclared success tolerance is $1/65536$. The stored status and message remain even when the exact certificate succeeds; an unsuccessful optimizer

would retain its feasible fallback and its actual gap rather than disappear from the catalogue.

TABLE AS.II

COMPLETE NEW CONSTRUCTION WORK AND OBSERVATION SCOPE

| $n$ | Basis | Action values | Feature means | Seconds | Cost rows |
|---|---|---|---|---|---|
| 8 | ReLU | 8448 | 38016 | 1.174 | 0 |
| 8 | Bernstein | 8448 | 38016 | 3.174 | 0 |
| 16 | ReLU | 33792 | 152064 | 4.793 | 0 |
| 16 | Bernstein | 33792 | 152064 | 10.617 | 0 |

Action values include continuous-action covering and whole-cell feasible proposals at both dates. Feature means count evaluations for the fitted future, not terminal formulas. Seconds include both fits, qualification, matched least-squares ablations and the service output; the independent checker has a separate clock. There are no recursive optimal-value queries and no new independent policy-cost observations. Single-service times are descriptive.

The service clock includes both backward constructions, all exact action and feature-mean evaluations, the qualification optimizer, its rationalization, the matched bounded least-squares ablation and the service output. The separate clock record identifies its endpoint. These are single recorded construction times; no estimate of mean runtime, clock stability or a universal neural speed ratio is inferred. The independent audit has its own time and never replaces a scientific service time. Exact arithmetic-operation counts are not a bound on rational bit complexity.

The independent action-value evaluator does not invoke the production positive-part moment expansion. It partitions the innovation interval at every ReLU breakpoint and integrates affine pieces. For the degree-four conventional continuation it uses the positive Boole weights on five equally spaced points; the result is exact for this polynomial degree. Final-date values are checked against the retained original rational investment primitive. It also reconstructs every feasible node action, the original continuous-action lower enclosure, the robust acquired action and its whole-cell gap. Thus the newly computed residual account is tied to original primitives rather than only to a stored hash.

The current construction does not sample a new expected policy-cost difference. The support supplied by its fitted continuation is available to Theorem A.33, but no new stopped investment comparison is asserted. The separately executed fixed-sample comparisons, two-control workloads and cache processes retain their original observations, contracts and coverage accounts.

## APPENDIX AT: EXACT SEQUENTIAL-LEDGER REGRESSIONS

The new regression suite checks conditional means, range reservation, target changes and failed numerical episodes. Its adaptive binary-tree diagnostic enumerates all $2^{12} = 4096$ leaves and every prefix, with exact probabilities and changing predictable ranges. The event count is a finite-model check of the implemented boundary, not an empirical substitute for the supermartingale proof. Rational logarithm bounds and coefficient duals are also tested independently of floating solver stopping flags.

A support failure and a numerical failure are deliberately different branches. An empty intersection or a changed fixed target is rejected. A numerical failure records the prequalified full support, advances the prefix and charges its work. The interface cannot prove that a caller fixed the support before observing the episode; that temporal and probabilistic premise remains part of the algorithm's contract. A cache carried across episodes is included in the target definition or is handled as varying conditional cost.

All prior scientific arrays and sources remain unchanged. The ordinary-source build runs the inherited and current tests and recomputes the published tables. The clean archive includes the transitive R62 input sources, including the original service kernel omitted by the interrupted R72 archive. The reproduction report identifies the suites run inside that archive and compares actual PDF text and page counts after deleting generated PDFs and tables. Visual inspection is recorded separately against the output hashes. None of those publication checks is an external mathematical acceptance decision.

## APPENDIX AU: TRANSPORTING PURCHASED BELLMAN WITNESSES

A second way to control the descendant cost is to reuse a calculation already purchased. Exact memoization only recognizes the same state. The following rule transfers a lower Bellman bound and its feasible upper witness to a nearby state, after repairing capacity and accounting for the full displacement. It is compatible with either a neural or a conventional generator of the original certified record.

### AU.1. *An original-optimum transport certificate*

At remaining horizon $r$, a record at $y$ contains an action $a_y \in A(y)$ and numbers satisfying

$$L_y \le V_r^*(y) \le Q_r^*(y, a_y) \le U_y.$$

It also identifies the economic primitives, remaining horizon and numerical contract under which the record was obtained. A file hash detects changed bytes; validity of the inequalities is a separate premise. For the original scalar and two-control economies, let

$$L_V = G_r, \qquad L_Q = \frac{8}{d} + \frac{11\beta}{16} G_{r-1}, \qquad L_A = \frac{13}{16} + \frac{3\beta d}{8} G_{r-1}.$$

These bound state variation of $V_r^*$ and state and action variation of $Q_r^*$. The action comparison may be made on the ambient simplex $a \ge 0$, $\sum_i a_i \le 1/4$: the original transition remains in the state cube there, even if a particular action violates the smaller capacity at one endpoint.

At a new state $x$, repair $a_y$ to an actual lattice action $\widetilde{a} \in A(x)$. Set $D = \|x - y\|_1$ and $\Delta_a = \|\widetilde{a} - a_y\|_1$. Define

$$L(x) = \max\{0, L_y - L_V D\}, \qquad U(x) = U_y + L_Q D + L_A \Delta_a. \tag{A.93}$$

The truncation at zero uses the nonnegative costs of this economy. It is not part of a general signed-cost transfer rule.

THEOREM A.47—Reuse after capacity repair: *Under the stated record and modulus conditions,*

$$L(x) \le V_r^*(x) \le Q_r^*(x,\widetilde{a}) \le U(x).$$

*Hence outward endpoints with $\overline{U}(x) - \underline{L}(x) \le \delta$ permit the actual repaired action to be returned without a new Bellman query. An identity mismatch, failed feasibility check or excessive width requires a new verified calculation; proximity alone licenses no reuse.*

*The same assertion holds after any history-dependent selection of valid records. At each future date, suppose the selected record or fresh query supplies the original local-gap and acquisition premises of Theorem A.51. The resulting, possibly history-dependent controller retains that theorem's original-optimum loss sum. It need not coincide with the action returned by a fresh search at the same tolerance.*

The proof is in Supplement, Section BY.3.

The implemented repair first rounds the new capacity downward on the $2^{-32}$ lattice, scales a vector action proportionally when necessary, and rounds each coordinate downward. The certificate charges its exact resulting $\ell^1$ displacement, not merely the change in capacity. Retrieval may use approximate distances or a learned index as an ordering heuristic. Every returned hit must still satisfy (A.93) under outward arithmetic.

## AU.2. *A finite-cover account and its premises*

A small number of hits does not establish a uniform reuse complexity bound. Such a bound requires a fill precision, a coverage argument and a retention rule. Let the action quantum be $q$, let $m \in \{1,2\}$, and suppose every fresh record is purchased with gap at most $\gamma\delta$, where $0 < \gamma < 1$. The repair just described obeys

$$\Delta_a \le \kappa D + (m+1)q, \qquad \kappa = 1/(8d).$$

Consider the exact-endpoint transport rule, or an outward rule with a known total additional width allowance $\xi \ge 0$. Define

$$\rho_* = \frac{(1-\gamma)\delta - L_A(m+1)q - \xi}{L_V + L_Q + \kappa L_A}. \tag{A.94}$$

PROPOSITION A.48—Paid fills and complete reuse work: *Suppose $\rho_* > 0$, all valid records are retained, the context is fixed, and every stored candidate is checked before a miss is declared. A record within distance $\rho_*$ certifies a hit. On the unit state cube, the number $M_r$ of fresh records at remaining horizon $r$ is at most*

$$M_r \le \left(1 + \frac{d}{\rho_*}\right)^d. \tag{A.95}$$

*If $C_r(\gamma\delta)$ bounds the complete cost of a fresh fill, a workload of $B$ requests obeys*

$$W_B \le W_{\text{setup}} + M_r C_r(\gamma\delta) + \sum_{b=1}^{B} \left(W_b^{\text{lookup}} + W_b^{\text{check}} + W_b^{\text{load/store}}\right). \tag{A.96}$$

*The lookup and check costs include rejected candidates. Naive retrieval uses $O(M_r d)$ arithmetic operations per request, apart from rational bit costs. Storage, eviction, invalidation and context changes require their own additional fill accounts when the stated retention premises do not hold.*

The proof is in Supplement, Section BY.4.

This bound makes the dimensional price of uniform reuse explicit. It is not a dimension-free caching theorem. It also separates a first expensive Bellman fill from later transport checks. Caching at the workload root can avoid all descendants on a hit, but does not accelerate the descendants of a miss. The executed bounded cache uses 128 records and the requested fill tolerance, not the tighter $\gamma\delta$ premise. Its operation counts below are direct observations, not an application of (A.95). A cache at recursive descendants would need a separate execution and ledger.

## APPENDIX AV: CONTINUATION EVALUATION AND PAID QUERY REUSE

The evidence separates three questions: whether a certified readout returns an admissible policy, whether its actual expected cost meets a declared margin, and whether a purchased calculation can be reused profitably. The preceding training and inference catalogue remains unchanged. The new execution fixes localized query workloads and exact conditional-moment checks before running them. It changes neither the economic primitives nor the continuous feasible-action target.

### AV.1. *Retained trained policies and economic precision*

The retained 188-process catalogue is reproduced in Appendix, Section AK. Hat–ReLU and Bernstein readouts purchase the same teacher information and have exactly $5d$ coefficients; the achieved objective and box dual gaps are recomputed rationally. Independent validation and sixty-four-block reuse receive separate work accounts. Both their favorable and unfavorable repetitions remain visible. In particular, the eight-state task does not identify a learned complete-work winner.

The direct policy-cost experiment fixes its sample count and numerical precision before execution. All twelve original-law paired intervals lie within the declared equivalence region $[-1/256, 1/256]$, and all contain zero. The two adaptive-minus-hat–ReLU intervals are approximately $[-0.00068263, 0.00070764]$ and $[-0.00160291, 0.00160253]$. Their shared inference services cost about 9.03 and 408.63 seconds. These are informative quality comparisons with expensive qualification, not an identified neural cost ranking or a free inference improvement. The complete precision and adoption tables are retained in the appendix.

The frozen-continuation theorem supplies a different way to evaluate that expected-cost estimand. Its moment component is executed here; a new trained critic meeting the old equivalence margin at lower complete qualification cost is not presumed. The earlier optimal-advantage observations are not counted again as samples of the new formula.

### AV.2. *A fixed original-law query catalogue*

The new design has five tasks: two scalar-control tasks at $(d, T) = (2,2), (8,3)$ and three two-control tasks at $(d, T) = (2,2), (2,3), (2,4)$. Each task has three rules: fresh calculation, exact-key reuse and geometric witness transport. The source commitment fixes the task list, tolerances, repetition counts, 128-record capacity, invalidation schedule, workload order and 900-second per-process limit. There are 36 prescribed complete processes. None trains a new action predictor or loads a precomputed conventional policy.

A workload starts at alternating coordinates $0.4, 0.6$. Odd requests repeat that state. Predetermined even requests perturb one coordinate by $2^{-20}$. Except in the

four-date stress case, a late distant request has every coordinate equal to 0.75, and a specified earlier block invalidates the context. The first four tasks have 32, 16, 24 and 8 requests; the four-date task has four. The first three tasks have three timing repetitions, the three-date vector task two, and the four-date task one. Rules are executed in a fixed cyclic order across repetitions.

Every miss calls the unchanged original recursive verifier at the declared tolerance. Every hit uses the exact state-transfer and capacity-repair test. The cache is serialized, hashed, reloaded and validated after every block. The receipt begins before process creation and ends after exit, including imports, all descendant work, file output and shutdown; writing the parent receipt itself is separate publication overhead. A single CPU is pinned. Available start and end frequency information is recorded, but frequency is not locked. Repetitions support descriptive comparisons, not a hardware-uniform or population timing theorem.

The table counts Bellman action evaluations at all descendants, not only workload roots. Geometric hits at a nonidentical state distinguish transport from exact memoization. Their capacity-repair count is separately recorded, including downward repairs that change the returned action. The distant request and context invalidation test whether a cheaper rule refuses an unjustified reuse rather than forcing a hit. An unchanged numerical law with a new context identifier still triggers the declared conservative invalidation.

The four-date task retains the first expensive tree. A hit avoids that tree on a later request, whereas a miss pays for it again. This is a local repeated-query improvement under the original optimum contract, not a disappearance of the long-horizon first-fill cost. The trace records every source witness, state displacement, repaired action and outward endpoint. The absence of a state lattice does not justify omitting either the descendant count or resident memory.

The full repetition and storage ledger is in Appendix, Section AR. Prefix clocks there begin after module loading and are labeled as body clocks; they are not substituted for complete process receipts. The bounded cache and requested fill tolerance do not satisfy the stronger retained-cover premises of Proposition A.48, so that proposition’s uniform packing count is not presented as an empirical result.

TABLE AV.I

COMPLETE ORIGINAL-LAW QUERY WORKLOADS WITH AND WITHOUT REUSE

| Task | Rule | Fills | Hits | Bellman queries | Seconds |
|---|---|---|---|---|---|
| S2/2 | Fresh | 32 | 0 | 490 | 1.166 |
| S2/2 | Exact | 7 | 25 | 115 | 0.765 |
| S2/2 | Geometric | 3 | 29 | 55 | 0.669 |
| S8/3 | Fresh | 16 | 0 | 1,610 | 2.740 |
| S8/3 | Exact | 9 | 7 | 959 | 1.867 |
| S8/3 | Geometric | 3 | 13 | 401 | 1.167 |
| V2/2 | Fresh | 24 | 0 | 397 | 2.471 |
| V2/2 | Exact | 7 | 17 | 125 | 1.216 |
| V2/2 | Geometric | 3 | 21 | 61 | 0.916 |
| V2/3 | Fresh | 8 | 0 | 25,534 | 22.047 |
| V2/3 | Exact | 5 | 3 | 17,023 | 14.380 |
| V2/3 | Geometric | 3 | 5 | 11,349 | 9.591 |
| V2/4 | Fresh | 4 | 0 | 5,085,947 | 347.972 |
| V2/4 | Exact | 2 | 2 | 2,542,973 | 168.527 |
| V2/4 | Geometric | 1 | 3 | 1,271,487 | 84.069 |

S and V denote scalar and two-control tasks; numbers are state dimension and horizon. Queries include every recursive descendant. Seconds are medians of complete processes: three repetitions for S2/2, S8/3 and V2/2, two for V2/3, and one stress observation for V2/4. All prescribed records are retained. These are localized workload comparisons, not population policy-cost samples.

## AV.3. *Independent conditional-moment and transport checks*

The rational integration catalogue uses four $(d, m)$ specifications, namely $(2,1), (8,1), (2,2), (8,2)$, and 64 state–action contexts per specification. Each context has sixteen rational neurons. The original bilinear state drift, action exposure and uniform innovation projections are evaluated exactly. An independent checker clips the corresponding innovation rectangle and integrates the affine neuron through polygon area and first moments. It imports no production moment, cache or interval-query routine.

TABLE AV.II

FOUR-DATE, TWO-CONTROL FIRST-FILL AND COMPLETE WORKLOAD ACCOUNT

| Rule | First-fill queries | All queries | Seconds | RSS (KiB) |
|---|---|---|---|---|
| Fresh | 1,271,487 | 5,085,947 | 347.972 | 99,212 |
| Exact | 1,271,487 | 2,542,973 | 168.527 | 99,216 |
| Geometric | 1,271,487 | 1,271,487 | 84.069 | 99,212 |

Four requested root states, two independent innovations, and common local tolerance 512/14911. The first-fill column counts its full descendant tree. Exact reuse recognizes identical states; geometric reuse additionally transfers a valid witness to a nearby state. RSS is total process peak memory, not logical cache bytes. One repetition is descriptive only.

TABLE AV.III

INDEPENDENT CHECKS OF THE NEW OPERATIONS

| Quantity | Count |
|---|---|
| Prescribed complete processes | 36 |
| Returned / failed | 36 / 0 |
| Checked root returns | 708 |
| Checked reuse returns | 357 |
| Different-state reuse returns | 127 |
| Repaired-action reuse returns | 18 |
| Independent exact neuron integrals | 4096 |
| New independent policy-cost rows | 0 |

The checker imports no production numerical kernel. It verifies each new transfer against a prior paid fill and independently integrates each rational neuron on the original innovation support. It does not independently regenerate the internal optimal-value trees of inherited fills.

The same independent checker verifies every new hit from its earlier paid fill, including feasibility, action lattice, context, original primitive moduli, exact transport endpoints and outward conversion. This is independent reconstruction of the new transfer operation. The inherited recursive fill endpoints remain a disclosed dependency; their entire internal trees are not independently regenerated by this checker. The preceding independent full-bin cash-path calculations and the histor-

ical accumulation-aliasing counterexample remain in the retained appendix and companions.

The moment tests establish exact conditional integration, not the accuracy of an unobserved fitted critic. The query workload establishes a localized certificate-preserving reuse comparison, not independent expected policy-cost observations. A history-dependent transported policy is not identified with the frozen policies in the old cost-inference study. Its economic adoption therefore requires its own eligible cost account or a decision maker's declared worst-case accuracy requirement, together with the measured fill, reuse and qualification charges.

## AV.4. *Constructed continuations and attained qualification*

The readout theorem is executed in four predeclared backward constructions in the original two-state, two-date, scalar-control economy. Two midpoint covers, with eight or sixteen cells per coordinate, are crossed with additive shallow ReLU and degree-four Bernstein features. Each basis has nine coefficients bounded by eight. At each date the future is the continuation actually constructed at the next date. Exact conditional moments supply thirty-three continuous feasible action values per node and thirty-three robust, downward-quantized acquired-action proposals. The original state-dependent capacity and continuous innovation law are unchanged.

TABLE AV.IV

ATTAINED CONTINUATION QUALIFICATION IN THE ORIGINAL INVESTMENT ECONOMY

| $n$ | Basis | $\Omega_h$ upper | Regret upper | Max. dual gap |
|---|---|---|---|---|
| 8 | ReLU | 5.39764 | 8.46716 | 0.000000014 |
| 8 | Bernstein | 5.36565 | 8.11941 | 0.000000000 |
| 16 | ReLU | 3.73299 | 5.28939 | 0.000000014 |
| 16 | Bernstein | 4.43474 | 5.81890 | 0.000000000 |

Each row is one predeclared, complete two-date construction. Bounds are rounded upward; the exact fractions and dates are retained. Regret upper denotes the proved original-optimum account, not an estimated loss or an attained economic equivalence margin. The maximum dual gap concerns only the declared fixed-feature, fixed-future readout objectives.

The table reports the attained two-date residual account $\Omega_h$, the original-optimum bound $\Omega_h + H_h^\pi$, and the largest exact one-step optimization gap. A smaller qualification objective is not itself an identified expected-cost gain. The matched least-squares ablation in the appendix uses each service's identical features and fixed-future labels; it tests the fitting objective, not a separately propagated controller. Both fitting and uniform qualification are paid, and the explicit state cover is tensor based.

All eight datewise optimization gaps are below $1.4 \times 10^{-8}$, against the predeclared tolerance $1/65536$. Their qualified residual bounds are strictly below the matched label-loss bounds; the reductions range from about 18 to 63 percent. The resulting global regret bounds remain conservative, as the table makes explicit. Those objective reductions do not establish a narrow economic equivalence interval or cheaper qualification at the earlier $1/256$ margin.

Every new action value is independently reconstructed using the retained original rational terminal formula or a different exact integration method for the fitted continuation. Continuous-action lower covers, whole-cell choices and primal–dual bounds are then recomputed. These constructions add actual qualified continuations, not new independent cost observations. The stopping theorem has separate exact finite-model regressions; the earlier fixed-sample equivalence intervals are not retroactively given an anytime interpretation.

## APPENDIX AW: COMPLETE CONSTRUCTED-POLICY COST RECORDS

This appendix reports the entire prospectively specified cost catalogue. The policies are the four acquired actors already returned by the qualification-optimal backward constructions, together with zero investment. The constant action on a cell is read from the original integer action array. No proposed readout is retrained on these paths, and no callback to the optimal Bellman recursion replaces its actual fitted future. At a fixed acquisition resolution, the ReLU and Bernstein constructions have the same nine-coefficient budget and state-and-action information. Cross-resolution comparisons answer a different question and retain their distinct observation contracts.

TABLE AW.I: Every marginal original-law cost interval

| Observation | Controller | Cost interval | Full width |
|---|---|---|---|
| C | Zero | [0.773560, 0.780120] | 0.006559 |
| C | R8 | [0.607702, 0.613094] | 0.005391 |
| C | B8 | [0.684025, 0.690150] | 0.006125 |
| C | R16 | [0.607008, 0.612371] | 0.005362 |
| C | B16 | [0.682901, 0.689013] | 0.006112 |
| Y | Zero | [0.773561, 0.780120] | 0.006559 |
| Y | R8 | [0.607716, 0.613107] | 0.005390 |
| Y | B8 | [0.684034, 0.690159] | 0.006125 |
| Y | R16 | [0.607022, 0.612384] | 0.005361 |
| Y | B16 | [0.682910, 0.689022] | 0.006112 |

C is full cash cost and Y integrates the last innovation. Both estimate the same policy cost. All ten intervals belong to the declared joint family. Marginal intervals need not meet the target assigned to paired differences.

TABLE AW.II: Every paired policy-cost interval and declared decision

| Obs. | First minus second | Difference interval | Sign | Width | Equiv. |
|---|---|---|---|---|---|
| C | B8 − B16 | [0.000662, 0.001599] | positive | Yes | Yes |
| C | B8 − R16 | [0.076496, 0.078301] | positive | Yes | No |
| C | R16 − B16 | [-0.077164, -0.075372] | negative | Yes | No |
| C | R8 − B16 | [-0.076436, -0.074682] | negative | Yes | No |
| C | R8 − B8 | [-0.077573, -0.075805] | negative | Yes | No |
| C | R8 − R16 | [0.000227, 0.001191] | positive | Yes | Yes |
| C | Zero − B16 | [0.090176, 0.091590] | positive | Yes | No |
| C | Zero − B8 | [0.089052, 0.090453] | positive | Yes | No |
| C | Zero − R16 | [0.166007, 0.168293] | positive | Yes | No |
| C | Zero − R8 | [0.165317, 0.167565] | positive | Yes | No |
| Y | B8 − B16 | [0.000661, 0.001599] | positive | Yes | Yes |

| Obs. | First minus second | Difference interval | Sign | Width | Equiv. |
|---|---|---|---|---|---|
| Y | B8 − R16 | [0.076491, 0.078296] | positive | Yes | No |
| Y | R16 − B16 | [-0.077159, -0.075368] | negative | Yes | No |
| Y | R8 − B16 | [-0.076432, -0.074678] | negative | Yes | No |
| Y | R8 − B8 | [-0.077568, -0.075801] | negative | Yes | No |
| Y | R8 − R16 | [0.000227, 0.001191] | positive | Yes | Yes |
| Y | Zero − B16 | [0.090168, 0.091580] | positive | Yes | No |
| Y | Zero − B8 | [0.089044, 0.090443] | positive | Yes | No |
| Y | Zero − R16 | [0.165995, 0.168280] | positive | Yes | No |
| Y | Zero − R8 | [0.165305, 0.167552] | positive | Yes | No |

Width and Equiv. are exact-endpoint decisions at the separate predeclared thresholds, each 1/256. Positive zero-minus-controller means a gain from replacing the zero incumbent. Common-path dependence is retained; the twenty intervals are not twenty independent samples.

The width and equivalence columns are different decisions. A precisely positive gain over the installed zero-investment controller fails equivalence because it identifies a material improvement, not because its computation failed. Conversely, a small point estimate with an interval spanning the chosen margin cannot justify equivalence. All decisions use unrounded rational endpoints. The printed intervals are rounded outwards, so a reader need not infer a sign from a decimal that understates uncertainty.

### AW.1. *Acquisition, bin integration and arithmetic*

Each path row contains two initial-state and two innovation bin indices. A forty-bit bin represents its full original-law continuous cell. The implementation does not replace the conditional distribution within the bin by a midpoint. At each date, a rectangle enclosing the true state can intersect several acquired cells. The selected action range then includes every cell in that rectangle. Closed upper boundaries are included conservatively. The endpoint calculation propagates that entire range through the stage cost and next transition.

A zero action and every stored constructed action are feasible on the full cell. In particular, the action quantum is checked against the minimum state-dependent capacity on the cell before costs are evaluated. State uncertainty is therefore not silently removed by using the lower-cell center as the true state. Acquisition ambiguity, continuous-bin variation and arithmetic expansion all contribute to the retained numerical width. They can interact, so a decomposition into three separately additive causal effects is not inferred from a single interval run. The exact aggregate numerical term is reported alongside independent ambiguity counts and the two arithmetic implementations' endpoint differences.

TABLE AW.III: Exact sources of paired interval width

| Obs. | Contrast | Numerical | Variance terms | Support term |
|---|---|---|---|---|
| C | B8 − B16 | 0.000000001 | 0.000020 | 0.000917 |
| C | B8 − R16 | 0.000000001 | 0.000888 | 0.000917 |
| C | R16 − B16 | 0.000000001 | 0.000874 | 0.000917 |
| C | R8 − B16 | 0.000000001 | 0.000837 | 0.000917 |
| C | R8 − B8 | 0.000000001 | 0.000850 | 0.000917 |
| C | R8 − R16 | 0.000000001 | 0.000046 | 0.000917 |
| C | Zero − B16 | 0.000000001 | 0.000497 | 0.000917 |
| C | Zero − B8 | 0.000000001 | 0.000483 | 0.000917 |
| C | Zero − R16 | 0.000000001 | 0.001368 | 0.000917 |
| C | Zero − R8 | 0.000000001 | 0.001331 | 0.000917 |
| Y | B8 − B16 | 0.000000001 | 0.000020 | 0.000917 |
| Y | B8 − R16 | 0.000000001 | 0.000887 | 0.000917 |
| Y | R16 − B16 | 0.000000001 | 0.000874 | 0.000917 |
| Y | R8 − B16 | 0.000000001 | 0.000837 | 0.000917 |
| Y | R8 − B8 | 0.000000001 | 0.000850 | 0.000917 |
| Y | R8 − R16 | 0.000000001 | 0.000046 | 0.000917 |
| Y | Zero − B16 | 0.000000001 | 0.000495 | 0.000917 |
| Y | Zero − B8 | 0.000000001 | 0.000482 | 0.000917 |
| Y | Zero − R16 | 0.000000001 | 0.001367 | 0.000917 |
| Y | Zero − R8 | 0.000000001 | 0.001330 | 0.000917 |

Unrounded components sum exactly to the recorded width. The numerical term is the mean retained full-bin endpoint width, including acquisition and arithmetic effects; these are not asserted to be separately additive causes. Variance terms use exact endpoint sample variances and upward rational square roots.

The first evaluator uses centered polynomial expressions with binary64 outward operations. The second expands the cost and transition polynomials and uses integer floor and ceiling on a $2^{-48}$ grid. It obtains acquired-action extrema by directly enumerating each distinct intersected rectangle rather than reading the first evaluator's range tables. Thus it shares neither floating arithmetic nor cost, transition or lookup routines with the production evaluator. The stored endpoints use the hull, not the more optimistic intersection, of these two evaluations. Intersection with the global primitive support is justified separately.

Every new full-bin path is recomputed by both implementations during replay. A further exact-rational midpoint calculation checks 512 predetermined rows for each policy, including both estimators. Synthetic boundary tests exercise bins at the state and innovation extremes and across acquired-cell boundaries. These pointwise tests supplement the interval-propagation proof; they are not called full-bin verification by themselves. The historical aliasing counterexample and its corrected R67 execution remain unchanged.

## AW.2. *Statistical allocation*

There are 30 estimands and 60 one-sided endpoint events in one fixed family. The empirical-Bernstein radii use the common log allowance ten. The union bound is at most $120e^{-10} < 1/100$, verified independently by a rational lower partial sum for $e^{10}$. All endpoint sums and sums of squares are integer calculations, so no floating sample-variance cancellation enters the confidence endpoints. Upward rational square roots enlarge rather than shrink the radii.

The new family is distinct from the earlier R69 optimal-advantage-centered inference family. Combining conclusions from different families requires adding their error allowances. The R73 construction records contain no new independent cost rows, and replay of the R77 rows adds none. The sample stream, row count, bin precision, policy hashes, estimands and margin are fixed before execution. A

failure of the declared width target is retained; it does not authorize more samples after inspecting the signs. No optional-stopping interpretation is assigned to this fixed-sample experiment.

## AW.3. *Paid work and adoption*

TABLE AW.IV

MEASURED CONSTRUCTION AND COMPLETE INDEPENDENT EVALUATION PHASES

| Policy | Prior build (s) | Float (s) | Integer (s) | Through output (s) | Ambiguities |
|---|---|---|---|---|---|
| Zero | – | 1.890 | 14.528 | 18.184 | 0 |
| R8 | 1.174 | 1.747 | 15.673 | 19.407 | 0 |
| B8 | 3.174 | 1.729 | 14.998 | 18.581 | 0 |
| R16 | 4.793 | 1.692 | 15.463 | 19.353 | 0 |
| B16 | 10.617 | 1.690 | 14.931 | 18.656 | 0 |

Prior construction clocks are inherited unchanged and include the original matched fitting ablations. Through-output time includes lookup preparation, both evaluators, exact midpoint checks and compressed arrays. The two evaluation columns are only components of that total. Ambiguities count endpoint rectangles spanning more than one actor cell. These are inference-service timings, not new generator cold-start competitions.

*The parent process.* The complete five-controller service took 105.049 seconds from process launch through shutdown and log flushing. Exact statistical postprocessing took 8.204 seconds; process peak resident memory was 189128 KiB. Each arithmetic implementation evaluated 2621440 complete policy trajectories, and the additional rational midpoint check covered 2560 predetermined policy trajectories. The recorded number of original-optimum value queries is zero. CPU frequency is observed where available, not claimed locked.

The acquired-action rectangle tables are paid preprocessing, not a free neural operation. Their simple exhaustive construction is practical at the declared small covers but is not claimed to scale linearly in state dimension. The independent integer implementation deliberately pays for another complete calculation. The parent service also pays for reading and hashing controller records, both evaluations, exact

midpoint checks, statistical inference, compression, durable output and shutdown. These costs qualify the returned policies; they do not retime the original backward construction services or establish a new cold-start generator ranking.

Corollary A.36 provides a priced decision for every declared deployment count and monetary scale. In particular, the gain interval in Table Z.I yields an entire replacement-fee frontier, rather than a favorable isolated fee. The zero-investment incumbent is a feasible installed policy in the original economy. An improvement over it does not establish dominance over an adaptive Bellman search or over every conventional basis. The matched feature comparison and the retained conventional-query catalogue must be read separately.

The experiment's units are those of the theoretical investment loss. A currency calibration, an empirically estimated installation fee and a deployment-frequency model would be additional economic assumptions. The mathematical adoption interval states their role explicitly and remains valid for all such nonnegative prices on the same coverage event. It therefore supports a conditional purchase calculation without manufacturing an empirical welfare estimate.

## APPENDIX AX: SIMULTANEOUS CHOICE AMONG CONSTRUCTED CONTROLLERS

The preceding experiment qualifies the same complete controllers returned by the NBO fits. A decision maker must also choose between them. Marginal upper costs give the valid rule in Corollary A.36, but discard the more precise paired observations. We retain all available comparisons on their original simultaneous event. No additional simulation, independence assumption between estimators, or allocation of statistical error is required.

Let $J_i$ be the expected original-law cost of constructed controller $i$, for $i = 1, \dots, q$. Introduce an anchor $\theta_0 = 0$. Every reported marginal interval $[L_i, U_i]$ supplies constraints $\theta_i \le U_i$ and $-\theta_i \le -L_i$. Every reported paired interval $[l_{ij}, u_{ij}]$ supplies

$$\theta_i - \theta_j \le u_{ij}, \qquad \theta_j - \theta_i \le -l_{ij}.$$

Use both estimators when their intervals belong to the joint family of Theorem A.35. Denote the resulting bounded set of cost vectors by $\mathcal{C}$. An edge $a \to b$ of weight $c$

represents $\theta_b - \theta_a \le c$, and $d(a,b)$ is its shortest-path distance. These are classical difference constraints; the computational reduction is not a new graph algorithm (Floyd, 1962). Its role here is to turn paid NBO cost observations into an attainable, price-uniform choice certificate.

THEOREM A.49—Sharp catalogue choice: *Suppose all marginal intervals are finite and $\mathcal{C} \neq \varnothing$. For every $i, j$, the tight upper bound implied by the retained inequalities is*

$$\max_{\theta \in \mathcal{C}} (\theta_i - \theta_j) = d(j,i). \tag{A.97}$$

*For any declared resource charges $w_1, \dots, w_q \ge 0$, put*

$$\Gamma_i(w) = \max_{1 \le j \le q} \{d(j,i) + w_i - w_j\}. \tag{A.98}$$

*Then $\Gamma_i(w)$ is exactly the largest possible cost regret of controller $i$ over $\mathcal{C}$. A minimizer $\widehat{i}$ of $\Gamma_i(w)$ minimizes that worst-case regret among the constructed controllers. On the original simultaneous coverage event,*

$$J_{\widehat{i}} + w_{\widehat{i}} - \min_j (J_j + w_j) \le \min_i \Gamma_i(w).$$

*The inequality holds for all charges on that same event, including charges chosen after inspecting the intervals. Controller $i$ is optimal for every cost vector in $\mathcal{C}$ exactly when $\Gamma_i(w) = 0$.*

The proof in the Numerical and Development Appendix constructs a feasible cost vector attaining each bound. Sharpness refers to the retained interval inequalities, not to all information in the raw paths or to a statistically optimal procedure. The comparator is the qualified finite catalogue, not the original continuous Bellman optimum; the earlier original-optimum guarantees remain necessary. A negative constraint cycle means inconsistent inputs and stops selection rather than authorizing a silently repaired interval. Closure uses $O(q^3)$ rational additions and comparisons and $O(q^2)$ storage; rational bit complexity and all prior qualification work remain additional costs.

*An identified ordering and a priced refinement.* In the fixed constructed-policy catalogue, the joint intervals identify

$$J_{\mathrm{ReLU},16} < J_{\mathrm{ReLU},8} < J_{\mathrm{Bernstein},16} < J_{\mathrm{Bernstein},8} < J_{\mathrm{zero}}.$$

Thus the finest ReLU controller is the unique zero-charge cost minimizer in this catalogue. At matched resolutions, its per-deployment gain over Bernstein is enclosed by $[0.0758058, 0.0775680]$ for $n = 8$ and $[0.0753723, 0.0771589]$ for $n = 16$, rounded outward. These compare actual attained continuations and their complete feasible actors, not optional query proposals. They do not reverse the adverse cold-start and reuse findings for the different recursive-query controllers.

Let $[l, u]$ be the closed interval for $J_{\mathrm{ReLU},8} - J_{\mathrm{ReLU},16}$. Suppose only the finer controller bears an incremental per-deployment charge $f \geq 0$, all other charges being equal. The other three policies are already dominated by the coarser ReLU controller. The two remaining regret certificates are therefore

$$\Gamma_{16}(f) = (f - l)_+, \qquad \Gamma_8(f) = (u - f)_+.$$

The fine controller is universally optimal for $f \leq l$; the coarse one is universally optimal for $f \geq u$. Between those thresholds, the regret-minimizing choice changes at $(l + u)/2$. Here $[l, u] \subset [0.0002277, 0.0011901]$ and the exact switch is approximately 0.00070894. Exact endpoints, all adjacent gaps and attaining cost vectors are deposited. Transitive closure does not further tighten the paired bounds in this particular catalogue; no such improvement is claimed.

For $m$ deployments and monetary scale $v > 0$, replace $f$ by the incremental total charge divided by $mv$. This gives a complete conditional fee region without assigning a monetary value to normalized losses or inferring a deployment horizon from the sample count. Acquiring finer information is an economic choice only after its actual incremental costs are specified.

## APPENDIX AY: PROOF AND IMPLEMENTATION OF SIMULTANEOUS CONTROLLER CHOICE

PROOF OF THEOREM A.49: If $\theta \in \mathcal{C}$, summing its constraints along a path from $j$ to $i$ gives $\theta_i - \theta_j$ at most the path's length. Hence it is at most $d(j,i)$. Feasibility rules out a negative cycle, because summing the constraints around a cycle has left side zero. Every vertex has a finite path to and from the anchor, by the finite marginal bounds. All shortest distances are therefore finite and $d(j,j) = 0$.

Fix $j$ and define a cost vector, including the anchor, by

$$\theta_k^{(j)} = d(j,k) - d(j,0), \qquad k = 0, \dots, q.$$

For each edge $a \to b$ of weight $c$, the shortest-path triangle inequality gives $d(j,b) \leq d(j,a) + c$. Thus $\theta_b^{(j)} - \theta_a^{(j)} \leq c$, and $\theta_0^{(j)} = 0$. This vector belongs to $\mathcal{C}$ and attains $\theta_i^{(j)} - \theta_j^{(j)} = d(j,i)$. This proves (A.97). The same construction proves that a graph with no negative cycle and finite anchor paths has a feasible cost vector; hence the cycle check is an exact consistency test.

For a fixed controller $i$, its cost regret at $\theta$ is

$$\theta_i + w_i - \min_j(\theta_j + w_j) = \max_j\{\theta_i - \theta_j + w_i - w_j\}.$$

The maximum is finite, so taking its supremum over $\mathcal{C}$ commutes with the maximum over $j$. The preceding attainment result gives exactly (A.98). The term $j = i$ is zero, so $\Gamma_i(w) \geq 0$. For a maximizing $j$, the vector $\theta^{(j)}$ attains this regret. Minimizing over $i$ proves the claimed deterministic minimax property. On the coverage event the true cost vector belongs to $\mathcal{C}$; inserting it proves the statistical conclusion without another probability bound. All subsequent arguments are deterministic for every $w$, which establishes price-uniformity. Finally, nonnegative regret is identically zero exactly when its maximum is zero. *Q.E.D.*

The ordinary Floyd recurrence evaluates the all-pairs distances with $(q+1)^3$ candidate relaxations. The implementation uses exact fractions, including the rational confidence endpoints, rather than shortest paths through rounded printed entries. Its independent cross-check runs a separate single-source Bellman–Ford recurrence

from every vertex. Each returned attaining vector is checked against every original edge and against every claimed attained difference. Negative self-loops and longer negative cycles are regression cases; an inconsistent graph cannot pass by resetting its diagonal.

### AY.1. *Refinement charges and the finite comparison*

Let the fine and coarse ReLU controllers be denoted by $A$ and $B$, with $J_B - J_A \in [l, u]$, where these endpoints are the tight implications of $\mathcal{C}$. Suppose all policies except $A$ have a common charge and $A$ has an additional charge $f$. The retained inequalities identify $B$ as less costly than every other controller in the catalogue, so those controllers cannot be a best-cost comparator or a regret-minimizing choice. Equation (A.98) reduces to

$$\Gamma_A(f) = \max\{0, f - l\}, \qquad \Gamma_B(f) = \max\{0, u - f\}.$$

For $f \le l$, only $A$ can have positive strict cost advantages against all alternatives; its worst-case regret is zero. For $f \ge u$, $B$ has zero worst-case regret. On $[l, u]$, the difference of the two regrets is $2f - l - u$, proving the switching formula. At a boundary, universal weak optimality, rather than strict optimality, is the appropriate statement. The independently checked exact fee probes include both boundaries, the switching point, and points outside the ambiguity interval.

TABLE AY.I

ADJACENT GAPS IN THE CONSTRUCTED-POLICY COST ORDERING

| Higher minus lower cost | Lower | Upper |
|---|---|---|
| ReLU, $n = 8$ minus ReLU, $n = 16$ | 0.0002277 | 0.0011901 |
| Bernstein, $n = 16$ minus ReLU, $n = 8$ | 0.0746820 | 0.0764312 |
| Bernstein, $n = 8$ minus Bernstein, $n = 16$ | 0.0006622 | 0.0015984 |
| Installed zero minus Bernstein, $n = 8$ | 0.0890522 | 0.0904430 |

All endpoints are rounded outward. Both original estimators and all marginal and paired intervals enter one consistent difference system. These are the same frozen actors and the same 524,288 rows, not a new experiment. Charges are set to zero only for this statistical cost ordering.

TABLE AY.II

PER-DEPLOYMENT GAINS AVAILABLE TO PAY INCREMENTAL CHARGES

| Replacement | Lower | Upper |
|---|---|---|
| ReLU versus Bernstein, $n = 8$ | 0.0758058 | 0.0775680 |
| ReLU versus Bernstein, $n = 16$ | 0.0753723 | 0.0771589 |
| Refinement from ReLU $n = 8$ to $n = 16$ | 0.0002277 | 0.0011901 |

A positive lower gain licenses any smaller incremental total charge in normalized cost units. For $m$ deployments and monetary scale $v$, multiply the endpoints by $mv$. Construction, acquisition, qualification and installation charges remain explicit; seconds are not assigned an unobserved monetary price.

Both original cash and conditional-estimator intervals qualify the same expected costs and are covered by the same thirty-estimand event. Intersecting their implications therefore needs no independence between estimators and no new multiplicity allowance. It does not convert two estimators on one stream into two independent datasets. The exact constraint ledger, closure, attaining cost vectors, zero-charge regrets and fee probes are in the machine-readable decision record.

All ten unordered contrasts and five marginal costs from each estimator enter the computation. In this particular dataset, transitive paths do not improve the direct paired endpoint intersections. The new decision result is consequently not presented as a new precision experiment. It uses the already purchased joint information to establish an identified catalogue ordering and sharp conditional choice regions. A separate exact test provides a graph in which transitivity strictly tightens an indirect comparison; that test is a mathematical regression, not another observation from the investment economy.

The qualification process is part of the cost of learning these regions. Its historical complete service clock is retained, not replaced by the much shorter graph-processing clock. Controller construction, acquisition, qualification, deployment and installation charges are distinct. A common sunk cost cancels from comparisons only under a decision problem in which it is indeed common and sunk. The arbitrary-charge theorem does not authorize dropping a method-specific cost. Like-

wise, the catalogue includes the actually attained ReLU and Bernstein actors; it does not include the different recursive adaptive-search controller merely because that controller was evaluated in an earlier experiment.

## APPENDIX AZ: PROOF, INSTALLATION COSTS, AND FINITE DEPLOYMENT COHORTS

PROOF OF THEOREM A.37: For fixed $j$, each feasible cost vector satisfies $\theta_i - \theta_j \le d(j,i)$. Nonnegative allocation weights therefore give

$$\sum_i p_i(\theta_i - \theta_j) \le \sum_i p_i d(j,i).$$

The potential $\theta_i^{(j)} = d(j,i) - d(j,0)$ is feasible and attains all of these differences at once. Hence equality is attained for every weighted sum in this display, not just for each pair separately. Commuting the maximum over $\mathcal{C}$ with the finite maximum over $j$ gives (A.78); a maximizing row supplies an attaining potential. The argument would not hold for arbitrary separately sharp pairwise bounds lacking a common attaining vector.

Minimizing the largest row payoff is the program

$$\min_{p,z} z \quad \text{subject to } p \ge 0, \quad \mathbf{1}^\top p = 1, \quad Hp \le z\mathbf{1}.$$

Its dual maximizes $y$ subject to $\lambda \ge 0$, $\mathbf{1}^\top \lambda = 1$ and $H^\top \lambda \ge y\mathbf{1}$. Both simplexes are compact and the matrix is finite, so linear-programming duality gives equal attained values. Independently of optimality, every feasible pair satisfies

$$\min_i (H^\top \lambda)_i \le \lambda^\top H p \le \max_j (Hp)_j.$$

Taking the appropriate infimum or supremum proves the computable gap bound. For rational input, exact rational vertices exist, and the certificate only requires feasibility and these inequalities, not a floating-point solver status. Finally, on the original coverage event the true cost vector belongs to $\mathcal{C}$. All preceding inequalities are deterministic for every $p, w$ on that event. The allocation draw is independent of the deployment trajectory and takes place before information acquisition, so

its expected cost is $p^\top J$. This proves the stated economic interpretation. It is not a pathwise guarantee for a particular random assignment. *Q.E.D.*

### AZ.1. *Installed subsets rather than free simultaneous availability*

Suppose a decision maker serves $m \geq 1$ deployments and may install a nonempty subset $S$ at total normalized cost $K(S)$. Write $K_i = K(\{i\})$, and assume $K(S) \geq K_i$ for every $i \in S$. The benchmark installs the best single controller, paying its own $K_j$. For $p$ supported on $S$, the exact worst-case average regret is

$$\mathcal{R}_S(p, w) = \max_j \left\{ \sum_i p_i H_{ji}(w) + \frac{K(S) - K_j}{m} \right\}. \tag{A.99}$$

Indeed, the installation term is independent of $\theta$, and the same attaining potential proves the formula. The assumption on $K$ makes the cost of an allocation at least the weighted average of its corresponding singleton costs, and hence at least their minimum; the regret is nonnegative. For each subset, (A.99) is a linear program on $\Delta_S$. Enumerating the nonempty subsets and retaining the least optimum yields the exact installation-and-allocation decision. It requires at most $2^q - 1$ programs and is not advertised as dimension-free optimization. Lower bounds from the dual of every subset, together with an attaining primal allocation for the chosen subset, certify a global solution. A controller outside the installed subset remains a possible oracle comparator and must not be removed from the rows.

### AZ.2. *Two-controller formula and extra setup threshold*

With only the fine controller charged $f$, put $a = f - l$ and $b = u - f$. Within the ambiguity region both are positive and $a + b = u - l$. If $p$ is the fine share, the two worst-case comparator regrets are $pa$ and $(1 - p)b$. Their maximum is minimized at their intersection, giving $p = b/(a + b)$ and value $ab/(a + b)$. Outside that region a uniformly optimal pure controller has zero regret. If $l = u$, the gain is known exactly: use the fine controller below $l$, the coarse one above it, and any allocation at equality; no division by a zero width is needed.

When both controllers incur additional cost $\kappa$ only for $0 < p < 1$, their optimal nontrivial allocation is unchanged and its value is $ab/(a + b) + \kappa$. The best pure

value is $\min\{a,b\}$. Subtracting gives the exact saving $\min\{a,b\}^2/(a+b)$ and proves (A.80). Equality gives a tie. A positive extra cost can make the pure rule optimal; the deployment analysis never assumes that loading or operating an additional controller has zero cost. Formula (A.99) handles unequal singleton installation charges as well.

### AZ.3. *A deterministic allocation to a finite cohort*

Random assignment is unnecessary when $m$ homogeneous deployments with the same expected cost law can be assigned in advance. Deploy $A$ on exactly $k$ tasks and $B$ on the others, without choosing tasks according to unqualified outcome information. The average expected cost is the mixture with $p = k/m$; independence between tasks is not needed for this expectation identity. For $l < f < u$, a minimizing integer $k$ is one of $\lfloor mp_A^* \rfloor$ and $\lceil mp_A^* \rceil$. This follows because the piecewise-linear regret decreases up to $p_A^*$ and increases afterwards. Nearest rounding also supplies

$$0 \le V_m(f) - V(f) \le \frac{\max\{f-l, u-f\}}{2m}.$$

An extra two-controller setup charge is included only for $0 < k < m$; both pure endpoints must then also be tested. Nonhomogeneous deployment laws require separately qualified cost vectors and are not covered by relabeling the existing sample as a cohort.

The deposited implementation checks exact probability feasibility, all original cost inequalities at each attaining potential, and equality between primal and dual payoffs. Two-policy formulas are checked against a separate exhaustive rational vertex solver on the full five-controller catalogue. Finite-cohort choices are checked against every integer allocation. The original thirty cost estimands, 524,288 evaluation rows, fitted coefficients, scientific clocks and adverse selector comparisons remain unchanged. The optimization tables contain no new simulation or timing evidence. The sharpness claim is relative to the retained interval system; neither an exact decision solution nor a positive finite comparison establishes a population-wide neural ranking.

TABLE AZ.I

QUALIFIED ACQUISITION-FEE CHOICES AND DEPLOYMENT ALLOCATIONS

| Fee position | Fine share | Pure regret upper | Mixed regret upper | Setup saving lower |
|---|---|---|---|---|
| 0 | 1 | 0.00000000 | 0.00000000 | 0.00000000 |
| 1/4 | 3/4 | 0.00024058 | 0.00018044 | 0.00006014 |
| 1/2 | 1/2 | 0.00048115 | 0.00024058 | 0.00024057 |
| 3/4 | 1/4 | 0.00024058 | 0.00018044 | 0.00006014 |
| 1 | 0 | 0.00000000 | 0.00000000 | 0.00000000 |

Fee position $t$ means $f = l + t(u - l)$. Both controllers were qualified under the original law; a share selects an entire controller before information acquisition. Regret is worst-case expected loss relative to the best single catalogue controller. The last column is the largest extra normalized cost of maintaining both controllers compatible with strict improvement; equality gives a tie. Values are outward rounded. No row is a new simulation or a calibrated installation fee.

## APPENDIX BA: NEURAL BELLMAN QUERIES WITHOUT A STATE LATTICE

The preceding construction makes numerical error accountable, but a complete tensor reference can absorb the benefit of a fitted continuation. We therefore change the computational organization, not the economic problem. A Neural Bellman selector proposes an action at the state actually acquired. A recursive interval query then bounds the original Bellman value at that state and checks the proposed action against the entire continuous feasible interval. No conventional policy is precomputed at all states. The controller is the resulting callable algorithm, not the unverified proposal alone.

The distinction matters for attribution. Learning may reduce the work needed to locate a useful action. It does not supply the inequalities used to certify that action. The adaptive conventional comparator receives the same inequalities and the same original-optimum target. The trained value-fit policies in the preceding study remain distinct from the action-query predictor introduced here.

### BA.1. *A curvature lower bound on an adaptive action partition*

Use the original scalar economy of Section BT, with $r$ dates remaining. Write $V_r$ for its optimal value and $Q_r(x, a)$ for the value of a first action followed by the

optimal future. Let $G_r, M_r$ be the primitive constants in Theorem A.60, indexed by remaining horizon, and put

$$H_r = \frac{21}{4} + \beta M_{r-1}\frac{5d}{32}, \qquad \Lambda_r = \frac{13}{16} + \beta G_{r-1}\frac{3d}{8}.$$

Thus $H_r$ is an upper curvature bound in the scalar action, and $\Lambda_r$ bounds the absolute action derivative. Neither depends on a fitted model.

LEMMA A.50—Parabolic minorants: *Suppose $f$ is $H$-semiconcave on $[l,u]$, and $L_l \le f(l)$, $L_u \le f(u)$. For $s \in [0,1]$,*

$$f(l + s(u-l)) \ge (1-s)L_l + sL_u - \frac{H}{2}(u-l)^2 s(1-s) =: p(s).$$

*Consequently $\min_{[0,1]} p$ is a lower bound for the minimum of $f$ on the interval. With $K = H(u-l)^2/2$ and $D = L_u - L_l$, this lower bound is $L_l$ if $D \ge K$, $L_u$ if $D \le -K$, and otherwise*

$$\frac{L_l + L_u}{2} - \frac{K}{4} - \frac{D^2}{4K}.$$

*The endpoints $L_l, L_u$ need not themselves belong to a convex interpolant. Increasing $K$ preserves the lower-bound property.*

The proof is in Supplement, Section 1.6.

At an acquired state, retain an ordered partition of the feasible action interval and an outward value bracket at every partition point. The minimum of the parabolic minorants bounds the continuous-action infimum from below. The smallest upper endpoint supplies an actual feasible witness. A neural or polynomial proposal inserts another point into this partition; it cannot remove the endpoints or any part of the feasible interval. The running lower bound is the maximum of all previously valid global lower bounds, so new information cannot weaken the certificate. The upper bound is the minimum over every evaluated feasible witness.

### BA.2. *Recursive queries under the original continuous law*

For an endpoint action with $r > 1$, divide the scalar innovation into $q_r$ equiprobable bins and query $V_{r-1}$ at each conditional-mean next state. If each child returns

width at most $\eta_r$, the endpoint bracket has width at most

$$\beta\eta_r + \frac{\beta M_{r-1} d}{6144 q_r^2} + \omega_r. \tag{A.100}$$

Here $\omega_r$ includes outward primitive arithmetic and the Lipschitz expansion from an interval-valued next state to its queried center. Convexity provides the lower conditional-mean inequality; semiconcavity and the exact conditional variance provide the upper allowance. The innovation remains continuous. There is no state-interpolation term because a child value is queried rather than read from a state lattice.

For a requested local gap $\delta$, the implementation sets $\eta_r \le \delta/(4\beta)$ and chooses a dyadic $q_r$ so that the second term in (A.100) is at most $\delta/4$. A separate endpoint-width guard charges arithmetic and the small capacity-rounding remainder. Failure of that guard is a precision failure, not a successful return. The terminal query uses analytic integration instead of recursive quadrature.

THEOREM A.51—A state-grid-free NBO service: *Consider the original scalar investment economy. At every recursive query, suppose the outward endpoint and capacity-transfer accounts give width at most $5\delta/8$. Let the lower bound be the running maximum described above. Refine an interval attaining the smallest current minorant whenever the global upper-minus-lower gap exceeds $\delta$. Each new point must lie between one fifth and four fifths of that interval. Arbitrary additional learned proposals may be inserted initially. Then:*

*(i) every returned interval $[L_r(x), U_r(x)]$ contains $V_r(x)$, and its returned feasible action $a_r(x)$ satisfies*

$$Q_r(x, a_r(x)) - V_r(x) \le U_r(x) - L_r(x) \le \delta;$$

*(ii) the real-arithmetic refinement terminates. With at most one interior learned proposal, a sufficient point budget is*

$$6 + \left\lceil 5c(x)\sqrt{H_r/(3\delta)} \right\rceil; \tag{A.101}$$

*(iii) for the policy that calls this service at every subsequent date, the original-law loss satisfies*

$$J_T^{\pi}(x) - V_T(x) \le \sum_{t=0}^{T-1} \beta^t \left[ \delta_t + \left( 8 + \frac{11\beta d G_{T-t-1}}{16} + d G_{T-t} \right) 2^{-b} \right], \tag{A.102}$$

*when the state is acquired downward at b bits per coordinate and every returned action is feasible at that acquired state. All action rounding must precede its endpoint evaluation. The inequalities hold uniformly in the true initial state, not only at the states used in an experiment.*

The proof is in Supplement, Section 1.7.

The finite-precision contract is explicit. The executed service uses the width guard, a 512-point cap, and a 32-bit downward action lattice. A precision or cap failure withholds the floating certificate and invokes a separately counted exact-rational enumeration. This fallback constructs values only at the required descendants; it does not load a precomputed conventional policy. The rational fallback chooses dyadic action and quadrature counts until each cover or quadrature allowance is at most one quarter of the request, and requests child width at most one quarter divided by the discount. Provided $\delta > 8\Lambda_r 2^{-32}$ at every recursive level, its exact brackets have width at most three quarters of the request. This proves a finite return for every acquired state in the declared tasks, including states absent from the workload. Rational bit-operation cost is additional work and is not replaced by a unit-cost floating clock. The mathematical point budget is checked against the declared tolerance range; termination in an arbitrary floating implementation is not inferred merely from the real-arithmetic argument. Fitting convergence is unnecessary for validity: any finite prediction, and even its clipped replacement when numerically invalid, is only a proposed query.

### BA.3. *A certified analytic terminal kernel*

The last decision admits an especially strong common comparator. Let $y = F(x,0,0)$, $b_0 = 1/2 - 2\,\mathrm{mean}(y)$, and $\gamma_j = B_j$. Define

$$D(x) = \frac{8}{d}\sum_j \gamma_j(y_j - 5/8) + \frac{1}{2d}\sum_j(\gamma_j - \gamma_{j+1})(y_j - y_{j+1}).$$

For every even $d$, the exact terminal action derivative is

$$16a^3 + \left(2 + \frac{41\beta}{32}\right)a + \beta D(x) - 3\beta(b_0 - 3a/4)_+. \tag{A.103}$$

Each branch is a strictly increasing cubic. Its floating closed-form root supplies only a candidate. The independently evaluated outward derivative and exact-law objective certify that candidate; a safeguarded derivative bracket remains available if the candidate is inaccurate.

For completeness, if $f$ is $\mu$-strongly convex and an outward interval encloses $f'(a)$, the largest compatible projected derivative residual $R$ gives

$$f(a) - \min_{[0,c]} f \le R^2/(2\mu).$$

At zero, a nonnegative derivative has residual zero; at capacity, a nonpositive derivative has residual zero. Otherwise the absolute derivative bound is sufficient. This follows by minimizing the strong-convexity supporting quadratic over the feasible interval and, conservatively, over the line. Here $\mu = 2$. Equation (A.103), its $41/32$ and $9/4$ coefficients, analytic shock moments, and outward derivatives are independently checked against rational formulas. Every method uses this kernel; it is not credited to the neural predictor.

### BA.4. *Work, prediction, and the economic object*

Let $A_r(\delta)$ be a bound on action endpoints at a query. Excluding training, a scalar operation account obeys

$$W_1(\delta) = O(dA_1), \qquad W_r(\delta) \le CdA_r(\delta) + A_r(\delta)q_r W_{r-1}(\delta/(4\beta)).$$

The constants $dG_r, dM_r$ are bounded independently of $d$, so the declared endpoint and quadrature budgets need not grow exponentially with the state dimension. Primitive evaluations still cost $O(d)$. This removes the state lattice, not the horizon or shock-tree cost. The executed vectorization batches a query's innovation descendants. Its largest live state batch is reported; it does not claim the smaller memory of a fully streamed implementation. Increasing the horizon or the shock dimension may still be expensive. The new execution is scalar-control; the earlier full two-control result is retained rather than relabeled as a query-service experiment.

A useful prediction can reduce the endpoint count or return its own action from the initial partition. An inaccurate prediction can instead add work. This is a prediction-dependent algorithm, not a theorem of neural dominance. Learned warm starts have their own literature, including Sakaue and Oki (2023); their discrete-convex setting differs from the original-law Bellman intervals here. Convex control policies and learned optimization parameters are studied by Agrawal et al. (2020). Our comparison identifies the extra proposal cost and the resulting query transcript rather than claiming priority for learning a warm start.

The purchase is now a complete callable controller with a declared error contract. Its training cost, every subsequent certified query, and its return-time cost belong to the same service. A reduction in queried state storage is economically relevant for repeated decisions only after these costs are counted. The original cost normalization is maintained. Workload seconds and normalized expected-cost loss are reported separately; a price per unit of computation can convert the former to the latter only as an explicitly specified decision-maker parameter.

## APPENDIX BB: OPERATIONAL PREDICTION CERTIFICATES AND READOUT TRAINING

Proposition A.42 separates witness accuracy from lower-certificate localization. Its distance to an unknown minimizer is useful for analysis but is not a stopping observable. We now replace that distance by an upper envelope computed from information already purchased. This gives a certificate before an additional Bell-

man query, a training criterion measured in unresolved verification work, and a constructive neural readout with a computable optimization gap.

### BB.1. *A decision before the next query*

Fix a state, a date, and the current action partition. Write $f(a) = Q_t^*(x,a)$, and suppose $f$ is $\mu$-strongly convex on the feasible interval. Let $\overline{f}_l, \overline{f}_u$ be verified upper endpoints at two consecutive actions $l < u$. Let $L$ be the current lower bound for the original feasible infimum, including the capacity and arithmetic allowances. For $a = l + s(u-l)$, define

$$C(a) = (1-s)\overline{f}_l + s\overline{f}_u - \frac{\mu}{2}(a-l)(u-a), \qquad 0 \le s \le 1. \tag{A.104}$$

Every symbol in this expression is either a primitive bound or an existing endpoint. No derivative or action at the unknown optimum occurs.

THEOREM A.52—Pre-query closure: *Under the preceding conditions, $f(a) \le C(a)$ throughout $[l,u]$. Let $a_\Delta \in [l,u]$ be the actual feasible rounded proposal. An outward upper endpoint $\overline{C}(a_\Delta)$ satisfying*

$$\overline{C}(a_\Delta) - L \le \delta \tag{A.105}$$

*certifies $f(a_\Delta) - \inf_{a \in A(x)} f(a) \le \delta$ without another Bellman query. If the screen does not close, the protected refinement of Theorem A.41 retains its validity and finite query bound. The complete acquired policy therefore retains the original-optimum account of Theorem A.51.*

*Equivalently, the union over partition intervals of the sublevel sets $\{a : C(a) - L \le \delta\}$ is a computable certified action set. Each constituent is a quadratic sublevel interval intersected with its partition cell. Endpoints obtained numerically must be rounded inward, or the final action must be checked by* (A.105).

*Proof.* See Supplement, Section BW.1.

The screen is shared by every comparator. In particular, minimizing the quadratic upper envelope is an available conventional proposal. Prediction is valuable only to the extent that its proposed location or evaluation cost improves the observed

service. The executed neural rule screens its feasible prediction on the partition interval containing it, even when that location is outside the protected part of the currently critical interval. A failed screen does not authorize an unprotected split.

### BB.2. *A computable loss for unresolved work*

A *certificate context* $z$ consists of the state, remaining horizon, tolerance, partition endpoints and their verified bounds. A proposal $a_\theta(z)$ is feasible and is computed without another Bellman query. Define its verified score and a margin loss by

$$s_\theta(z) = \overline{C}_z(a_\theta(z)) - L_z, \tag{A.106}$$

$$\ell_\tau(z,\theta) = \min\left\{1, \frac{(s_\theta(z) - \delta_z + \tau)_+^2}{\tau^2}\right\}, \qquad \tau > 0. \tag{A.107}$$

Let $K(z)$ be a valid cap on further endpoint queries if the screen fails and the protected fallback is followed. This is a count of endpoint queries at the current state. The recursive cost of each endpoint is charged through the earlier descendant-work recursion. On contexts satisfying the declared precision contract, the real-arithmetic and rational-recovery bounds supply finite such caps.

THEOREM A.53—Certificate loss and additional work: *If $N_\theta(z)$ is the number of additional endpoint queries after the screen, then*

$$N_\theta(z) \le K(z)\mathbf{1}\{s_\theta(z) > \delta_z\} \le K(z)\ell_\tau(z,\theta). \tag{A.108}$$

*Consequently, an achieved margin loss bounds the average additional queries on any specified finite context catalogue, with no unknown minimizing action.*

*For a distributional assertion, condition on construction of a finite family of $M$ predictors. Suppose $z_1, \dots, z_n$ are independent validation contexts from a declared predictor-independent law, and $0 \le K(z) \le K_{\max}$. With probability at least $1 - \alpha$, simultaneously for that family,*

$$\mathbb{E}N_\theta(z) \le \frac{1}{n}\sum_{i=1}^{n} K(z_i)\ell_\tau(z_i,\theta) + K_{\max}\sqrt{\frac{\log(M/\alpha)}{2n}}. \tag{A.109}$$

*The cost of obtaining and validating contexts and of evaluating the predictor is additional work, not part of $N_\theta$.*

*Proof.* See Supplement, Section BW.2.

The context-law requirement matters. A stream of recursively generated contexts can depend on earlier predictions; it is not automatically an independent validation sample for (A.109). The deterministic inequality (A.108) remains valid context by context. The recorded action-label mean squared error is not substituted for (A.107). The new loss measures a certificate's unresolved work rather than resemblance to an approximate action label.

### BB.3. *A constructive ReLU readout with an optimization certificate*

The work loss can be trained without invoking a successful hidden-layer optimizer. Fix a nonnegative affine–ReLU feature map $\phi(z) \in \mathbb{R}^p$ with $\sum_j \phi_j(z) = 1$. Linear hat functions on a fixed coordinate partition are one exact construction; averages of such maps give multivariate features. A box-constrained readout $w \in [0,1]^p$ proposes

$$a_w(z) = l_z + (u_z - l_z)\phi(z)^\top w.$$

The partition interval and features are fixed before optimizing $w$. Let $\eta_i \geq 0$ cover action rounding and evaluation of the score, so that the computed score of the actual rounded action is at most $C_i(a_w(z_i)) + \eta_i - L_i$. For example, on lattice-aligned endpoints an action Lipschitz bound gives $\eta_i = \Lambda_i \Delta_a$ for downward rounding. Define

$$F(w) = \frac{1}{n}\sum_i K_i\big(C_i(a_w(z_i)) + \eta_i - L_i - \delta_i + \tau\big)_+^2 + \frac{\lambda}{2}\|w\|_2^2, \qquad \lambda \geq 0. \tag{A.110}$$

Unlike an action-label objective, this loss directly upper-bounds the additional-work account. It trains the upper witness against the current global lower certificate.

PROPOSITION A.54—Certified readout training: *The function $F$ is convex and continuously differentiable on the box. For $g = \nabla F(w)$, put*

$$D(w) = g^\top w - \sum_{j=1}^{p} \min\{0, g_j\}. \tag{A.111}$$

*Then $0 \le F(w) - \min_{v\in[0,1]^p} F(v) \le D(w)$. On the training contexts the average additional endpoint-query count is at most $F(w)/\tau^2$. If $\lambda > 0$, the minimizer $w^*$ is unique and*

$$\|w - w^*\|_2 \le \sqrt{2D(w)/\lambda}.$$

*All quantities in these conclusions can be bounded by outward arithmetic; they are exact rational quantities when the contexts, features and iterate are rational.*

*Proof.* See Supplement, Section BW.3.

This is a fixed-feature neural construction, not a convergence assertion for unrestricted nonconvex network fitting. The deposited rational implementation uses finite conditional-gradient steps and reports its achieved loss and dual gap. Its regression examples test the proposition; they are not new investment-policy observations. The preceding R67 service comparison retained its preregistered action-label fits. The new comparison in Section AK separately executes this certificate-aware objective, with equal-information Bernstein and label-loss controls; no historical fit is relabeled as its minimizer. Prediction-dependent optimization has an established literature (Sakaue and Oki, 2023, Sambharya et al., 2023); the additional object here is a computable original-law Bellman certificate that can license an action before another query and supplies a directly auditable training criterion.

## APPENDIX BC: TWO CONTROLS AND TWO CONTINUOUS INNOVATIONS

The absence of a state lattice does not remove action or innovation dimensions. We extend the query implementation to the previously specified two-control economy of Section BT: $a_1, a_2 \ge 0$, $a_1 + a_2 \le c(x)$, the interaction cost is $a_1 a_2/4$, and the second innovation is an independent common-direction uniform shock. The original scalar model and its experiments are unchanged.

At a queried state, triangulate the feasible action simplex. Let $v_0, v_1, v_2$ be a triangle's vertices with verified lower endpoints $l_0, l_1, l_2$. If $\theta_j(a)$ are its barycentric coordinates, define

$$p(a) = \sum_{j=0}^{2} \theta_j(a) l_j - \frac{H}{2} \sum_{j=0}^{2} \theta_j(a) \|v_j - a\|_2^2. \tag{A.112}$$

This is the quadratic $H\|a\|_2^2/2 + b^\top a + c$ interpolating the lower endpoints. Its approximate minimizer is only a proposed support point. For any point $y \in \mathbb{R}^2$, including a numerically displaced point outside the triangle,

$$p(y) + \min_j \nabla p(y)^\top (v_j - y) \le \min_{a \in \mathrm{co}(v_0, v_1, v_2)} p(a). \tag{A.113}$$

Thus an outward supporting-plane calculation, not an uncertified quadratic solver, supplies the lower endpoint.

THEOREM A.55—Simplicial Bellman query: *For the two-control primitives, (A.112) is a lower minorant of the true Bellman objective. Taking the smallest valid triangle lower endpoint and the best feasible-action upper endpoint gives a lower–upper certificate for the full continuous simplex. Include the omitted capacity strip and actual action-rounding allowances. Returning when this gap is at most $\delta$ gives the same acquired-policy loss recursion as Theorem A.51.*

*A failed floating triangulation can be replaced by an exact-rational simplex enumeration. Under the stated positive precision slack, this recovery is finite at every finite horizon. With two innovations divided into $q$ bins each, a valid local quadrature error is*

$$\zeta = \frac{5\beta M_{t+1} d}{24576 q^2}. \tag{A.114}$$

*If $A_r$ bounds the number of action queries at a state with $r$ dates remaining, a work account has the form*

$$W_r \le C_d A_r + A_r q_r^2 W_{r-1}, \qquad C_d = O(d), \tag{A.115}$$

*with separate charges for triangulation, arithmetic and output. There is no tensor state-table factor, but the descendant tree remains.*

*Proof.* See Supplement, Section BW.4.

The executed algorithm uses longest-edge refinement of the lowest triangle and an outward terminal convexity-gap check. Its floating cap is followed by the stated recovery rule, not by an uncertified returned action. The implementation batches descendants but can allocate a descendant array before slicing it into query batches; a reported maximum query batch is therefore not a bound on total live state memory. We report process memory separately. The two-control execution is a verified extension of the NBO query backend, not evidence for a trained two-output selector.

## APPENDIX BD: DUAL VERIFICATION AND POLICY-CLASS TRANSFER

This appendix proves the numerical certificate used in Theorem A.26 and states the condition needed to extend its coupling argument beyond two dates. The implemented two-date calculation remains the one in the article. No unexecuted longer-horizon optimization problem is counted as part of that experiment.

### BD.1. *A lower bound independent of solver success*

Use the notation of (A.48). The switch problem can be written as

$$\begin{aligned} \inf_{\theta,u,l,L} \quad & u - l + 2\rho G + \alpha L, \\ \text{subject to} \quad & y_i^+ - \phi_i^\top \theta \le u, \quad l \le y_i^- - \phi_i^\top \theta, \\ & s_j^\top \theta \le L, \quad -s_j^\top \theta \le L, \quad L \ge 0, \\ & c_{Cb} + d_{Cb}^\top \theta \le 0, \quad |\theta_k| \le B_k, \qquad \alpha = 2\rho. \end{aligned}$$

The minimum over $(u,l,L)$ at any feasible $\theta$ equals $F(\theta)$. For any feasible point, probability vectors $p,q$ give

$$u - l \ge p^\top y^+ - q^\top y^- + \sum_i (q_i - p_i)\phi_i^\top \theta.$$

For nonnegative $\xi_j^+, \xi_j^-$ with total mass at most $\alpha$,

$$\alpha L \geq \sum_j (\xi_j^+ - \xi_j^-) s_j^\top \theta.$$

The unused mass is nonnegative and $L \geq 0$. Moreover, $\nu(c_{Cb} + d_{Cb}^\top \theta) \leq 0$ for $\nu \geq 0$. Adding this nonpositive expression to the right side of the preceding lower bound cannot invalidate it. Finally,

$$\inf_{|\theta_k| \leq B_k} z^\top \theta = -\sum_k B_k |z_k|.$$

Combining these inequalities proves (A.48). It proves weak duality for every rationally feasible tuple, including tuples obtained from an unsuccessful numerical optimizer. Strong duality or an attained numerical optimum is unnecessary for the action-stability conclusion. The lower bound may be uninformative, but it cannot assert a successful certificate without the strict comparison in (A.46).

The independent verifier reconstructs the derivative rows directly. It computes each expected polynomial feature by integrating its expanded monomials through

$$\mathbb{E} Y^p = \frac{v^{p+1} - u^{p+1}}{(p+1)(v-u)}, \qquad Y \sim \mathrm{Unif}[u, v],$$

and each hinge by the integral of its positive-part antiderivative. This differs from the construction's even-moment expansion. Immediate state costs cancel in an action contrast, leaving $b^2 - a^2 + 4(b^4 - a^4)$ and the discounted expected-feature difference. Every coefficient, primitive and multiplier in this replay is rational. The verifier neither imports the construction's LP routines nor calls its feature-expectation or dual-bound functions.

### BD.2. *First-disagreement extension*

Let two finite-horizon policies have a common terminal cost and common transition primitives. Suppose they choose the same action outside measurable sets $E_t$ at each date. Assume a common-innovation realization of the actual kernels is available, so trajectories agree while states and actions agree. Let the tail cash cost

from date $t$ lie in $[m_t, M_t]$ under either policy, uniformly over feasible paths, and write $H_t = M_t - m_t$. Put $b_0 = 1$ and $b_t = \prod_{j<t} \beta_j$.

PROPOSITION A.56—First-disagreement cost bound: *If $\widehat{\pi}$ is the reference policy and $D_t$ is the event that the coupled policies first choose different actions at date $t$, then*

$$|J^{\pi} - J^{\widehat{\pi}}| \leq \sum_{t=0}^{T-1} b_t H_t \Pr(D_t) \leq \sum_{t=0}^{T-1} b_t H_t \Pr_{\widehat{\pi}}(X_t \in E_t). \tag{A.116}$$

*The first events are disjoint. Any independent bound on the entire cash-cost difference may be intersected with this bound. To apply it to an optimization class, the action agreement and bad sets must hold uniformly for every policy in that class.*

PROOF: On $D_t$, the histories, states and costs before $t$ coincide. The absolute difference of the remaining discounted costs is at most $b_t H_t$. On the event of no disagreement the costs coincide throughout. Summing over the disjoint first-disagreement events and taking expectations proves the first inequality. Up to the first disagreement, the common state is the state of the reference policy driven by the same innovations. A first disagreement at date $t$ requires this state to lie in $E_t$, proving the second inequality. *Q.E.D.*

This proposition is a cost-transfer identity, not a substitute for constructing longer-horizon stable sublevels. In particular, changing a terminal readout changes labels upstream. Independently small optimization gaps at different fitted label problems do not automatically establish the uniform action-agreement premise. In the executed two-date experiment the last-date actor depends only on the common terminal payoff; Theorem A.26 therefore needs the first-date switch problems alone.

### BD.3. *Confidence and economic interpretation*

The original cost event concerns a finite list of frozen attained policies. On that event all deterministic stability enlargements are valid simultaneously, including for a coefficient vector selected after seeing the records but still belonging to a certified fixed-label sublevel. The probability statement does not change because

the optimizer class is uncountable. A new unqualified model or newly fitted label problem is not covered by that argument.

A failed action-stability certificate is a numerical statement about the available lower bound, not a witness of policy variation. Conversely, policy invariance of an exact optimization class does not assert that an arbitrary practical optimizer reaches that class. The deposited feasible iterate and dual gap describe what the construction attained; the switch lower bounds describe what every member of a stated sublevel must do. These are distinct quantifiers, and neither is inferred from a collection of fitting seeds.

For the two calibrated $n = 16$ controllers let $[l, u]$ be the positive interval for ReLU cost minus Bernstein cost. Let $\kappa$ be the additional per-deployment charge for Bernstein, after all actual installation and computation charges have been converted into the same normalized cost units. The corresponding net difference is $[l - \kappa, u - \kappa]$. Bernstein is strictly less costly for every $\kappa < l$; ReLU is strictly less costly for every $\kappa > u$; neither strict ordering is certified in the intermediate region. This is an exact conditional price region, not a monetary calibration. It remains valid for the fully action-stable optimizer classes, and it does not use construction times with different service boundaries as interchangeable monetary prices.

## APPENDIX BE: COMPLETE FACTORIAL AND PRECISION RECORDS

The following tables report both estimators, all marginal costs, paired contrasts and predeclared interactions. A paired-width target is not a marginal-cost target. The two estimators share the same underlying rows and target the same expected costs; no independent sample size is gained by computing both. The interval width is decomposed into the two endpoint sampling-variance allowances, the bounded-support term and the mean numerical enclosure width. Arithmetic, continuous-bin width, and any acquisition branching enter the latter through actual endpoint propagation. The implementation does not claim a uniquely identifiable additive attribution of that combined numerical width to each intermediate operation.

TABLE BE.I

OBSERVED WIDTH COMPONENTS FOR THE COMPLETE FROZEN FAMILY

| Estimator | Type | Count | Max width | Support | Variance | Numerical |
|---|---|---|---|---|---|---|
| Cash | Marginal | 9 | 0.006901 | 0.000505 | 0.006397 | 1.40e-11 |
| Cash | Pair | 36 | 0.002443 | 0.001009 | 0.001434 | 2.79e-11 |
| Cash | Interaction | 2 | 0.002913 | 0.002018 | 0.000895 | 5.52e-11 |
| Conditional | Marginal | 9 | 0.006901 | 0.000505 | 0.006397 | 1.35e-11 |
| Conditional | Pair | 36 | 0.002441 | 0.001009 | 0.001433 | 2.70e-11 |
| Conditional | Interaction | 2 | 0.002912 | 0.002018 | 0.000895 | 5.34e-11 |

Maxima are taken within each row and need not occur in the same estimand. Support and variance are the two-sided finite-sample allowances. Numerical is the mean endpoint-enclosure width, inclusive of propagated arithmetic and bin effects. The predeclared paired width target is 1/256; marginal intervals are not governed by that paired target. Components are not counted twice.

TABLE BE.II

CONSTRUCTION OPERATION COUNTS AND RECORDED SERVICE BOUNDARIES

| Actor | Action values | Feature values | Seconds | Label ablation |
|---|---|---|---|---|
| CB16 | 33792 | 152064 | 6.269 | No |
| CB8 | 8448 | 38016 | 1.547 | No |
| CR16 | 33792 | 152064 | 2.699 | No |
| CR8 | 8448 | 38016 | 0.653 | No |
| LB16 | 33792 | 152064 | 10.617 | Yes |
| LB8 | 8448 | 38016 | 3.174 | Yes |
| LR16 | 33792 | 152064 | 4.793 | Yes |
| LR8 | 8448 | 38016 | 1.174 | Yes |

Legacy clocks include a matched fixed-label loss ablation; calibrated clocks do not. Hence this table does not compare identical end-to-end timing services. All reported counts and clocks come from their original construction records. The shared cost-inference service, independent replay, barrier verification and publication work have separate clocks and are not assigned zero cost.

| Estimator | Estimand | Interval | Width target |
|---|---|---|---|

TABLE BE.III: Every original-law cost estimand in the R80 simultaneous family

| Estimator | Estimand | Interval | Width target |
|---|---|---|---|
| Cash | CB16 | [0.598996, 0.604724] | – |
| Cash | CB8 | [0.601755, 0.607492] | – |
| Cash | CR16 | [0.600489, 0.606198] | – |
| Cash | CR8 | [0.605889, 0.611618] | – |
| Cash | LB16 | [0.682148, 0.688581] | – |
| Cash | LB8 | [0.683269, 0.689716] | – |
| Cash | LR16 | [0.606271, 0.611918] | – |
| Cash | LR8 | [0.606970, 0.612648] | – |
| Cash | Z | [0.772792, 0.779694] | – |
| Cash | CB8 minus CB16 | [0.002241, 0.003286] | Pass |
| Cash | CB8 minus CR16 | [0.000748, 0.001812] | Pass |
| Cash | CR16 minus CB16 | [0.000955, 0.002012] | Pass |
| Cash | CR8 minus CB16 | [0.006337, 0.007450] | Pass |
| Cash | CR8 minus CB8 | [0.003572, 0.004689] | Pass |
| Cash | CR8 minus CR16 | [0.004855, 0.005965] | Pass |
| Cash | LB16 minus CB16 | [0.082602, 0.084406] | Pass |
| Cash | LB16 minus CB8 | [0.079844, 0.081638] | Pass |
| Cash | LB16 minus CR16 | [0.081108, 0.082933] | Pass |
| Cash | LB16 minus CR8 | [0.075693, 0.077528] | Pass |
| Cash | LB8 minus CB16 | [0.083723, 0.085541] | Pass |
| Cash | LB8 minus CB8 | [0.080964, 0.082773] | Pass |
| Cash | LB8 minus CR16 | [0.082229, 0.084068] | Pass |
| Cash | LB8 minus CR8 | [0.076814, 0.078663] | Pass |
| Cash | LB8 minus LB16 | [0.000613, 0.001643] | Pass |
| Cash | LB8 minus LR16 | [0.076429, 0.078368] | Pass |
| Cash | LR16 minus CB16 | [0.006635, 0.007834] | Pass |
| Cash | LR16 minus CB8 | [0.003870, 0.005072] | Pass |
| Cash | LR16 minus CR16 | [0.005159, 0.006342] | Pass |

| Estimator | Estimand | Interval | Width target |
|---|---|---|---|
| Cash | LR16 minus CR8 | [-0.000271, 0.000952] | Pass |
| Cash | LR16 minus LB16 | [-0.077233, -0.075307] | Pass |
| Cash | LR8 minus CB16 | [0.007365, 0.008532] | Pass |
| Cash | LR8 minus CB8 | [0.004602, 0.005769] | Pass |
| Cash | LR8 minus CR16 | [0.005889, 0.007041] | Pass |
| Cash | LR8 minus CR8 | [0.000457, 0.001653] | Pass |
| Cash | LR8 minus LB16 | [-0.076498, -0.074612] | Pass |
| Cash | LR8 minus LB8 | [-0.077633, -0.075734] | Pass |
| Cash | LR8 minus LR16 | [0.000186, 0.001243] | Pass |
| Cash | Z minus CB16 | [0.173223, 0.175544] | Pass |
| Cash | Z minus CB8 | [0.170464, 0.172776] | Pass |
| Cash | Z minus CR16 | [0.171729, 0.174071] | Pass |
| Cash | Z minus CR8 | [0.166315, 0.168664] | Pass |
| Cash | Z minus LB16 | [0.090114, 0.091644] | Pass |
| Cash | Z minus LB8 | [0.088993, 0.090509] | Pass |
| Cash | Z minus LR16 | [0.165927, 0.168370] | Pass |
| Cash | Z minus LR8 | [0.165232, 0.167636] | Pass |
| Cash | Interaction 16 | [-0.079210, -0.076297] | Pass |
| Cash | Interaction 8 | [-0.082255, -0.079372] | Pass |
| Conditional | CB16 | [0.598987, 0.604714] | – |
| Conditional | CB8 | [0.601746, 0.607483] | – |
| Conditional | CR16 | [0.600481, 0.606189] | – |
| Conditional | CR8 | [0.605880, 0.611609] | – |
| Conditional | LB16 | [0.682138, 0.688570] | – |
| Conditional | LB8 | [0.683259, 0.689705] | – |
| Conditional | LR16 | [0.606262, 0.611908] | – |
| Conditional | LR8 | [0.606961, 0.612638] | – |
| Conditional | Z | [0.772780, 0.779682] | – |
| Conditional | CB8 minus CB16 | [0.002241, 0.003286] | Pass |
| Conditional | CB8 minus CR16 | [0.000748, 0.001812] | Pass |
| Conditional | CR16 minus CB16 | [0.000955, 0.002013] | Pass |
| Conditional | CR8 minus CB16 | [0.006337, 0.007450] | Pass |
| Conditional | CR8 minus CB8 | [0.003571, 0.004688] | Pass |

| Estimator | Estimand | Interval | Width target |
|---|---|---|---|
| Conditional | CR8 minus CR16 | [0.004854, 0.005965] | Pass |
| Conditional | LB16 minus CB16 | [0.082601, 0.084405] | Pass |
| Conditional | LB16 minus CB8 | [0.079843, 0.081637] | Pass |
| Conditional | LB16 minus CR16 | [0.081107, 0.082932] | Pass |
| Conditional | LB16 minus CR8 | [0.075693, 0.077527] | Pass |
| Conditional | LB8 minus CB16 | [0.083722, 0.085540] | Pass |
| Conditional | LB8 minus CB8 | [0.080964, 0.082772] | Pass |
| Conditional | LB8 minus CR16 | [0.082228, 0.084067] | Pass |
| Conditional | LB8 minus CR8 | [0.076814, 0.078662] | Pass |
| Conditional | LB8 minus LB16 | [0.000613, 0.001643] | Pass |
| Conditional | LB8 minus LR16 | [0.076428, 0.078367] | Pass |
| Conditional | LR16 minus CB16 | [0.006635, 0.007834] | Pass |
| Conditional | LR16 minus CB8 | [0.003869, 0.005072] | Pass |
| Conditional | LR16 minus CR16 | [0.005159, 0.006341] | Pass |
| Conditional | LR16 minus CR8 | [-0.000271, 0.000952] | Pass |
| Conditional | LR16 minus LB16 | [-0.077232, -0.075307] | Pass |
| Conditional | LR8 minus CB16 | [0.007365, 0.008532] | Pass |
| Conditional | LR8 minus CB8 | [0.004602, 0.005768] | Pass |
| Conditional | LR8 minus CR16 | [0.005889, 0.007041] | Pass |
| Conditional | LR8 minus CR8 | [0.000457, 0.001653] | Pass |
| Conditional | LR8 minus LB16 | [-0.076497, -0.074612] | Pass |
| Conditional | LR8 minus LB8 | [-0.077633, -0.075733] | Pass |
| Conditional | LR8 minus LR16 | [0.000186, 0.001243] | Pass |
| Conditional | Z minus CB16 | [0.173220, 0.175540] | Pass |
| Conditional | Z minus CB8 | [0.170461, 0.172772] | Pass |
| Conditional | Z minus CR16 | [0.171726, 0.174067] | Pass |
| Conditional | Z minus CR8 | [0.166313, 0.168660] | Pass |
| Conditional | Z minus LB16 | [0.090113, 0.091641] | Pass |
| Conditional | Z minus LB8 | [0.088991, 0.090506] | Pass |
| Conditional | Z minus LR16 | [0.165925, 0.168367] | Pass |
| Conditional | Z minus LR8 | [0.165230, 0.167633] | Pass |
| Conditional | Interaction 16 | [-0.079209, -0.076297] | Pass |

| Estimator | Estimand | Interval | Width target |
|---|---|---|---|
| Conditional | Interaction 8 | [-0.082254, -0.079371] | Pass |

Codes: Z is zero control; L and C denote legacy and calibrated qualification, R and B denote ReLU and Bernstein, and the final number is the acquisition resolution. Display endpoints are rounded outward. Full exact endpoints, moments, equivalence flags and both numerical implementations remain in the machine-readable records.

## APPENDIX BF: ACTIONWISE ECONOMIC TRANSFER AND ITS COMPLETE NUMERICAL ACCOUNT

Theorem A.27 separates two numerical tasks. The action-switch dual identifies actions that cannot be selected by a qualified readout. Original-law integration prices the actions not excluded by that dual. Neither calculation substitutes a fitted critic for the cost of the implemented continuation policy.

### BF.1. *Conditional contrasts from the original primitives*

For a fixed initial cell $C$, reference action $a$, and alternative action $b$, couple the two trajectories with the same first innovation $W$. Write $Y^b, Y^a$ for their next states and $B = \tau_1(Y^b)$, $A = \tau_1(Y^a)$ for the fixed last-date actor. The action pair need not agree on the two next states. Let $m(y,d)$ denote the original transition with innovation zero and action $d$. Its opposite-signed last innovation leaves the state sum unchanged, so the shortage term in the terminal cost is unaffected. The terminal quadratic terms give

$$\mathbb{E}[g(m(y,d) + (W', -W')) \mid y, d] = g(m(y,d)) + 5\mathbb{E}[(W')^2] = g(m(y,d)) + \frac{5}{3072}. \tag{A.117}$$

The correction is the same for both trajectories and cancels in their difference. It cannot be omitted from an individual conditional cost. Feasibility and the inherited transition bounds ensure that the actual states remain in $[0,1]^2$; intersecting an enclosing box with this domain does not clip the economic transition law.

Consequently the paired conditional cash cost is

$$\begin{aligned} Z_{Cb}(X,W) = c_0(X,b) - c_0(X,a) \\ + \frac{15}{16}\{c_1(Y^b,B) - c_1(Y^a,A)\} \\ + \frac{225}{256}\{g(m(Y^b,B)) - g(m(Y^a,A))\}. \end{aligned} \tag{A.118}$$

Its expectation conditional on $C$ is $\Delta_C(b)$. The first difference cancels the common state cost exactly. The remaining terms retain a separately propagated state difference rather than subtracting independently widened state boxes.

For example, $\delta = Y^b - Y^a = ((b-a)/2, (b-a)/4)$. The bilinear difference satisfies

$$Y_1^b Y_2^b - Y_1^a Y_2^a = \delta_1 Y_2^b + Y_1^a \delta_2.$$

This identity gives the difference of the next mean states, including the separate term $B - A$. The square identity $u^2 - v^2 = (u-v)(u+v)$ and the quartic identity $u^4 - v^4 = (u-v)(u+v)(u^2+v^2)$ preserve analogous cancellation in costs. For the shortage term, whenever an interval encloses $d = u - v$,

$$u_+ - v_+ \in ([u]_+ - [v]_+) \cap [\min\{0, \underline{d}\}, \max\{0, \overline{d}\}].$$

Multiplication by an enclosure of $u_+ + v_+$ encloses the difference of their squares. An empty intersection stops verification.

PROPOSITION A.57—Complete-bin enclosure: *Partition each initial cell into $r^2$ equal rectangles and the first innovation into $q$ equal bins, with dyadic $r, q$. Evaluate* (A.118) *on every complete subbox by outward interval operations. On each next-state rectangle, enclose the fixed actor by the minimum and maximum of its action indices over every intersected acquired cell, including boundary cells. Let $[l_s, u_s]$ be the resulting paired interval on subbox $s$, intersected with the valid signed support $[-H, H]$. Then*

$$\frac{1}{r^2 q}\sum_s l_s \le \Delta_C(b) \le \frac{1}{r^2 q}\sum_s u_s.$$

*The hull of two independently valid interval implementations is valid. Intersecting independently valid integrated bands across refinement levels is valid as well.*

PROOF: The transition and centered identities are algebraic identities for every state and innovation in the subbox. The rectangle extrema contain the actual actor action even when the rectangle crosses an actor discontinuity. Interval inclusion, propagated through each arithmetic operation and the valid intersections, therefore contains (A.118) pointwise. Equation (A.117) accounts for the remaining innovation under its actual continuous marginal law. All subboxes have exact conditional probability $1/(r^2q)$, so integration preserves the endpoint inequalities. Taking a hull preserves inclusion, as does intersecting two bands that separately contain the same integral. *Q.E.D.*

The two deposited implementations do not import each other's interval, transition, cost, partition or actor routines. The binary64 implementation uses outward next-representable-number operations and a precomputed rectangle-extremum table. The integer implementation uses Python-integer endpoints in units $2^{-48}$, directed integer division, expanded cost polynomials and direct enumeration of distinct acquisition rectangles. Their endpoint streams must overlap on every subbox. A separately written rational trajectory calculation checks four deterministic midpoint trajectories per action and rung. Those checks are regressions; the proof concerns every point in each box, not only its midpoint.

The complete records retain exact integer endpoint sums, their common denominator, the digest of all four independently computed endpoint streams, the largest endpoint discrepancy and the number of processed boxes. They do not claim to deposit an additional random sample or to count the two arithmetic implementations as independent statistical observations. Full replay recomputes those endpoint streams, not only the saved sums.

## BF.2. *Refinement and resource requirements*

PROPOSITION A.58—Refinement with a discontinuous acquired actor: *In the original two-date model, fix the finite acquired actor, initial cell and feasible first-action pair. In exact interval arithmetic, the integrated width in Proposition A.57 tends to zero as $r, q \to \infty$ along dyadic partitions. The same conclusion holds with outward arithmetic*

*whose uniform propagated padding tends to zero. The executed fixed-precision calculation retains its nonzero numerical width rather than invoking this limiting statement.*

*If $P$ alternatives remain at the largest qualification tolerance, a level uses exactly $Pr^2q$ paired subboxes in each implementation. Its binary64 actor tables occupy $O(n^4)$ entries and the direct preprocessing used here takes at most $O(n^6)$ operations. After that preprocessing each rectangle query is constant work. The independent enumerator instead stores the $n^2$ actor indices, groups queried rectangles by sorting their codes, and can inspect up to $n^2$ indices per distinct rectangle. Its conservative worst-case work is therefore*

$$O\Big(Pr^2q\,[n^2+\log(r^2q)]\Big).$$

*Processing one alternative at a time uses $O(n^4+r^2q)$ numerical working storage, apart from retained integrated records.*

PROOF: For a fixed first action, each next-state coordinate is a polynomial of the initial state plus either $W$ or $-W$. For any terminal acquisition boundary $Y_j = k/n$, its preimage has conditional innovation measure zero for every initial state, because the coefficient of $W$ is nonzero. There are finitely many boundaries and two coupled trajectories. Fubini's theorem therefore gives a measure-zero union of their preimages in the integration domain.

At every other interior point, sufficiently fine boxes intersect only the fixed actor cell containing each next state. On such boxes, (A.118) is a finite composition of continuous polynomial and positive-part operations with constant actor actions. Its exact interval width tends to zero. The support intersection bounds all widths by $2H$. Boundary sets of the initial partition also have measure zero. Dominated convergence now gives convergence of the integrated width. Vanishing outward padding preserves the argument. It does not justify removing the actual fixed-precision remainder from a finite calculation.

There are $r^2q$ subboxes per action and $P$ retained actions. The direct binary64 precomputation enumerates $O(n^4)$ rectangles and scans at most $n^2$ entries each. A stored rectangle extremum then uses its four indices. For $m = r^2q$ boxes, the independent algorithm spends $O(m \log m)$ work grouping codes and at most $O(mn^2)$ work enumerating their actor indices. Finally, keeping only one alternative's in-

terval arrays at a time gives the stated working storage. Summing the displayed counts over every executed rung, rather than only the last one, gives the complete refinement work. *Q.E.D.*

These are counts for the actual two-date procedure. They are not a reduction of the retained vector recursion's horizon growth, and table bytes are not substituted for process peak memory. The completed R83 recovery records its verification clock and peak resident memory separately from the interrupted R82 phase clocks. Development pilots informed the dyadic resolution before the source freeze; the computation is described as deterministic verification rather than a prospectively selected new economic experiment.

### BF.3. *A classwide adoption decision on the original event*

Let the signed optimizer-sublevel interval for $J^{\pi_j} - J^{\pi_k}$ be $[L, U]$ after applying (A.53). If deployment of class $k$ rather than class $j$ has an additional total charge $\tau$, the net advantage of $k$ belongs to $[L - \tau, U - \tau]$ for every allowed pair of readouts on the same original event. Thus $\tau < L$ supports $k$ uniformly and $\tau > U$ supports the opposite decision uniformly. The intermediate region remains unresolved. This statement prices construction, verification and installation only when their conversion into economic cost units is explicitly supplied. A measured second of verification is not itself such a price.

The extension uses a deterministic outer set of possible acquired actions. It permits the qualification tolerance or the comparison to be selected after observing the original simultaneous cost event, provided the original attained controllers and event are not changed and the stated deterministic enclosures hold. There is no new probability expenditure over an uncountable set of readouts. Conversely, a newly fitted hidden-feature model, a changed terminal actor, altered transition law or new statistical estimand is not covered merely because its displayed objective resembles the one here.

## BF.4. *Complete refinement catalogue*

The tables retain all four calibrated constructions, all three objective allowances and all four refinement rungs. ReLU and Bernstein are each evaluated at both resolutions, including the coarse ReLU construction whose exact-optimizer action certificate remains partly unresolved. The actionwise radii are compared with their binary uncertified-mass bounds. Every calibrated pair and both own-controller references are retained in the machine-readable records; no unresolved or adverse entry is removed.

TABLE BF.I: Every model, allowance and integration level

| Actor | $\epsilon$ | $r/q$ | Actions | $Hq_\epsilon$ | $v^-$ | $v^+$ |
|---|---|---|---|---|---|---|
| R8 | 0 | 2/8 | 826 | 2.897233 | -0.044714 | 0.046545 |
| R8 | $2^{-16}$ | 2/8 | 826 | 2.897233 | -0.044714 | 0.046545 |
| R8 | $2^{-12}$ | 2/8 | 832 | 2.897233 | -0.044719 | 0.046677 |
| R8 | 0 | 4/16 | 826 | 2.897233 | -0.024598 | 0.027002 |
| R8 | $2^{-16}$ | 4/16 | 826 | 2.897233 | -0.024598 | 0.027002 |
| R8 | $2^{-12}$ | 4/16 | 832 | 2.897233 | -0.024598 | 0.027140 |
| R8 | 0 | 8/32 | 826 | 2.897233 | -0.014780 | 0.017316 |
| R8 | $2^{-16}$ | 8/32 | 826 | 2.897233 | -0.014780 | 0.017316 |
| R8 | $2^{-12}$ | 8/32 | 832 | 2.897233 | -0.014780 | 0.017436 |
| R8 | 0 | 16/64 | 826 | 2.897233 | -0.010033 | 0.012550 |
| R8 | $2^{-16}$ | 16/64 | 826 | 2.897233 | -0.010033 | 0.012550 |
| R8 | $2^{-12}$ | 16/64 | 832 | 2.897233 | -0.010033 | 0.012680 |
| B8 | 0 | 2/8 | 0 | 0.000000 | 0.000000 | 0.000000 |
| B8 | $2^{-16}$ | 2/8 | 0 | 0.000000 | 0.000000 | 0.000000 |
| B8 | $2^{-12}$ | 2/8 | 9 | 0.724309 | -0.009862 | 0.009662 |
| B8 | 0 | 4/16 | 0 | 0.000000 | 0.000000 | 0.000000 |
| B8 | $2^{-16}$ | 4/16 | 0 | 0.000000 | 0.000000 | 0.000000 |
| B8 | $2^{-12}$ | 4/16 | 9 | 0.724309 | -0.004296 | 0.004111 |
| B8 | 0 | 8/32 | 0 | 0.000000 | 0.000000 | 0.000000 |
| B8 | $2^{-16}$ | 8/32 | 0 | 0.000000 | 0.000000 | 0.000000 |
| B8 | $2^{-12}$ | 8/32 | 9 | 0.724309 | -0.002097 | 0.001917 |
| B8 | 0 | 16/64 | 0 | 0.000000 | 0.000000 | 0.000000 |
| B8 | $2^{-16}$ | 16/64 | 0 | 0.000000 | 0.000000 | 0.000000 |
| B8 | $2^{-12}$ | 16/64 | 9 | 0.724309 | -0.001080 | 0.000900 |
| R16 | 0 | 2/8 | 0 | 0.000000 | 0.000000 | 0.000000 |

| Actor | $\epsilon$ | $r/q$ | Actions | $Hq_\epsilon$ | $v^-$ | $v^+$ |
|---|---|---|---|---|---|---|
| R16 | $2^{-16}$ | 2/8 | 8 | 0.160958 | -0.001425 | 0.001457 |
| R16 | $2^{-12}$ | 2/8 | 78 | 1.227300 | -0.009666 | 0.009816 |
| R16 | 0 | 4/16 | 0 | 0.000000 | 0.000000 | 0.000000 |
| R16 | $2^{-16}$ | 4/16 | 8 | 0.160958 | -0.000708 | 0.000737 |
| R16 | $2^{-12}$ | 4/16 | 78 | 1.227300 | -0.004801 | 0.004936 |
| R16 | 0 | 8/32 | 0 | 0.000000 | 0.000000 | 0.000000 |
| R16 | $2^{-16}$ | 8/32 | 8 | 0.160958 | -0.000345 | 0.000373 |
| R16 | $2^{-12}$ | 8/32 | 78 | 1.227300 | -0.002382 | 0.002507 |
| R16 | 0 | 16/64 | 0 | 0.000000 | 0.000000 | 0.000000 |
| R16 | $2^{-16}$ | 16/64 | 8 | 0.160958 | -0.000165 | 0.000193 |
| R16 | $2^{-12}$ | 16/64 | 78 | 1.227300 | -0.001168 | 0.001291 |
| B16 | 0 | 2/8 | 0 | 0.000000 | 0.000000 | 0.000000 |
| B16 | $2^{-16}$ | 2/8 | 13 | 0.261556 | -0.002388 | 0.002391 |
| B16 | $2^{-12}$ | 2/8 | 57 | 1.126702 | -0.010598 | 0.010616 |
| B16 | 0 | 4/16 | 0 | 0.000000 | 0.000000 | 0.000000 |
| B16 | $2^{-16}$ | 4/16 | 13 | 0.261556 | -0.001197 | 0.001195 |
| B16 | $2^{-12}$ | 4/16 | 57 | 1.126702 | -0.005268 | 0.005271 |
| B16 | 0 | 8/32 | 0 | 0.000000 | 0.000000 | 0.000000 |
| B16 | $2^{-16}$ | 8/32 | 13 | 0.261556 | -0.000596 | 0.000592 |
| B16 | $2^{-12}$ | 8/32 | 57 | 1.126702 | -0.002628 | 0.002628 |
| B16 | 0 | 16/64 | 0 | 0.000000 | 0.000000 | 0.000000 |
| B16 | $2^{-16}$ | 16/64 | 13 | 0.261556 | -0.000299 | 0.000295 |
| B16 | $2^{-12}$ | 16/64 | 57 | 1.126702 | -0.001311 | 0.001311 |

R and B denote calibrated ReLU and Bernstein; 8 and 16 are acquisition resolutions. All 48 inherited model–allowance–level bands are shown. These are deterministic refinements, not independent observations.

TABLE BF.II: Every directed optimizer-class comparison

| Pair | $\epsilon$ | $r/q$ | Lower | Upper | Sign |
|---|---|---|---|---|---|
| R8–B8 | 0 | 2/8 | -0.041142 | 0.051233 | unresolved |
| R8–B8 | $2^{-16}$ | 2/8 | -0.041142 | 0.051233 | unresolved |
| R8–B8 | $2^{-12}$ | 2/8 | -0.050809 | 0.061226 | unresolved |
| R8–B8 | 0 | 4/16 | -0.021026 | 0.031690 | unresolved |
| R8–B8 | $2^{-16}$ | 4/16 | -0.021026 | 0.031690 | unresolved |

| Pair | $\epsilon$ | $r/q$ | Lower | Upper | Sign |
|---|---|---|---|---|---|
| R8–B8 | $2^{-12}$ | 4/16 | -0.025136 | 0.036124 | unresolved |
| R8–B8 | 0 | 8/32 | -0.011208 | 0.022004 | unresolved |
| R8–B8 | $2^{-16}$ | 8/32 | -0.011208 | 0.022004 | unresolved |
| R8–B8 | $2^{-12}$ | 8/32 | -0.013124 | 0.024221 | unresolved |
| R8–B8 | 0 | 16/64 | -0.006461 | 0.017237 | unresolved |
| R8–B8 | $2^{-16}$ | 16/64 | -0.006461 | 0.017237 | unresolved |
| R8–B8 | $2^{-12}$ | 16/64 | -0.007360 | 0.018447 | unresolved |
| R8–R16 | 0 | 2/8 | -0.039859 | 0.052510 | unresolved |
| R8–R16 | $2^{-16}$ | 2/8 | -0.041316 | 0.053934 | unresolved |
| R8–R16 | $2^{-12}$ | 2/8 | -0.049680 | 0.062306 | unresolved |
| R8–R16 | 0 | 4/16 | -0.019743 | 0.032966 | unresolved |
| R8–R16 | $2^{-16}$ | 4/16 | -0.020479 | 0.033674 | unresolved |
| R8–R16 | $2^{-12}$ | 4/16 | -0.024678 | 0.037905 | unresolved |
| R8–R16 | 0 | 8/32 | -0.009925 | 0.023280 | unresolved |
| R8–R16 | $2^{-16}$ | 8/32 | -0.010297 | 0.023625 | unresolved |
| R8–R16 | $2^{-12}$ | 8/32 | -0.012431 | 0.025782 | unresolved |
| R8–R16 | 0 | 16/64 | -0.005178 | 0.018514 | unresolved |
| R8–R16 | $2^{-16}$ | 16/64 | -0.005370 | 0.018679 | unresolved |
| R8–R16 | $2^{-12}$ | 16/64 | -0.006468 | 0.019812 | unresolved |
| R8–B16 | 0 | 2/8 | -0.038377 | 0.053995 | unresolved |
| R8–B16 | $2^{-16}$ | 2/8 | -0.040767 | 0.056382 | unresolved |
| R8–B16 | $2^{-12}$ | 2/8 | -0.048998 | 0.064724 | unresolved |
| R8–B16 | 0 | 4/16 | -0.018261 | 0.034452 | unresolved |
| R8–B16 | $2^{-16}$ | 4/16 | -0.019455 | 0.035648 | unresolved |
| R8–B16 | $2^{-12}$ | 4/16 | -0.023531 | 0.039857 | unresolved |
| R8–B16 | 0 | 8/32 | -0.008443 | 0.024766 | unresolved |
| R8–B16 | $2^{-16}$ | 8/32 | -0.009034 | 0.025361 | unresolved |
| R8–B16 | $2^{-12}$ | 8/32 | -0.011071 | 0.027514 | unresolved |
| R8–B16 | 0 | 16/64 | -0.003695 | 0.019999 | unresolved |
| R8–B16 | $2^{-16}$ | 16/64 | -0.003989 | 0.020298 | unresolved |
| R8–B16 | $2^{-12}$ | 16/64 | -0.005006 | 0.021441 | unresolved |
| B8–R16 | 0 | 2/8 | 0.000748 | 0.001812 | positive |
| B8–R16 | $2^{-16}$ | 2/8 | -0.000709 | 0.003237 | unresolved |
| B8–R16 | $2^{-12}$ | 2/8 | -0.018930 | 0.021139 | unresolved |
| B8–R16 | 0 | 4/16 | 0.000748 | 0.001812 | positive |
| B8–R16 | $2^{-16}$ | 4/16 | 0.000011 | 0.002519 | positive |
| B8–R16 | $2^{-12}$ | 4/16 | -0.008484 | 0.010722 | unresolved |

| Pair | $\epsilon$ | $r/q$ | Lower | Upper | Sign |
|---|---|---|---|---|---|
| B8–R16 | 0 | 8/32 | 0.000748 | 0.001812 | positive |
| B8–R16 | $2^{-16}$ | 8/32 | 0.000375 | 0.002157 | positive |
| B8–R16 | $2^{-12}$ | 8/32 | -0.003855 | 0.006109 | unresolved |
| B8–R16 | 0 | 16/64 | 0.000748 | 0.001812 | positive |
| B8–R16 | $2^{-16}$ | 16/64 | 0.000555 | 0.001977 | positive |
| B8–R16 | $2^{-12}$ | 16/64 | -0.001622 | 0.003879 | unresolved |
| B8–B16 | 0 | 2/8 | 0.002241 | 0.003286 | positive |
| B8–B16 | $2^{-16}$ | 2/8 | -0.000149 | 0.005673 | unresolved |
| B8–B16 | $2^{-12}$ | 2/8 | -0.018237 | 0.023545 | unresolved |
| B8–B16 | 0 | 4/16 | 0.002241 | 0.003286 | positive |
| B8–B16 | $2^{-16}$ | 4/16 | 0.001047 | 0.004483 | positive |
| B8–B16 | $2^{-12}$ | 4/16 | -0.007326 | 0.012664 | unresolved |
| B8–B16 | 0 | 8/32 | 0.002241 | 0.003286 | positive |
| B8–B16 | $2^{-16}$ | 8/32 | 0.001650 | 0.003881 | positive |
| B8–B16 | $2^{-12}$ | 8/32 | -0.002484 | 0.007830 | unresolved |
| B8–B16 | 0 | 16/64 | 0.002241 | 0.003286 | positive |
| B8–B16 | $2^{-16}$ | 16/64 | 0.001947 | 0.003585 | positive |
| B8–B16 | $2^{-12}$ | 16/64 | -0.000149 | 0.005497 | unresolved |
| R16–B16 | 0 | 2/8 | 0.000955 | 0.002013 | positive |
| R16–B16 | $2^{-16}$ | 2/8 | -0.002860 | 0.005857 | unresolved |
| R16–B16 | $2^{-12}$ | 2/8 | -0.019326 | 0.022426 | unresolved |
| R16–B16 | 0 | 4/16 | 0.000955 | 0.002013 | positive |
| R16–B16 | $2^{-16}$ | 4/16 | -0.000947 | 0.003945 | unresolved |
| R16–B16 | $2^{-12}$ | 4/16 | -0.009116 | 0.012216 | unresolved |
| R16–B16 | 0 | 8/32 | 0.000955 | 0.002013 | positive |
| R16–B16 | $2^{-16}$ | 8/32 | 0.000019 | 0.002980 | positive |
| R16–B16 | $2^{-12}$ | 8/32 | -0.004054 | 0.007147 | unresolved |
| R16–B16 | 0 | 16/64 | 0.000955 | 0.002013 | positive |
| R16–B16 | $2^{-16}$ | 16/64 | 0.000496 | 0.002503 | positive |
| R16–B16 | $2^{-12}$ | 16/64 | -0.001523 | 0.004614 | unresolved |

All 72 comparisons use the existing R80 joint event plus deterministic actionwise transfer. Inconclusive coarse ReLU classes and adverse comparisons remain in the catalogue.

TABLE BF.III: All fixed-reference classwide gains

| Class | Reference | $\epsilon$ | Lower gain | Upper gain |
|---|---|---|---|---|
| R8 | Zero | 0 | 0.153764 | 0.178692 |
| R8 | Zero | $2^{-16}$ | 0.153764 | 0.178692 |
| R8 | Zero | $2^{-12}$ | 0.153633 | 0.178692 |
| R8 | Legacy R8 | 0 | -0.012092 | 0.011685 |
| R8 | Legacy R8 | $2^{-16}$ | -0.012092 | 0.011685 |
| R8 | Legacy R8 | $2^{-12}$ | -0.012222 | 0.011685 |
| B8 | Zero | 0 | 0.170461 | 0.172772 |
| B8 | Zero | $2^{-16}$ | 0.170461 | 0.172772 |
| B8 | Zero | $2^{-12}$ | 0.169561 | 0.173851 |
| B8 | Legacy B8 | 0 | 0.080964 | 0.082772 |
| B8 | Legacy B8 | $2^{-16}$ | 0.080964 | 0.082772 |
| B8 | Legacy B8 | $2^{-12}$ | 0.080064 | 0.083851 |
| R16 | Zero | 0 | 0.171726 | 0.174067 |
| R16 | Zero | $2^{-16}$ | 0.171533 | 0.174232 |
| R16 | Zero | $2^{-12}$ | 0.170435 | 0.175234 |
| R16 | Legacy R16 | 0 | 0.005159 | 0.006341 |
| R16 | Legacy R16 | $2^{-16}$ | 0.004967 | 0.006506 |
| R16 | Legacy R16 | $2^{-12}$ | 0.003869 | 0.007509 |
| B16 | Zero | 0 | 0.173220 | 0.175540 |
| B16 | Zero | $2^{-16}$ | 0.172926 | 0.175838 |
| B16 | Zero | $2^{-12}$ | 0.171910 | 0.176851 |
| B16 | Legacy B16 | 0 | 0.082601 | 0.084405 |
| B16 | Legacy B16 | $2^{-16}$ | 0.082307 | 0.084703 |
| B16 | Legacy B16 | $2^{-12}$ | 0.081291 | 0.085716 |

Fixed reference cost minus each calibrated class cost. All 24 final-level zero and same-resolution legacy comparisons are retained. No reference is renamed by dropping an identity prefix or refitted.

TABLE BF.IV: Executed R83 independent recovery work

| Actor | $r/q$ | Pairs | Boxes | Seconds | Float queries | Integer queries |
|---|---|---|---|---|---|---|
| R8 | 2/8 | 832 | 26624 | 4.068 | 53248 | 53248 |
| R8 | 4/16 | 832 | 212992 | 9.149 | 425984 | 425984 |
| R8 | 8/32 | 832 | 1703936 | 50.399 | 3407872 | 3407872 |

| Actor | $r/q$ | Pairs | Boxes | Seconds | Float queries | Integer queries |
|---|---|---|---|---|---|---|
| R8 | 16/64 | 832 | 13631488 | 378.921 | 27262976 | 27262976 |
| B8 | 2/8 | 9 | 288 | 0.045 | 576 | 576 |
| B8 | 4/16 | 9 | 2304 | 0.100 | 4608 | 4608 |
| B8 | 8/32 | 9 | 18432 | 0.558 | 36864 | 36864 |
| B8 | 16/64 | 9 | 147456 | 4.333 | 294912 | 294912 |
| R16 | 2/8 | 78 | 2496 | 0.387 | 4992 | 4992 |
| R16 | 4/16 | 78 | 19968 | 0.868 | 39936 | 39936 |
| R16 | 8/32 | 78 | 159744 | 4.608 | 319488 | 319488 |
| R16 | 16/64 | 78 | 1277952 | 32.504 | 2555904 | 2555904 |
| B16 | 2/8 | 57 | 1824 | 0.286 | 3648 | 3648 |
| B16 | 4/16 | 57 | 14592 | 0.639 | 29184 | 29184 |
| B16 | 8/32 | 57 | 116736 | 3.490 | 233472 | 233472 |
| B16 | 16/64 | 57 | 933888 | 23.956 | 1867776 | 1867776 |

Each box is reevaluated by both independent arithmetic implementations. These are the first completed R83 recovery-run clocks, including endpoint hashing and rational checks within each level. They do not replace the failed R82 integration clocks or measure a matched training-speed comparison. Subsequent verification has separately named clocks.

## APPENDIX BG: EXACT INVERSION, BOUNDARY CHECKS AND SOURCE-QUALIFIED RECOVERY

### BG.1. *A finite algorithm for the entire tolerance interval*

For one construction, let $P$ be the number of integrated nonreference actions and $N$ the number of acquisition cells. Sort the action events by $D_{Cb} - U$. Events with nonpositive threshold are active at zero. Maintain the lowest lower endpoint and highest upper endpoint on each cell, initially including the exact reference interval $[0, 0]$. When an event is activated, update those two extrema and their probability-weighted sums. Equal thresholds are processed together. The resulting records contain every knot, the two class endpoints, the number of admitted alternatives, and the open or closed status of the stratum's right endpoint.

PROPOSITION A.59—Exact finite inversion: *Suppose the switch bounds, attained objectives, cell masses and integrated endpoints are rational. The complete class band is*

*computed with $O(P \log P + N)$ comparisons and arithmetic operations after integration, using $O(P + N)$ storage. Let $K_j, K_k$ denote the numbers of tolerance strata for two constructions. For a fixed replacement charge or equivalence margin, the sufficient region in Theorem A.28 can be represented by at most $K_j$ horizontal strips and found using at most $K_j + K_k$ endpoint-predicate evaluations. The same bound applies to a certified ordering in the opposite direction.*

PROOF: Sorting costs $O(P \log P)$ comparisons. Each event changes at most two cell extrema and two weighted sums; reference initialization takes $O(N)$ work. Storing events, cells and resulting strata uses the stated memory. These counts exclude the already paid action integration, and rational bit complexity depends on the input denominators.

Order both stratum lists from smaller to larger tolerance. Start at the largest second-coordinate stratum. For the current first-coordinate stratum, decrease the second index until the desired endpoint test holds or the index is exhausted. Record the resulting maximal admissible second-coordinate prefix. Move to the next first-coordinate stratum and continue from the same second index. Monotonicity implies that this index never needs to increase. Each successful test advances the first coordinate, and each failed test decreases the second coordinate. There are at most $K_j + K_k$ tests. Each recorded strip has the first stratum's exact endpoint convention and the upper boundary convention of its last admitted second stratum. Exhaustion leaves no later feasible strip. Thus the algorithm returns precisely the largest region certified by these fixed endpoint tests, not merely an inner grid of isolated tolerance points. *Q.E.D.*

The implementation also enumerates the finite product grid independently and checks every returned and excluded stratum pair. Separately, it rebuilds the possible-action set directly from the original switch predicates at every knot, every adjacent-stratum midpoint and the final endpoint. This check does not reuse the incremental cell-extremum update. It includes negative and repeated switch thresholds, a switch at the maximum allowed tolerance, and strict sign or equivalence tests at equality.

An economic margin and an objective excess are not interchangeable numbers. The plotted or tabulated tolerance region is an implication from one to the other

for the specified construction. It neither assumes that the nonlinear ReLU and Bernstein classes coincide nor attributes a larger allowable objective excess to universally better numerical conditioning. A valid stopping certificate is sufficient for an individual output; it is not a probability that a randomly initialized algorithm attains that certificate within a given time.

## BG.2. *Finite numerical and statistical domains*

The integrated catalogue is complete up to $e_j = 1/4096$. Actions whose switch lower bound exceeds $U_j + e_j$ cannot be selected within that objective sublevel and require no economic integration for this result. The calculation does not extrapolate the band beyond $e_j$. Each class has its original labels, feature map, coefficient box, acquisition grid, robust action menu and fixed terminal actor. Changing any of these objects requires new qualification and, where relevant, a new integration account.

There is no new empirical-Bernstein event in this inversion. The original nine-controller simultaneous event supplies the attained marginal and paired costs. The actionwise bounds extend it deterministically to all admissible readouts and tolerances. A data-selected comparison or stopping tolerance is covered because the event already implies every stated comparison. An unlisted controller or an additional statistical estimand does not acquire coverage merely by receiving a similar display name.

For a common tolerance the sufficient region is an initial interval, possibly empty or the entire declared domain. When a switch destroys the endpoint test, its exact knot is the supremum and is excluded. Decimal summaries must therefore not be used as executable stopping thresholds. The deposited rational boundaries and their inclusion flags are authoritative. A deployment fee is handled by comparing the lower gain endpoint with the supplied fee, not by converting a measured second into a monetary amount without a declared exchange rate.

## BG.3. *Recovery of the incomplete predecessor*

The earlier actionwise run completed the four model files but stopped while looking up a legacy-controller contrast. Its string operation removed the prefix

`calibrated-` without restoring `legacy-`. The resulting name was absent from the frozen policy roster. The failure log, source freeze and four completed model files are retained unchanged. The present recovery reproduces that exception as a negative regression and resolves the comparison through an explicit roster-checked identity map, with construction and callable-actor hashes bound to the existing cost record.

The source-qualified replay recomputes every paired subbox with both original independent arithmetic implementations and requires exact agreement of the resulting model records. It then computes the previously missing 72 paired and 24 own-controller transfers and the new full tolerance regions. It does not relabel the failed run as successful, invent its missing final service clock, refit a controller or add a cost-observation row. The old logged integration-phase durations and the new complete replay durations are reported separately. All inherited full-bin cost calculations and exact action-switch duals remain part of the release verification.

## BG.4. *Complete tolerance-frontier results*

The six calibrated pairs each receive three endpoint tests: positive difference, negative difference and equivalence within $1/256$. Empty regions and the coarse ReLU class are retained. The machine-readable result gives all horizontal strips and their rational boundaries; the following table gives every common-tolerance slice.

TABLE BG.I: All eighteen tolerance-frontier targets

| Pair | Target | Strips | Guaranteed common range | Exact end included |
|---|---|---|---|---|
| R8–B8 | $\lvert J_1 - J_2\rvert < 2^{-8}$ | 0 | $\varnothing$ | – |
| R8–B8 | $J_1 < J_2$ | 0 | $\varnothing$ | – |
| R8–B8 | $J_1 > J_2$ | 0 | $\varnothing$ | – |
| R8–R16 | $\lvert J_1 - J_2\rvert < 2^{-8}$ | 0 | $\varnothing$ | – |
| R8–R16 | $J_1 < J_2$ | 0 | $\varnothing$ | – |
| R8–R16 | $J_1 > J_2$ | 0 | $\varnothing$ | – |
| R8–B16 | $\lvert J_1 - J_2\rvert < 2^{-8}$ | 0 | $\varnothing$ | – |
| R8–B16 | $J_1 < J_2$ | 0 | $\varnothing$ | – |
| R8–B16 | $J_1 > J_2$ | 0 | $\varnothing$ | – |
| B8–R16 | $\lvert J_1 - J_2\rvert < 2^{-8}$ | 10 | $[0, 2^{-12}]$ | yes |

| Pair | Target | Strips | Guaranteed common range | Exact end included |
|---|---|---|---|---|
| B8–R16 | $J_1 < J_2$ | 0 | $\varnothing$ | – |
| B8–R16 | $J_1 > J_2$ | 6 | $[0, 0.000049063]$ | no |
| B8–B16 | $\lvert J_1 - J_2 \rvert < 2^{-8}$ | 6 | $[0, 0.000034382]$ | no |
| B8–B16 | $J_1 < J_2$ | 0 | $\varnothing$ | – |
| B8–B16 | $J_1 > J_2$ | 10 | $[0, 0.000213074]$ | no |
| R16–B16 | $\lvert J_1 - J_2 \rvert < 2^{-8}$ | 44 | $[0, 0.000177245]$ | no |
| R16–B16 | $J_1 < J_2$ | 0 | $\varnothing$ | – |
| R16–B16 | $J_1 > J_2$ | 31 | $[0, 0.000067491]$ | no |

The strip count concerns the full two-dimensional sufficient region, not just its equal-tolerance slice. The last column refers to the exact rational supremum, not the inward-rounded displayed endpoint. Every returned and excluded pair of tolerance strata is independently checked.

TABLE BG.II: Uniform gains at the largest integrated allowance

| Class | Reference | Gain for $0 \le \epsilon \le 2^{-12}$ |
|---|---|---|
| R8 | Zero | [0.153633, 0.178692] |
| R8 | Legacy R8 | [-0.012222, 0.011685] |
| B8 | Zero | [0.169561, 0.173851] |
| B8 | Legacy B8 | [0.080064, 0.083851] |
| R16 | Zero | [0.170435, 0.175234] |
| R16 | Legacy R16 | [0.003869, 0.007509] |
| B16 | Zero | [0.171910, 0.176851] |
| B16 | Legacy B16 | [0.081291, 0.085716] |

Every optimizer-class output in the displayed tolerance interval is included. All four zero-controller gains have positive lower endpoints. The coarse ReLU-versus-legacy comparison remains inconclusive. The full exact step function for every reference is deposited.

## APPENDIX BH: ECONOMIC COMPARISONS BEYOND EXACT ACTION STABILITY

We apply Theorem A.27 to the same four calibrated constructions in Section AA. No controller is refitted, and the original-law cost observations are unchanged. The additional computation evaluates the economic consequences of actions not excluded by the exact optimization barriers. It therefore answers a different relia-

bility question from whether one numerical fitting attempt returned its displayed coefficient vector.

The objective allowances are 0, 1/65536 and 1/4096. At the largest allowance, the complete 20,480-switch catalogue leaves 976 alternative actions for economic evaluation: 832 for coarse ReLU, nine for coarse Bernstein, 78 for fine ReLU and 57 for fine Bernstein. Each is evaluated on all four dyadic levels $(r,q) = (2,8),(4,16),(8,32),(16,64)$, where $r$ subdivides each initial-cell coordinate and $q$ subdivides the first innovation. No action is discarded because its cost effect is adverse or its switch bound is uninformative. Both whole-bin arithmetic implementations evaluate every level, and the final innovation retains its exact conditional expectation.

TABLE BH.I

FINE-GRID OPTIMIZER-CLASS COST DIFFERENCES

| $\epsilon$ | Binary-mass band | Actionwise band | Sign |
|---|---|---|---|
| 0 | [0.000955, 0.002013] | [0.000955, 0.002013] | positive |
| $2^{-16}$ | [-0.421558, 0.424526] | [0.000496, 0.002503] | positive |
| $2^{-12}$ | [-2.353046, 2.356014] | [-0.001523, 0.004614] | unresolved |

ReLU minus Bernstein costs for all qualified readout pairs with fixed labels and terminal actors. Every earlier dyadic level is retained. Cost endpoints are rounded outward; allowances are exact.

At the standard objective allowance $2^{-16}$, the ReLU-minus-Bernstein class interval is $[0.000496, 0.002503]$, compared with the binary-mass interval $[-0.421558, 0.424526]$. The positive actionwise interval applies uniformly to the stated approximate-optimizer classes. No new attained-policy sample is used.

The comparison concerns all readouts in the two attained-objective sublevels, including every 1/65536-optimal readout of their respective fixed-label problems. It does not require the same fitted coefficients or complete action agreement. By contrast, an arbitrary fitting attempt with no verified objective guarantee does not enter the class merely because it returns a feasible policy. The original fallback and fitting-warning results retain their separate interpretation.

For an additional Bernstein deployment charge $\tau$, subtracting $\tau$ from the two endpoints gives the uniform net-benefit interval for choosing Bernstein over ReLU. A charge below a positive lower endpoint is covered by the classwide policy-cost saving; a charge above the upper endpoint favors the opposite decision. The intermediate region is unresolved. The comparison uses normalized model costs and an explicitly supplied charge, rather than an invented processor price or empirical installation calibration.

The complete numerical appendix includes all 48 model–allowance–level bands, all 72 pairwise transfers and all 24 final-level comparisons with the zero or corresponding legacy controller. It retains the coarse ReLU sublevels and the larger-allowance outcomes alongside the fine-grid comparison. The cellwise extrema are outer bounds: separately possible actions need not be jointly selectable by a single coefficient vector. Their residual width is not characterized as a population fitting error.

The completed recovery reevaluates 18,270,720 paired boxes per arithmetic implementation and verifies 15,616 exact rational midpoint differences. Its independently recorded time through the verification calculations is 515.246 seconds; peak resident memory is 88,652 KiB. The failed R82 run retains its own partial integration-phase clocks and exception record. This recovery is not a new policy-cost sample. The completed R83 recovery accounts for all refinement levels, both arithmetic implementations and rational checks; its output-accounting boundary is recorded explicitly. Subsequent replay has separate clocks. The interrupted R82 phase clocks are preserved, not replaced or promoted to a missing complete-service clock. The computation adds neither an independent cost-observation row nor a new success-frequency sample. Development pilots informed the deterministic refinement resolution before its source freeze; it is not presented as a newly randomized prospective economic experiment. The unchanged prospective R80 statistical family and the new deterministic integration together support the classwide statements on their original event.

## APPENDIX BI: WINNER AND COMMON-READOUT SUPPORT CERTIFICATES

The coefficient vector is shared across the acquisition cells. This appendix gives the exact certificates used to retain that restriction in the main economic bound. All coefficients in the executed fixed-feature problem are rational. Floating optimization proposes multipliers; the formulas below, evaluated in rational arithmetic, decide whether those proposals are admissible.

### BI.1. *A full-menu lower qualification bound*

Write $\alpha = 2\rho$, let $\phi_i$ be the frozen feature rows, and let $r_j$ be the rows defining the calibrated derivative bound. The objective is

$$F(\theta) = \max_i (y_i^+ - \phi_i^\mathsf{T}\theta) + \max_i (\phi_i^\mathsf{T}\theta - y_i^-) + \alpha G + \alpha \max_j |r_j^\mathsf{T}\theta|.$$

For one winner or a collection of winners, assemble the inequalities $d_k^\mathsf{T}\theta + c_k \le 0$. These include every competing menu action for each specified winner. Take probability vectors $p, q$, nonnegative derivative multipliers $\xi_j^+, \xi_j^-$ satisfying $\sum_j (\xi_j^+ + \xi_j^-) \le \alpha$, and arbitrary nonnegative event multipliers $\nu_k$. Define

$$s = \sum_i (q_i - p_i)\phi_i + \sum_j (\xi_j^+ - \xi_j^-) r_j + \sum_k \nu_k d_k,$$

$$D = \sum_i p_i y_i^+ - \sum_i q_i y_i^- + \alpha G + \sum_k \nu_k c_k - \sum_m B_m |s_m|.$$

Then $D$ is a lower bound on the infimum of $F$ over the event and the coefficient box.

To see this, each of the first two maxima dominates its corresponding probability-weighted average. The derivative term dominates the signed derivative combination because its total nonnegative multiplier mass is at most $\alpha$. For a feasible event, adding $\sum_k \nu_k (d_k^\mathsf{T}\theta + c_k)$ cannot increase the resulting lower expression. Minimizing its linear part over $|\theta_m| \le B_m$ gives $-\sum_m B_m |s_m|$. This proves the bound without strong duality, a constraint qualification or a successful numerical minimizer. Taking the maximum with an inherited lower bound preserves validity.

The numerical implementation scales the event rows before proposing multipliers. It then undoes that scaling exactly. Probability multipliers are projected to a rational

simplex; derivative multipliers are made nonnegative and rescaled only when their total mass exceeds the exact derivative budget. Event multipliers are nonnegative rationals obtained from the proposed binary values and the exact row scales. The independent checker reconstructs every feature integral from the separate antiderivative implementation, every menu comparison, and the final coefficient-box support value. It does not call the production optimizer or production certificate routine.

### BI.2. *Empty events require a separate proof*

For nonnegative weights $\lambda_k$, set

$$\eta = \sum_k \lambda_k c_k - \sum_m B_m \left| \sum_k \lambda_k d_{km} \right|.$$

If $\eta > 0$, the event is empty. Indeed, the left side is the minimum over the box of $\sum_k \lambda_k (d_k^\top \theta + c_k)$, whereas an event-feasible point would make this sum nonpositive. The executed slack problem additionally imposes $\sum_k \lambda_k \le 1$ when proposing the weights; this normalization is convenient but is not needed for the implication once strict positivity has been checked exactly.

An infeasible numerical solver status is not a proof of $\eta > 0$. When no positive exact separation is obtained, the event receives only its valid ordinary objective lower bound and remains possible whenever that bound does not exceed the stated sublevel. The numerical catalogue preserves every such solver status. In the executed study, all 104 nonsuccess statuses correspond to positive independently checked separation certificates, rather than to silently discarded observations or failed fitting attempts.

### BI.3. *Exact economic assignment duals*

For a chosen tolerance, the pair-conflict relaxation has nonnegative variables $z_i$, one unit of mass per acquisition cell, and $z_i + z_j \le 1$ for each verified incompatible pair. Given nonnegative rational pair multipliers $\lambda_{ij}$ and weights $w_i$, the

construction

$$\alpha_C = \max\left\{0, \max_{i \text{ in } C}\left(w_i - \sum_{j:\{i,j\} \text{ is a cut}} \lambda_{ij}\right)\right\}$$

satisfies every assignment-dual inequality. Its value is $\sum_C \alpha_C + \sum_{\{i,j\}} \lambda_{ij}$. The zero reference action explains why nonnegative $\alpha_C$ is a harmless conservative choice. The lower cost endpoint uses the negative conditional lower-cost weights and has a separately verified dual.

The pair multipliers are rounded to nonnegative dyadic rationals. Every remaining residual is paid by the exact cell maximum; no floating primal optimum is used as a certified upper bound. Setting all pair multipliers to zero recovers the full-menu cellwise extrema. Whenever a proposed dual is less useful, that valid fallback is retained. The independent checker verifies every variable's covering inequality, all signs, all objective sums, and every final interval transfer.

### BI.4. *Why both refinements matter*

For three action scores $0, -1, -2$, the second action defeats the reference score zero but cannot win the full menu. A reference-only barrier can therefore remain inconclusive even though the complete winner event is empty. This is a distinction about the implemented argmin, not a change in the economic action set.

For the shared-coefficient restriction, take $\theta \in [-1, 1]$ and two cells. An adverse action in the first cell requires $\theta \geq 1$; one in the second requires $\theta \leq -1$. Both events are separately feasible but their intersection is empty. If each adverse action contributes $1/2$ to aggregate cost, independent cell bounds allow a cost increase of one. The compatible bound is $1/2$, and it is attained by one endpoint of the parameter interval. This supplies a strict example for the improvement claimed by the theorem.

Pair restrictions need not characterize every feasible policy. With three unit-weight alternatives and all three pair conflicts, an assignment relaxation permits mass $1/2$ on each alternative, even though a zero–one assignment can select at most one. The corresponding fractional support can therefore be larger than the true worst cost. More generally, pairwise intersection does not imply joint intersection

of all winner regions. The manuscript consequently calls the computed object an enclosure, not the exact attainable policy set or the exact worst optimizer output. Additional certified joint events may strengthen it without invalidating any existing bound.

### BI.5. *Work, storage and continuation scope*

Let $P$ be the number of admitted alternatives, $K$ the number of declared joint events, $m$ the maximum menu size, $d$ the readout dimension, $N$ the number of cells, and $E$ the number of retained pair cuts at a tolerance. A singleton event adds at most $m-1$ rows and a pair event at most $2(m-1)$ rows to the same fixed qualification problem. The experiment executes exactly $P+K$ qualification problems. It does not claim a dimension-independent bound on an LP solver's arithmetic or bit complexity.

Once multipliers have been obtained, checking a qualification certificate takes a number of rational operations linear in its feature, derivative and event coefficients. Checking one assignment dual takes $O(P+N+E)$ rational operations and storage linear in its retained variables, cells and cuts. Constructing and solving the assignment LP is separate work. The executed pair budget limits qualification work without assuming that unchecked intersections are feasible or infeasible. Previously purchased conditional integration remains an explicit prerequisite, not free computation.

The identity transferring conditional contrasts to aggregate policy cost holds because the policies share the stated future controller and only their first acquired action varies. A longer horizon can use the same identity when its entire fixed-continuation conditional cost is enclosed. Jointly refitting that future controller changes the estimand and requires another qualification argument. The results therefore do not claim that the present pair graph certifies arbitrary changes at every date or regenerates the large vector-control descendant tree.

## APPENDIX BJ: COMPLETE COMMON-READOUT NUMERICAL RECORDS

TABLE BJ.I: Every declared optimizer-class tolerance and its three nested bounds

| Model | $\epsilon$ | Allowed | Cuts | Actionwise | Full menu | Compatible |
|---|---|---|---|---|---|---|
| R8 | 0 | 689 | 351 | $[-0.010033, 0.012550]$ | $[-0.010033, 0.008534]$ | $[-0.010023, 0.008257]$ |
| R8 | $2^{-16}$ | 689 | 351 | $[-0.010033, 0.012550]$ | $[-0.010033, 0.008534]$ | $[-0.010023, 0.008257]$ |
| R8 | $2^{-15}$ | 689 | 351 | $[-0.010033, 0.012550]$ | $[-0.010033, 0.008534]$ | $[-0.010023, 0.008257]$ |
| R8 | $2^{-14}$ | 689 | 351 | $[-0.010033, 0.012550]$ | $[-0.010033, 0.008534]$ | $[-0.010023, 0.008257]$ |
| R8 | $2^{-13}$ | 691 | 351 | $[-0.010033, 0.012589]$ | $[-0.010033, 0.008570]$ | $[-0.010023, 0.008293]$ |
| R8 | $2^{-12}$ | 691 | 351 | $[-0.010033, 0.012680]$ | $[-0.010033, 0.008570]$ | $[-0.010023, 0.008293]$ |
| B8 | 0 | 0 | 0 | $[0.000000, 0.000000]$ | $[0.000000, 0.000000]$ | $[0.000000, 0.000000]$ |
| B8 | $2^{-16}$ | 0 | 0 | $[0.000000, 0.000000]$ | $[0.000000, 0.000000]$ | $[0.000000, 0.000000]$ |
| B8 | $2^{-15}$ | 0 | 0 | $[0.000000, 0.000000]$ | $[0.000000, 0.000000]$ | $[0.000000, 0.000000]$ |
| B8 | $2^{-14}$ | 3 | 0 | $[-0.000428, 0.000318]$ | $[-0.000428, 0.000318]$ | $[-0.000428, 0.000318]$ |
| B8 | $2^{-13}$ | 6 | 0 | $[-0.000876, 0.000680]$ | $[-0.000876, 0.000680]$ | $[-0.000876, 0.000680]$ |
| B8 | $2^{-12}$ | 9 | 14 | $[-0.001080, 0.000900]$ | $[-0.001080, 0.000900]$ | $[-0.001032, 0.000792]$ |
| R16 | 0 | 0 | 0 | $[0.000000, 0.000000]$ | $[0.000000, 0.000000]$ | $[0.000000, 0.000000]$ |
| R16 | $2^{-16}$ | 8 | 0 | $[-0.000165, 0.000193]$ | $[-0.000165, 0.000193]$ | $[-0.000165, 0.000193]$ |
| R16 | $2^{-15}$ | 14 | 0 | $[-0.000287, 0.000343]$ | $[-0.000287, 0.000343]$ | $[-0.000287, 0.000343]$ |
| R16 | $2^{-14}$ | 18 | 2 | $[-0.000363, 0.000425]$ | $[-0.000363, 0.000425]$ | $[-0.000345, 0.000404]$ |
| R16 | $2^{-13}$ | 34 | 10 | $[-0.000621, 0.000621]$ | $[-0.000621, 0.000621]$ | $[-0.000582, 0.000576]$ |
| R16 | $2^{-12}$ | 61 | 140 | $[-0.001168, 0.001291]$ | $[-0.001162, 0.001249]$ | $[-0.001009, 0.001053]$ |

| Model | $\epsilon$ | Allowed | Cuts | Actionwise | Full menu | Compatible |
|---|---|---|---|---|---|---|
| B16 | 0 | 0 | 0 | [0.000000, 0.000000] | [0.000000, 0.000000] | [0.000000, 0.000000] |
| B16 | $2^{-16}$ | 13 | 7 | [−0.000299, 0.000295] | [−0.000299, 0.000295] | [−0.000251, 0.000241] |
| B16 | $2^{-15}$ | 16 | 6 | [−0.000380, 0.000371] | [−0.000380, 0.000371] | [−0.000328, 0.000318] |
| B16 | $2^{-14}$ | 25 | 18 | [−0.000590, 0.000570] | [−0.000590, 0.000570] | [−0.000516, 0.000490] |
| B16 | $2^{-13}$ | 33 | 51 | [−0.000773, 0.000759] | [−0.000773, 0.000759] | [−0.000650, 0.000627] |
| B16 | $2^{-12}$ | 56 | 204 | [−0.001311, 0.001311] | [−0.001309, 0.001311] | [−0.001012, 0.001000] |

Closed weak winner regions retain ties conservatively. Strict qualification inequalities alone exclude actions or joint events; threshold equality is retained. A zero interval follows from the absence of alternative actions in the covered class, not from rounding a small nonzero interval. Each interval is displayed on two lines, with its lower endpoint above its upper endpoint; no endpoint digit or catalogue row is omitted.

TABLE BJ.II: Complete simultaneous class-cost catalogue

| First minus second | $\epsilon$ | Actionwise | Compatible | Sign | 1/256 eq. |
|---|---|---|---|---|---|
| R8–B8 | 0 | [−0.006461, 0.017237] | [−0.006451, 0.012945] | unresolved | no |
| R8–B8 | $2^{-16}$ | [−0.006461, 0.017237] | [−0.006451, 0.012945] | unresolved | no |
| R8–B8 | $2^{-15}$ | [−0.006461, 0.017237] | [−0.006451, 0.012945] | unresolved | no |
| R8–B8 | $2^{-14}$ | [−0.006779, 0.017665] | [−0.006769, 0.013373] | unresolved | no |
| R8–B8 | $2^{-13}$ | [−0.007140, 0.018152] | [−0.007131, 0.013856] | unresolved | no |
| R8–B8 | $2^{-12}$ | [−0.007360, 0.018447] | [−0.007242, 0.014012] | unresolved | no |
| R8–R16 | 0 | [−0.005178, 0.018514] | [−0.005168, 0.014222] | unresolved | no |

| First minus second | $\epsilon$ | Actionwise | Compatible | Sign | $1/256$ eq. |
|---|---|---|---|---|---|
| R8–R16 | $2^{-16}$ | $[-0.005370, 0.018679]$ | $[-0.005360, 0.014387]$ | unresolved | no |
| R8–R16 | $2^{-15}$ | $[-0.005520, 0.018800]$ | $[-0.005510, 0.014508]$ | unresolved | no |
| R8–R16 | $2^{-14}$ | $[-0.005602, 0.018877]$ | $[-0.005571, 0.014567]$ | unresolved | no |
| R8–R16 | $2^{-13}$ | $[-0.005798, 0.019174]$ | $[-0.005744, 0.014839]$ | unresolved | no |
| R8–R16 | $2^{-12}$ | $[-0.006468, 0.019812]$ | $[-0.006221, 0.015267]$ | unresolved | no |
| R8–B16 | 0 | $[-0.003695, 0.019999]$ | $[-0.003686, 0.015707]$ | unresolved | no |
| R8–B16 | $2^{-16}$ | $[-0.003989, 0.020298]$ | $[-0.003927, 0.015957]$ | unresolved | no |
| R8–B16 | $2^{-15}$ | $[-0.004065, 0.020379]$ | $[-0.004003, 0.016035]$ | unresolved | no |
| R8–B16 | $2^{-14}$ | $[-0.004265, 0.020588]$ | $[-0.004175, 0.016223]$ | unresolved | no |
| R8–B16 | $2^{-13}$ | $[-0.004454, 0.020811]$ | $[-0.004312, 0.016393]$ | unresolved | no |
| R8–B16 | $2^{-12}$ | $[-0.005006, 0.021441]$ | $[-0.004685, 0.016754]$ | unresolved | no |
| B8–R16 | 0 | $[0.000748, 0.001812]$ | $[0.000748, 0.001812]$ | positive | yes |
| B8–R16 | $2^{-16}$ | $[0.000555, 0.001977]$ | $[0.000555, 0.001977]$ | positive | yes |
| B8–R16 | $2^{-15}$ | $[0.000405, 0.002098]$ | $[0.000405, 0.002098]$ | positive | yes |
| B8–R16 | $2^{-14}$ | $[-0.000104, 0.002493]$ | $[-0.000083, 0.002475]$ | unresolved | yes |
| B8–R16 | $2^{-13}$ | $[-0.000748, 0.003112]$ | $[-0.000704, 0.003074]$ | unresolved | yes |
| B8–R16 | $2^{-12}$ | $[-0.001622, 0.003879]$ | $[-0.001336, 0.003612]$ | unresolved | yes |
| B8–B16 | 0 | $[0.002241, 0.003286]$ | $[0.002241, 0.003286]$ | positive | yes |
| B8–B16 | $2^{-16}$ | $[0.001947, 0.003585]$ | $[0.002000, 0.003536]$ | positive | yes |
| B8–B16 | $2^{-15}$ | $[0.001871, 0.003665]$ | $[0.001924, 0.003614]$ | positive | yes |

| First minus second | $\epsilon$ | Actionwise | Compatible | Sign | 1/256 eq. |
|---|---|---|---|---|---|
| B8–B16 | $2^{-14}$ | [0.001244, 0.004193] | [0.001324, 0.004120] | positive | no |
| B8–B16 | $2^{-13}$ | [0.000607, 0.004738] | [0.000739, 0.004616] | positive | no |
| B8–B16 | $2^{-12}$ | [−0.000149, 0.005497] | [0.000210, 0.005089] | positive | no |
| R16–B16 | 0 | [0.000955, 0.002013] | [0.000955, 0.002013] | positive | yes |
| R16–B16 | $2^{-16}$ | [0.000496, 0.002503] | [0.000549, 0.002455] | positive | yes |
| R16–B16 | $2^{-15}$ | [0.000299, 0.002734] | [0.000352, 0.002683] | positive | yes |
| R16–B16 | $2^{-14}$ | [0.000023, 0.003026] | [0.000120, 0.002932] | positive | yes |
| R16–B16 | $2^{-13}$ | [−0.000424, 0.003405] | [−0.000254, 0.003239] | unresolved | yes |
| R16–B16 | $2^{-12}$ | [−0.001523, 0.004614] | [−0.001054, 0.004077] | unresolved | no |

Equivalence requires the exact rational interval to be strictly inside the predeclared margin. It is distinct from a signed ranking; both can hold for a small nonzero cost difference. The displayed six-decimal endpoints are descriptive and do not decide a boundary test. Each interval is displayed on two lines, with its lower endpoint above its upper endpoint; no endpoint digit or catalogue row is omitted. Every alternative previously admitted up to $2^{-12}$ is included in the singleton roster. The fixed pair-selection rule, including its deterministic cost-based ordering and tie handling, is part of the source lock. Its execution contributes no additional independent cost row or fit. The 24 model–tolerance records and 36 class comparisons use the already covered original-law cost family. Source, output and execution-clock hashes are retained with the exact multipliers, all 104 infeasibility certificates, and all 16,548 independently checked assignment inequalities.

TABLE BJ.III

INCREMENTAL CERTIFICATE WORK FOR THE COMPLETED CATALOGUE

| Model | Single | Joint | Single (s) | Joint (s) | Assign (s) | Total (s) |
|---|---|---|---|---|---|---|
| R8 | 832 | 512 | 10.960 | 3.981 | 0.251 | 15.224 |
| B8 | 9 | 36 | 0.269 | 0.322 | 0.012 | 0.603 |
| R16 | 78 | 512 | 6.101 | 37.707 | 0.068 | 43.892 |
| B16 | 57 | 512 | 5.547 | 39.088 | 0.080 | 44.728 |

Times belong to one recorded ARM execution, not a cross-basis speed experiment or population timing estimate. The complete interpreter boundary is 104.699 seconds and includes imports, input reads, all calculations, durable records and shutdown. Its child peak resident set is 106588 KiB. Previously paid fitting, integration and cost inference retain their original clocks; independent checking and publication overhead have separate records.

## APPENDIX BK: COMPLETE PROSPECTIVE EVIDENCE AND VERIFICATION BOUNDARIES

The fixed evaluation program reconstructs the R85 streams from the protocol and the nine unchanged acquired action tables. The source, bin arrays, raw endpoint arrays and exact integer moments have separate identities. Both arithmetic implementations cover every evaluated bin; midpoint checks supplement rather than replace that full-bin calculation. A replay is not an additional independent sample.

### BK.1. *Complete interval family and width components*

For each of two streams, two estimators and 36 unordered pairs, the numerical tables report both directed endpoints and the exact decomposition

$$\begin{aligned}\text{width} = {} & \text{endpoint enclosure} \\ & + \text{sampling variance allowance} \\ & + \text{finite support allowance}\,.\end{aligned}$$

The endpoint enclosure combines original-law bin propagation, acquisition branching and directed arithmetic. It is not split into fictitiously independent subcomponents. Acquisition ambiguity counts, integer rectangle evaluations, midpoint checks

and endpoint-disagreement magnitudes are reported separately in the machine-readable record. The cash and conditioned estimators use the same underlying rows and are not independent replicates.

TABLE BK.I: Complete original-law pairwise evaluation and interval widths

| Set | First–second | Lower | Upper | Enclosure | Variance | Support |
|---|---|---|---|---|---|---|
| CC | Zero–LR8 | 0.128344 | 0.204068 | 2.72e-11 | 0.011165 | 0.064559 |
| CC | Zero–LB8 | 0.055344 | 0.123956 | 2.67e-11 | 0.004052 | 0.064559 |
| CC | Zero–LR16 | 0.128864 | 0.204906 | 2.72e-11 | 0.011483 | 0.064559 |
| CC | Zero–LB16 | 0.056412 | 0.125127 | 2.67e-11 | 0.004156 | 0.064559 |
| CC | Zero–CR8 | 0.129790 | 0.205047 | 2.70e-11 | 0.010698 | 0.064559 |
| CC | Zero–CB8 | 0.133972 | 0.208944 | 2.72e-11 | 0.010413 | 0.064559 |
| CC | Zero–CR16 | 0.135167 | 0.210379 | 2.71e-11 | 0.010654 | 0.064559 |
| CC | Zero–CB16 | 0.136695 | 0.211744 | 2.71e-11 | 0.010489 | 0.064559 |
| CC | LR8–LB8 | -0.112401 | -0.040711 | 2.75e-11 | 0.007131 | 0.064559 |
| CC | LR8–LR16 | -0.031794 | 0.033152 | 2.80e-11 | 0.000387 | 0.064559 |
| CC | LR8–LB16 | -0.111228 | -0.039645 | 2.75e-11 | 0.007024 | 0.064559 |
| CC | LR8–CR8 | -0.031819 | 0.034245 | 2.77e-11 | 0.001505 | 0.064559 |
| CC | LR8–CB8 | -0.027666 | 0.038171 | 2.79e-11 | 0.001279 | 0.064559 |
| CC | LR8–CR16 | -0.026295 | 0.039428 | 2.78e-11 | 0.001164 | 0.064559 |
| CC | LR8–CB16 | -0.024906 | 0.040933 | 2.79e-11 | 0.001280 | 0.064559 |
| CC | LB8–LR16 | 0.041231 | 0.113239 | 2.75e-11 | 0.007449 | 0.064559 |
| CC | LB8–LB16 | -0.031239 | 0.033478 | 2.70e-11 | 0.000159 | 0.064559 |
| CC | LB8–CR8 | 0.042138 | 0.113399 | 2.73e-11 | 0.006702 | 0.064559 |
| CC | LB8–CB8 | 0.046334 | 0.117283 | 2.74e-11 | 0.006390 | 0.064559 |
| CC | LB8–CR16 | 0.047529 | 0.118717 | 2.74e-11 | 0.006628 | 0.064559 |
| CC | LB8–CB16 | 0.049057 | 0.120081 | 2.74e-11 | 0.006465 | 0.064559 |
| CC | LR16–LB16 | -0.112067 | -0.040164 | 2.75e-11 | 0.007344 | 0.064559 |
| CC | LR16–CR8 | -0.032612 | 0.033679 | 2.78e-11 | 0.001731 | 0.064559 |
| CC | LR16–CB8 | -0.028491 | 0.037638 | 2.79e-11 | 0.001570 | 0.064559 |
| CC | LR16–CR16 | -0.027099 | 0.038875 | 2.78e-11 | 0.001415 | 0.064559 |
| CC | LR16–CB16 | -0.025715 | 0.040383 | 2.79e-11 | 0.001539 | 0.064559 |

| Set | First–second | Lower | Upper | Enclosure | Variance | Support |
|---|---|---|---|---|---|---|
| CC | LB16–CR8 | 0.041073 | 0.112225 | 2.73e-11 | 0.006593 | 0.064559 |
| CC | LB16–CB8 | 0.045270 | 0.116108 | 2.74e-11 | 0.006280 | 0.064559 |
| CC | LB16–CR16 | 0.046464 | 0.117543 | 2.74e-11 | 0.006519 | 0.064559 |
| CC | LB16–CB16 | 0.047993 | 0.118907 | 2.74e-11 | 0.006355 | 0.064559 |
| CC | CR8–CB8 | -0.028664 | 0.036743 | 2.77e-11 | 0.000848 | 0.064559 |
| CC | CR8–CR16 | -0.027326 | 0.038034 | 2.76e-11 | 0.000801 | 0.064559 |
| CC | CR8–CB16 | -0.025891 | 0.039492 | 2.77e-11 | 0.000824 | 0.064559 |
| CC | CB8–CR16 | -0.031183 | 0.033812 | 2.78e-11 | 0.000436 | 0.064559 |
| CC | CB8–CB16 | -0.029663 | 0.035186 | 2.78e-11 | 0.000290 | 0.064559 |
| CC | CR16–CB16 | -0.031022 | 0.033916 | 2.77e-11 | 0.000379 | 0.064559 |
| CA | Zero–LR8 | 0.128357 | 0.204076 | 2.63e-11 | 0.011160 | 0.064559 |
| CA | Zero–LB8 | 0.055356 | 0.123959 | 2.59e-11 | 0.004044 | 0.064559 |
| CA | Zero–LR16 | 0.128875 | 0.204913 | 2.63e-11 | 0.011479 | 0.064559 |
| CA | Zero–LB16 | 0.056424 | 0.125130 | 2.59e-11 | 0.004148 | 0.064559 |
| CA | Zero–CR8 | 0.129803 | 0.205056 | 2.61e-11 | 0.010694 | 0.064559 |
| CA | Zero–CB8 | 0.133985 | 0.208952 | 2.63e-11 | 0.010408 | 0.064559 |
| CA | Zero–CR16 | 0.135183 | 0.210391 | 2.62e-11 | 0.010649 | 0.064559 |
| CA | Zero–CB16 | 0.136710 | 0.211754 | 2.62e-11 | 0.010485 | 0.064559 |
| CA | LR8–LB8 | -0.112404 | -0.040714 | 2.66e-11 | 0.007131 | 0.064559 |
| CA | LR8–LR16 | -0.031795 | 0.033151 | 2.70e-11 | 0.000387 | 0.064559 |
| CA | LR8–LB16 | -0.111231 | -0.039648 | 2.66e-11 | 0.007025 | 0.064559 |
| CA | LR8–CR8 | -0.031819 | 0.034245 | 2.68e-11 | 0.001505 | 0.064559 |
| CA | LR8–CB8 | -0.027667 | 0.038171 | 2.70e-11 | 0.001279 | 0.064559 |
| CA | LR8–CR16 | -0.026291 | 0.039432 | 2.69e-11 | 0.001164 | 0.064559 |
| CA | LR8–CB16 | -0.024904 | 0.040935 | 2.69e-11 | 0.001280 | 0.064559 |
| CA | LB8–LR16 | 0.041232 | 0.113241 | 2.66e-11 | 0.007450 | 0.064559 |
| CA | LB8–LB16 | -0.031239 | 0.033478 | 2.61e-11 | 0.000158 | 0.064559 |
| CA | LB8–CR8 | 0.042142 | 0.113403 | 2.64e-11 | 0.006702 | 0.064559 |
| CA | LB8–CB8 | 0.046336 | 0.117285 | 2.65e-11 | 0.006390 | 0.064559 |
| CA | LB8–CR16 | 0.047536 | 0.118723 | 2.65e-11 | 0.006628 | 0.064559 |
| CA | LB8–CB16 | 0.049062 | 0.120086 | 2.65e-11 | 0.006465 | 0.064559 |
| CA | LR16–LB16 | -0.112069 | -0.040165 | 2.66e-11 | 0.007344 | 0.064559 |
| CA | LR16–CR8 | -0.032610 | 0.033681 | 2.69e-11 | 0.001732 | 0.064559 |

| Set | First–second | Lower | Upper | Enclosure | Variance | Support |
|---|---|---|---|---|---|---|
| CA | LR16–CB8 | -0.028491 | 0.037639 | 2.70e-11 | 0.001571 | 0.064559 |
| CA | LR16–CR16 | -0.027094 | 0.038879 | 2.69e-11 | 0.001415 | 0.064559 |
| CA | LR16–CB16 | -0.025711 | 0.040387 | 2.70e-11 | 0.001539 | 0.064559 |
| CA | LB16–CR8 | 0.041077 | 0.112229 | 2.64e-11 | 0.006593 | 0.064559 |
| CA | LB16–CB8 | 0.045272 | 0.116111 | 2.65e-11 | 0.006280 | 0.064559 |
| CA | LB16–CR16 | 0.046471 | 0.117549 | 2.65e-11 | 0.006519 | 0.064559 |
| CA | LB16–CB16 | 0.047998 | 0.118912 | 2.65e-11 | 0.006355 | 0.064559 |
| CA | CR8–CB8 | -0.028664 | 0.036742 | 2.68e-11 | 0.000847 | 0.064559 |
| CA | CR8–CR16 | -0.027322 | 0.038037 | 2.67e-11 | 0.000800 | 0.064559 |
| CA | CR8–CB16 | -0.025889 | 0.039494 | 2.68e-11 | 0.000823 | 0.064559 |
| CA | CB8–CR16 | -0.031178 | 0.033815 | 2.69e-11 | 0.000434 | 0.064559 |
| CA | CB8–CB16 | -0.029660 | 0.035188 | 2.69e-11 | 0.000289 | 0.064559 |
| CA | CR16–CB16 | -0.031024 | 0.033914 | 2.68e-11 | 0.000378 | 0.064559 |
| EC | Zero–LR8 | 0.162747 | 0.169569 | 2.72e-11 | 0.002787 | 0.004035 |
| EC | Zero–LB8 | 0.087126 | 0.092174 | 2.67e-11 | 0.001012 | 0.004035 |
| EC | Zero–LR16 | 0.163417 | 0.170317 | 2.72e-11 | 0.002866 | 0.004035 |
| EC | Zero–LB16 | 0.088239 | 0.093314 | 2.67e-11 | 0.001040 | 0.004035 |
| EC | Zero–CR8 | 0.163832 | 0.170543 | 2.70e-11 | 0.002676 | 0.004035 |
| EC | Zero–CB8 | 0.168031 | 0.174669 | 2.71e-11 | 0.002603 | 0.004035 |
| EC | Zero–CR16 | 0.169279 | 0.175977 | 2.71e-11 | 0.002664 | 0.004035 |
| EC | Zero–CB16 | 0.170792 | 0.177448 | 2.71e-11 | 0.002621 | 0.004035 |
| EC | LR8–LB8 | -0.079416 | -0.073601 | 2.75e-11 | 0.001780 | 0.004035 |
| EC | LR8–LR16 | -0.001357 | 0.002774 | 2.79e-11 | 0.000096 | 0.004035 |
| EC | LR8–LB16 | -0.078275 | -0.072489 | 2.75e-11 | 0.001752 | 0.004035 |
| EC | LR8–CR8 | -0.001175 | 0.003234 | 2.77e-11 | 0.000374 | 0.004035 |
| EC | LR8–CB8 | 0.003017 | 0.007367 | 2.79e-11 | 0.000315 | 0.004035 |
| EC | LR8–CR16 | 0.004310 | 0.008630 | 2.78e-11 | 0.000286 | 0.004035 |
| EC | LR8–CB16 | 0.005787 | 0.010138 | 2.78e-11 | 0.000316 | 0.004035 |
| EC | LB8–LR16 | 0.074270 | 0.080163 | 2.75e-11 | 0.001858 | 0.004035 |
| EC | LB8–LB16 | -0.000912 | 0.003164 | 2.70e-11 | 0.000041 | 0.004035 |
| EC | LB8–CR8 | 0.074681 | 0.080394 | 2.73e-11 | 0.001678 | 0.004035 |
| EC | LB8–CB8 | 0.078884 | 0.084516 | 2.74e-11 | 0.001598 | 0.004035 |
| EC | LB8–CR16 | 0.080132 | 0.085825 | 2.73e-11 | 0.001658 | 0.004035 |

| Set | First–second | Lower | Upper | Enclosure | Variance | Support |
|---|---|---|---|---|---|---|
| EC | LB8–CB16 | 0.081645 | 0.087296 | 2.74e-11 | 0.001616 | 0.004035 |
| EC | LR16–LB16 | -0.079023 | -0.073158 | 2.75e-11 | 0.001830 | 0.004035 |
| EC | LR16–CR8 | -0.001911 | 0.002552 | 2.78e-11 | 0.000428 | 0.004035 |
| EC | LR16–CB8 | 0.002273 | 0.006694 | 2.79e-11 | 0.000386 | 0.004035 |
| EC | LR16–CR16 | 0.003570 | 0.007952 | 2.78e-11 | 0.000347 | 0.004035 |
| EC | LR16–CB16 | 0.005046 | 0.009461 | 2.79e-11 | 0.000380 | 0.004035 |
| EC | LB16–CR8 | 0.073569 | 0.079254 | 2.73e-11 | 0.001650 | 0.004035 |
| EC | LB16–CB8 | 0.077772 | 0.083376 | 2.74e-11 | 0.001569 | 0.004035 |
| EC | LB16–CR16 | 0.079020 | 0.084684 | 2.74e-11 | 0.001630 | 0.004035 |
| EC | LB16–CB16 | 0.080533 | 0.086155 | 2.74e-11 | 0.001587 | 0.004035 |
| EC | CR8–CB8 | 0.002037 | 0.006288 | 2.77e-11 | 0.000216 | 0.004035 |
| EC | CR8–CR16 | 0.003322 | 0.007559 | 2.76e-11 | 0.000203 | 0.004035 |
| EC | CR8–CB16 | 0.004810 | 0.009055 | 2.77e-11 | 0.000209 | 0.004035 |
| EC | CB8–CR16 | -0.000795 | 0.003351 | 2.78e-11 | 0.000110 | 0.004035 |
| EC | CB8–CB16 | 0.000717 | 0.004824 | 2.78e-11 | 0.000072 | 0.004035 |
| EC | CR16–CB16 | -0.000574 | 0.003558 | 2.77e-11 | 0.000097 | 0.004035 |
| EA | Zero–LR8 | 0.162743 | 0.169563 | 2.63e-11 | 0.002784 | 0.004035 |
| EA | Zero–LB8 | 0.087126 | 0.092170 | 2.59e-11 | 0.001009 | 0.004035 |
| EA | Zero–LR16 | 0.163413 | 0.170311 | 2.63e-11 | 0.002863 | 0.004035 |
| EA | Zero–LB16 | 0.088238 | 0.093309 | 2.59e-11 | 0.001036 | 0.004035 |
| EA | Zero–CR8 | 0.163826 | 0.170535 | 2.61e-11 | 0.002673 | 0.004035 |
| EA | Zero–CB8 | 0.168028 | 0.174662 | 2.63e-11 | 0.002600 | 0.004035 |
| EA | Zero–CR16 | 0.169274 | 0.175969 | 2.62e-11 | 0.002661 | 0.004035 |
| EA | Zero–CB16 | 0.170788 | 0.177441 | 2.62e-11 | 0.002618 | 0.004035 |
| EA | LR8–LB8 | -0.079413 | -0.073598 | 2.66e-11 | 0.001779 | 0.004035 |
| EA | LR8–LR16 | -0.001356 | 0.002774 | 2.70e-11 | 0.000095 | 0.004035 |
| EA | LR8–LB16 | -0.078272 | -0.072486 | 2.66e-11 | 0.001751 | 0.004035 |
| EA | LR8–CR8 | -0.001177 | 0.003232 | 2.68e-11 | 0.000374 | 0.004035 |
| EA | LR8–CB8 | 0.003017 | 0.007367 | 2.70e-11 | 0.000315 | 0.004035 |
| EA | LR8–CR16 | 0.004308 | 0.008629 | 2.69e-11 | 0.000285 | 0.004035 |
| EA | LR8–CB16 | 0.005786 | 0.010137 | 2.69e-11 | 0.000316 | 0.004035 |
| EA | LB8–LR16 | 0.074268 | 0.080160 | 2.66e-11 | 0.001857 | 0.004035 |
| EA | LB8–LB16 | -0.000912 | 0.003164 | 2.61e-11 | 0.000041 | 0.004035 |

| Set | First–second | Lower | Upper | Enclosure | Variance | Support |
|---|---|---|---|---|---|---|
| EA | LB8–CR8 | 0.074676 | 0.080389 | 2.64e-11 | 0.001678 | 0.004035 |
| EA | LB8–CB8 | 0.078881 | 0.084514 | 2.65e-11 | 0.001597 | 0.004035 |
| EA | LB8–CR16 | 0.080127 | 0.085820 | 2.65e-11 | 0.001658 | 0.004035 |
| EA | LB8–CB16 | 0.081642 | 0.087292 | 2.65e-11 | 0.001616 | 0.004035 |
| EA | LR16–LB16 | -0.079020 | -0.073156 | 2.66e-11 | 0.001830 | 0.004035 |
| EA | LR16–CR8 | -0.001913 | 0.002550 | 2.68e-11 | 0.000428 | 0.004035 |
| EA | LR16–CB8 | 0.002273 | 0.006694 | 2.70e-11 | 0.000386 | 0.004035 |
| EA | LR16–CR16 | 0.003569 | 0.007950 | 2.69e-11 | 0.000346 | 0.004035 |
| EA | LR16–CB16 | 0.005046 | 0.009460 | 2.69e-11 | 0.000379 | 0.004035 |
| EA | LB16–CR8 | 0.073564 | 0.079249 | 2.64e-11 | 0.001649 | 0.004035 |
| EA | LB16–CB8 | 0.077769 | 0.083373 | 2.65e-11 | 0.001568 | 0.004035 |
| EA | LB16–CR16 | 0.079016 | 0.084680 | 2.65e-11 | 0.001629 | 0.004035 |
| EA | LB16–CB16 | 0.080530 | 0.086152 | 2.65e-11 | 0.001587 | 0.004035 |
| EA | CR8–CB8 | 0.002039 | 0.006290 | 2.68e-11 | 0.000216 | 0.004035 |
| EA | CR8–CR16 | 0.003322 | 0.007560 | 2.67e-11 | 0.000203 | 0.004035 |
| EA | CR8–CB16 | 0.004812 | 0.009056 | 2.67e-11 | 0.000209 | 0.004035 |
| EA | CB8–CR16 | -0.000796 | 0.003349 | 2.69e-11 | 0.000110 | 0.004035 |
| EA | CB8–CB16 | 0.000716 | 0.004823 | 2.69e-11 | 0.000072 | 0.004035 |
| EA | CR16–CB16 | -0.000573 | 0.003559 | 2.68e-11 | 0.000097 | 0.004035 |

Set: C/E denotes calibration/evaluation; C/A denotes cash/analytically conditioned estimation. Controller codes C/L denote calibrated/legacy qualification, R/B denote ReLU/Bernstein, and 8/16 is the acquisition size. Directed endpoints and allowances round outwards; enclosure widths are displayed in scientific notation. Exact rational records determine all conclusions. These are reconstructions of the fixed R85 streams, not additional independent observations.

## BK.2. *Selected-service price and fee breakpoints*

Every comparison is selected minus comparator. If the upper cost endpoint is $u$, then the guaranteed extra-fee ceiling at zero resource price is $-\lambda u$. With resource prices it becomes $-\lambda u - p \cdot (w_{\hat{j}} - w_k)$. Negative ceilings are retained: they do not support a positive fee based solely on policy quality. A declared normalized loss margin does not calibrate $\lambda$ or a monetary fee.

TABLE BK.II

EVERY SELECTED-SERVICE QUALITY AND EXTRA-FEE BREAKPOINT

| Comparator | Cost lower | Cost upper | Fee ceiling |
|---|---|---|---|
| CB8 | -0.004823 | -0.000716 | 0.000716 |
| CR16 | -0.003559 | 0.000573 | -0.000573 |
| CR8 | -0.009056 | -0.004812 | 0.004812 |
| LB16 | -0.086152 | -0.080530 | 0.080530 |
| LB8 | -0.087292 | -0.081642 | 0.081642 |
| LR16 | -0.009460 | -0.005046 | 0.005046 |
| LR8 | -0.010137 | -0.005786 | 0.005786 |
| Zero | -0.177441 | -0.170788 | 0.170788 |

Selected service: calibrated Bernstein with acquisition size 16. At zero computation price and unit value of normalized loss, the sufficient extra-fee ceiling is minus the exact upper cost endpoint. Its display rounds down. At general prices, subtract the priced resource difference and multiply the quality term by the declared loss value. Negative ceilings remain in the table; the normalized units are not a monetary calibration.

## BK.3. *Diagnostics and independently checked kernels*

The new exact fixtures compare path enumeration with forward occupation recursion, test the candidate-occupation counterexample, retain ties, reject impossible shared-parameter assignments, and exhibit a fractional outer bound that is valid but unattained. Separate tests check the ReLU/min-plus identity, label-noise enclosure, persistent-exploration recursion, sufficient screen closure, correct lower-versus-upper work comparison, dated allocation, and price-region boundaries.

The frozen evaluation records zero new fits. Predictor warning rates, fallback frequencies and zero-progress query probes are not identified by that fact. Earlier fitting warnings and their affected services remain in the inherited experiment records. The R70 audit covers 188 service processes and records its prediction-replay differences; it is not a 336-service manifest. The source-specific diagnostic index records these distinctions instead of assigning unsupported aggregate counts to the current evaluation.

The large inherited vector descendant tree, a derivative branch-and-bound comparator, energy calibration, and a population experiment for a newly trained cross-regime proposal require their own numerical evidence. Exact finite-state fixtures, whole-bin evaluation of the nine attained controllers, or a successful PDF build do not silently certify those different objects. This evidence distinction does not remove the original algorithms, economic applications or full-horizon theorems from the manuscript.

### BK.4. *Publication identity*

Current document hashes, source hashes, test counts, exact replay status and clean reconstruction status are recorded under the current revision's audit directory. A publication gate rejects absent current PDFs, bibliographies or source-matched checks. Visual inspection is a separately signed-off observation against actual PDF hashes; neither a compiler exit code nor an inherited release record is such an observation. The full preservation edition retains every preceding active substantive section. Its actual page count is reported rather than concealed by reducing type size or deleting mathematical content.

## APPENDIX BL: SHARED-PROPOSAL TIMING AND PREDICTION DIAGNOSTICS

The prospective protocol and work comparison are described in Section AD. The following tables retain every recorded architecture, repetition and service. The experiment has 256 training states, 128 calibration states and 384 evaluation states. Five services, three repetitions and a second architecture do not multiply these numbers. The two fitting attempts belong to that original experiment; no new fitting or independent economic observation is introduced by this reconstruction.

TABLE BL.I: Complete service-process timing on two architectures

| Machine | $d$ | Service | Minimum | Median | Maximum | Peak RSS |
|---|---|---|---|---|---|---|
| ARM | 2 | Adaptive | 2.2281 | 2.2445 | 2.2543 | 37460 |
| ARM | 2 | Bisection | 2.2802 | 2.2922 | 2.2979 | 37460 |

| Machine | $d$ | Service | Minimum | Median | Maximum | Peak RSS |
|---|---|---|---|---|---|---|
| ARM | 2 | Analytic | 2.3019 | 2.3020 | 2.3199 | 37460 |
| ARM | 2 | ReLU | 2.2852 | 2.2859 | 2.2931 | 37460 |
| ARM | 2 | Quadratic | 2.3140 | 2.3301 | 2.3325 | 37460 |
| ARM | 4 | Adaptive | 3.1327 | 3.1382 | 3.1494 | 37460 |
| ARM | 4 | Bisection | 3.1498 | 3.1511 | 3.1646 | 37460 |
| ARM | 4 | Analytic | 3.2099 | 3.2199 | 3.2265 | 37460 |
| ARM | 4 | ReLU | 3.2162 | 3.2180 | 3.2431 | 37460 |
| ARM | 4 | Quadratic | 3.1811 | 3.1826 | 3.2101 | 37460 |
| ARM | 8 | Adaptive | 4.7303 | 4.7375 | 4.7521 | 37460 |
| ARM | 8 | Bisection | 4.7055 | 4.7124 | 4.7267 | 37460 |
| ARM | 8 | Analytic | 4.7668 | 4.7677 | 4.7800 | 37460 |
| ARM | 8 | ReLU | 4.8055 | 4.8185 | 4.8345 | 37460 |
| ARM | 8 | Quadratic | 4.7369 | 4.7394 | 4.7595 | 37460 |
| x86 | 2 | Adaptive | 1.4946 | 1.5136 | 1.5391 | 38872 |
| x86 | 2 | Bisection | 1.5285 | 1.5755 | 1.6262 | 38872 |
| x86 | 2 | Analytic | 1.5503 | 1.5977 | 1.6500 | 38872 |
| x86 | 2 | ReLU | 1.5320 | 1.6187 | 1.6403 | 38872 |
| x86 | 2 | Quadratic | 1.5615 | 1.5779 | 1.6423 | 38940 |
| x86 | 4 | Adaptive | 2.0980 | 2.1648 | 2.1755 | 38872 |
| x86 | 4 | Bisection | 2.1669 | 2.1774 | 2.1983 | 38872 |
| x86 | 4 | Analytic | 2.1923 | 2.2253 | 2.2426 | 38872 |
| x86 | 4 | ReLU | 2.2509 | 2.2590 | 2.3218 | 38872 |
| x86 | 4 | Quadratic | 2.1730 | 2.2414 | 2.2586 | 38940 |
| x86 | 8 | Adaptive | 3.2329 | 3.3182 | 3.3426 | 38872 |
| x86 | 8 | Bisection | 3.1925 | 3.2403 | 3.2974 | 38896 |
| x86 | 8 | Analytic | 3.2210 | 3.3071 | 3.3646 | 38872 |
| x86 | 8 | ReLU | 3.2258 | 3.3676 | 3.4170 | 38884 |
| x86 | 8 | Quadratic | 3.2354 | 3.2788 | 3.3248 | 38936 |

Times are seconds from process creation through shutdown, including loading and serialization, for all 128 states. RSS is the largest recorded process capacity in KiB, not an additive cost. All three repetitions are retained. Architecture and image differences are not a randomized hardware treatment.

TABLE BL.II: Every complete-process timing repetition

| Machine | $d$ | Service | Run | Seconds | RSS (KiB) |
|---|---|---|---|---|---|
| ARM | 2 | Adaptive | 1 | 2.254329 | 37460 |
| ARM | 2 | Adaptive | 2 | 2.228087 | 37460 |
| ARM | 2 | Adaptive | 3 | 2.244527 | 37460 |
| ARM | 2 | Bisection | 1 | 2.280225 | 37460 |
| ARM | 2 | Bisection | 2 | 2.297854 | 37460 |
| ARM | 2 | Bisection | 3 | 2.292237 | 37460 |
| ARM | 2 | Analytic | 1 | 2.319868 | 37460 |
| ARM | 2 | Analytic | 2 | 2.302035 | 37460 |
| ARM | 2 | Analytic | 3 | 2.301861 | 37460 |
| ARM | 2 | ReLU | 1 | 2.285942 | 37460 |
| ARM | 2 | ReLU | 2 | 2.285214 | 37460 |
| ARM | 2 | ReLU | 3 | 2.293085 | 37460 |
| ARM | 2 | Quadratic | 1 | 2.330145 | 37460 |
| ARM | 2 | Quadratic | 2 | 2.313985 | 37460 |
| ARM | 2 | Quadratic | 3 | 2.332537 | 37460 |
| ARM | 4 | Adaptive | 1 | 3.149352 | 37460 |
| ARM | 4 | Adaptive | 2 | 3.132699 | 37460 |
| ARM | 4 | Adaptive | 3 | 3.138180 | 37460 |
| ARM | 4 | Bisection | 1 | 3.149823 | 37460 |
| ARM | 4 | Bisection | 2 | 3.164596 | 37460 |
| ARM | 4 | Bisection | 3 | 3.151078 | 37460 |
| ARM | 4 | Analytic | 1 | 3.219859 | 37460 |
| ARM | 4 | Analytic | 2 | 3.226512 | 37460 |
| ARM | 4 | Analytic | 3 | 3.209879 | 37460 |
| ARM | 4 | ReLU | 1 | 3.216248 | 37460 |
| ARM | 4 | ReLU | 2 | 3.243123 | 37460 |
| ARM | 4 | ReLU | 3 | 3.217992 | 37460 |
| ARM | 4 | Quadratic | 1 | 3.181105 | 37460 |
| ARM | 4 | Quadratic | 2 | 3.182628 | 37460 |
| ARM | 4 | Quadratic | 3 | 3.210054 | 37460 |
| ARM | 8 | Adaptive | 1 | 4.737509 | 37460 |

| Machine | $d$ | Service | Run | Seconds | RSS (KiB) |
|---|---|---|---|---|---|
| ARM | 8 | Adaptive | 2 | 4.730321 | 37460 |
| ARM | 8 | Adaptive | 3 | 4.752148 | 37460 |
| ARM | 8 | Bisection | 1 | 4.726693 | 37460 |
| ARM | 8 | Bisection | 2 | 4.705514 | 37460 |
| ARM | 8 | Bisection | 3 | 4.712397 | 37460 |
| ARM | 8 | Analytic | 1 | 4.780037 | 37460 |
| ARM | 8 | Analytic | 2 | 4.766775 | 37460 |
| ARM | 8 | Analytic | 3 | 4.767667 | 37460 |
| ARM | 8 | ReLU | 1 | 4.818480 | 37460 |
| ARM | 8 | ReLU | 2 | 4.805483 | 37460 |
| ARM | 8 | ReLU | 3 | 4.834531 | 37460 |
| ARM | 8 | Quadratic | 1 | 4.759465 | 37460 |
| ARM | 8 | Quadratic | 2 | 4.736929 | 37460 |
| ARM | 8 | Quadratic | 3 | 4.739444 | 37460 |
| x86 | 2 | Adaptive | 1 | 1.539052 | 38872 |
| x86 | 2 | Adaptive | 2 | 1.513580 | 38872 |
| x86 | 2 | Adaptive | 3 | 1.494582 | 38872 |
| x86 | 2 | Bisection | 1 | 1.528538 | 38872 |
| x86 | 2 | Bisection | 2 | 1.626167 | 38872 |
| x86 | 2 | Bisection | 3 | 1.575515 | 38872 |
| x86 | 2 | Analytic | 1 | 1.550318 | 38872 |
| x86 | 2 | Analytic | 2 | 1.597653 | 38872 |
| x86 | 2 | Analytic | 3 | 1.649967 | 38872 |
| x86 | 2 | ReLU | 1 | 1.532015 | 38872 |
| x86 | 2 | ReLU | 2 | 1.618734 | 38872 |
| x86 | 2 | ReLU | 3 | 1.640346 | 38872 |
| x86 | 2 | Quadratic | 1 | 1.561463 | 38872 |
| x86 | 2 | Quadratic | 2 | 1.577949 | 38872 |
| x86 | 2 | Quadratic | 3 | 1.642332 | 38940 |
| x86 | 4 | Adaptive | 1 | 2.097984 | 38872 |
| x86 | 4 | Adaptive | 2 | 2.175456 | 38872 |
| x86 | 4 | Adaptive | 3 | 2.164793 | 38872 |
| x86 | 4 | Bisection | 1 | 2.177404 | 38872 |

| Machine | $d$ | Service | Run | Seconds | RSS (KiB) |
|---|---|---|---|---|---|
| x86 | 4 | Bisection | 2 | 2.166911 | 38872 |
| x86 | 4 | Bisection | 3 | 2.198307 | 38872 |
| x86 | 4 | Analytic | 1 | 2.192295 | 38872 |
| x86 | 4 | Analytic | 2 | 2.225293 | 38872 |
| x86 | 4 | Analytic | 3 | 2.242576 | 38872 |
| x86 | 4 | ReLU | 1 | 2.250892 | 38872 |
| x86 | 4 | ReLU | 2 | 2.321841 | 38872 |
| x86 | 4 | ReLU | 3 | 2.258990 | 38872 |
| x86 | 4 | Quadratic | 1 | 2.258650 | 38872 |
| x86 | 4 | Quadratic | 2 | 2.241449 | 38940 |
| x86 | 4 | Quadratic | 3 | 2.173017 | 38872 |
| x86 | 8 | Adaptive | 1 | 3.318157 | 38872 |
| x86 | 8 | Adaptive | 2 | 3.342642 | 38872 |
| x86 | 8 | Adaptive | 3 | 3.232907 | 38872 |
| x86 | 8 | Bisection | 1 | 3.297421 | 38872 |
| x86 | 8 | Bisection | 2 | 3.240277 | 38872 |
| x86 | 8 | Bisection | 3 | 3.192528 | 38896 |
| x86 | 8 | Analytic | 1 | 3.307147 | 38872 |
| x86 | 8 | Analytic | 2 | 3.364552 | 38872 |
| x86 | 8 | Analytic | 3 | 3.221046 | 38872 |
| x86 | 8 | ReLU | 1 | 3.416991 | 38872 |
| x86 | 8 | ReLU | 2 | 3.367554 | 38872 |
| x86 | 8 | ReLU | 3 | 3.225830 | 38884 |
| x86 | 8 | Quadratic | 1 | 3.324847 | 38872 |
| x86 | 8 | Quadratic | 2 | 3.278846 | 38936 |
| x86 | 8 | Quadratic | 3 | 3.235404 | 38872 |

No repetition is selected or deleted. These 90 process runs evaluate the same 384 held-out states with five services; they are not 90 independent population experiments.

TABLE BL.III: Distinct prediction, protection and service events

| $d$ | Service | No model | Screens | Close | Refine | Invalid | Repair | Zero |
|---|---|---|---|---|---|---|---|---|
| 2 | Adaptive | 59 | 69 | 0 | 69 | 0 | 0 | 0 |
| 2 | Bisection | 59 | 78 | 2 | 76 | 0 | 0 | 0 |
| 2 | Analytic | 59 | 77 | 0 | 77 | 0 | 0 | 0 |
| 2 | ReLU | 59 | 73 | 0 | 73 | 0 | 1 | 0 |
| 2 | Quadratic | 59 | 78 | 0 | 78 | 0 | 0 | 0 |
| 4 | Adaptive | 53 | 75 | 0 | 75 | 0 | 0 | 0 |
| 4 | Bisection | 53 | 78 | 1 | 77 | 0 | 0 | 0 |
| 4 | Analytic | 53 | 82 | 0 | 82 | 0 | 0 | 0 |
| 4 | ReLU | 53 | 83 | 0 | 83 | 0 | 18 | 0 |
| 4 | Quadratic | 53 | 78 | 0 | 78 | 0 | 15 | 0 |
| 8 | Adaptive | 61 | 67 | 0 | 67 | 0 | 0 | 0 |
| 8 | Bisection | 61 | 67 | 0 | 67 | 0 | 0 | 0 |
| 8 | Analytic | 61 | 69 | 0 | 69 | 0 | 0 | 0 |
| 8 | ReLU | 61 | 73 | 0 | 73 | 0 | 7 | 0 |
| 8 | Quadratic | 61 | 67 | 0 | 67 | 0 | 4 | 0 |

No model: prediction-free service returns. Close: screen closures. Refine: protected refinements. Invalid and repair count prediction diagnostics, not failed certificates. Zero counts probes improving neither bound. Learned-model calls and dense arithmetic are separate from these events. Logical network multiplication/addition/division counts are in the exact source-bound summary; interval-operation counts and energy are not measured.

The disjoint action-value query account is in Table AD.I; it is not a second addition of terminal work to a total that already includes it. The archived dense-model arithmetic is an implementation-level algebraic count. Peak resident memory is capacity. The timing observations are complete processes, not an estimated mapping from every operation to energy. Missing interval-operation or energy measurements remain missing rather than being assigned zero. Any cross-architecture numerical disagreement is retained by the reconstruction audit before the tables are accepted.

## APPENDIX BM: EXACT TRANSPORT AND COMPLETE-POLICY DIAGNOSTICS

The singular-transition example of Section X separates a genuine original-law regret from an incomplete numerical residual account. Its regret and transport charge converge with the approximation distance, although the total-variation discrepancy does not. All calculations below are exact rational arithmetic.

TABLE BM.I

EXACT SINGULAR-TRANSITION DIAGNOSTIC

| $h$ | True regret | Nominal bound | Transport bound | TV distance |
|---|---|---|---|---|
| 1/4 | 1/8 | 0 | 1/2 | 1 |
| 1/16 | 1/32 | 0 | 1/8 | 1 |
| 1/64 | 1/128 | 0 | 1/32 | 1 |
| 1/256 | 1/512 | 0 | 1/128 | 1 |
| 1/1024 | 1/2048 | 0 | 1/512 | 1 |

All entries are exact rational numbers. The zero nominal bound omits the transition charge and is invalid. These five fixtures add no independent economic observations.

A separate three-date, two-state fixture enumerates all 64 deterministic complete binary policies. Independent backward dynamic programming evaluates each under the original transition kernel and computes its original optimum. The proposed certificate is formed from a different numerical kernel, arbitrary stored continuations, their exact Lipschitz moduli, the full-menu residual span and the implemented action gap. The test checks both signs of the resulting policy sandwich at each initial state. It also tests constant shifts, terminal error, discount indexing, exact versus unqualified arithmetic, probability normalization and invalid inputs. The exhaustive fixture is a check of the implementation and does not replace the general proof in Supplement CD.

The complete-pattern and coupled-resource fixtures are retained in their ordinary source and audit records. In particular, the true shared-sign class and its pattern-disaggregated bound are zero while the earlier pair-conflict upper bound is one half. The allocation example charges a product of adjacent-date inverse precisions; it does not hold the successor circuit cost constant. None of these algebraic fixtures

is reported as an additional investment sample, trained network or large-vector runtime experiment.

## APPENDIX BN: CURRENT-SOURCE REPRODUCTION AND RETAINED SCOPE

The current reproduction command requires the ordinary article, technical supplement, numerical appendix and response sources, rather than substituting an older compiled article. It checks the preserved active input list, reconstructs the required tables, executes the selected mathematical and numerical checks, and records their exact source and output identities. Each executed check has its own exit status and log. A clean typesetting reconstruction removes inherited auxiliary and bibliography outputs. The final release gate additionally requires a rendered-page inspection record tied to the actual four PDF hashes.

A compiled candidate, a passed arithmetic test and a visually inspected package are distinct delivery states. The current manifest determines which states have actually been attained. The full matched large-vector comparator study, measured energy and application-specific monetary calibration remain separate evidence obligations; they are not inferred from the new transport theorem or a successful build. The original diffusion, recursive-preference, temporal-self and game results remain in their existing active sections and retained companions under their original hypotheses.

## APPENDIX BO: EXACT DIAGNOSTICS FOR SIGNED RECURSIVE QUALIFICATION

The new diagnostics isolate the qualifications in Theorem A.3. They are deterministic finite-state examples, not new investment observations or fitting attempts. The evaluation has three decision dates, two states at each date and two feasible actions per state. Every one of the $2^6 = 64$ complete policies is evaluated at both initial states for each of 27 configurations: three numerical transition laws, risk levels zero, one half and three quarters, and discounts zero, one half and one. This gives 1,728 complete-policy evaluations and 3,456 initial-state inequalities. Numerical action values use exact minimization of the average-value-at-risk threshold expression. A

separately written capped-density dual computes the original policy and optimal values without importing that implementation.

Table BO.I retains every configuration. Its largest actual loss and largest certificate are separate maxima over the 64 policies and two initial states; they are not presented as an attained worst-case pair. Minimum slack is computed for the corresponding policy and initial state before minimization. The diagnostic checks that every actual loss is nonnegative and below its own certificate. General continuous-state, endogenous-context and entropic assertions rest on their displayed analytic proofs, not on this finite enumeration.

TABLE BO.I: Every exact signed-risk qualification configuration

| Case | $\alpha$ | $\beta$ | Policies | Max. loss | Max. bound | Min. slack |
|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 64 | $\frac{1}{8}$ | $\frac{7}{16}$ | $\frac{5}{16}$ |
| 1 | 0 | $\frac{1}{2}$ | 64 | $\frac{173}{1024}$ | $\frac{45}{64}$ | $\frac{511}{1024}$ |
| 1 | 0 | 1 | 64 | $\frac{205}{512}$ | $\frac{119}{64}$ | $\frac{311}{256}$ |
| 1 | $\frac{1}{2}$ | 0 | 64 | $\frac{1}{8}$ | $\frac{7}{16}$ | $\frac{5}{16}$ |
| 1 | $\frac{1}{2}$ | $\frac{1}{2}$ | 64 | $\frac{11}{64}$ | $\frac{101}{128}$ | $\frac{71}{128}$ |
| 1 | $\frac{1}{2}$ | 1 | 64 | $\frac{17}{32}$ | $\frac{61}{32}$ | $\frac{5}{4}$ |
| 1 | $\frac{3}{4}$ | 0 | 64 | $\frac{1}{8}$ | $\frac{7}{16}$ | $\frac{5}{16}$ |
| 1 | $\frac{3}{4}$ | $\frac{1}{2}$ | 64 | $\frac{11}{64}$ | $\frac{45}{64}$ | $\frac{31}{64}$ |
| 1 | $\frac{3}{4}$ | 1 | 64 | $\frac{1}{4}$ | $\frac{3}{2}$ | $\frac{9}{8}$ |
| 2 | 0 | 0 | 64 | $\frac{1}{8}$ | $\frac{7}{16}$ | $\frac{5}{16}$ |
| 2 | 0 | $\frac{1}{2}$ | 64 | $\frac{173}{1024}$ | $\frac{297}{256}$ | $\frac{759}{1024}$ |
| 2 | 0 | 1 | 64 | $\frac{205}{512}$ | $\frac{279}{64}$ | $\frac{1281}{512}$ |
| 2 | $\frac{1}{2}$ | 0 | 64 | $\frac{1}{8}$ | $\frac{7}{16}$ | $\frac{5}{16}$ |
| 2 | $\frac{1}{2}$ | $\frac{1}{2}$ | 64 | $\frac{11}{64}$ | $\frac{145}{128}$ | $\frac{759}{1024}$ |
| 2 | $\frac{1}{2}$ | 1 | 64 | $\frac{17}{32}$ | $\frac{67}{16}$ | $\frac{77}{32}$ |
| 2 | $\frac{3}{4}$ | 0 | 64 | $\frac{1}{8}$ | $\frac{7}{16}$ | $\frac{5}{16}$ |
| 2 | $\frac{3}{4}$ | $\frac{1}{2}$ | 64 | $\frac{11}{64}$ | $\frac{75}{64}$ | $\frac{49}{64}$ |
| 2 | $\frac{3}{4}$ | 1 | 64 | $\frac{1}{4}$ | $\frac{17}{4}$ | $\frac{41}{16}$ |

| Case | $\alpha$ | $\beta$ | Policies | Max. loss | Max. bound | Min. slack |
|---|---|---|---|---|---|---|
| 3 | 0 | 0 | 64 | $\frac{1}{8}$ | $\frac{7}{16}$ | $\frac{5}{16}$ |
| 3 | 0 | $\frac{1}{2}$ | 64 | $\frac{173}{1024}$ | $\frac{45}{64}$ | $\frac{511}{1024}$ |
| 3 | 0 | 1 | 64 | $\frac{205}{512}$ | $\frac{119}{64}$ | $\frac{311}{256}$ |
| 3 | $\frac{1}{2}$ | 0 | 64 | $\frac{1}{8}$ | $\frac{7}{16}$ | $\frac{5}{16}$ |
| 3 | $\frac{1}{2}$ | $\frac{1}{2}$ | 64 | $\frac{11}{64}$ | $\frac{57}{64}$ | $\frac{637}{1024}$ |
| 3 | $\frac{1}{2}$ | 1 | 64 | $\frac{17}{32}$ | $\frac{41}{16}$ | $\frac{105}{64}$ |
| 3 | $\frac{3}{4}$ | 0 | 64 | $\frac{1}{8}$ | $\frac{7}{16}$ | $\frac{5}{16}$ |
| 3 | $\frac{3}{4}$ | $\frac{1}{2}$ | 64 | $\frac{11}{64}$ | $\frac{7}{8}$ | $\frac{19}{32}$ |
| 3 | $\frac{3}{4}$ | 1 | 64 | $\frac{1}{4}$ | $\frac{9}{4}$ | $\frac{27}{16}$ |

All quantities are exact rational values. Case 2 uses deterministic numerical successors; Cases 1 and 3 move successor mass in opposite directions. These are diagnostic configurations, not independent economic samples.

The adversarial tests also retain atoms at a quantile, zero-probability outcomes, exact ties, malformed probability inputs, infeasible actions, negative or inconsistent allowances, terminal error and correct date weights. A one-date singular-law example attains the signed discrepancy charge while its symmetric counterpart is twice as large. A two-date example rejects cancelling a discrepancy that is action independent only within each state. A four-outcome example separates static from nested tail risk. The tail-transport factor is attained as the risk level approaches one. These tests guard the original economic objective rather than selecting a more convenient replacement.

The new exact code does not refit the shared proposal, append an independent cost sample, or replace an earlier warning, timing or unresolved ranking. The source-bound publication manifest reports actual test exits, table hashes and current-document hashes separately from scientific scope. The largest vector investment reimplementation, a new matched vector tournament, measured energy and application-specific monetary calibration are not supplied by these diagnostics. The corresponding original sections and evidence remain active.

## APPENDIX BP: INDEPENDENT DIAGNOSTICS FOR CENTERED RISK QUALIFICATION

This section reports deterministic diagnostics of Section 4. They use finite rational models with three states, two feasible actions and original nested average value at risk. They are not additional investment observations, new fitting attempts, or a rerun of the inherited largest-vector problem. The production calculation uses the vertices of a capped probability simplex. The independent evaluator minimizes the scalar threshold representation of average value at risk and performs a separately coded original-value recursion. It does not import the centered-risk implementation.

The declared horizons are $2,3,4,8,16,32,64$, risk levels are $0,1/2,3/4$, and the four shared continuation parameters are $-1,0,1,2$. At date $t$, state $s \in \{0,1,2\}$ and action $a \in \{0,1\}$, transition weights are

$$(1 + (t+s+a) \bmod 3, \quad 1 + (2s+a) \bmod 4, \quad 1 + (t+2a) \bmod 3),$$

normalized by their sum. Costs are $((s+2a+t) \bmod 5 - 2)/7 + a/5$, the discount is $3/4$, and the terminal vector is $(0,1/3,-1/4)$. These fully specified diagnostic primitives avoid a hidden change in objective between implementations. They are separate from the paper's economic calibration.

A single parameter determines the stored circuit throughout a configuration:

$$f_{t,\theta}(s) = (T-t)\theta/20 + \frac{\theta+1}{7}(s/2 - 1/2)_+ + \frac{t \bmod 3}{11}(1/2 - s/2)_+.$$

The implemented policy is greedy for this circuit under the original one-step evaluation, with smaller-action tie breaking. No training claim is attached to these prescribed circuits. Residual intervals use radii $0,1/64,1/8$ times $(1+s)/3$; terminal intervals use the same statewise radius. The exact configuration records retain the complete input, every centered residual and risk penalty, lower and upper potentials, a feasible maximizing risk occupation, and all strict kernel-exclusion certificates. Unpruned potential bounds are computed before any exclusion, preventing circular error boxes.

TABLE BP.I

CENTERED-RISK DIAGNOSTICS AT RESIDUAL RADIUS 1/8

| Horizon | Cases | Scalar bound | State bound | Cuts | Bits |
|---|---|---|---|---|---|
| 2 | 12 | 1.225359 | 0.481250 | 22 | 22 |
| 3 | 12 | 1.240733 | 0.590358 | 20 | 26 |
| 4 | 12 | 1.468255 | 0.643092 | 11 | 32 |
| 8 | 12 | 1.617102 | 0.745949 | 22 | 49 |
| 16 | 12 | 1.727700 | 0.791032 | 11 | 98 |
| 32 | 12 | 1.737229 | 0.796780 | 22 | 185 |
| 64 | 12 | 1.737321 | 0.796836 | 11 | 359 |

Each horizon contains all three risk levels and four shared parameters. Bounds are maxima over that block, rounded upwards; cuts count strictly excluded risk pieces for the displayed radius. Bits is the maximum numerator or denominator bit length in the retained exact records. The complete catalogue includes all three radii, including zero. These are deterministic diagnostics, not economic observations or measured speedups.

The independent audit checks original complete-policy values, original optimal values, both sides of the state-dependent certificate, every risk-flow equality, primal–dual equality for the exact fixed-policy centered recursion, and each excluded affine piece against its retained witness. It also enumerates all complete Markov policies at horizons two and three, retaining every initial-state comparison. Long-horizon records compare two distinct backward implementations rather than expanding a large scenario tree. Their operation counts and complete process duration are reported as different objects; rational bit length is retained. They do not measure energy or identify a neural speed advantage.

The adversarial tests include a risk kernel that is inactive at the stored continuation but becomes maximizing after perturbation, equality at the exclusion threshold, incorrect joint minimization over action and risk, zero discounts, signed terminal discrepancies, arbitrary datewise constant shifts, state-dependent bias, corrupted occupation masses, and reversed residual intervals. The absorbing-state example in the proof is executed for each declared horizon: its state-dependent certificate is

zero while its scalar bound is the horizon. The comparison is a strict mathematical diagnostic, not an empirical policy-cost gain.

All inherited independent cost records, warnings, held-out proposal comparisons and cross-machine observations retain their own identities. The current source and execution manifests specify exactly which checks have run. The new diagnostic catalogue never increases the denominators of an inherited statistical table.

## APPENDIX BQ: CONTINUOUS-RISK EVIDENCE, UNCERTAINTY, AND PRESERVATION

This section records the continuous investment execution separately from the retained finite-risk catalogue and the earlier expected-cost experiments. All new scientific calculations use the unchanged R94 protocol and numerical source. The R95 additions supply the primitive proofs, the integrated manuscript, the complete diagnostic summary and source-bound validation of the resulting evidence.

### BQ.1. *Complete comparisons and screened work*

The full retained files contain 2,048 rows for each of four arms at each risk level, for 16,384 arm-state records and 4,096 independent initial-state observations. Every first action, gap, original-optimum bound, cost enclosure and failed-target count is retained. There are six fitting attempts, all in preparation. Calibration uses 128 separate states per risk level. Supplementary terminal calculations at dimensions four and eight use 16 prespecified states each and have no primary inferential weight. They are not a replication of the earlier vector-control experiment.

The empirical-Bernstein intervals have one family confidence level of 95% for twelve prespecified contrasts. Their numerical endpoints and sampling radii are computed from every row, using a proved support rather than observed extrema. Conservative Hoeffding intervals are retained as well; a favorable rule is not selected after evaluation. The fixed sample size, margin and confidence rule are those in the pre-evaluation protocol.

*Numerical components.* The original innovation is uniform on $[-1/32, 1/32]$. The one-date original terminal risk is integrated analytically and enclosed outwards.

TABLE BQ.I

ALL PRESPECIFIED SIMULTANEOUS PAIRED POLICY-COST COMPARISONS

| $\alpha$ | Contrast | 95% interval | Sampling | Equiv. |
|---|---|---|---|---|
| 0.5 | Online risk-DP − Quadratic | [-0.029854, 0.029614] | 0.022867 | Yes |
| 0.5 | Online risk-DP − ReLU | [-0.030569, 0.029054] | 0.022944 | Yes |
| 0.5 | Online risk-DP − Spline | [-0.029898, 0.029582] | 0.022872 | Yes |
| 0.5 | Quadratic − ReLU | [-0.030446, 0.029172] | 0.022941 | Yes |
| 0.5 | Quadratic − Spline | [-0.029773, 0.029697] | 0.022867 | Yes |
| 0.5 | ReLU − Spline | [-0.029211, 0.030410] | 0.022943 | Yes |
| 0.75 | Online risk-DP − Quadratic | [-0.029882, 0.029592] | 0.022869 | Yes |
| 0.75 | Online risk-DP − ReLU | [-0.030770, 0.028896] | 0.022965 | Yes |
| 0.75 | Online risk-DP − Spline | [-0.029880, 0.029595] | 0.022870 | Yes |
| 0.75 | Quadratic − ReLU | [-0.030622, 0.029038] | 0.022962 | Yes |
| 0.75 | Quadratic − Spline | [-0.029729, 0.029734] | 0.022864 | Yes |
| 0.75 | ReLU − Spline | [-0.029037, 0.030625] | 0.022963 | Yes |

The unit is an independent initial state under the declared dyadic uniform distribution. There are twelve fixed comparisons in one family. Equivalence uses the prespecified margin 1/16 in normalized cost units. Every interval includes zero: none identifies strict superiority. The calibration selector is online risk-DP at both risk levels.

First-date integration uses full-bin bounds. A half-bin contributes the explicit continuation allowance $15/(32n)$; a complete terminal controller contributes at most $2^{-20}$ before first-date discounting. State observation and terminal optimization are bounded together by $2^{-21} + 30\,2^{-40} < 2^{-20}$. Arithmetic rounding is propagated through the interval operations rather than replaced by an unverified final decimal tolerance. These allowances overlap with conservative bin bounds in the code and must not be added again to the final cost interval. State covers qualify residuals; they do not change the transition law. There is no numerical entropic evaluation in this experiment.

*What the independent implementation checks.* A separately written Fraction implementation reconstructs the original quadratic shock polynomial from primitive evaluations, computes AVaR through a primal threshold integral, and solves its own one-date action problems. It imports neither the production interval kernel nor

TABLE BQ.II

STRICT CONTINUUM-BIN EXCLUSIONS WITH THEIR CONSTRUCTION COSTS

| $\alpha$ | Continuation | Excluded/candidate | No-cut rows | Full (ms) | Pruned (ms) |
|---|---|---|---|---|---|
| 0.5 | Online risk-DP | 744/4,096 | 14 | 44.800 | 44.940 |
| 0.5 | Quadratic | 746/4,096 | 14 | 44.776 | 44.899 |
| 0.5 | ReLU | 738/4,096 | 14 | 43.996 | 44.116 |
| 0.5 | Spline | 742/4,096 | 14 | 46.796 | 46.912 |
| 0.75 | Online risk-DP | 2,058/4,096 | 5 | 44.074 | 44.211 |
| 0.75 | Quadratic | 2,056/4,096 | 5 | 43.904 | 44.015 |
| 0.75 | ReLU | 2,053/4,096 | 5 | 42.972 | 43.083 |
| 0.75 | Spline | 2,062/4,096 | 5 | 43.789 | 43.903 |

The first 32 prespecified rows of every arm are retained, with 128 bins each. Full includes unpruned bound construction and sorting. Pruned additionally pays for the strict proof and retained-bin sorting. Both return identical upper-tail values. Repeated sorts are timing diagnostics, not new economic observations; these measurements show no end-to-end gain from screening.

the production rational terminal fallback. The audit checks the six serialized fits, eighteen exact feature evaluations, all 16,384 raw row identities and hashes, and twelve complete paired-inference replays. For sixteen prespecified policy fixtures, a finer 512-bin independent continuous-policy enclosure must be contained in the reported production enclosure, not merely intersect it. All these checks pass in the local execution. The inference replay shares the production interval formula and is identified as replay, not a second independently derived statistical implementation. This is kernel-level independent implementation, not external authorship or a second full learning and largest-vector stack.

## BQ.2. *The full inherited tightening distribution*

The new summary uses all 252 finite-risk configurations, including equality and no-cut configurations. Absolute improvements and relative-improvement quartiles are stored alongside every exact rational scalar and state-dependent bound. These calculations reorganize previously generated mathematical evidence and create no new fits or independent observations. The absorbing example continues to mean

TABLE BQ.III

COMPLETE DISTRIBUTION AUDIT OF THE INHERITED FINITE-RISK CATALOGUE

| Diagnostic | Count |
|---|---|
| All configurations | 252 |
| Strictly tighter state-dependent bound | 252 |
| Equal bounds | 0 |
| Looser state-dependent bound | 0 |
| No-cut configurations | 118 |
| Candidate / retained pieces | 91,260 / 73,549 |

Recomputed from every one of the 252 retained configurations, not from an improved subset. These are deterministic finite-risk diagnostics, not new fits, continuous-law experiments or independent economic observations. Detailed absolute and relative quantiles are retained in the source-bound audit.

zero policy loss with a nonzero continuation bias; it is not an exact continuation fit. The existing rational bit-growth records remain separate from a measured exact-versus-interval arithmetic crossover, which has not been run here.

## BQ.3. *Source, execution and release boundaries*

The controlling advisory report is the R92 review deposited on 13 October 2026. Its reviewed manuscript is the complete R92 article. The R94 development branch is a descendant of that manuscript; its paper entry point had not yet incorporated its new continuous-risk implementation. R95 retains the original title and author, adds the continuous primitive proof and executed comparison, and relocates fifteen complete historical main-text sections to this appendix. Each moved file is byte-identical to the R92 source. The only inherited scientific-section prose amendment clarifies that the earlier experiments use expectation, while the new explicitly specified experiment uses nested AVaR. No original mathematical claim or adverse observation is removed.

The cloud R94 attempt stopped before scientific execution when creation of its source branch was rejected. The available source-and-evidence artifact therefore contains sources but no completed primary run. R95 executed those sources locally.

A serialized actor and calibration hash was written before evaluation; it was not remotely timestamped or published before that local evaluation. The source checks in the evaluator reject changed models or changed frozen numerical files. Two tool-interrupted local starts are retained as interrupted attempts, and the completed run is identified separately. Repeated evaluations of the same pseudorandom stream do not increase the number of independent observations.

New-science tests, inherited-test replay, independent continuous-kernel validation, clean typesetting and actual-PDF inspection are separate release gates. The downloaded R92 publication artifact intentionally omits several older numerical datasets and external historical code directories. Missing-fixture errors in replay of those inherited suites are recorded rather than skipped or reclassified as passed. The full repository materialization procedure preserves those historical dependencies; the local export is not evidence that their full replay has passed. The publication continuation restores the original repository dependencies and records its new source-bound replay and delivery separately under `audit/remote95`. Historical local-only receipts remain unchanged. An ARM64 execution, energy measurement or full largest-vector replication is reported only with its own record.

## APPENDIX BR: R96 COMPLETE WORK AND EVIDENCE IDENTITIES

The primary numerical record identifies its exact source commit and the hash of the remotely committed model and calibration freeze. The table manifest binds every generated table to that record. Subsequent manuscript edits do not change the source attribution of previously generated scientific tables. Repeated architecture runs use the same actors and states and do not create independent observations.

The original R95 ledgers and failed preservation gate remain on their original branch and in the inherited record. The current map classifies the earlier operator-scope clarification as an amendment, retains the old source and hash, and checks its current identity. Unchanged sections still require byte identity and exactly one current document assignment. Reclassifying an amendment therefore does not remove its preservation test.

TABLE BR.I

FRESH-PROCESS WORKLOAD TIMES, INCLUDING STARTUP AND SHUTDOWN

| Risk | Service | Median seconds | Observed range |
|---|---|---|---|
| 0.50 | Online risk-DP | 1.53020 | [1.51997, 1.55074] |
| 0.50 | Quadratic | 0.98127 | [0.97867, 0.98524] |
| 0.50 | ReLU | 1.29428 | [1.28720, 1.30073] |
| 0.50 | Adaptive spline | 1.26250 | [1.25920, 1.26526] |
| 0.75 | Online risk-DP | 1.52729 | [1.52572, 1.61732] |
| 0.75 | Quadratic | 0.98531 | [0.97848, 0.99030] |
| 0.75 | ReLU | 1.27456 | [1.27109, 1.30314] |
| 0.75 | Adaptive spline | 1.26657 | [1.25655, 1.26992] |

Five fresh processes per arm–risk pair on the same fixed 32-state workload. CPU frequency is recorded, not controlled. Ranges are descriptive, not confidence intervals for other workloads or hardware. Primary evaluation has a different, larger sample and is not timed by scaling these rows.

TABLE BR.II

MATCHED CONSTRUCTION AND ITS PAID COMPONENTS

| Risk | Representation | Design rank | Active nonlinear | Search seconds | Cover seconds |
|---|---|---|---|---|---|
| 0.50 | Quadratic | 6 | 3 | 0.06044 | 0.05545 |
| 0.50 | ReLU | 6 | 3 | 0.04043 | 0.05394 |
| 0.50 | Adaptive spline | 6 | 3 | 0.03258 | 0.05377 |
| 0.75 | Quadratic | 6 | 3 | 0.05156 | 0.05419 |
| 0.75 | ReLU | 6 | 3 | 0.04127 | 0.05383 |
| 0.75 | Adaptive spline | 6 | 3 | 0.03261 | 0.05395 |

Every executable model uses 128 bytes. The full record additionally reports common-label construction, each calibration, serialization/reload, condition numbers, every candidate solver call, and process memory. The table columns are components, not an assertion that they sum to the entire preparation campaign. The independent endpoint audit is a separate paid campaign.

The source revision distinguishes the full inherited test replay, new scientific tests, exact paired inference, random independent endpoints, timing, clean document builds and visual inspection. The final source-bound release record reports which of these were completed, and any incomplete or failed record remains visible. Neither

submission of a workflow nor successful compilation establishes an unexecuted numerical comparison.

## APPENDIX BS: COMPLETE R97 BUDGET CATALOGUE AND VERIFICATION MAP

This section retains all matched rank–label comparisons. Each LS–CA pair uses the same feature geometry and available labels. The two risk levels, three ranks, two label budgets, and three stored families give 36 pairs. Rejection counts are out of 64 paired diagnostic states per cell. No population inference is obtained by treating rows aggregated across different budgets as repeated observations of one fixed policy.

TABLE BS.I: All matched-budget residual and admission comparisons

| $\alpha$ | Rank | Labels | Family | LS span | CA span | LS rej. | CA rej. |
|---|---|---|---|---|---|---|---|
| 0.5 | 4 | 256 | Quadratic | 1.3346 | 0.9689 | 24 | 24 |
| 0.5 | 4 | 256 | Relu | 1.4264 | 1.1004 | 26 | 24 |
| 0.5 | 4 | 256 | Spline | 1.2277 | 0.9726 | 30 | 24 |
| 0.5 | 4 | 1024 | Quadratic | 1.0825 | 0.8654 | 14 | 17 |
| 0.5 | 4 | 1024 | Relu | 1.3183 | 1.1004 | 17 | 17 |
| 0.5 | 4 | 1024 | Spline | 1.1883 | 1.0186 | 14 | 15 |
| 0.5 | 5 | 256 | Quadratic | 0.8991 | 0.6284 | 0 | 23 |
| 0.5 | 5 | 256 | Relu | 1.0052 | 0.7463 | 15 | 22 |
| 0.5 | 5 | 256 | Spline | 0.7968 | 0.6181 | 28 | 21 |
| 0.5 | 5 | 1024 | Quadratic | 0.8736 | 0.6284 | 1 | 23 |
| 0.5 | 5 | 1024 | Relu | 0.9686 | 0.7463 | 15 | 19 |
| 0.5 | 5 | 1024 | Spline | 0.7780 | 0.6145 | 20 | 15 |
| 0.5 | 6 | 256 | Quadratic | 0.6136 | 0.3294 | 0 | 2 |
| 0.5 | 6 | 256 | Relu | 0.5305 | 0.4231 | 36 | 37 |
| 0.5 | 6 | 256 | Spline | 0.6347 | 0.4589 | 24 | 22 |
| 0.5 | 6 | 1024 | Quadratic | 0.6467 | 0.3294 | 0 | 12 |

| $\alpha$ | Rank | Labels | Family | LS span | CA span | LS rej. | CA rej. |
|---|---|---|---|---|---|---|---|
| 0.5 | 6 | 1024 | Relu | 0.7046 | 0.5697 | 18 | 16 |
| 0.5 | 6 | 1024 | Spline | 0.6983 | 0.5150 | 19 | 19 |
| 0.75 | 4 | 256 | Quadratic | 1.0525 | 0.8649 | 32 | 30 |
| 0.75 | 4 | 256 | Relu | 0.9287 | 0.8893 | 43 | 36 |
| 0.75 | 4 | 256 | Spline | 1.0605 | 0.9174 | 32 | 30 |
| 0.75 | 4 | 1024 | Quadratic | 1.0755 | 0.8649 | 13 | 20 |
| 0.75 | 4 | 1024 | Relu | 1.3232 | 1.1010 | 16 | 19 |
| 0.75 | 4 | 1024 | Spline | 1.1276 | 0.9801 | 20 | 21 |
| 0.75 | 5 | 256 | Quadratic | 0.8610 | 0.6178 | 7 | 22 |
| 0.75 | 5 | 256 | Relu | 0.5696 | 0.5681 | 30 | 30 |
| 0.75 | 5 | 256 | Spline | 0.6532 | 0.5528 | 23 | 23 |
| 0.75 | 5 | 1024 | Quadratic | 0.8825 | 0.6178 | 0 | 24 |
| 0.75 | 5 | 1024 | Relu | 0.6339 | 0.5681 | 30 | 35 |
| 0.75 | 5 | 1024 | Spline | 0.7657 | 0.6074 | 14 | 19 |
| 0.75 | 6 | 256 | Quadratic | 0.6277 | 0.3312 | 0 | 5 |
| 0.75 | 6 | 256 | Relu | 0.4689 | 0.4620 | 20 | 13 |
| 0.75 | 6 | 256 | Spline | 0.8129 | 0.6080 | 21 | 28 |
| 0.75 | 6 | 1024 | Quadratic | 0.6479 | 0.3312 | 0 | 8 |
| 0.75 | 6 | 1024 | Relu | 0.7148 | 0.5714 | 20 | 19 |
| 0.75 | 6 | 1024 | Spline | 0.8030 | 0.5940 | 22 | 23 |

*State regions and component work.* The machine-readable catalogue partitions the initial square into sixteen fixed rectangular cells and records proposal counts and rejection reasons for every arm–budget cell. Empty regions are retained. Stage clocks and primitive-call counts are attached to the same source and model identity as their raw rows; they do not exclude rejected proposals. Whole-process timings, not sums of selected stage clocks, are used for the workload comparison. Reproduction software verifies the complete configuration and row counts before generating any table.

*Preservation.* The preceding investment, risk, screening, approximation, and vector-control studies retain their original data and assumptions. The original R96 manuscript sources remain under the pinned historical revision, and the current revision retains unmodified scientific section files. The current article uses a workload-corrected copy of the precision section; the uncorrected historical file is not overwritten. Primitive investment accuracy, horizon allocation, and intervening proof sections are reproduced below as complete sections, rather than removed to shorten the active article.

*Audit interpretation.* A binary64 independent proposal replay checks an implementation at the declared action-quantization tolerance; it is not an exact-real decision proof. Exact rational arithmetic checks the statistical contrasts and selected difficult original-law endpoint cases. The original random 64-case audit and the new deterministic extremal audit have different selection rules. Neither is mislabeled as a rational reconstruction of every endpoint, independent authorship, or a fully independent numerical-optimization stack. The source and discrepancy receipts allow each claim to be inspected at its actual scope.

## APPENDIX BT: ORIGINAL-OPTIMUM ACCURACY IN THE INVESTMENT ECONOMY

The exact witness theorem answers an implementation question conditional on a continuation. An economist also needs to know how close the complete implemented policy is to the original optimum. We now establish that connection in the investment economy itself. The argument exploits a property not used by the earlier cell-range calculation: despite the bilinear transition, the Bellman objective remains jointly convex. This permits a second-order interpolation account and transfers verified nodal decisions to an implementable policy over the entire state domain. It does not assume that a fitted neural continuation is convex.

*Relation to the preceding two-control query.* The earlier query experiment used the fixed shared capacity $1/8$. The complete dynamic extension below uses the original state-dependent capacity $c(x)$, with the retained exposure columns, action-cost interaction and second shock. Its nesting claim concerns the original scalar

economic model; it is not an assertion that a full dynamic policy was already present in the earlier isolated query.

### BT.1. *Primitives and Bellman regularity*

Let $d \geq 2$ be even, let indices be cyclic, and set $\beta = 15/16$. Write

$$S(x) = \frac{2}{d}\sum_j (x_j - 5/8)^2 + \frac{1}{4d}\sum_j (x_j - x_{j+1})^2 + 2\left(\frac{1}{2} - \frac{2}{d}\sum_j x_j\right)_+^2,$$

$$g(x) = S(x) + \frac{2}{d}\sum_j (x_j - 5/8)^2, \qquad c(x) = \frac{1}{8} + \frac{1}{8d}\sum_j x_j.$$

The original scalar control has $0 \leq a \leq c(x)$, running cost $S(x) + a^2 + 4a^4$, and transition

$$F_j(x, a, z) = \frac{1}{16} + \frac{x_j}{2} + \frac{x_{j+1}}{8} + \frac{x_j(1 - x_{j+1})}{16} + B_j a + s_j z,$$

where $(B_j, s_j) = (1/2, 1)$ at alternating coordinates and $(1/4, -1)$ at the others, and $z$ is uniform on $[-1/32, 1/32]$.

For the retained two-control extension, $a_1, a_2 \geq 0$, $a_1 + a_2 \leq c(x)$, the second exposure column interchanges $1/2$ and $1/4$, and the action cost is

$$R(a) = a_1^2 + a_2^2 + 4(a_1^4 + a_2^4) + a_1 a_2/4.$$

An independent common-direction uniform innovation of radius $1/64$ is added. This is the previously specified extension, nested at $a_2 = 0$ when that innovation is disabled. Both models preserve $[0,1]^d$. Let $m \in \{1, 2\}$ denote the number of controls and define

$$Q_t^*(x, a) = S(x) + R(a) + \beta \mathbb{E} V_{t+1}^*(F(x, a, Z)),$$

$$V_t^*(x) = \min_{a \in A(x)} Q_t^*(x, a), \qquad V_T^* = g.$$

All costs and laws in this display are the economic primitives, not a fitted surrogate.

THEOREM A.60—Regularity of the original Bellman problem: *For either control specification and every finite horizon, $V_t^*$ is convex and continuously differentiable in the*

*interior, with the corresponding one-sided extension on the cube. It is $G_t$-Lipschitz in the $\ell^1$ norm and $M_t$-semiconcave in the Euclidean norm, where valid constants are*

$$G_T = 10/d, \qquad G_t = \frac{1037}{128d} + \beta\frac{47}{64}G_{t+1},$$
$$M_T = 26/d, \qquad M_t = \frac{22}{d} + \frac{21}{256d} + \beta\left(\frac{9}{16}M_{t+1} + \frac{1}{8}G_{t+1}\right).$$

*In particular, $G_t \le 27/d$ and $M_t \le 54/d$. The function $Q_t^*$ is jointly strongly convex on its convex feasible graph, with state curvature at least $107/(128d)$ and action curvature at least $7/4$.*

*A valid semiconcavity constant along each coordinate line is*

$$D_T = \frac{9}{d} + \frac{16}{d^2},$$
$$D_t = \frac{5}{d} + \frac{16}{d^2} + \frac{21}{256d^2} + \beta M_{t+1}\left(\frac{85}{256} + \frac{23}{256d}\right).$$

*Consequently, for the positive multilinear interpolant $I_n$ on a mesh of spacing $1/n$,*

$$0 \le I_n V_t^*(x) - V_t^*(x) \le \kappa_t, \qquad \kappa_t = \frac{dD_t}{8n^2}. \tag{A.119}$$

The proof is in Supplement, Section 1.4.

The theorem is a structural statement about these economic primitives. It is not a claim that every controlled diffusion, recursive preference specification, or game in the complete development edition has a convex Bellman problem. Nor does it require a signed-weight neural fit to preserve convexity. Convex approximation of dynamic programs has an established literature; the additional work here is to verify the needed curvature for the original nonlinear capacity-constrained model and connect it to the actual NBO actor and complete error account.

### BT.2. *Certified nodal Bellman bounds*

Suppose nodal arrays satisfy $l_{t+1} \le V_{t+1}^* \le u_{t+1}$ at all grid vertices. Equation (A.119) gives the functions $I_n l_{t+1} - \kappa_{t+1}$ and $I_n u_{t+1}$ as valid lower and upper continuations. The lower function need not itself be convex; convexity is used for

the unknown true value and for the comparison argument, not asserted of arbitrary endpoint arrays.

Divide each uniform innovation component into $q$ equal bins and use its conditional mean. Convexity gives a lower quadrature bound for the true continuation. Semiconcavity gives an upper error

$$\zeta_{t+1} = \frac{M_{t+1}}{6q^2} \sum_{r=1}^{s} \rho_r^2 \|s_r\|_2^2, \tag{A.120}$$

where $s = 1$ or $2$, $\rho_1 = 1/32$, and $\rho_2 = 1/64$ in the extension. The expectation remains the original continuous law. At the terminal date $g$ is evaluated directly, so no terminal interpolation allowance is needed.

At a vertex $v$, let $\mathcal{A}_A(v)$ be the capacity simplex lattice with spacing $c(v)/A$, including all its faces. For $m = 1, 2$, a minimizer can be represented by neighboring lattice vertices whose coordinate variances sum to at most $mc(v)^2/(4A^2)$. A valid action smoothness constant is

$$H_t = \frac{21}{4} + \beta M_{t+1} \|B\|_2^2, \quad \|B\|_2^2 = \frac{5d}{32} \ (m = 1), \quad \|B\|_2^2 = \frac{9d}{32} \ (m = 2).$$

Taylor's inequality and that barycentric representation imply

$$0 \le \min_{a \in \mathcal{A}_A(v)} Q_t^*(v, a) - V_t^*(v) \le \delta_t := \frac{mH_t}{128A^2}. \tag{A.121}$$

The linear Taylor terms cancel even when the optimum lies on a face. This is an error against the continuous feasible optimum, not only against the finite action menu.

Let $q_t^-(v, a)$ and $q_t^+(v, a)$ include all interpolation, quadrature and arithmetic errors and enclose $Q_t^*(v, a)$. Then

$$l_t(v) = \max\{0, \min_{a \in \mathcal{A}_A(v)} q_t^-(v, a) - \delta_t\}, \qquad u_t(v) = \min_{a \in \mathcal{A}_A(v)} q_t^+(v, a) \tag{A.122}$$

are valid nodal Bellman bounds. The zero truncation uses the nonnegative economic costs. Extra neural or conventional proposals may improve the upper side but do not remove the lower action-cover allowance. Positive interpolation, averaging and minimum maps are monotone and nonexpansive; the separate allowances in

(A.119)–(A.121) are therefore propagated, rather than being discarded because an endpoint calculation contains the truth.

BT.3. *From a fitted witness to the complete implemented policy*

Let $a_t(v) \in A(v)$ be the actual action returned at each vertex by any generator. It may be selected by the exact neural witness, a conventional fitted continuation, or their common verified fallback. Write $w_v(x)$ for the multilinear weights and set $\bar{\pi}_t(x) = \sum_v w_v(x) a_t(v)$. Affineness of capacity makes this action feasible. Let $q_t^{\mathrm{sel},+}(v)$ be a verified upper bound on its original-optimum continuation value $Q_t^*(v, a_t(v))$, using the same lower and upper Bellman reference for all generators.

THEOREM A.61—Actual-policy, whole-domain certificate: *Set*

$$e_t = \max_v \{ q_t^{\mathrm{sel},+}(v) - l_t(v) \} + \kappa_t.$$

*Suppose the implemented policy $\widehat{\pi}_t$ is feasible and differs from $\bar{\pi}_t$ by at most $b_t$ in action $\ell^1$ norm, uniformly over the state cube. Define*

$$\Lambda_t = \frac{13}{16} + \beta \frac{3d}{8} G_{t+1}, \qquad D_T^\pi = 0, \quad D_t^\pi = e_t + \Lambda_t b_t + \beta D_{t+1}^\pi.$$

*Then, at every true state and date,*

$$0 \le J_t^{\widehat{\pi}}(x) - V_t^*(x) \le D_t^\pi.$$

*This statement is conditional on the actual recorded policy, not on success of its fitting algorithm. It applies equally to a pure fitted policy and to a policy augmented by verified common candidates.*

The proof is in Supplement, Section 1.5.

In the execution, observation is rounded down at forty bits. An acquired state $y \le x$ has no greater capacity than the true state. Each interpolated action is then rounded down at twenty bits using the lower endpoint of its arithmetic enclosure. If $L_{ij}$ bounds the derivative of action coordinate $i$ with respect to state coordinate $j$,

and $\eta$ bounds native interpolation error, a valid implementation allowance is

$$b_t \le 2^{-40} \sum_{i,j} L_{ij} + m(2^{-20} + 2\eta + \eta_{\mathrm{out}}).$$

Here $L_{ij}$ is computed from the maximum adjacent nodal difference times $n$, and $\eta_{\mathrm{out}}$ covers endpoint subtraction. All these numbers are recorded outward. This new barycentric acquired actor is explicitly distinguished from the earlier discontinuous cell-constant actor; an old actor is not silently replaced during a purported fixed-policy revalidation.

### BT.4. *Arithmetic, work, and scope*

The native interpolation kernel forms nonnegative tensor weights and a weighted sum of at most $2^d$ stored binary64 endpoints. It is compiled without reassociation or floating-point contraction. For $K = (3d+4)2^d + 32$, an explicit conservative absolute allowance is

$$\eta = \frac{K2^{-53}}{1 - K2^{-53}} \max_v |z_v| + K2^{-1022}.$$

The final term conservatively covers underflow. Input coordinates, affine indexing, finite endpoints and the binary64 format are checked. Roundoff expansion and the ordinary interval cost arithmetic remain separate from the analytic approximation bounds. Exact rational fixtures compare the native sum with its mathematical tensor interpolant.

With $N = (n+1)^d$ vertices, $K_A = A + 1$ scalar actions or $K_A = (A+1)(A+2)/2$ two-control actions, the reference calculation uses $O(TNK_A q^s 2^d d)$ elementary work under the explicit tensor implementation. Stored reference values use $O(TN)$ words and a policy uses $O(TNm)$ words. Streaming batches avoid a state-pair table. Proposal fitting, exact rational witness work, actor transfer, serialization, independent cost inference and compilation are additional costs, not absorbed into this bound. The analytic errors are second order in state, action and innovation resolution; the state-count and tensor-evaluation factors still depend exponentially on dimension. The result supplies a matched original-optimum comparison without asserting that these costs have disappeared.

## APPENDIX BU: HORIZON COMPLEXITY AND THE PRICE OF ACCURACY

The inherited recursive query experiment exposes descendant growth. The own-future constructive backend has a different dependency graph: once $f_{t+1}$ is stored, a query evaluates that representation and does not solve the remaining Bellman problem again. This distinction changes the horizon account without removing any of the recursive benchmarks.

For each date let $S_t$ be its state-context cover size, $J_{ti}$ its feasible action-net size at node $i$, $M_t$ its certified integration work in primitive representatives, and $E_{t+1}$ the cost of evaluating its stored successor. Construction uses

$$Q_t = M_t \sum_{i \le S_t} J_{ti}, \qquad W \le \sum_{t<T} \{Q_t(C_t^{\mathrm{prim}} + E_{t+1}) + C_t^{\mathrm{compile}}\} + C_T^{\mathrm{terminal}} + C^{\mathrm{output}}. \tag{A.123}$$

The formula counts a directed acyclic sequence of stored continuations. A literal descendant solver instead multiplies successor solves by its action and shock branching factors. Neither expression is a lower bound for every possible algorithm. The parameters in (A.123) may themselves grow with horizon if the primitive regularity deteriorates.

PROPOSITION A.62—Optimal dated error allocation: *Suppose a certified constructive implementation has a resource upper account $a_t \eta_t^{-p}$ for attaining total date-$t$ residual allowance $\eta_t > 0$, with common $p > 0$. Each $a_t$ includes action search, integration, feasibility, compilation and the cost of evaluating the stored successor; it cannot hide another recursive Bellman solve. Put $w_t = B_t > 0$ and*

$$S = \sum_t a_t^{1/(p+1)} w_t^{p/(p+1)}.$$

*Among allocations satisfying $\sum_t w_t \eta_t \le \varepsilon$, the minimum of this resource upper model and its minimizer are*

$$W^*_{\mathrm{model}} = S^{p+1} \varepsilon^{-p}, \qquad \eta_t^* = \frac{\varepsilon}{S} (a_t / w_t)^{1/(p+1)}. \tag{A.124}$$

*This optimizes the declared upper model, not the unknown fastest algorithm. Rounding each allowance down to a dyadic number within a factor two preserves the error constraint and increases that model by at most $2^p$.*

*If $a_t \le A$ and $w_t = \beta^t$ for $0 < \beta < 1$, then*

$$W^*_{\text{model}} \le A\varepsilon^{-p}(1-\beta^{p/(p+1)})^{-(p+1)}.$$

*With a maximum permissible allowance $\eta_{\max}$ and positive per-date setup costs, clipping the displayed allocation adds at most $\sum_t a_t \eta_{\max}^{-p}$ to this upper bound, plus the explicit setup account. For $w_t = 1$, the un-clipped bound is $A(T+1)^{p+1}\varepsilon^{-p}$. Thus uniform regularity and reusable stored continuations give polynomial, rather than descendant-tree, horizon dependence at fixed dimensions. They do not give dimension-free computation.*

The power model follows, for example, when effective state, action and innovation covers and their certified errors have polynomial work bounds, successor evaluation has the separately proved compiled bound, and all moduli are uniformly controlled. The exponent must include all these factors. Tensor compilation in Theorem 3 has a $2^{p_t}$ query factor and cell-location work; a dense neural cone has an $S_{t+1}$ factor. Those are different implementation accounts and must not be interchanged.

### BU.1. *Operation vectors, arithmetic and reliability*

We retain the resource vector

$$(N_{\text{primitive}}, N_{\text{descendant}}, N_{\text{interval}}, N_{\text{rational}}, N_{\text{network}}, B_{\text{read}}, B_{\text{write}}, M_{\text{peak}}).$$

Counts are reported before scalar prices are applied. Rational operations additionally record operand bit lengths; an arithmetic-operation count is not a bit-complexity bound. Multiplication and division costs at those lengths enter a bit-cost account. Loading, validation, serialization and shutdown belong to complete process time. Peak memory is a capacity requirement, not an additive flow cost unless its duration or rental convention is specified.

A fitting warning, invalid proposal, zero-progress probe, successful screen, protected fallback and terminal service status are different events. In particular, zero new fits cannot estimate fitting reliability, and a successful fallback cannot be counted as a successful prediction. The retained 336-service recursive catalogue and the nine-controller attained-cost experiment have different units of observation. New mathematical fixtures below establish the operation identities and allocation

formula; they are not relabeled as a re-execution of the inherited eight-million-query vector tree or as a hardware calibration. The complete proof of Proposition A.62 is in Supplement G.

## APPENDIX BV: POLICY-COST INFERENCE AT THE SCALE OF VERIFIED REGRET

The query certificate and a direct cost experiment answer different questions. Nevertheless, a certificate can improve the experiment without replacing its estimand. The relevant observation is a sum of optimal Bellman advantages along the actual policy trajectory. Its expectation is policy loss, while its range is controlled by the local accuracy contract rather than by the range of economic costs. All conditional expectations used to construct this observation must themselves be evaluated and charged.

Write $V_t$ for the original Bellman optimum, $V_T = g$, $b_0 = 1$, and $b_t = \prod_{j<t} \beta_j$. For a feasible frozen policy $\pi$, define

$$a_t^\pi(x) = c_t(x, \pi_t(x)) + \beta_t P_t^{\pi_t(x)} V_{t+1}(x) - V_t(x), \qquad Z^\pi = \sum_{t<T} b_t a_t^\pi(X_t^\pi).$$

The symbol $a_t^\pi$ denotes an advantage, not an action. Assume a verified local bound $0 \le a_t^\pi(x) \le e_t^\pi$ for every true state, including the acquisition allowance. Put $G_\pi = \sum_{t<T} b_t e_t^\pi$. A sample contains the actual trajectory of each compared controller under the original law; common initial states and common innovations may couple the controllers. Different sample rows are independent conditional on the frozen construction.

THEOREM A.63—Bellman-centered cost comparison: *For two such controllers $\pi, \nu$ with a common initial law $\lambda$, let $D = Z^\pi - Z^\nu$. Then*

$$\mathbb{E}D = \mathbb{E}_\lambda[J_0^\pi - J_0^\nu], \qquad -G_\nu \le D \le G_\pi. \tag{A.125}$$

*Suppose deterministic computation encloses $D_i$ by $[L_i, U_i] \subseteq [-G_\nu, G_\pi]$ on every sampled continuous bin, including every compatible acquired action. For a conditionally fixed finite family of $M$ contrasts, put $R = G_\pi + G_\nu$ and*

$$h_n = R\sqrt{\frac{\log(2M/\alpha)}{2n}}.$$

*With probability at least* $1-\alpha$*, simultaneously for the family,*

$$\mathbb{E}_\lambda[J_0^\pi - J_0^\nu] \in \left[\frac{1}{n}\sum_i L_i - h_n, \frac{1}{n}\sum_i U_i + h_n\right]. \tag{A.126}$$

*The interval may be intersected with* $[-G_\nu, G_\pi]$*. Its width is the mean numerical enclosure width plus* $2h_n$*. More generally, any valid endpoint-sample concentration bound may replace the bounded-sum radius.*

*Proof.* See Supplement, Section BW.5.

The construction is related to martingale control variates and sequential value-based evaluation, including Jiang and Li (2016). It is not an off-policy importance-weighting argument and does not assume that a fitted value is correct. The additional requirements here are an original-optimum advantage enclosure, a true-state local regret bound, and a paid numerical evaluation of the conditional value. The target in (A.126) is the original expected cost difference, not the difference between two regret certificates. The realized centered observation generally differs from the realized cash-cost difference; their expectations agree by the displayed identity.

### BV.1. *Numerical construction and a target-precision rule*

The returned action is fixed before evaluation. At a center $\bar{x}$ of a true-state box, let $[Q_L, Q_U]$ enclose $Q_t^*(\bar{x}, \pi_t(x))$ for that fixed action and let $[V_L, V_U]$ enclose $V_t(\bar{x})$. If $L_Q, L_V$ are state Lipschitz bounds and $\rho$ bounds $\|x - \bar{x}\|$, then

$$[Q_L - V_U - (L_Q + L_V)\rho, Q_U - V_L + (L_Q + L_V)\rho] \cap [0, e_t^\pi] \tag{A.127}$$

is a valid advantage enclosure. An empty intersection stops verification. Actions must be feasible at every state used by the endpoint queries. If the box crosses an acquisition boundary, one may enumerate all compatible acquired actions, or use $[0, e_t^\pi]$ for the unresolved part. The latter is conservative and its width remains in the reported numerical account. It is not permissible to discard that trajectory.

For the scalar investment economy, at $r$ remaining dates we use

$$L_Q = \frac{8}{d} + \frac{11\beta}{16}G_{r-1}, \qquad L_V = G_r, \qquad e_t^\pi = \delta_t + d(L_Q + L_V)2^{-b},$$

from the original acquisition argument. The evaluator queries the optimum with its own, smaller tolerance; it does not retrain the policy or change the action being evaluated. Original-law quadrature, action-cover error and arithmetic are already included in $[Q_L, Q_U]$ and $[V_L, V_U]$. They must not be omitted or counted again as independent random noise.

COROLLARY A.64—Precision and noninferiority: *Fix a numerical half-width allocation $v > 0$ and a sampling allocation $s > 0$ before observing the new contrasts. If $n^{-1}\sum_i (U_i - L_i) \le 2v$ and*

$$n \ge \frac{R^2 \log(2M/\alpha)}{2s^2},$$

*then the interval in (A.126) has half-width at most $v + s$. For a prespecified loss margin $m > 0$, an upper endpoint at most $m$ establishes noninferiority at that margin; containment in $[-m, m]$ establishes the corresponding two-sided equivalence statement on the same event. Neither statement identifies a strict advantage.*

*Proof.* See Supplement, Section BW.6.

Both $n$ and evaluator precision have costs. The deposited design fixes them prospectively and reports failure to meet the width allocation rather than increasing the sample after inspecting a favorable sign. The actual numerical width is decomposed into the conditional Bellman enclosure, the state-bin Lipschitz expansion, unresolved acquisition branches and endpoint arithmetic. The conditional Bellman component includes its action-cover and continuous-shock quadrature work. A smaller statistical support is therefore not advertised as a free runtime improvement.

### BV.2. *The adoption decision after inference*

Let $[l, u]$ enclose $\mathbb{E}[J^\pi - J^\nu]$, let $W_\pi, W_\nu$ be measured complete deployment work under a declared boundary, and let $p \ge 0$ be a decision maker's price per unit of that work. Replacing $\nu$ by $\pi$ at fee $f \ge 0$ has net-cost difference in

$$[l + p(W_\pi - W_\nu) + f,\ u + p(W_\pi - W_\nu) + f].$$

A negative upper endpoint licenses the replacement under those declared prices. A noninferiority finding permits a separate assessment of computational savings; it does not convert a normalized loss margin into an estimated monetary value. This keeps the original economic purchase question while making its statistical precision and its required price inputs explicit.

## APPENDIX BW: PROOFS OF THE OPERATIONAL AND NEW RESULTS

### BW.1. *Proof of the result A.52*

PROOF: Strong convexity gives $f((1-s)l+su) \le (1-s)f(l)+sf(u)-\mu s(1-s)(u-l)^2/2$. Replacing the endpoint values by their upper bounds proves (A.104). Subtraction of a valid lower bound proves (A.105). A failed screen adds no endpoint to the partition and alters no lower bound. Subsequent protected refinement is therefore covered by the earlier termination proof. A successful screen supplies exactly the local accuracy premise used in the acquired-policy recursion. Finally, $C$ is a convex quadratic in $a$, proving the sublevel-set assertion. *Q.E.D.*

### BW.2. *Proof of the result A.53*

PROOF: A successful screen returns without an endpoint query. A failed screen invokes a procedure bounded by $K(z)$, and its margin loss equals one. This proves (A.108). For each fixed predictor the variable $K(z)\ell_\tau(z,\theta)$ lies in $[0,K_{\max}]$. The bounded-sum inequality of Hoeffding (1963) and a union bound over the conditionally fixed family give (A.109). *Q.E.D.*

### BW.3. *Proof of the result A.54*

PROOF: The chord is a convex quadratic in its normalized action, which is affine in $w$. Squaring the positive part preserves convexity and gives a continuously differentiable function. The nonnegative weighted sum and quadratic penalty preserve these properties. Convexity gives $F(w)-F(v) \le g^\top(w-v)$, and the minimum of $g^\top v$ on the box is $\sum_j \min\{0,g_j\}$. This proves (A.111). Every failed screen has squared positive-part residual at least $\tau^2$, while every successful screen uses zero

further endpoint queries. Summing the corresponding $K_i$ bounds proves the work claim. Strong convexity with modulus $\lambda$ proves the distance bound. *Q.E.D.*

### BW.4. *Proof of the result A.55*

PROOF: Semiconcavity means that $f(a) - H\|a\|_2^2/2$ is concave. Its barycentric inequality, with vertex values replaced by lower endpoints, gives (A.112). The tangent inequality for the convex quadratic $p$ proves (A.113); a linear function attains its simplex minimum at a vertex. Taking minima over a cover preserves the lower bound. Each feasible upper witness is an action of the actual model. The local gap premise therefore implies the earlier policy recursion, irrespective of how the triangle or support point was proposed.

For finiteness, use simplex lattice denominator $A$, with $(A+1)(A+2)/2$ vertices. On triangles of diameter proportional to $c(x)/A$, the curvature error tends to zero as $A^{-2}$. The endpoint error is reduced by finite descendant accuracy and quadrature refinements. The retained exact-rational recovery contract requires $\delta > 16\Lambda\, 2^{-32}$ at each recursive call, leaving positive slack after rounding; dyadic enlargement of $A$ and $q$ then terminates. This is a finite arithmetic procedure, not a uniform bound on rational bit cost. For (A.114), the independent innovation variances after conditioning on the bins are $\rho_j^2/(3q^2)$, their direction norms squared are $d$, and the semiconcavity Taylor remainder is $M_{t+1}/2$ times their sum. Substitution of $\rho_1 = 1/32$, $\rho_2 = 1/64$ gives the coefficient. Finally, each action query has $q_r^2$ descendants, proving (A.115). *Q.E.D.*

### BW.5. *Proof of the result A.63*

PROOF: Conditional expectation and $b_{t+1} = b_t\beta_t$ imply

$$\mathbb{E}\sum_{t<T} b_t a_t^{\pi}(X_t^{\pi}) = \mathbb{E}\left[\sum_{t<T} b_t c_t(X_t^{\pi}, \pi_t(X_t^{\pi})) + b_T g(X_T^{\pi}) - V_0(X_0^{\pi})\right].$$

The intermediate value terms cancel. Subtracting the two expectations cancels the initial value under the common initial law, proving the identity. Summing the pointwise local bounds proves the range. Applying the bounded-sum inequality to each lower and upper endpoint sample, with failure allocation $\alpha/(2M)$, and

taking a union bound proves (A.126). Endpoint containment supplies the comparison between endpoint expectations and the target. The width identity follows by subtraction. No pathwise identity between $D$ and the realized cumulative-cost difference is asserted. *Q.E.D.*

### BW.6. *Proof of the result A.64*

PROOF: The sample-size condition implies $h_n \le s$. Insert this and the numerical-width condition into the width identity. The final claims are set inclusions for the simultaneous confidence interval. *Q.E.D.*

### BW.7. *Proof of the result A.45*

PROOF: The bounded-sum argument in Theorem A.53 gives $Pf_\theta \le u_\theta$ for every $\theta$ on one event of probability at least $1-\alpha$. For $0 \le f \le K_{\max}$, the layer-cake representation gives $|Pf - Qf| \le K_{\max}\|P - Q\|_{\mathrm{TV}}$. Hence, on that event,

$$\mathbb{E}[N_{\theta_b}(Z_b) \mid \mathcal{F}_{b-1}] \le \min\{K_{\max}, u_{\theta_b} + K_{\max}\rho_b\}.$$

The variables $c_b N_{\theta_b}(Z_b)$ have conditional ranges of length at most $c_b K_{\max}$. Conditional Hoeffding bounds, multiplied successively along the filtration and followed by the exponential Markov inequality, show that the sum exceeds its conditional means by at most the final term of (A.91), except on an event of probability $\gamma$. Union with the validation failure event proves the result. Every selected readout remains a proposal to the same verified gate. Its statewise certificate, and thus the earlier acquired-policy recursion, is unchanged. *Q.E.D.*

### BW.8. *Proof of the result A.46*

PROOF: The expected cost difference in a normal block is $m\Delta$; it is zero in a declined block. Linearity of expectation gives $m(B-D)\Delta$, without requiring block independence. Multiply the cost interval by its nonnegative price, and add the work difference and fee. All price choices use the same event, so no additional probability allocation is needed. *Q.E.D.*

# APPENDIX BX: RETAINED INTERVENING TRAINING RESULTS

## BX.1. *Proof of Proposition A.43*

PROOF: A successful screen incurs no recovery queries. On a failed screen the clipped loss equals one, and recovery work is at most $WK$. The pointwise bound therefore holds for the bounded variable $WK\ell_\tau$. The bounded-sum inequality and a union bound give the claim. None of the separately named service costs is included in the definition of recovery work. *Q.E.D.*

## BX.2. *Proof of Proposition A.44*

PROOF: The lower cover still covers the entire feasible set, so its unchanged lower bound remains valid. Each additional upper endpoint bounds the true cost of its own feasible action, hence is an upper bound for the infimum. Their minimum preserves this property. The returned action's true cost is no greater than its own upper endpoint, proving the regret claim. *Q.E.D.*

# APPENDIX BY: PROOFS FOR FROZEN CONTINUATIONS AND PURCHASED WITNESSES

## BY.1. *Frozen-continuation comparison*

PROOF OF THEOREM A.22: Fix the construction information. Conditional expectation at date $t$ gives

$$\mathbb{E}[b_t \beta_t P_t^{\pi_t} h_{t+1}(X_t)] = \mathbb{E}[b_{t+1} h_{t+1}(X_{t+1})].$$

Summing these equalities cancels all intermediate continuation terms in (A.33); the terminal correction gives

$$\mathbb{E} Z_h^\pi = \mathbb{E} J_0^\pi - \mathbb{E} h_0(X_0^\pi).$$

The common initial law cancels the last expectation when policies are compared. Their trajectory coupling otherwise does not matter for this identity.

Pointwise, $Q_t^h(x, \pi_t(x)) - h_t(x) = r_t^h(x) + a_t^{h,\pi}(x)$. For any two trajectory realizations, the difference of their residual terms is between $-\omega_t$ and $\omega_t$. Their greedy-gap

difference is between $-\eta_t^{\nu}$ and $\eta_t^{\pi}$. The same oscillation argument applies to $g - h_T$ at the terminal states. Multiplication by the nonnegative discounts and summation proves (A.35).

For the regret assertion, choose constants $\underline{r}_t, \overline{r}_t$ enclosing $r_t^h$ with difference at most $\omega_t$; an arbitrarily small enlargement handles unattained infima and suprema. Likewise enclose the terminal correction by $\underline{z}, \overline{z}$. For every feasible policy $\mu$, $a_t^{h,\mu} \geq 0$. Telescoping therefore gives, at each initial state,

$$J_0^{\mu} \geq h_0 + \sum_{t<T} b_t \underline{r}_t + b_T \underline{z}, \qquad J_0^{\pi} \leq h_0 + \sum_{t<T} b_t (\overline{r}_t + \eta_t^{\pi}) + b_T \overline{z}.$$

Taking the infimum over $\mu$ in the first inequality, subtracting it from the second, and then removing the enlargement proves (A.36). No minimizing action or optimal-policy attainment is needed. If $h_t$ is replaced by $h_t + k_t$, then $Q_t^h - h_t$ increases by $\beta_t k_{t+1} - k_t$, and the terminal correction decreases by $k_T$. Their discounted sum changes by $-k_0$, identically on both trajectories. Residual oscillations and greedy gaps are unchanged. With $h = V^*$ all residual and terminal oscillations vanish. *Q.E.D.*

For completeness, the confidence statement is conditional on both policy and continuation construction. After intersection with the true support, each endpoint sample lies in an interval of length $R_h$. A lower-tail or upper-tail bounded-sum bound with failure allocation $\alpha/(2M)$ gives the two ends of (A.37). Endpoint containment implies $\mathbb{E}L_i \leq \mathbb{E}D_h \leq \mathbb{E}U_i$. A union bound proves simultaneous coverage. Dependence within a paired trajectory is permitted; independence is required across the sampled rows under this construction. A predictor or cache modified in response to earlier inference rows is not conditionally fixed by the original training information.

### BY.2. *Exact conditional moments and uniform qualification*

PROOF OF PROPOSITION A.23: If $U$ is uniform on $[-A, A]$, integration of $(z + u)_+$ gives

$$\frac{1}{2A} \int_{-A}^{A} (z + u)_+ \, du = \frac{(z + A)_+^2 - (z - A)_+^2}{4A}.$$

A second integration with respect to an independent uniform variable on $[-B, B]$ integrates each positive-part square to a positive-part cube divided by three. Dividing by both uniform densities gives (A.39). Vanishing radii correspond to deterministic projections and must be removed before division. Symmetry of the innovation permits each nonzero projection coefficient to be replaced by its absolute value.

In a rational shallow network every projected center and radius is rational. Positive parts, powers, sums and divisions by checked nonzero rationals are exact rational operations. The affine part has its usual conditional mean, and linearity of integration allows signed readout coefficients. Computing all $K$ projections at a state and action takes $O(Kd)$ arithmetic operations; each subsequent one- or two-shock neuron integral takes a fixed number of operations. Summation along $n$ trajectories of length $T$ gives the stated count. This says nothing about the size of the numerators and denominators or about the work used to construct and qualify $h$. *Q.E.D.*

*Proof of the residual qualification account.* Let $v_t^h(x) = \inf_{a \in A(x)} Q_t^h(x, a)$. Given an action $a \in A(y)$, proportional reduction to $A(x)$ has $\ell^1$ displacement $(\sum_i a_i - c(x))_+$, at most $(c(y) - c(x))_+ \le \kappa \|x - y\|_1$. Hence

$$v_t^h(x) \le Q_t^h(y, a) + (L_{x,t} + \kappa L_{a,t}) \|x - y\|_1.$$

Taking the infimum over $a$, and reversing $x, y$, proves the asserted modulus of $v_t^h$. Subtracting $h_t$ proves that of $r_t^h$. Every point of a radius-$\rho$ state cover is within $\rho$ of a cover point, so its endpoint expansion yields (A.40). For an action mesh with fill radius $s$, approximate any feasible action by a feasible mesh action within $s$. The Lipschitz inequality gives $\min_{\text{mesh}} Q_t^h - L_{a,t} s \le v_t^h \le \min_{\text{mesh}} Q_t^h$. Outward action endpoints preserve both inequalities. There are $N_t A_t$ such conditional evaluations per date. Actor-specific greedy gaps and the terminal correction are additional certificates, not consequences of checking only the residual at the cover points.

## BY.3. *Transport and actual-policy validity*

PROOF OF THEOREM A.47: The value modulus gives $V_r^*(x) \ge L_y - L_V D$. Nonnegative economic costs give the zero lower bound. Variation first in state and then

in action yields

$$Q_r^*(x,\widetilde{a}) \le Q_r^*(y,a_y) + L_Q D + L_A\|\widetilde{a} - a_y\|_1 \le U(x).$$

Both state endpoints and the repaired action lie in the ambient state–action domain on which these moduli are valid. Feasibility of $\widetilde{a}$ supplies $V_r^*(x) \le Q_r^*(x,\widetilde{a})$. Outward arithmetic preserves the two end inequalities; testing the resulting width, rather than an inward-rounded width, gives the local gap.

At each history and date, the returned action has original-optimum advantage no greater than its certified local gap and the stated acquisition allowance. Conditional expectation of the next value therefore gives the same one-step regret recursion used in Theorem A.51. Backward induction proves the full discounted loss sum, even when the action selected at $x$ depends on which valid records earlier requests purchased. No equality with the fresh-search action is used. In particular, a population policy-cost interval for a different, frozen controller cannot be borrowed merely because both controllers have the same local-gap upper bound. *Q.E.D.*

The moduli can also be checked directly from the inherited primitive proof. The state cost has $\ell^1$ Lipschitz constant $8/d$; the transition's induced $\ell^1$ state norm is at most $11/16$. The action cost on the ambient nonnegative simplex of total mass at most $1/4$ has component derivatives bounded by $13/16$, and each exposure column has $\ell^1$ norm $3d/8$. Composition with the $G_{r-1}$-Lipschitz true continuation gives $L_Q, L_A$. The value modulus $G_r$ is Theorem A.60. These constants do not depend on a fitted predictor.

### BY.4. *Capacity rounding and the fill bound*

PROOF OF PROPOSITION A.48: Let $c_q(x) = q\lfloor c(x)/q \rfloor$. If an already lattice-aligned $a_y$ has total mass no greater than $c_q(x)$, it is unchanged. Otherwise proportional reduction removes total mass $\sum_i (a_y)_i - c_q(x)$, and downward rounding removes less than $mq$ more. Since $c_q(x) > c(x) - q$ and $a_y \in A(y)$, the total displacement is at most $\kappa D + (m+1)q$.

Before the favorable zero truncation of the lower bound, the expanded width is no greater than

$$\gamma\delta + (L_V + L_Q + \kappa L_A)D + L_A(m+1)q + \xi.$$

It is at most $\delta$ for $D \le \rho_*$. Because all records are checked and retained, two centers at which fresh fills are purchased must consequently be more than $\rho_*$ apart in $\ell^1$. Partition each coordinate of the unit cube into $n = \lceil d/\rho_* \rceil$ equal intervals, with a fixed boundary convention. The diameter of each resulting box is at most $d/n \le \rho_*$. It can contain at most one fresh center. Thus $M_r \le n^d \le (1 + d/\rho_*)^d$. Each fill is paid once, whereas every query incurs its actual lookup, verification and record handling. Adding these operations proves (A.96). If records are evicted or invalidated, the separation argument applies only within a period in which the required earlier records remain available; lost coverage must be repurchased and charged. *Q.E.D.*

## APPENDIX BZ: PROOFS FOR QUALIFICATION-OPTIMAL READOUTS AND SEQUENTIAL COSTS

### BZ.1. *The readout program and its dual*

PROOF OF THEOREM A.24: The readout has Lipschitz constant at most $L(\theta) = \sum_k L_k|\theta_k|$. At a cover point the residual is in $[y_j^- - F_j\theta, y_j^+ - F_j\theta]$. The Lipschitz constant of that residual is at most $C + L(\theta)$. For any state choose a cover point at distance at most $\rho$. Its lower endpoint minus $\rho(C + L(\theta))$ is a valid lower bound there, and the upper endpoint plus that quantity is a valid upper bound. Taking the minimum and maximum over the cover gives the asserted oscillation bound. No distribution of states, interpolation of a discontinuous actor, or convexity of the fitted continuation is used.

Introduce $u, l \in \mathbb{R}$ and $s_k \ge 0$ and minimize

$$u - l + 2\rho C + \sum_k \lambda_k s_k$$

subject to

$$u \ge y_j^+ - F_j\theta, \qquad l \le y_j^- - F_j\theta \qquad (j = 1, \dots, N),$$

$$s_k \geq \theta_k, \qquad s_k \geq -\theta_k \qquad (k = 1, \ldots, K),$$

and $-B_k \leq \theta_k \leq B_k$. The stated dimensions follow. For any $\theta$ the tight choices of $u, l$ and the choice $s_k = |\theta_k|$ attain (A.41); conversely every feasible point has objective at least that expression. Continuity on the compact coefficient box gives attainment.

The maximum of finitely many real numbers is the maximum of their averages over probability vectors. Consequently the objective can be written as

$$2\rho C + \max_{p,q \in \Delta_N} \left\{ p^\top y^+ - q^\top y^- + [F^\top (q - p)]^\top \theta + \sum_k \lambda_k |\theta_k| \right\}.$$

For a scalar $z$,

$$\min_{|v| \leq B} \{zv + \lambda |v|\} = -B(|z| - \lambda)_+.$$

Indeed zero is minimizing when $|z| \leq \lambda$, whereas the appropriate endpoint of the coefficient interval is minimizing otherwise. Applying this identity coordinatewise proves weak duality in (A.42). The finite program is feasible and has a finite attained optimum. Linear-programming duality, or equivalently minimax on the compact coefficient box and the two probability simplexes, interchanges the minimum and maximum in the last display. It gives equality for some $p, q$. For any explicitly supplied feasible triple, subtracting its dual lower bound from its primal objective bounds its excess over the optimum. *Q.E.D.*

*Rational verification and the backward qualification chain.* The empirical implementation uses rational feature values, rational label endpoints and rational moduli. A numerical optimizer proposes coefficients and dual multipliers. Coefficients are rounded to a fixed dyadic lattice and clipped to the declared box. Nonnegative dual proposals are rounded and normalized to sum exactly to one; a fixed unit vector is the valid fallback if their mass vanishes. The objective, support function and optimization gap are then recomputed in rational arithmetic. Solver failure or a gap exceeding the declared tolerance is recorded rather than converted into an assertion of optimal fitting.

At date $t$ this argument conditions on the actual $h_{t+1}$ produced by the next date's computation. Repeating backwards gives valid bounds $\omega_t \leq \mathcal{W}_t$ simultaneously be-

cause they are deterministic qualifications, not confidence statements. Substituting them into the frozen-continuation proof gives

$$\Omega_h \le \sum_{t<T} b_t \mathcal{W}_t + b_T \omega_T, \qquad J_0^\pi - V_0^* \le \sum_{t<T} b_t \mathcal{W}_t + b_T \omega_T + H_h^\pi.$$

The same substitution is valid in the cost-contrast support. This does not compare the future continuations generated by different backward optimization choices, and hence makes no joint optimality assertion.

### BZ.2. *Whole-cell policy qualification in the executed construction*

For the new two-state construction, the original terminal payoff is retained exactly. Each subsequent continuation is additive with nine coefficients: a constant and four features per state coordinate. The ReLU family uses $x_j$ and $(x_j - k/4)_+$, $k = 1,2,3$. The conventional family uses the four nonconstant degree-four Bernstein terms $\binom{4}{k} x_j^k (1-x_j)^{4-k}$, $k = 1,\dots,4$. Feature Lipschitz bounds are respectively one and four. The latter follows by differentiating a Bernstein term: its derivative is four times the difference of two nonnegative degree-three Bernstein terms whose sum over the full basis is one. Both families have the same coefficient box and node information; their spanning spaces and modulus bounds need not coincide.

The midpoint cover with $n$ cells per coordinate has radius $\rho = 1/n$ in $\ell^1$. If the actual future has Lipschitz constant $L_+$, the inherited primitive inequalities give

$$L_x = 4 + \beta \frac{11}{16} L_+, \qquad L_a = \frac{13}{16} + \beta \frac{3}{4} L_+, \qquad C = L_x + \frac{L_a}{16}, \quad \beta = \frac{15}{16}.$$

At the terminal continuation $L_+ = 5$; at an earlier date it is the verified coefficient-weighted feature modulus. The 33-point action mesh at a midpoint $x$ has exact fill radius $c(x)/64$. Thus subtracting $L_a c(x)/64$ from its minimum action value gives the lower continuous-action label, while the minimum itself is the upper label. All those evaluations are rational conditional expectations under the original uniform innovation law.

For the implemented actor, a second 33-point menu is rounded down to the $1/4096$ lattice and bounded by the minimum capacity over the whole observation

cell. Let $a_j$ minimize its exact midpoint action value and let $y_j^-$ be the continuous-action lower label. At any state $x$ in that cell,

$$Q^h(x,a_j) - \inf_a Q^h(x,a) \le Q^h(x_j,a_j) - y_j^- + (L_x + C)\|x - x_j\|_1$$
$$\le Q^h(x_j,a_j) - y_j^- + (L_x + C)\rho.$$

Taking the maximum over all cells gives the recorded uniform greedy-gap allowance. Replacing $a_j$ by zero gives the comparator allowance. The selected action is feasible throughout its cell, including its closed boundary, and a fixed boundary convention gives a well-defined actor. This proves the continuous-state, continuous-action qualification; midpoint evaluations alone would not do so.

For a ReLU future, exact conditional integration follows Proposition A.23. The independent checker instead splits the innovation interval at every hinge crossing and integrates each affine piece by its endpoint trapezoid. For a Bernstein future, the production calculation expands the uniform moments. The independent checker uses the positive five-point weights $(7,32,12,32,7)/90$ on equally spaced points from $-1/32$ to $1/32$. Substitution of each monomial of degree zero through five verifies this rule directly, so it integrates the conditional degree-four polynomial exactly. The checker separately uses the retained original rational terminal primitive at every final-date action value. These checks concern all new construction nodes and actions, not the entire historical recursive Bellman computation.

### BZ.3. *A time-uniform endpoint bound*

PROOF OF THEOREM A.33: Condition on $\mathcal{F}_{i-1}$. The frozen-continuation telescoping proof applies to the two complete controllers selected for this episode; it gives $\mathbb{E}[D_i \mid \mathcal{F}_{i-1}] = \Delta_i$. History-dependent actions cause no change in the tower property provided the actual conditional transition kernels and all required greedy-gap qualifications remain valid. The common initial law cancels the initial continuation term. The terminal correction is still required.

Write $R_i = b_i - a_i$. Since $L_i \le D_i \le U_i$ and the endpoints lie in $[a_i, b_i]$,

$$\mathbb{E}[L_i \mid \mathcal{F}_{i-1}] \le \Delta_i \le \mathbb{E}[U_i \mid \mathcal{F}_{i-1}].$$

Conditional Hoeffding's lemma therefore gives, for each fixed $\lambda > 0$,

$$\mathbb{E}\left[\exp\left\{\lambda(L_i - \Delta_i) - \frac{\lambda^2 R_i^2}{8}\right\}\middle|\mathcal{F}_{i-1}\right] \le 1.$$

To see the constant directly, a variable supported on an interval of length $R$ has tilted variance at most $R^2/4$. The second derivative of its centered log moment-generating function is this tilted variance. Integrating twice from zero bounds that log moment-generating function by $\lambda^2 R^2/8$. The nonpositive conditional mean of $L_i - \Delta_i$ can only decrease the exponential bound. The same reasoning applies to $\Delta_i - U_i$.

Products of these conditional factors are nonnegative supermartingales starting at one. Ville's inequality bounds the probability that the lower-endpoint product for $\lambda_j$ ever exceeds $2/(\alpha w_j)$ by $\alpha w_j/2$. The same bound holds for the upper-endpoint product. A union bound over both signs and all $j$ gives simultaneous validity with failure probability at most $\alpha \sum_j w_j \le \alpha$. Outside that exceptional event every partial sum is bounded by every corresponding linear boundary. Taking their minimum gives (A.70) and (A.71). The predictable support also bounds each $\Delta_i$, justifying the additional intersection.

The event already covers every finite index, so evaluating it at a finite stopping time does not consume additional error probability. With a fixed complete controller pair and fixed initial-law/reset contract, its conditional expected original-law contrast is the same number $\Delta$ at every episode. Predictable changes of a valid critic alter the observations and ranges, not this identity. Without that fixed target, the preceding calculation identifies only the displayed conditional average. *Q.E.D.*

The implementation need not evaluate exponentials. For $x \ge 1$, write $x = 2^k y$ with $1 \le y < 2$, and use

$$\log y = 2\sum_{r=0}^{m-1} \frac{z^{2r+1}}{2r+1} + \mathcal{R}_m, \qquad 0 \le \mathcal{R}_m \le \frac{2z^{2m+1}}{(2m+1)(1-z^2)}, \quad z = \frac{y-1}{y+1}.$$

Use the same expression at $z = 1/3$ for $\log 2$. It gives an exact rational upper bound on each logarithm in the boundary; replacing the logarithm by that bound enlarges the interval. Arithmetic operation counts do not bound rational bit complexity.

The finite set of bets is deliberate: a finite minimum is directly computable. It remains time-uniform for arbitrarily many episodes, although a finite bet set can leave a positive asymptotic radius. Increasing the predeclared set improves its supported scales but is a new design choice, not a free after-the-fact selection. One may instead preallocate a countable summable family; an implementation using any finite subset retains validity with its assigned weights.

### BZ.4. *Stopping, failed episodes and the replacement decision*

PROOF OF COROLLARY A.34: On the simultaneous event, $l_N \leq \Delta \leq u_N$. Adding the declared incremental charge and the one-time charge divided by the deployment count gives the claimed interval. Its upper endpoint being negative implies strict net cost reduction. Inclusion in an equivalence band is a different two-sided assertion. The event supports every such deterministic algebraic implication, including a stopping rule that inspects the interval repeatedly.

For a new epoch, condition on the history at its start, which includes the construction and selection of its new controller pair. Apply the theorem with that epoch's previsible error allocation $\alpha_e$. A conditional union bound followed by expectation bounds the probability of any erroneous qualification by $\mathbb{E}\sum_e \alpha_e$, no larger than the total declared budget. No independence between epochs is required. *Q.E.D.*

A support reserved before an episode remains valid when its numerical evaluation fails. Recording the full interval $[a_i, b_i]$ then preserves endpoint containment and the martingale argument. Deleting the episode after an outcome-dependent failure need not preserve the conditional law. All attempted work is retained irrespective of which interval is recorded. An invalid support certificate is different: it supplies no admissible substitute observation and must not be treated as a harmless numerical timeout.

Finally, a conditional kernel-bias bound can be accommodated only when it is itself justified: if $|\mathbb{E}[D_i \mid \mathcal{F}_{i-1}] - \Delta_i| \leq \zeta_i$ with previsible certified $\zeta_i$, enlarge both ends by $n^{-1}\sum_{i\leq n}\zeta_i$. This follows by applying the proved bound to the actual conditional mean and then using the triangle inequality. An empirical model discrepancy without such a bound is not set to zero.

## APPENDIX CA: PROOFS FOR CONSTRUCTED-POLICY COST QUALIFICATION

PROOF OF (A.74) AND THE OBSERVATION IDENTITIES: Write the final state as $(m_1 + w, m_2 - w)$, where $\mathbb{E}w = 0$ and $\mathbb{E}w^2 = 1/3072$. The sum of the coordinates, and hence the shortage term, is constant in $w$. The two state-square terms in $g$ contribute $4\mathbb{E}w^2$; the coordinate-difference square contributes $\mathbb{E}w^2$. All linear terms vanish in expectation. This proves (A.74). Since the first two actions are measurable with respect to the initial state and the first innovation, conditional expectation of (A.72) with respect to those variables is (A.73). The tower property gives the equal expected costs. Applying the conditional-variance decomposition to the paired cash difference gives the stated variance inequality, including when the two actors have different acquisition resolutions.

For the primitive support, the lower next-state bound is $1/32$. Termwise upper bounds give $29/32$ in the first coordinate and $27/32$ in the second, so the unit cube is invariant for every feasible action. On that cube, $|x_j - 5/8| \le 5/8$, $|x_1 - x_2| \le 1$, the positive shortage is at most $1/2$, and $a \le 1/4$. Substitution into the nonnegative cost terms gives $c \le 103/64$ and $g \le 37/16$. Discounting gives (A.75); conditional expectation preserves the same support. *Q.E.D.*

PROOF OF THEOREM A.35: Condition on all information used to construct the fixed policies. Lower and upper endpoint variables are bounded functions of independent bin rows. Their expectations bracket the original estimand, by pointwise enclosure or by integration within each bin. Apply the one-sided bounded empirical-Bernstein inequality to $U_i$ and to $-L_i$, after affine rescaling to an interval of length one. Each failure probability is at most $2e^{-\ell}$, with the radius in the theorem. A union bound over two endpoint events for each of $K$ estimands gives $4Ke^{-\ell}$. It requires no independence among estimands on a shared row. Subtracting the lower confidence endpoint from the upper one yields (A.77). Enlarging either radius and rounding confidence endpoints outward cannot remove the target from a covering interval. Integrating over the conditioning information proves the unconditional assertion when the construction itself was random. *Q.E.D.*

*Validity of the two numerical realizations.* For binary64 operations, each computed elementary lower endpoint is moved to the adjacent lower representable number and each upper endpoint to the adjacent upper number. For integer-directed operations, multiplication on the $2^{-48}$ grid uses the floor of the smallest endpoint product divided by $2^{48}$ and the ceiling of the largest. Division by a positive integer uses the corresponding integer floor and ceiling. Addition and negation are exact on this grid. Squaring includes zero when its argument interval straddles zero, and the positive-part map is monotone. These rules enclose the result of each elementary operation.

Induction through either polynomial formula therefore encloses the cost and transition of every point in the input box. At an acquisition step, each possible true state belongs to one of the enumerated intersected cells, so its deployed action lies between the enumerated extrema. A second induction through the two dates proves the full-bin cost enclosure. Conditional final integration is valid by the exact identity already proved. The state and cost intersections use established primitive supports. Taking the hull of two valid intervals is conservative even when one is wider. Neither interval intersection with a sampled value nor agreement at a midpoint is used to establish this full-bin claim.

In the statistical implementation, sample sums and sums of squares are integers. Their rational combination gives the unbiased variance exactly. For a rational $v \geq 0$, the least $k$ satisfying $k^2 \geq 2^{120} v$ gives the upward square root $k/2^{60}$. Each radius enlargement is less than $2^{-60}$, except that it is zero for an exactly represented root. The recorded width decomposition uses these enlarged roots and is an exact rational identity. The independent midpoint trajectories and synthetic boundary cases are regression checks supplementary to, not substitutes for, the enclosure argument.

## APPENDIX CB: PROOF OF PATTERN-CONSISTENT TRANSFER

### CB.1. *Embedding and comparison with the earlier relaxation*

Fix an actual qualified parameter $\theta$ and its complete pattern $\sigma$. Give that pattern weight one and all others weight zero. Under the original law of its complete policy, set $m^{\sigma}_{tC} = \Pr(X_t \in C)$ and $z^{\sigma}_{tCD} = \Pr(X_t \in C, X_{t+1} \in D)$. Integrating the whole-cell

transition bounds gives (A.62). The same occupation identity proved in Supplement E bounds the true cost difference between the two objectives in (A.63). This proves validity for the entire parameter class. There is no additional probability union over its members.

To compare the two relaxations, take any disaggregated feasible point and define

$$y_{tCa} = \sum_{\sigma:\sigma(t,C)=a} \lambda_\sigma, \qquad q_{tCa} = \sum_{\sigma:\sigma(t,C)=a} m^\sigma_{tC},$$

$$z_{tCaD} = \sum_{\sigma:\sigma(t,C)=a} z^\sigma_{tCD}, \qquad m_{tC} = \sum_\sigma m^\sigma_{tC}.$$

Conservation implies $\sum_C m^\sigma_{tC} = \lambda_\sigma$ at every date, hence $0 \le m^\sigma_{tC} \le \lambda_\sigma$ and $q_{tCa} \le y_{tCa}$. All assignment, flow and transition constraints of (A.4) follow by summation. Each retained pattern satisfies every certified conflict, so their convex combination satisfies its corresponding inequality in $y$. Both advantage objectives are unchanged by this projection. Consequently the disaggregated minimum is no smaller, and its maximum is no larger, than the earlier pair. Deleting only patterns certified impossible preserves coverage and shrinks the feasible set. Nonempty compactness follows from the actual-policy embedding, bounded total mass and closed linear constraints. This also proves existence of the displayed extrema and justifies ordinary linear-programming dual certificates.

### CB.2. *Quantitative comparison with an attained pattern*

Now assume every retained pattern is actual and has a qualification witness. Consider one pattern with positive weight and divide its flow variables by that weight. Its row sums define a time-inhomogeneous stochastic matrix $\widetilde{P}_t$ on the dated cells. At a row with zero mass choose any stochastic row within the same transition intervals; such a row exists because the original transition at any state in the nonempty cell supplies one.

Let $m_t$ be the true cell law of this pattern's original continuous-state policy, and $P_t$ its conditional cell-to-cell probabilities given the date-$t$ cell under that law. The law inside a cell can depend on the complete earlier history. Nevertheless, conditional expectation and the whole-cell hypothesis put every row of $P_t$ inside the same

interval box. At an unvisited cell choose an arbitrary original-law row. This choice does not affect $m_t P_t = m_{t+1}$. No Markov assumption on the observed cell process has been imposed.

Two stochastic rows in this box differ in $\ell^1$ norm by at most $\delta_t$. Writing $\widetilde{m}_t$ for the normalized relaxed law, stochastic contraction gives

$$\begin{aligned}\|\widetilde{m}_{t+1} - m_{t+1}\|_1 &\le \|(\widetilde{m}_t - m_t)\widetilde{P}_t\|_1 + \|m_t(\widetilde{P}_t - P_t)\|_1 \\ &\le \|\widetilde{m}_t - m_t\|_1 + \delta_t.\end{aligned}$$

The initial laws agree. Since the distance between two probability vectors is at most two, induction bounds this distance by $e_t$.

Let $a_{tC}$ denote the true conditional expected reference advantage under this pattern at date $t$, with an arbitrary value in the enclosure on unvisited cells. Then $\ell_{tC\sigma(t,C)} \le a_{tC} \le u_{tC\sigma(t,C)}$. For the upper objective at that date,

$$\begin{aligned}\sum_C u_{tC\sigma(t,C)}\widetilde{m}_{tC} - \sum_C a_{tC}m_{tC} &= \sum_C u_{tC\sigma(t,C)}(\widetilde{m}_{tC} - m_{tC}) \\ &\quad + \sum_C (u_{tC\sigma(t,C)} - a_{tC})m_{tC} \\ &\le M_t e_t + \omega_t.\end{aligned}$$

The lower objective is treated with the reverse inequality. Summing with $B_t$ bounds any normalized pattern-flow objective within $\mathcal{E}$ of that pattern's true cost difference. Weighting by $\lambda_\sigma$ and summing proves that every feasible upper objective is at most $j_+ + \mathcal{E}$ and every feasible lower objective at least $j_- - \mathcal{E}$. The actual-policy embeddings give the other inequalities in (A.64).

When the transitions and advantages are exact on a finite state space, $\delta_t = \omega_t = 0$. Each fixed pattern has a unique occupation sequence; the convex mixture of their costs has the same minimum and maximum as this finite set. A retained qualification witness therefore attains each extremal policy value. More generally the quantitative inequalities prove the stated convergence whenever $\mathcal{E}$ tends to zero uniformly. This is not a claim that arbitrary partitions, discontinuous deterministic transitions or incomplete pattern searches satisfy that condition.

### CB.3. *Strict improvement in the shared-sign example*

Let $\theta = 1$ select the right-hand initial action and the free action at the right-hand successor. Let $\theta = -1$ select the left-hand initial action and the costly action at the right-hand successor. All other costs are zero. Both complete policies cost zero. In the conflict relaxation, set the right-hand initial assignment and occupation to $1/2$, and the right-hand costly assignment and occupation also to $1/2$. Put the remaining initial mass on the left-hand action; choose the free action in the left-hand successor. The forbidden pair's assignments sum to one, so its conflict inequality holds. Its cost is $1/2$. Conversely that cost cannot exceed either member of the forbidden pair, whose assignments sum to one, and thus is at most $1/2$. Pattern disaggregation has only the two zero-cost occupation sequences and gives upper bound zero. The numerical test computes the three bounds by independent path enumeration and linear programming.

## APPENDIX CC: PROOF OF COUPLED ACCURACY ALLOCATION

Write $f_j(\eta) = c_j \prod_t \eta_t^{-a_{jt}}$. For any vector $v$ and positive $\eta$, direct differentiation yields

$$v^\top \nabla^2 f_j(\eta) v = f_j(\eta) \left\{ \left( \sum_t a_{jt} \frac{v_t}{\eta_t} \right)^2 + \sum_t a_{jt} \left( \frac{v_t}{\eta_t} \right)^2 \right\} \geq 0.$$

Because every coordinate appears with positive exponent somewhere, summing these forms is strictly positive for every nonzero $v$. Hence $F$ is strictly convex on the positive orthant. Its feasible set is convex and nonempty. Every coordinate is bounded above by its cap and by $\varepsilon / w_t$. If a feasible sequence sends a coordinate to zero, a term containing that coordinate tends to infinity: all other coordinates in that term have fixed finite upper bounds. A finite sublevel set containing any feasible allocation is consequently bounded away from zero and compact. Continuity gives attainment; strict convexity gives uniqueness.

After $\eta_t = \exp(x_t)$, every nonconstant term is an exponential of an affine function. The budget becomes $\sum_t w_t \exp(x_t) \leq \varepsilon$ and each cap becomes $x_t \leq \log \bar{\eta}_t$. Both the

objective and constraints are convex. This proves the alternative representation, without asserting that a floating-point optimization status is a proof.

At any feasible candidate, the supporting-hyperplane inequality is

$$F(\eta) \geq F(\eta^0) + g \cdot (\eta - \eta^0).$$

If $r_t = g_t + \lambda w_t + \nu_t \geq 0$, positivity of $\eta$ and its budget and caps imply

$$\begin{aligned} g \cdot \eta &= \sum_t r_t \eta_t - \lambda \sum_t w_t \eta_t - \sum_t \nu_t \eta_t \\ &\geq -\lambda \varepsilon - \sum_t \nu_t \bar{\eta}_t. \end{aligned}$$

Substitution proves (A.31). A sufficiently small positive vector satisfies all inequalities strictly, so Slater's condition holds. At the optimum the usual stationarity and complementary-slackness equations provide nonnegative $\lambda, \nu$ with $r = 0$, $\lambda(\sum_t w_t \eta_t - \varepsilon) = 0$ and $\nu_t(\eta_t - \bar{\eta}_t) = 0$. They make the lower certificate equal to the feasible objective. Equivalently, these conditions themselves give a complete sufficiency proof by the supporting-hyperplane inequality. Since $g_t < 0$, the stated maximum-ratio choice with $\nu = 0$ is always a nonnegative feasible certificate.

For integer exponents the objective, derivatives, constraints and certificate at rational arguments use only rational operations. The proof does not license substituting unqualified decimal derivatives when real exponents are used; outward enclosures must then make the claimed inequalities valid. For example, a rational affine lower support with verified slope bounds on the feasible box can replace the exact tangent. The finite fixtures use integer exponents and independently recompute the gradients, so no such analytic-rounding issue is hidden in their reported gaps.

If $\eta_t^0/2 < \eta_t^{\mathrm{dyad}} \leq \eta_t^0$, then

$$f_j(\eta^{\mathrm{dyad}}) \leq 2^{\sum_t a_{jt}} f_j(\eta^0) \leq 2^P f_j(\eta^0).$$

Summation proves the rounding bound, while decreasing coordinates preserves both the error budget and all upper caps. Finally, the equal allocation $\eta_t = \varepsilon/T$ is feasible under the corollary's hypotheses. Since $T/\varepsilon \geq 1$, every monomial is at most

$C(T/\varepsilon)^P$. There are at most $KT$ of them. Adding $F_0$ proves Corollary A.19. Neither this feasible upper construction nor the dual lower bound on its resource model is an information-theoretic lower bound for all Bellman algorithms.

## APPENDIX CD: PROOFS FOR ORIGINAL-LAW TRANSPORT QUALIFICATION

### CD.1. *One-step comparison*

For probability measures $\mu, \nu$ and a $K$-Lipschitz function $f$, every coupling $\Gamma$ of the measures satisfies

$$\left| \int f \, d\mu - \int f \, d\nu \right| = \left| \int (f(y) - f(z)) \, \Gamma(dy, dz) \right| \le K \int d(y,z) \, \Gamma(dy, dz).$$

Taking an infimum over couplings proves the required $KW_1$ bound. The argument does not require densities or attainment of the infimum. Boundedness or the assumed finite first moments makes the integrals well defined.

Fix a qualified $\theta$ and suppress its subscript. Define the original-law action operator

$$Q_t^P f_{t+1}(x,a) = c_t(x,a) + \beta_t P_t^a f_{t+1}(x).$$

The preceding coupling inequality and (A.65) give, for every original feasible action,

$$|Q_t^P f_{t+1}(x,a) - \widehat{Q}_t(x,a)| \le b_t. \tag{A.128}$$

If two functions on a nonempty action set differ uniformly by at most $b$, their infima differ by at most $b$: each function is everywhere at least the infimum of the other minus $b$, and the reverse argument gives the upper bound. No differentiability, uniqueness or continuity of the minimizing action is needed. It follows that

$$\begin{aligned} \ell_t - b_t &\le T_t f_{t+1} - f_t \le u_t + b_t, \\ \ell_t - b_t &\le T_t^\pi f_{t+1} - f_t \le u_t + \gamma_t + b_t. \end{aligned} \tag{A.129}$$

The lower policy inequality uses feasibility: the approximate action value at the implemented action is at least its infimum. The upper inequality uses the certified approximate greediness. All inequalities hold on the whole original state domain.

### CD.2. *Backward induction and the original optimum*

For a sequence $r_t$, define $S_t(r,d) = \sum_{s=t}^{T-1}(\prod_{j=t}^{s-1}\beta_j)r_s + (\prod_{j=t}^{T-1}\beta_j)d$. Empty products equal one. The claim

$$S_t(\ell - b, d^-) \le V_t^* - f_t \le S_t(u + b, d^+),$$
$$S_t(\ell - b, d^-) \le V_t^\pi - f_t \le S_t(u + \gamma + b, d^+)$$

follows by backward induction. At $T$, it is the terminal enclosure. Suppose the first inequality holds at $t+1$. Order preservation and the identity $T_t(v+k) = T_t v + \beta_t k$ give its lower bound at $t$ from the lower bound in (A.129). The upper bound follows in the same way. The fixed-policy operator has the same order and constant-shift properties, so the second line follows separately. This induction is under the original kernels; the numerical kernels do not determine the economic value being bounded.

Subtract the lower bound for $V_0^* - f_0$ from the upper bound for $V_0^\pi - f_0$. The result is exactly $\mathcal{R}_\theta$. Feasibility gives $V_0^\pi \ge V_0^*$ and proves (A.67). The proof holds at each initial state and hence after averaging under any initial distribution supported on the declared domain. It also holds for every parameter for which the uniform hypotheses were certified. If the hypotheses are established on a single simultaneous data event, the conclusion holds on that event without an additional union over parameter vectors. Pointwise certificates obtained only at observed states do not create that simultaneous event.

The parameter may determine a different future continuation and a different future action at every date. We fixed it only to prove an inequality for its complete policy; the proof never substitutes another parameter's future. Discontinuous acquisition or deterministic tie-breaking therefore causes no problem as long as the resulting policy is measurable, feasible and satisfies the whole-domain residual inequalities. A transition bound applying only under the reference policy would not establish (A.128) and is insufficient.

### CD.3. *State-independent shifts and singular laws*

Replace $f_t$ by $f_t + k_t$ and $\widehat{Q}_t$ by $\widehat{Q}_t + \beta_t k_{t+1}$. Since both kernels have total mass one, the arithmetic discrepancy, transport modulus and greedy gap are unchanged.

The approximate Bellman residual is shifted by $\beta_t k_{t+1} - k_t$. Its certified lower and upper endpoints shift by that same amount, so their difference is unchanged. The terminal enclosure endpoints both decrease by $k_T$. Every term of $\mathcal{R}_\theta$ is consequently invariant. Using absolute levels of the continuation in place of its oscillation would forfeit this property.

If $\widehat{P}_t$ is the law of $q_t(Y)$ when $Y$ has law $P_t$, the joint law of $(Y, q_t(Y))$ is a coupling. Its expected distance is at most $h_t$. Thus $W_1(P_t, \widehat{P}_t) \le h_t$. Substitution into $2\sum_t B_t b_t$ gives (A.68). In particular, $\delta_h$ and $\delta_0$ have transport distance $h$ but total-variation distance $\sup_A |\delta_h(A) - \delta_0(A)| = 1$ when $h > 0$. Mutual singularity therefore does not obstruct this continuation-based calculation. The conclusion concerns the transport contribution only; a nonvanishing residual span or greedy error still has to be paid.

### CD.4. *Stationary discounted bound*

Let $T$ and $T^\pi$ be the bounded discounted Bellman and policy operators. They preserve order, translate constants by $\beta$ and are sup-norm contractions with modulus $\beta$. Iterating (A.129) from the bounded function $f$, geometric summation yields

$$\frac{\ell - b}{1-\beta} \le V^* - f \le \frac{u + b}{1-\beta}, \qquad V^\pi - f \le \frac{u + \gamma + b}{1-\beta}.$$

The contraction limits are the original optimal and policy values. Alternatively, apply the finite-horizon induction and let the bounded discounted terminal difference tend to zero. Subtraction proves Corollary A.32. This argument requires a certified modulus of the stored $f$; it does not infer a uniform modulus for every optimal value from Lipschitz primitives. That distinction matters when transition dynamics expand distances.

### CD.5. *Resource allocation and the other economic objectives*

With $\eta_t \ge u_t - \ell_t + \gamma_t$, the sufficient budget (A.69) is obtained by rearranging the nonnegative terms in $\mathcal{R}_\theta$. After subtracting fixed terminal and arithmetic contributions, it is a linear budget in the residual and transport allowances. On positive-weight coordinates, any resource upper model satisfying Theorem A.18 has that

theorem's convexity, attainment, dyadic-rounding and dual-certificate properties. This is a direct application with additional accuracy coordinates, not a newly asserted lower bound on all numerical methods. Caps and certificates for the moduli are part of the input. Optimizing a modulus jointly requires retaining its parameter dependence in the resource and error constraints.

For a cash-invariant monotone one-step risk operator with discount $\beta_t$, a certified numerical discrepancy $b_t$ gives the same two residual inequalities and the same constant-shift induction. Neither the coupling proof for linear expectation nor cash invariance alone supplies that numerical discrepancy for a nonlinear risk functional. For a general Lipschitz aggregator, the separate absolute-error induction in Proposition A.2 supplies the valid replacement. Applying an original-optimum bound to each fixed-opponent best-response problem gives the stated unilateral incentive account; it does not prove existence of a game equilibrium. These arguments preserve the distinct hypotheses of the original applications.

### CD.6. *Diagnostic with a falsely free transition approximation*

In the one-date example, both numerical action values are constants: zero for action 0 and $h/2$ for action 1. Hence $\widehat{T}f_1 - f_0 = 0$, the numerical greedy gap is zero, and the terminal enclosure is exact. The true values of the two actions are $h$ and $h/2$, respectively. The numerical greedy policy therefore has regret $h/2$. The stored terminal function has modulus one and the largest transport discrepancy is $h$, so Theorem A.31 gives $2h$. The omitted-charge expression is zero and is invalid. All assertions are exact for rational $h$; no sampling or floating-point limit is used.

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